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IMPROVING VISUAL BASIC .NET PROFICIENCY THROUGH INTEGRATION OF 'DEVELOPING APPLICATIONS WITH VISUAL BASIC .NET: BEST PRACTICES AND TECHNIQUES' AMONG STUDENTS

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ABSTRACT

This research delves into the assessment of Visual Basic .NET proficiency enhancement among student cohorts facilitated by incorporating the e-book 'Developing Applications with Visual Basic .NET: Best Practices and Techniques.' Employing a quasi-experimental design featuring a non-equivalent control group, the study aims to evaluate the efficacy of the e-book's principles in augmenting students' proficiency levels in Visual Basic .NET programming. The research methodology involves stratifying participants into distinct groups: a treatment group, immersed in the methodologies advocated within the e-book, and a control group adhering to the traditional curriculum. Comparative analyses indicated the superiority of the proposed e-book approach over conventional pedagogical methods in fostering enhanced proficiency levels among students. The findings of this study provide compelling evidence supporting the efficacy of integrating 'Developing Applications with Visual Basic .NET: Best Practices and Techniques' as an instrumental resource for elevating students' competence in Visual Basic .NET programming. This research contributes valuable insights that can inform educational strategies and curriculum development in similar academic contexts.
Keywords: Visual Basic .NET programming, quasi-experimental design, non-equivalent control group, e-book

1. Introduction

Advancements in technology and programming languages continually shape educational methodologies in preparing students for dynamic professional landscapes. Amidst these evolutions, the study of Visual Basic .NET (VB.NET) remains foundational within the realm of software development [1]. This study delves into an innovative educational approach centered on the integration of the e-book 'Developing Applications with Visual Basic .NET: Best Practices and Techniques,' aiming to elevate students' proficiency in VB.NET programming.

In recent years, educational institutions have sought effective means to enhance programming education, aligning it with industry demands. This research employs a quasi-experimental design, featuring a non-equivalent control group, to rigorously evaluate the impact of integrating the aforementioned e-book's principles. By stratifying participants into distinct treatment and control groups, the study meticulously assesses the efficacy of the e-book's methodologies in elevating students' proficiency levels in VB.NET programming.

The methodological framework involves the meticulous allocation of participants: the treatment group immersed in the principles advocated by the e-book, and the control group following a traditional curriculum. Notably, comparative analyses highlighted the pronounced advantages of the e-book's approach over conventional pedagogical methods. The outcomes underscore the pivotal role of the e-book 'Developing Applications with Visual Basic .NET:

Best Practices and Techniques' as an instrumental resource for augmenting students' competence in VB.NET programming.

This research offers compelling evidence, underscoring the potency of integrating innovative educational resources like the aforementioned e-book to enhance students' proficiency in VB.NET. The insights gleaned from this study have far-reaching implications, guiding the development of educational strategies and curricular frameworks tailored to contemporary demands within technology-focused academic domains.

2. Literature Review

Visual Basic .NET (VB.NET) is a programming language widely employed in software development, esteemed for its versatility and extensive utilization within the field. This literature review aims to offer a comprehensive overview and critical analysis of existing literature on VB.NET, encompassing scholarly articles, books, academic papers, and industry reports. The review endeavors to provide a thorough understanding of VB.NET in contemporary software engineering contexts, exploring its evolution, distinguishing features, practical applications, advantages, and challenges encountered within the programming landscape [1].

The lineage of Visual Basic .NET can be traced back to its predecessor, Visual Basic, renowned for its user-friendly approach to programming. Literature highlights a significant transition from the classic Visual Basic to VB.NET, emphasizing a paradigm shift towards a more robust and object-oriented framework. Research conducted by [7] delves into the motivations and technical advancements underlying this transition, emphasizing enhancements in performance, language structure, and integration with the .NET framework.

Scholarly works extensively document the distinctive features and functionalities of VB.NET. Studies by [6] delve into VB.NET's syntax, its support for object-oriented programming paradigms, and integration with the Common Language Runtime (CLR). This literature underscores VB.NET's prowess in leveraging the expansive .NET framework libraries, facilitating rapid application development (RAD), data access, GUI design, and its interoperability with other languages within the .NET ecosystem.

Various research articles and case studies exemplify VB.NET's diverse applications across multiple industries. Works by [6] highlight its utilization in enterprise-level software development, particularly emphasizing its role in crafting scalable and secure applications. Moreover, studies within the educational sector [3] underscore VB.NET's efficacy as a pedagogical tool, owing to its simplicity and versatility in teaching programming concepts.

3. Methodology

This study employs a quasi-experimental design to assess the impact of integrating principles from the e-book "Developing Applications with Visual Basic .NET: Best Practices and Techniques." The design involves both an intervention group, which receives the implementation of the e-book's principles, and a control group that follows the traditional curriculum without exposure to the e-book's methodologies. Pre-existing classes or groups within an educational setting will be selected as per the quasi-experimental design for comparability.

Table 1. The design of non-equivalent control group

Group	Pre-Test	Intervention	Post-Test
Control	O_1	Conventional	O_2
Treatment	O_1	e-Book	O_2

Students in both control and treatment groups sat for a pre-test (O1) prior to the VB.Net lesson. The control group learnt about VB.Net via the conventional method - slide materials. Meanwhile, the cloud training module was deployed for the treatment group. The study variables are the varied intervention methods; conventional and e-Book for control and treatment groups, respectively. A post-test (O2) was executed for both groups after the lesson.

4. Result and Discussion

This research endeavor was conducted within the academic precincts of one of the prominent Northern Polytechnics in Malaysia, employing a structured division of participants into control and treatment groups. The study cohort comprised a total of 71 respondents, meticulously distributed between the treatment and control groups. Notably, 36 individuals (51%) were strategically allocated to the treatment group, while 35 participants (49%) were assigned to the control group, as elucidated in Table 4. Each respondent was enrolled as a student in the DDT (Diploma in Information Technology (Digital Technology)) program, highlighting the homogeneity of the cohort in terms of academic pursuit. Additionally, all participants resided within the confines of the hostel located within the premises of one of the distinguished Northern Polytechnics in Malaysia.

4.1 Homogeneity Test among Student Groups

Homogeneity testing aimed to determine if the treatment and control groups exhibited comparable characteristics. Various demographic and academic variables, such as age, gender distribution, academic performance metrics, prior programming experience, and familiarity with Visual Basic .NET, were considered as potential factors influencing homogeneity.

Statistical tests, including chi-square tests for categorical variables and independent sample t-tests for continuous variables, were conducted to compare the distributions and means across the treatment and control groups. The significance level was set at $\alpha = 0.05$ to ascertain the statistical significance of any differences observed between the groups.
The outcomes of the homogeneity tests indicated no statistically significant differences between the treatment and control groups across the considered variables. Specifically, the analysis revealed comparable distributions in demographic characteristics, academic metrics, and prior programming experience between the two groups (refer to Table 2 for detailed statistics).

Table 2. Mean descriptive statistics of pre- Proficiency by group

Group	Number of Respondents	Mean	Std. Deviation
Control	35	1.69	.266
Treatment	36	1.60	.283

A homogeneity test was executed on treatment and control groups (significance level: 0.05) to identify the variance, if any, of Proficiency among the respondents prior to intervention. Table 3 presents the outcomes of the t-test by group.

Table 3. T-test of Proficiency among students by group

Group	Number of Respondents	Mean	Std. Deviation	t	df	Sig
Control	35	1.69	.266	1.453	89	0.150
Treatment	36	1.60	.283			

The investigation into the improvement of Visual Basic .NET Proficiency via the integration of 'Developing Applications with Visual Basic .NET: Best Practices and Techniques' showcased non-significant variance in mean scores between the treatment and control groups ($t = 1.453$

& df = 89, p > 0.05). These findings indicate a lack of statistically significant differences between the groups before the intervention, affirming homogeneity in students' proficiency levels towards VB.NET.

The rigorous statistical analysis conducted demonstrated a comparable baseline level of proficiency in VB.NET among participants across both groups. These results denote an equivalent starting point for both the treatment and control groups, bolstering the credibility and robustness of subsequent analyses evaluating the impact of the intervention.

4.2 Inference statistics

> H_1: There is a significant difference in the mean scores of students' proficiency levels towards VB.NET between the treatment and control groups

Hypothesis H1 was subjected to testing using an independent samples t-test. Preceding this analysis, Levene's test was conducted to ascertain the presence of variance in students' proficiency levels towards VB.NET among students after the intervention. Table 4 presents the outcomes obtained from Levene's test, aiming to determine the homogeneity variance for students' proficiency levels towards VB.NET in the post-test phase.

Table 4. Results of Levene's test after intervention (post-test)

Dependent Variable	F	df	Sig.
students' proficiency levels towards VB.	13.075	89	0.001

The obtained significance value from Levene's test for assessing students' proficiency levels towards VB.NET (post-test) was 0.001 ($p \leq 0.05$). This outcome indicated a statistically significant inequality in the variance between the treatment and control groups, leading to the dismissal of the assumption of equal variance. Subsequently, the results of the independent samples t-test, as presented in Table 5, delineate the outcomes derived from the comparison of means between the treatment and control groups following the intervention.

Table 5. Results of t-test by group (post-test)

Group	Number of Respondent	Mean	Std. Deviation	t	df	Sig
Control	35	4.03	.198	20.880	89	0.01
Treatment	36	5.26	.344			

This signifies that while both the conventional teaching method and the proposed e-book showed efficacy in enhancing students' proficiency levels, the comparative analysis demonstrated the superiority of the proposed e-book module over the conventional approach. This finding is supported by recent studies such as [8] which found that digital learning tools significantly improve student engagement and learning outcomes compared to traditional methods. Furthermore, the mean scores obtained from both the control and treatment groups exhibited statistical significance concerning the group and measurement tests, aligning with the results of [9] who highlighted the statistical relevance of digital tools in educational settings.

5. Conclusion

In conclusion, this study conducted a comprehensive assessment of the efficacy of 'Developing Applications with Visual Basic .NET: Best Practices and Techniques' in augmenting Visual Basic .NET proficiency among student cohorts. Employing a quasi-experimental design with a non-equivalent control group, the research revealed substantial advancements in Visual Basic .NET proficiency within the treatment group exposed to the e-book's methodologies. The comparative analysis between the treatment and control groups

demonstrated the superiority of the e-book's approach over traditional pedagogical methods. The discernible enhancement in proficiency levels among students immersed in the e-book's principles underscores its instrumental role in fostering enriched competence in Visual Basic .NET programming.

These findings present compelling evidence supporting the integration of 'Developing Applications with Visual Basic .NET: Best Practices and Techniques' as a pivotal resource for educational enhancement. The study contributes valuable insights that can significantly inform educational strategies and curriculum development within similar academic settings. This research advocates for the deliberate integration of innovative resources such as this e-book, emphasizing their potential to substantially elevate students' proficiency in Visual Basic .NET programming and, by extension, contribute to the continuous evolution and refinement of pedagogical practices in technology-driven academic domains.

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ALPHABET RECOGNITION FOR SIGN LANGUAGE USING DEEP LEARNING

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ABSTRACT

Sign language is very important for those with poor hearing and speech impairments. However, the development of sign language on its own is insufficient. The sign movements can be confusing and muddled up for someone who is unfamiliar with the sign language or has never learned it. However, a number of automated methods for recognising sign motions have made it possible to be narrowed in this communication gap. In this work, the study introduced a Convolutional Neural Network (CNN), that able to capture images of the hand gesture using the web camera and the system will identify and show the name of the picture once it has been taken. The proposed study is to create a system that can translate static movements into equivalent alphabets. A vision-based approach using a web camera is introduced to obtain the data from the signer and can be used offline. The user must be able to capture images of the hand gesture using a web camera, so that the developed system effectively displays the name of the taken image. The study achieved a remarkable accuracy of 98.7% to recognise 10 American sign gesture alphabets.
Keywords: Convolutional Neural Network (CNN), Sign Language Recognition, Hand Gesture Translation

1. Introduction

Sign languages provide a way to communicate without the use of voice. It is possible to communicate without using words by using sign language. This gives those who are hard of hearing or unable to speak another way to interact with others, which makes it extremely important [1] [2]. Furthermore, sign language is a universal language that facilitates communication between speakers of other languages, which is crucial in multicultural environments and during emergencies [3].

Meanwhile, since learning sign language demands memorization of several hand gestures and positions, it may be challenging to initiate communication with real-life sign language users. This shows that for anyone to comprehend sign language, there is a requirement for an automatic method for recognizing sign language [4] [5].

Prior research has employed wearable sensors-based and vision-based recognition as the two primary classification methods. In earlier work, it was done for hand gesture identification with 26 possible classes. The vision-based approach captures the hand's dynamic or static movements by using an RGB or RGB-D camera. The majority of the gestures, except for the letters "j" and "z," can be considered static since they do not need forearm movement [6].

Deep learning architecture has been applied in computer vision, audio recognition, speech recognition, sign language recognition, and medical image analysis, among other fields. Examples of these applications include deep neural networks (DNNs), convolutional neural networks, recurrent neural networks, deep belief networks, and restricted boltzman machines [7]. The main significance of deep learning is that it does not require hand-crafted features; yet, implementing deep learning techniques can be challenging because the method requires huge datasets.

A number of approaches have been put out to address the issue of hand gesture identification in sign language. In [8], Kim et al. demonstrated a way to utilize a deep neural network (DNN) to classify frames in image sequences of finger-spelled letters. Their work was capable of identifying hand positions from any user, even if they were not signers. They fed the deep neural networks with Histogram of Gradients (HoG) picture characteristics [9].

In, Tong et al. [10] without a colour image, they used depth photos as their input. They were also able to attain real-time rates by using a DNN for their classifier. With just colour images available to a small number of participants, the study was able to obtain real-time rates while utilising depth sensors.

In recent years, convolutional neural networks (CNN) have demonstrated great success in computer vision tasks such as image recognition. CNN were made popular starting with AlexNet [11]. This work aims to develop a reliable and real-time finger-spelt sign language classifier by applying the latest developments in deep learning algorithms.

A faster region-based convolutional neural network (R-CNN) [12] is used to translate hand motions from two-dimensional images into text. Their method identified the letters and mapped the positions of the hand motions. They made use of a dataset comprising of 15,360 phone camera photos with divergent backgrounds. The result shows a recognition rate of 93% for the collected American Sign Language (ASL) images dataset. The aim of this proposed study by [13] is to create a system that can translate static sign gestures into words.

In [14], the authors developed a dataset and a CNN sign language recognition system to interpret the American sign gesture alphabet and translate it to our natural language. To compare each dataset's accuracy and outcomes, three datasets were employed. The first dataset, which belongs to the authors, has 104,000 images for 26 letters of ASL; the second dataset of ASL has 52,000 images; and the third dataset contains 62,400 images. The training sets accounted for 70% of the datasets, while the testing and validation sets made up the remaining 15%. Based on the CNN model, all three datasets have an overall accuracy of 99%.

In this work, by utilising deep learning, specifically CNN which has proven to perform exceptionally well in image classification. The study was able to fulfil the demand for a reliable and excellent alphabet sign language recognition system [15] [16]. Since cameras are almost always present in laptops and cell phones, therefore this effort created a network that only uses flat images [17]. This also makes it simple to expand into other applications in the future, since no additional hardware or specialised sensors would be required. The research works on hand poses with the American Sign Language (ASL) as a reference [18] [19].

2. Methodology

This section will be discussing the datasets used and the approach employ to achieve the study aim and objectives. Moreover, it provides comprehensive details on image-based techniques and a brief overview of the dataset employed in ASL recognition systems. The proposed framework for alphabet detection and classification comprises the following steps: 1) data augmentation 2) image pre-processing and labelling, and 3) alphabet classification. This system conducted the comparison of the performance in term of accuracy and precision of alphabet using the deep CNN to detect the performance of system.

2.1. Data Augmentation

The raw image was captured by OPPO Reno 8 handphone camera with the front camera 32 MPixel where the handphone takes the picture between the ground and the camera with the approximate distance from 100cm to 120cm. A total of 500 RGB images consisting of 50 images each alphabet 1 to 10 in sign language were captured in different lighting and

background conditions of image. In order to increase the diversity of our dataset even more, this work added random rotations, random shifting in both horizontal and vertical dimensions, and random zooming. As a result, the study was able to obtain a total of 3000 photos as shown in Fig.

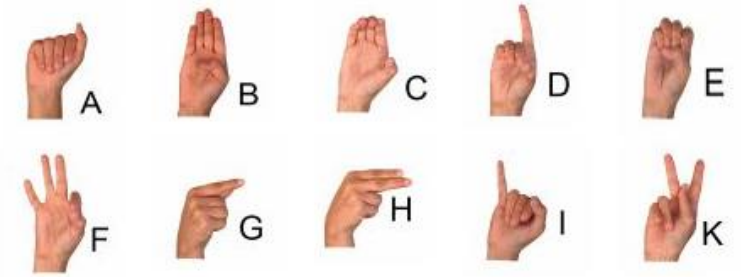


Fig. 1. Dataset collection of images

2.2 Training Dataset

The network prepares for training and testing dataset. The study employed training classifiers including *pickle, sklearn* and *numpy library*. The dataset used in training the network was composed of static hand poses of 10 letters starts with A to K, excluding the letters j. The 2500 images are chosen for the alphabet training dataset and another set of 500 images as a testing dataset. Fig. 1 shows some examples of training images. Table 1 tabulates the number of training and testing dataset for 10 alphabets of sign language.

Table 1. Number of training and testing dataset

	Letter hand poses
Training set	2500
Testing set	500

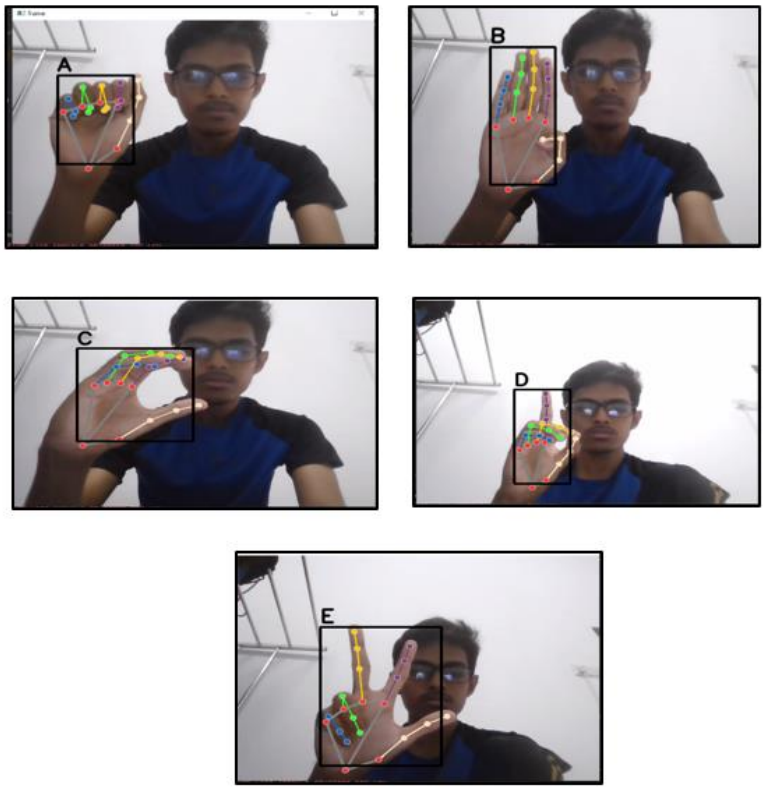


Fig. 1. Hand poses used as a basis in constructing the dataset for each letter

2.3 Deep Convolutional Neural Network (DCNN)

The architecture of DCNN is illustrated in Figure 3, consisting of three convolution layers, three pooling layers and two fully connected layers. The images of alphabet recognition are used as input for the network with 3 channels in DCNN. Convolution layers: Conv 1, Conv2 and Conv3 are equipped with filter sizes of 3x3 or 5×5 and stride of 1 respectively. According to previous study in DCNN, all filters are suggested to have a size not greater than 7x7 to avoid difficulty during the training process. In addition, convolutional layers are parameterized by the number of channels, filter size, number of strides, and number of padding [5]. Convolutional networks may include a pooling layer, hence the proposed network consists of max pooling (MaxPool) and average pooling (AvePool) which are performed with stride 2 over a 3 × 3 window. The stride defines how many columns and rows in the receptive field slide to produce new input width and height. Computational process may reduce if using a larger stride size which leads to produce fewer receptive field applications and a smaller output size [19].

In contrast, a larger stride size may also lose features of the input data. Another key aspect of the DCNN is a pooling layer. Max pooling takes the max values from an input array's sub-arrays, whereas average pooling takes the average values. Finally, the flattened output fed into multiple fully connected (FC) as the last layer to predict the images into ten different alphabet classification. The rectifier linear unit (ReLU) was introduced by Nair and Hinton [17] as a nonlinear activation function, adopted after the convolution layer for dealing with nonlinearity for multi classification.

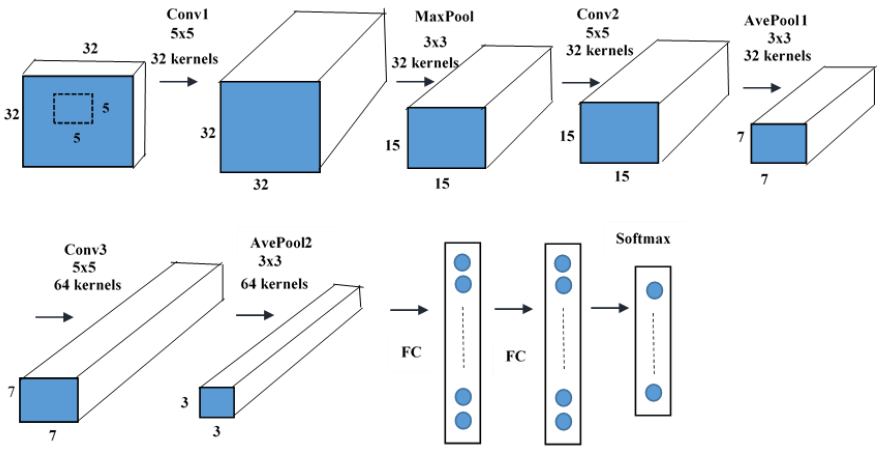


Fig. 3. DCNN architecture

2.4 Alphabet detection and classification using DCNN

A DCNN consists of many layers such as convolutional, pooling and fully connected layers [14]. In general, convolution layer extracts image features from different abstract level and made up of various changeable kernels, also known as filters to creates feature maps. There are two types of convolution parameters in this network which are filter size and filter numbers. The rectifier linear (ReLU) adopted to the output of every convolution and pooling layer to perform nonlinear transforms [12]. The pooling layer is a statistic for the non-overlapping regions so that it can reduce feature parameters and prevent over-fitting. The final layers of a CNN are composed of fully connected layers and a softmax layer. This layer is responsible for predicting the classification of the alphabet in sign language. The classification performance is measured for each type of pavement cracking by evaluating the classification accuracy defined as follows:

$$Classification\ accuracy = \frac{Number\ of\ alphabet\ samples\ correctly\ classified}{Numbers\ of\ alphabet\ samples}$$

3. Result and Discussion

In this study the size of filter and the layers of convolution were manually adjusted. The CNN was applied to the 500 testing images for analysis on network performance for alphabet classification using 2500 training dataset. The experimental result obtained a promising performance of 98.7% to classify into ten alphabet classifications. The proposed network is only used a low number of training and testing dataset compared to other researchers.

Table 2. Alphabet classification performance

Alphabet type	Percentage correctly identified
A	99.6%
B	99.8%
C	98.8%
D	98.8%
E	99.2%
F	98.9%
G	98.3%
H	98.1%
I	99.1%
K	99.2%
Average Accuracy	98.7%

From the Table 2 result, the classification of alphabet H have a lower accuracy compared to others due to the network failed to find the correct image. One of the reasons could be the feature extractor might be confused with the letter between G and H because they have the similarity image. Therefore, this method managed to provide high performance on alphabet detection and classification with acceptable accuracy.

4. Conclusion

The primary goal of the project was to create a system that could convert static sign language into the appropriate alphabet so that users may become acquainted with the basics of sign language. In this study, an efficient network architecture, based on CNN offers a highly promising method to replace the traditional method. The network achieved an overall accuracy of 98.7% on the 500 testing images. As a result, this technique has a lot of potential for use as a tool to assist those with speech and hearing impairments in sign language analysis. Although the study obtained a low number of training and testing dataset, the proposed system is still performed and well-matched with the existing systems. For future plans, the study will convert the trained image classification models for American sign language alphabet gestures into a real-time system and evaluate the performance with larger vocabularies.

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DIMENSI GURU DAN MURID DALAM KAEDAH PENDIDIKAN NABI SAW. : SATU PENELITIAN

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ABSTRAK

Kajian ini meneliti dimensi guru dan murid dalam memahami kaedah pendidikan Nabi SAW.. Dimensi guru merujuk kepada semua perihal dan keadaan yang berkait dengan diri Nabi SAW sebagai seorang guru kepada para Sahabat r.anhum. Manakala, dimensi murid pula merujuk kepada semua faktor dan situasi yang berkait dengan diri para Sahabat r.anhum sebagai murid. Kajian ini berbentuk kajian perpustakaan menggunakan kaedah analisis dokumen. Kajian ini penting untuk memahami dimensi guru dan murid disebalik kaedah pendidikan Nabi SAW. yang akan menjawab beberapa persoalan, antaranya ialah pertama, Nabi SAW. pernah melarang sesuatu perkara dilakukan tetapi di waktu yang lain membenarkan pula, kedua, Nabi SAW. tidak marah apabila ada seorang sahabat lelaki meminta izin untuk melakukan perkara keji dan ketiga, Nabi SAW. pernah melakukan perkara-perkara yang berbeza daripada kelaziman. Oleh itu, kajian ini bertujuan untuk pertama, menerangkan konsep dimensi guru dan dimensi murid dalam pendidikan Nabi SAW., kedua, menganalisis keberadaan dimensi guru dan murid melalui beberapa hadis pilihan merangkumi kaedah pendidikan dan rahsianya serta pengajaran yang diperolehi daripada hadis tersebut. Hasilnya, kajian ini mendapati bahawa dimensi guru ialah suatu konsep yang melibatkan semua perkara berkaitan Nabi SAW. selaku pendidik meliputi objektif pengajaran, sifat kasih sayang, strategi pendidikan, keterbatasan ilmu, reaksi spontan Nabi SAW. dan juga sifat-sifat peribadi Nabi SAW.. Manakala dimensi murid pula ialah suatu konsep yang melibatkan semua faktor berkaitan para sahabat r.anhum selaku murid meliputi bilangan, kemampuan ilmu dan kefahaman, kejahilan dan salah faham, minat dan kesungguhan untuk belajar dan penghormatan kepada Nabi SAW. Kedua-dua dimensi ini telah membentuk dan menentukan pemilihan kaedah pendidikan oleh Nabi SAW.

Kata Kunci: Hadis Pendidikan, Kaedah Pendidikan Nabi SAW., Dimensi Guru dan Murid

1. Pengenalan

Nabi Muhammad SAW adalah rasul yang diutuskan oleh Allah sebagai satu rahmat kepada seluruh alam (al-Anbiya,33:107)[1]. Bahkan Allah mengambarkan Nabi SAW adalah seorang sangat sayang dan kasih kepada orang-orang yang beriman (al-Taubah,33:21)[1]. Ini bertepatan dengan tugas Nabi SAW sebagai pembimbing umat atau guru kepada umat. Justeru, memahami rahsia di sebalik tindak tanduk Nabi SAW sebagai seorang guru adalah sesuatu yang sangat mustahak dan penting sesuai dengan janji Allah SWT bahawa Nabi SAW adalah contoh teladan yang terbaik bagi orang-orang yang mengharap rahmat Allah SWT dan juga hari Akhirat (Al-Ahzab,33:21)[1].

Dimensi guru iaitu semua perihal berkaitan diri Nabi SAW selaku pemilik hadis dan dimensi murid iaitu semua perihal berkaitan para sahabat yang ditujukan sesuatu hadis telah memberikan pengaruh yang besar kepada usaha untuk memahami konteks, sebab dan rahsia disebalik perbuatan mahupun perkataan Nabi SAW[2]. Oleh itu, dapat difahami bahawa memahami kedua-dua dimensi ini dengan tepat termasuklah yang berkaitan dengan pendidikan Nabi SAW dan kaedahnya adalah sangat penting dan mustahak. Ini kerana wujud beberapa kelainan dan persoalan berkaitan kaedah pendidikan Nabi SAW antaranya, pertama, Nabi SAW pernah melarang sesuatu perkara tetapi di waktu yang lain membenarkan

pula, kedua, Nabi SAW pernah bersoal jawab tentang sesuatu perkara tetapi pernah juga mengelak tentang sesuatu perkara, ketiga, Nabi SAW tidak marah apabila ada seorang sahabat lelaki meminta izin untuk melakukan perkara keji seperti zina, keempat, Nabi SAW tidak menjawab soalan yang telah diajukan kepadanya, sebaliknya membiarkan sahabat lain menjawabnya, kelima, Nabi SAW pernah melakukan perkara-perkara yang berbeza daripada kelazimannya sehingga menimbulkan tanda tanya kepada para sahabat.

Selain itu, kegagalan untuk memahami dimensi guru dan murid dengan berkesan akan menjejaskan kefahaman tentang gaya dan kaedah pendidikan Nabi SAW yang telah berjaya mendidik generasi sahabat sehingga menjadi umat terbaik sebagaimana firman Allah SWT : "*Kamu adalah sebaik-baik umat yang dilahirkan untuk umat manusia, kamu menyuruh kepada kebaikan, dan mencegah daripada kemungkaran, dan beriman kepada Allah. Sekiranya Ahli Kitab beriman, tentulah itu lebih baik bagi mereka, di antara mereka ada yang beriman, dan kebanyakan mereka adalah orang-orang yang fasik*". (Ali Imran,3:110)[1]. Selain itu, kegagalan ini telah membuka peluang kepada pihak tertentu untuk menolak kehujahan hadis seperti golongan anti hadis kerana mengganggap Nabi SAW tidak memiliki apa-apa keistimewaan berbanding manusia yang lain [3]. Kesannya, gaya dan kaedah pendidikan Nabi SAW tidaklah dianggap sebagai kaedah terbaik dan tidak perlu dicontohi oleh para guru sekarang. Seterusnya, didapati juga hampir tiada karya yang membicarakan kaedah pendidikan Nabi SAW secara tersusun yang memfokuskan kaitan kaedah pendidikan dengan hadis Nabi SAW[4]. Bahkan karya Abu Ghuddah yang berjudul *al-Rasul al-Mu'allim Wa Asalib Fi al-Ta'lim* hanya menggariskan kaedah-kaedah yang digunakan oleh Nabi SAW dalam pendidikan tetapi tidak mengaitkan keberadaan dimensi guru dan murid secara langsung.

Susulan itu, penulisan ini akan menerangkan, pertama, konsep dimensi guru (iaitu suatu perbincangan berkaitan dengan latar belakang dan diri Nabi SAW sebagai guru) dan dimensi murid (iaitu suatu perbincangan berkaitan keadaan para sahabat r.anhum sebagai murid) dalam pendidikan Nabi SAW, kedua, analisis keberadaan dimensi guru dan murid melalui beberapa hadis pilihan merangkumi kaedah pendidikan dan rahsianya serta pengajaran yang diperolehi daripada hadis tersebut.

2. Metodologi Kajian

Kajian ini merupakan kajian perpustakaan menggunakan metod analisis kandungan untuk meneliti keberadaan dimensi guru dan murid dalam kaedah pendidikan Nabi SAW. Kajian ini menumpukan kepada analisis 8 hadis pilihan melalui sokongan pelbagai dapatan daripada sumber buku, jurnal, dan artikel.

3. Perbahasan dan Perbincangan :

3.1 Dimensi Guru dalam Pendidikan Nabi SAW :

Dimensi guru dalam pendidikan Nabi SAW merujuk kepada semua keadaan berkaitan dengan Nabi SAW. selaku guru meliputi tatacara kehidupan, perbuatan, perkataan, rupa diri dan sebagainya. Dimensi ini boleh difahami melalui pertama, penelitian terhadap keadaan diri Nabi SAW selaku seorang guru, kedua, penelitian terhadap realiti zaman dan tempat yang dilalui oleh Nabi SAW selaku seorang guru[2]. Manakala Munawir pula menyatakan bahawa kajian berkaitan realiti masyarakat, budaya dan sejarah boleh digunakan untuk mengenalpasti rahsia disebalik terjadinya sesuatu hadis[5]. Ini bermakna, kaedah yang sama boleh digunakan untuk mengetahui kewujudan dimensi guru mahupun murid dalam pendidikan Nabi SAW. Antara dimensi guru yang terdapat dalam pendidikan Nabi SAW ialah:

Pertama, setiap guru memiliki sifat kemanusiaanya yang tersendiri seperti lupa, marah, sayang, benci dan seumpamanya. Sebagai contoh, Nabi SAW sebagai seorang guru juga pernah marah dan melaknat seseorang sebagaimana hadis berikut : "*Wahai Aisyah, tidak tahukah kamu apa yang pernah aku syaratkan kepada Tuhanku? Sesungguhnya aku telah*

memohon : Ya Allah, aku hanyalah seorang manusia jika ada seorang muslim yang aku laknat atau aku maki, maka jadikanlah perkara itu sebagai penghapus dosa dan pahala baginya" [HR Muslim, Bab : Barang siapa yang dilaknat atau dicela oleh Nabi SAW.][6]. Ini bermakna, walaupun sifat kemanusiaan ini bukan sebahagian daripada tugas rasmi seorang guru termasuklah Nabi SAW namun ianya akan memberi pengaruh kepada kaedah pendidikan termasuklah pendidikan Nabi SAW.

Kedua, setiap guru mempunyai keterbatasan ilmu mahupun fizikal. Keterbatasan ini juga wujud pada diri Nabi SAW. Sebagai contoh Nabi SAW tidak mengetahui tarikh kiamat, takdir dan lain-lain sebagaimana diceritakan oleh Umar bin al-Khattab r.a. : "*Bahawasanya Jibril AS pernah bertanya kepada Nabi : "Bilakah hari kiamat itu?. Baginda SAW menjawab : "Tidaklah orang yang ditanya itu lebih mengetahui daripada orang yang bertanya".* [HR al-Bukhari, Bab : Tanda-tanda Mencintai Allah SWT.][7]. Ini menunjukkan keterbatasan ilmu mahupun fizikal akan membentuk kepelbagaian kaedah dan konsep dalam pendidikan termasuklah pendidikan Nabi SAW kepada para sahabatnya.

Ketiga, setiap guru memiliki objektif pendidikannya melalui kata-katanya bukan perbuatannya. Sebagai contoh isu larangan puasa sambung kepada para sahabat sedangkan Baginda SAW sendiri melakukannya sebagaimana diceritakan oleh Aisyah r.anha: "*Rasulullah SAW melarang puasa sambung sebagai bentuk kasih sayang kepada mereka (para sahabat)*" [HR al-Bukhari, Bab : Puasa Wishal dan orang yang mengatakan "Pada malam hari tidak ada puasa"][7]. Ini bermakna, pendidikan Nabi SAW akan dibentuk dan dipengaruhi oleh tujuan Nabi SAW diutuskan iaitu sebagai rasul dan bukan ditentukan oleh perbuatannya sebagai seorang manusia.

Keempat, setiap guru memiliki kebolehan dan kepakaran tersendiri. Sebagai contoh, Nabi SAW adalah seorang paling fasih walaupun merupakan seorang yang ummi iaitu tidak boleh menulis dan membaca sebagaimana firmanNYA : "*Dialah yang mengutus kepada kaum yang buta huruf seorang Rasul di antara mereka, yang membacakan ayat-ayatNya kepada mereka, mensucikan mereka dan mengajarkan mereka Kitab dan Hikmah (al-Sunnah). Dan sesungguhnya mereka sebelumnya benar-benar dalam kesesatan yang nyata*" (al-Jumuah,62:2)[1]. Ini menjelaskan bahawa kebolehan seorang guru akan menentukan pemilihan sesuatu kaedah pendidikan dan gayanya termasuklah pendidikan Nabi SAW.

3.2 Dimensi Murid dalam Pendidikan Nabi SAW.

Dimensi murid dalam pendidikan Nabi SAW merujuk kepada sesiapa sahaja yang ditujukan kepadanya sesuatu hadis sama ada individu atau kumpulan daripada kalangan para sahabat r.anhum[2]. Murid disini juga boleh dirujuk sebagai *al-Sa'ilin* (Orang yang bertanya kepada Nabi SAW.) dan *al-Muta'alimin* (Orang yang belajar dengan Nabi SAW.)[4]. Ada dua cara yang boleh digunakan untuk memahami dimensi murid dalam pendidikan Nabi SAW. iaitu pertama, kajian terhadap ciri-ciri khusus indvidu yang berinteraksi dengan Nabi SAW. dan kedua, kajian lanjut tentang realiti zaman dan tempat yang melatari kehidupan para sahabat r.anhum selaku murid[2]. Antara dimensi murid yang terdapat dalam pendidikan Nabi SAW ialah :

Pertama, setiap murid memiliki perbezaan biodata merangkumi sifat, fizikal, umur dan sebagainya. Perbezaan ini akan menentukan pemilihan kaedah dan gaya pendidikan. Sebagai contoh, Nabi SAW. telah mengizinkan sahabat yang tua untuk mencium isterinya ketika sedang berpuasa tetapi tidak untuk sahabat yang masih muda. Rahsia perbezaan ini ialah perbezaan umur sebagaimana diceritakan oleh Abdullah bin Amru r.a bahawa Nabi SAW. bersabda, : "*Aku tahu kenapa kalian saling berpandangan satu sama lain; sesungguhnya orang yang sudah tua itu dapat menahan nafsu syahwatnya*".[HR. Ahmad][8]. Ini bermakna, perbezaan biodata murid akan menentukan pemilihan kaedah pendidikan termasuklah pendidikan Nabi SAW.

Kedua, setiap murid memiliki pengetahuan sedia ada dan kefahaman ilmu yang berbeza. Perbezaan ini akan menentukan pemilihan kaedah dan gaya pendidikan. Sebagai contoh, Nabi SAW telah mengajarkan Muaz bin Jabal r.a. suatu ilmu iaitu "*Barangsiapa berjumpa Allah dengan tidak menyekutukanNya dengan sesuatu apapun, maka dia akan masuk syurga*" tetapi tidak membenarkan Muaz menyampaikan ilmu itu kepada sahabat yang lain. Rahsia disebalik perbezaan ini ialah tahap kemampuan untuk memahami ilmu di antara para sahabat r.anhum sebagaimana jawapan Nabi SAW melalui sabdanya :"*Jangan, kerana aku bimbang mereka akan jadi malas (untuk beramal)*. [HR al-Bukhari, Bab : Mengkhususkan sebahagian ilmu kepada sebahagian orang kerana bimbang yang lain tidak dapat memahami][7]. Ini bermakna, perbezaan kemampuan murid akan menentukan pemilihan isi kandungan pendidikan termasuklah pendidikan Nabi SAW.

Ketiga, setiap murid memiliki peranan, minat, kelebihan dan kelemahan yang tersendiri. Semua ini akan menentukan pemilihan kaedah dan gaya pendidikan. Sebagai contoh, Nabi SAW telah memberi keringanan kepada Abdur Rahman bin Auf dan al-Zubair untuk menggunakan baju yang dibuat daripada sutera kerana penyakit kutu (gatal) yang dideritai oleh keduanya [HR. Al-Bukhari, Bab : Kain Sutera Dalam Peperangan][7] sedang Nabi SAW. pernah bersabda : "*Sesungguhnya kedua benda ini (Sutera dan emas) haram bagi kaum lelaki daripada umatku dan halal bagi kaum wanita*". [HR. Ibn Majah, Bab : Mengenakan Sutera dan Emas Bagi Wanita][9]. Ini menunjukkan bahawa kekurangan dan kelebihan murid akan membuka keringanan atau sebaliknya dan seterusnya akan menentukan kaedah pendidikan termasuk pendidikan Nabi SAW.

Keempat, setiap murid memiliki gaya pembelajaran, kaedah belajar dan kemahiran belajar yang berbeza seperti kemahiran menulis dan membaca serta mendengar dan menghafal. Semua ini akan menentukan pemilihan kaedah dan gaya pendidikan. Sebagai contoh, Abdullah bin Amr adalah seorang murid Nabi SAW yang sangat mahir menulis sehingga diberikan kebenaran khas oleh Nabi SAW. untuk menulis hadis. Beliau menulis adalah untuk membolehkannya menghafal semua yang beliau dengar daripada Nabi SAW. [HR Abu Daud, Bab : Mencatat ilmu][10]. Secara umum tidak semua sahabat mampu menulis dengan baik dan teliti dalam menulis semua yang didengari daripada Nabi SAW. sehingga telah menyebabkan Nabi melarang mereka untuk mencatat hadis sebagaimana diceritakan oleh Abu Said al-Khudri, bahawa Rasulullah ﷺ bersabda : "*Janganlah kalian menulis daripadaku, barangsiapa menulis daripadaku selain al-Quran hendaklah dihapuskan, dan ceritakanlah daripadaku dan tidak ada dosa. Barangsiapa berdusta atas namaku dengan sengaja, maka siapkanlah tempat duduknya daripada neraka*". [HR Muslim, Bab : Hukum menulis hadis][6]. Ini menunjukkan bahawa kemahiran belajar dan gaya pembelajaran akan menentukan kaedah pendidikan termasuk pendidikan Nabi SAW.

3.3 Analisis hadis

3.3.1 Hadis 1 : Kaedah Demonstrasi

Daripada Sahl bin Sa'd al-Sa'idi r.a., dia berkata, "Demi Allah, akulah orang yang paling mengetahui tentang masalah ini. Sesungguh aku telah melihat pada hari pertama mimbar tersebut dipasang dan ketika Rasulullah SAW duduk di atasnya. Lalu aku melihat Rasulullah SAW solat diatasnya. Baginda SAW bertakbir dalam posisi di atas mimbar lalu rukuk dalam posisi masih di atas mimbar. Kemudian Baginda SAW turun dengan melangkah ke belakang, lalu sujud di dasar mimbar, kemudian Baginda SAW mengulangi lagi hingga solat selesai. Setelah selesai, Baginda SAW menghadap kepada orang ramai lalu bersabda, "Wahai sekalian manusia, sesungguhnya aku berbuat seperti tadi supaya kalian mengikuti dan supaya kalian dapat mengambil pelajaran tentang tata cara solatku." [HR al-Bukhari, Bab : Khutbah diatas mimbar][7].

Analisis Hadis 1 :
Hadis ini menceritakan bahawa Nabi SAW. bersolat diatas mimbar daripada awal hingga akhir dan para sahabat r.anhum menjadi makmum di belakang Nabi SAW.. Tindakan Nabi SAW ini boleh dianggap sebagai kaedah demonstrasi untuk mengajarkan para sahabat r.anhum tentang tatacara solat. Namun persoalannya, kenapa Nabi SAW. melakukan demontrasi di atas mimbar? Berdasarkan keterangan daripada hadis ini, dapat difahami, pertama, kerana jumlah sahabat yang ramai yang tidak memungkinkan mereka diajar seorang demi seorang, kedua, solat diatas mimbar yang lebih tinggi adalah untuk membolehkan semua para sahabat yang bersolat di belakang Nabi SAW. dapat melihat dengan lebih jelas keseluruhan tatacara solat, dan ketiga, supaya para sahabat r.anhum faham dan seterusnya mampu mengajarkan kepada orang lain sebagaimana sabda Nabi SAW. dalam peristiwa yang lain : "*Pulanglah kepada keluarga kalian. Tinggallah bersama mereka dan ajari mereka serta perintahkan mereka dan solatlah kalian sebagaimana kalian melihatku solat*. [HR al-Bukhari, Bab : Azan Dan Iqamah Bagi Musafir Bila Solat Berjemaah][7]. Menurut Mohd Asri yang mengatakan bahawa tindakan ini untuk mengelak para sahabat tidak dapat melihat dengan sempurna tatacara solat Nabi SAW sesuai dengan tugas baginda sebagai seorang guru[2]. Kesimpulannya, dimensi guru iaitu objektif pengajaran dan isi kandungannya serta dimensi murid iaitu keadaan, kedudukan dan bilangan murid perlu diambil kira ketika menggunakan kaedah demontrasi sebagaimana yang telah ditunjukkan oleh Nabi SAW. ketika mengajar para sahabat tentang tatacara solat.

3.3.2 Hadis 2 : Kaedah Debat

Daripada Abu Umamah r.a, dia berkata ; Sesungguhnya seorang pemuda mendatangi Nabi ﷺ lalu berkata ; “Wahai Rasulullah! Izinkan aku untuk berzina”. Orang ramai mendatanginya lalu melarangnya, mereka berkata ; “Jangan, jangan”. Rasulullah SAW bersabda, "Mendekatlah". Dia mendekati Nabi SAW lalu duduk, kemudian Rasulullah SAW bersabda : "Apa kau menyukainya berzina dengan ibumu?" Pemuda itu menjawab ; Tidak, demi Allah wahai Rasulullah, semoga Allah menjadikanku sebagai penebus tuan. Nabi ﷺ bersabda; Orang-orang juga tidak menyukainya berzina dengan ibu-ibu mereka." Nabi SAW bersabda, "Apa kau menyukainya berzina dengan anak perempuanmu?" Tidak, demi Allah wahai Rasulullah semoga Allah menjadikanku sebagai penebus tuan. Nabi SAW bersabda ; Orang-orang juga tidak menyukai berzina dengan anak-anak perempuan mereka. " Rasulullah SWA bersabda, "Apa kau menyukai berzina dengan saudara perempuanmu?" Dia berkata : Tidak, demi Allah wahai Rasulullah semoga Allah menjadikanku sebagai penebus tuan. Nabi SAW bersabda; Orang-orang juga tidak menyukai berzina dengan saudara-saudara perempuan mereka. Rasulullah SAW bersabda, "Apa kau menyukainya berzina dengan saudara perempuan ayahmu?" Tidak, demi Allah wahai Rasulullah semoga Allah menjadikanku sebagai penebus Tuan. Nabi SAW bersabda; Orang-orang juga tidak menyukainya berzina dengan saudara perempuan ayah mereka." Rasulullah SAW bersabda, "Apa kau menyukainya berzina dengan saudara perempuan ibumu?" Tidak, demi Allah wahai Rasulullah semoga Allah menjadikanku sebagai penebus tuan. Nabi SAW bersabda; Orang-orang juga tidak menyukainya berzina dengan saudara perempuan ibu mereka." Kemudian Rasulullah SAW meletakkan tangan beliau pada pemuda itu dan berdoa, "Ya Allah! Ampunilah dosanya, bersihkan hatinya, jagalah kemaluannya." Setelah itu pemuda itu tidak pernah melirik apa pun.. [HR Ahmad][8].

Analisis Hadis 2 :
Hadis ini menceritakan bahawa Nabi SAW. tidak marah tetapi memilih untuk berdebat dan bersoal jawap dengan seorang pemuda yang telah datang meminta izin untuk melakukan zina. Justeru itu, persoalannya ialah apakah punca yang menyebabkan Nabi SAW. tidak marah bahkan cuba mendekati pemuda tersebut. Sedangkan para sahabat yang lain secara pantas telah melarang secara keras. Dengan merujuk kepada perbuatan Nabi SAW meliputi sikap, cara layanan, usaha untuk mendekati, bertanyakan soalan yang munasabah kepada pemuda tersebut, dapat difahami bahawa Nabi SAW. telah menggunakan dimensi seorang

guru iaitu cuba memahami dan membetulkan kesilapan seorang murid ketika berhadapan dengan permasalahan seumpama ini. Selain itu, dimensi murid iaitu jahil, tidak memahami konsep dan silap faham dalam beragama juga telah menyebabkan Nabi SAW. tidak marah bahkan telah mendoakan pemuda tersebut sebagaimana telah disebut secara jelas di dalam hadis. Bahkan kenyataan yang datang daripada perawi hadis iaitu "*Setelah itu pemuda itu tidak pernah melirik apa pun*" membuktikan bahawa segala tindakan pemuda tersebut adalah kerana kejahilan dan kebodohan semata-mata. Kesimpulannya, dengan merujuk kepada dimensi guru iaitu fokus guru, tindakan dan usaha untuk memahami murid, niat untuk membetulkan kesilapan murid dan juga dimensi murid iaitu jahil, salah faham dan kebodohan maka kaedah pendidikan Nabi SAW. secara debat dan soal jawab dapat difahami dan digunakan dengan lebih tepat. Ini kerana Nabi SAW juga pernah meninggalkan debat untuk sesuatu perkara yang tidak ada jawapan yang pasti sebagaimana telah diceritakan Umar bin al-Khattab r.a. : "*Bahawasanya Jibril a.s. pernah bertanya kepada Nabi SAW : "Bilakah hari Kiamat itu?. Baginda SAW. menjawab : "Tidaklah orang yang ditanya itu lebih mengetahui daripada orang yang bertanya".* [HR Muslim, Bab : Penjelasan Tentang Iman, Islam, Takdir dan Tanda-tanda al-Kiamat.][6]..

3.3.3 Hadis 3 : Kaedah Perbandingan

Daripada Ibn Abbas r.a. , dia berkata bahawa seorang wanita menemui Nabi SAW dan berkata : "Ibuku bernazar untuk melakukan haji, namun dia meninggal dunia sebelum sempat melakukan haji, bolehkah aku menggantikan hajinya?" Nabi ﷺ menjawab, "Silakan, berhajilah engkau untuk menggantikannya, bukankah engkau bersetuju sekiranya ibumu mempunyai hutang, bukankah engkau yang akan melunasinya?" Wanita itu menjawab, "Ya." Lantas Nabi SAW. berkata, "Penuhilah hutang Allah, sebab Allah lebih berhak untuk dilunasi hutangnya." [HR al-Bukhari, Bab : Perumpamaan Suatu Hukum Dengan Sesuatu Yang Allah Telah Jelaskan Hukumnya][7].

Analisis Hadis 3 :
Hadis ini menceritakan bahawa seorang sahabat r.anha bertanya hukum menggantikan nazar haji bagi pihak ibunya yang telah meninggal dunia. Namun, Nabi SAW tidak terus memberikan jawapan hukum secara jelas sebaliknya melontar satu persoalan yang sama sebagai satu perbandingan untuk dijawab orang sahabat yang bertanya. Setelah soalan itu dijawab oleh sahabat yang bertanya, barulah Rasulullah SAW. memberikan jawapan hukum dan seterusnya memberikan penegasan kepada jawapan tersebut. Ini bermakna, Nabi SAW. telah mendidik dan menjawab soalan sahabat tersebut menggunakan kaedah perbandingan. Namun persoalannya, apakah faktor yang menyebabkan Nabi SAW. memilih kaedah ini?. Dengan merujuk cerita lanjut di dalam hadis ini, maka boleh difahami bahawa Nabi SAW. mahu manggalakkan percambahan ilmu untuk muridnya, mengganjurkan perbincangan dan seterusnya bersama-sama membuat keputusan atas perkara yang ditanya. Disamping itu, dimensi Nabi SAW. sebagai guru yang bijak, tahu keadaan murid dan tahap ilmu murid telah menyebabkan Nabi SAW memilih untuk tidak menjawab terus pertanyaan itu walaupun tahu seorang guru perlu segera menghilangkan persoalan dan permasalahan murid. Selain itu, Nabi SAW juga telah mengambil kira dimensi murid iaitu kemampuan berfikir, bersoal balas dan ilmu sedia ada sebelum memilih untuk menggunakan kaedah perbandingan ini. Kesimpulannya, dengan merujuk kepada dimensi guru iaitu perancangan dan kemahuan untuk menggalakkan penglibatan murid ke dalam pengajaran dan juga dimensi murid iaitu kemampuan berfikir dan ilmu sedia maka rahsia di sebalik kaedah perbandingan yang dilakukan oleh Nabi SAW dapat difahami dengan lebih sempurna.

3.3.4 Hadis 4 : Kaedah Soal Jawab

Daripada Anas bin Malik r.a., dia berkata bahawa seorang lelaki bertanya kepada Nabi SAW, "Bilakah hari kiamat akan terjadi wahai Rasulullah?". Baginda SAW menjawab, "Apa yang telah kau persiapkan untuknya?". Lelaki itu menjawab, "Aku belum mempersiapkan banyak

untuk itu sama ada solat, puasa ataupun sedekah, namun aku hanya mencintai Allah dan rasul-Nya.". Baginda SAW. bersabda : "Kamu akan bersama dengan orang yang kamu cintai." [HR al-Bukhari, Bab : Tanda-tanda Mencintai Allah SWT.][7].

Analisis Hadis 4:
Hadis ini menceritakan bahawa seorang sahabat r.a bertanya bilakah kiamat akan tiba. Namun, Nabi SAW. tidak menjawab soalan itu sebaliknya bertanya soalan yang lebih utama iaitu persediaan untuk menghadapi hari kiamat. Kemudian Nabi SAW. memberikan respon yang sangat tegas kepada jawapan sahabat tersebut. Ini menunjukkan bahawa Nabi SAW. telah memilih untuk mendidik sahabat tersebut menggunakan kaedah soal jawab. Namun persoalannya, apakah faktor yang menyebabkan Nabi SAW. memilih kaedah ini?. Dengan merujuk dimensi Nabi SAW sebagai guru iaitu keterbatasan ilmu berkaitan tarikh berlaku kiamat dan objektif pengajaran guru iaitu persediaan menghadapi kiamat adalah lebih utama berbanding tarikhnya maka rahsia di sebalik pemilihan kaedah soal jawab ini lebih mudah untuk difahami. Selain itu, dimensi murid iaitu kesilapan memilih soalan dan keutamaan telah diambil kira oleh Nabi SAW. sebelum memilih untuk menggunakan kaedah soal jawab ini. Tambahan pula, Nabi SAW. pernah menjawab soalan yang ditanya berkaitan kiamat sesuai dengan tujuan orang yang bertanya sebagaimana telah diceritakan Umar bin al-Khattab r.a. : "*Bahawasanya Jibril a.s. pernah bertanya kepada Nabi SAW : "Bilakah hari Kiamat itu?. Baginda SAW. menjawab : "Tidaklah orang yang ditanya itu lebih mengetahui daripada orang yang bertanya".* Jibril a.s. bertanya lagi *: "Maka khabarkanlah kepadaku tentang tanda-tandanya?.* Baginda SAW. menjawab *: "Apabila seorang hamba melahirkan tuannya dan kamu melihat orang yang tidak beralas kaki, telanjang, miskin, pengembala kambing namun bermegah-megahan dalam membangunkan bangunan".* [HR Muslim, Bab : Penjelasan Tentang Iman, Islam, Takdir Dan Tanda-tanda Kiamat.][6].. Kesimpulannya, dengan merujuk kepada dimensi guru iaitu keterbatasan ilmu, tahap keilmuan guru dan objektif pengajaran seorang guru dan juga dimensi murid iaitu minat murid, bentuk soalan yang ditanya dan tujuannya maka rahsia di sebalik pemilihan kaedah soal jawab oleh Nabi SAW. dapat difahami dengan lebih baik.

3.3.5 Hadis 5 : Kaedah Perbincangan

Daripada Ibnu Abbas r.a., dia berkata : "Ada seorang lelaki menemui Rasulullah SAW, lalu dia berkata ; "Semalam aku bermimpi melihat segumpal awan yang menitiskan minyak samin dan madu, kemudian aku lihat orang ramai memintanya, ada yang meminta banyak dan ada yang meminta sedikit, tiba-tiba ada tali yang menghubungkan antara langit dan bumi, aku lihat engkau memegangnya kemudian engkau naik, kemudian ada orang lain memegangnya dan dia mengggunakannya untuk naik, kemudian ada orang yag mengambilnya dan menggunakannya untuk naik namun tali terputus, kemudian tali tersambung". Spontan Abu Bakar berkata ; "Wahai Rasulullah, ayah dan ibuku untuk tebusanmu, demi Allah, biarkan aku untuk mentakwilkannya!". "Takwilkanlah" kata Rasulullah SAW. Abu Bakar mengatakan; "Adapun awan, itulah Islam, adapun madu dan minyak samin yang menitis, itulah al-Quran, kerana al-Quran manisnya menitis, maka silakan ada yang memperbanyak atau mempersedikit, adapun tali yang menghubungkan langit dan bumi adalah kebenaran yang engkau pegang teguh sekarang ini, yang kerananya Allah SWT meninggikan kedudukanmu, kemudian ada seseorang sepeninggalanmu mengambilnya dan dia pun menjadi tinggi kedudukannya, lantas ada orang lain yang mengambilnya dan terputus, kemudian tali itu tersambung kembali sehingga ia menjadi tinggi kedudukannya kerananya, maka beritahulah aku ya Rasulullah, ayah dan ibuku sebagai tebusanmu, saya benar atau salah?". Nabi ﷺ menjawab : "Engkau benar sebahagian dan salah sebahagian!". Abu Bakar r.a. berkata ; "Demi Allah ya Rasulullah, tolong beritahukanlah kepadaku takwilku yang salah!". Nabi SAW menjawab, "Janganlah engkau bersumpah". [HR al-Bukhari, Bab : Ucapan orang yang tidak bermimpi kepada pentakwil mimpi jika dia tersalah][7].

Analisis Hadis 5:
Hadis ini menceritakan bahawa seorang sahabat r.a telah bertanya kepada Nabi SAW. perihal takwil mimpinya. Sebelum sempat dijawab oleh Nabi SAW. ada seorang sahabat yang lain meminta izin untuk memberikan pandangan dan jawapan kepada pertanyaan tadi dan dia diberikan keizinan oleh Nabi SAW. Kemudian Nabi SAW. memberikan pengesahan atas sebahagian jawapan dan memberikan teguran atas sebahagian jawapan sahabat tadi. Ini bermakna, Nabi SAW telah memilih untuk menggalakkan berlakunya perbincangan atas persoalan tadi. Maka, persoalannya ialah apakah faktor yang menyebabkan Nabi SAW. memilih kaedah perbincangan untuk menjawab pertanyaan tersebut. Dengan merujuk cerita bagaimana seorang sahabat meminta izin untuk menjawab soalan dan keizinan yang diberikan oleh Nabi SAW kepadanya sebagaimana hadis di atas, dapat difahami bahawa Nabi SAW telah mengambil kira perbincangan ilmu, tahap keyakinan diri dan ilmu muridnya sebelum memilih kaedah perbincangan ini. Selain itu, dimensi murid iaitu tahap keilmuan murid, kemampuan memahami isi pelajaran dan minat untuk mendalami ilmu sebagaimana yang telah dibuktikan oleh Abu Bakar r.a menyebabkan Nabi SAW. memilih untuk menggunakan kaedah ini. Kesimpulannya, merujuk kepada dimensi guru iaitu tahap pengetahuan tentang murid yang jelas dan perancangan untuk menggalakkan penglibatan murid ke dalam pengajaran dan juga dimensi murid iaitu minat dan ilmu sedia ada maka rahsia di sebalik kaedah perbincangan yang dilakukan oleh Nabi SAW. dapat difahami dengan lebih sempurna.

3.3.6 Hadis 6 : Kaedah Uji Minda

Daripada Muaz bin Jabal r.a., bahawa ketika Nabi SAW akan mengutuskan beliau ke Yaman, Baginda SAW. bersabda : "Bagaimana engkau akan memberikan keputusan apabila ada sebuah hukum dihadapkan kepadamu?". Muaz menjawab : "Saya akan memutuskan hukum menggunakan kitab Allah". Baginda SAW bersabda, "Seandainya engkau tidak mendapatkannya dalam kitab Allah?". Muaz menjawab : "Saya akan kembali kepada sunnah Rasulullah SAW". Baginda SAW bersabda : "Seandainya engkau tidak mendapatkannya dalam sunnah Rasulullah SAW serta dalam kitab Allah?". Muaz menjawab : "Saya akan berijtihad menggunakan pendapat saya, dan saya tidak akan mengurangi." Kemudian Rasulullah SAW menepuk dadanya dan berkata, "Segala puji bagi Allah yang telah memberikan petunjuk kepada utusan Rasulullah SAW untuk melakukan apa yang membuat senang Rasulullah SAW". [HR Abu Daud, Bab : Berijtihad Dengan Akal Untuk Memberi Keputusan Hukum][10].

Analisis hadis 6:
Hadis ini menceritakan bahawa Nabi SAW telah menguji tahap ilmu seorang sahabat iaitu Muaz bin Jabal r.a. sebelum beliau diutuskan sebagai guru untuk mengajarkan agama ke Yaman. Setelah uji minda ini dilalui dengan baik oleh Muaz bin Jabal, Nabi SAW memuji dan mendoakannya sesuai dengan perlakuan seorang guru kepada muridnya. Menurut Abu Ghuddah (1997) di antara tujuan Nabi SAW bertanya atau menguji para sahabat ialah untuk mengetahui sejauh mana kecerdasan dan penguasaan ilmu para sahabatnya. Kaedah uji minda pernah digunakan oleh Nabi SAW untuk menunjukkan rasa bangga dan pengiktirafan kepada sebahagian para sahabat r.anhum sebagaimana hadis berikut : "*Hai Abu Munzir! tahukah kamu, ayat manakah di antara ayat-ayat al-Quran yang ada padamu yang paling utama?" Abu Munzir berkata ; saya menjawab, "Allah dan rasul-Nya yang lebih mengetahui." Baginda SAW bertanya lagi, "Hai Abu Munzir, tahukah kamu, ayat manakah di antara ayat-ayat al-Quran yang ada padamu yang paling utama?" Abu Mundzir berkata ; Saya menjawab : "Allahu La Ilaaha Illa Huwa al-Hayyu al-Qayyum." Abu Munzir berkata; lalu Baginda SAW menepuk dadaku seraya bersabda : "Demi Allah, semoga dadamu dipenuhi dengan ilmu, wahai Abu Munzir*". [HR Muslim, Bab : Keutamaan surah al-Kahfi dan ayat kursi][6]. Ini bermakna, kaedah uji minda ini merupakan kaedah yang sesuai jika guru bermaksud untuk mengenalpasti tahap penguasaan murid atau memberikan pengiktirafan kepada murid sebagaimana yang telah dilakukan oleh Nabi SAW kepada Muaz ibn Jabal dan juga Ubai ibn

Kaab r.a. Kesimpulannya, dengan merujuk kepada dimensi guru iaitu perancangan, tujuan serta objektif pengajaran dan sikap guru kepada murid serta dimensi murid iaitu tugas khas, pengiktirafan kepada seorang murid dan bakat istimewa seorang murid, maka rahsia di sebalik kaedah uji minda yang dilaksanakan oleh Nabi SAW dapat diikuti dengan lebih tepat.

3.3.7 Hadis 7 : Kaedah Bercerita

Daripada Abu Hurairah, dia berkata Rasulullah SAW bersabda : "Pada suatu ketika ada seorang lelaki sedang berjalan melalui suatu jalan, lalu dia merasa sangat kehausan. Kebetulan dia menemui sebuah telaga, lalu dia turun ke telaga itu untuk minum. Setelah keluar daripada telaga, dia melihat seekor anjing menjulurkan lidahnya menjilat-jilat tanah kerana kehausan. Orang itu berkata dalam hatinya ; “Alangkah hausnya anjing itu, seperti yang baru aku alami”. Lalu dia turun kembali ke telaga, kemudian dia mengambil air dengan kasutnya, dibawa ke atas dan diberi minum anjing itu. Maka Allah berterima kasih kepadanya dan diampuni dosanya”. Para sahabat bertanya ; 'Ya, Rasulullah! Dapatkah pahala apabila kami menyayangi haiwan-haiwan ini?. Maka Nabi SAW menjawab : 'Ya, setiap perbuatan menyayangi makhluk hidup adalah berpahala". [HR al-Bukhari, Bab : Menyayangi Manusia dan Haiwan-haiwan][7].

Analisis Hadis 7:
Hadis ini menjelaskan bahawa Nabi SAW telah bercerita kepada para sahabat r.anhum tentang satu peristiwa seorang lelaki yang haus dan kebaikannya memberi minum kepada seekor anjing yang kehausan. Kemudian, Nabi SAW menceritakan lagi bahawa dengan kebaikan itu, Allah SWT telah merahmatinya dan mengampuni dosanya sehingga menarik pertanyaan para sahabat tentang kelebihan berbuat baik kepada binatang. Seterusnya, Nabi SAW memberikan jawapan kepada pertanyaan tadi dengan satu kaedah yang padat dan jelas. Justeru itu, persoalannya ialah kenapa Nabi SAW memilih kaedah bercerita berbanding kaedah lain untuk mengajar para sahabat kelebihan menyayangi makhluk bernyawa. Menurut Abu Ghuddah tujuan Nabi SAW bercerita adalah untuk menarik perhatian dan meneguhkan hati para sahabat r.anhum[4]. Manakala al-Quran pula menegaskan bahawa cerita-cerita orang terdahulu adalah untuk meneguhkan hati Nabi SAW dan juga memberi pengajaran kepada seluruh orang beriman sebagaimana firmanNya : “*Dan tiap-tiap berita dari berita rasul-rasul itu, Kami ceritakan kepadamu (Wahai Muhammad), untuk menguatkan hatimu dengannya, Dan telah datang kepadamu dalam berita ini kebenaran dan pengajaran serta peringatan bagi orang-orang yang beriman*” [Surah Hud,11:120)[1]. Ini bermakna, dimensi guru disebalik kaedah bercerita ialah kemampuan guru untuk meneguhkan hati dan menguatkan motivasi para pelajarnya selain kemampuan guru untuk menggalakkan pelajarnya mengambil dan mendalami sendiri hikmah di sebalik ceritanya. Di samping itu, kaedah bercerita juga menunjukkan seorang guru mampu menyusun semua isi pengajarannya dengan lebih padat dan menyeluruh. Selanjutnya, dimensi murid di sebalik kaedah bercerita ialah kesediaan murid untuk memberikan respon atau bertanyakan soalan yang sesuai sebagaimana yang telah ditunjukkan oleh para murid Nabi SAW. Kesimpulannya, merujuk kepada dimensi guru iaitu kemampuan untuk merancang isi dan objektif pengajaran dan dimensi murid iaitu tumpuan, perhatian dan pertanyaan seorang murid maka rahsia di sebalik kaedah bercerita yang dilaksanakan oleh Nabi SAW dapat diikuti dengan lebih berkesan.

3.3.8 Hadis 8 : Kaedah Syarahan

Daripada Abu Hurairah r.a., dia berkata ; Ketika Allah SWT memberikan kemenangan bagi rasulNya SAW keatas Kota Mekah, Baginda SAW. berdiri di hadapan manusia, maka Baginda SAW memuji Allah dan mensucikanNya kemudian bersabda, "Sesungguhnya Allah SWT telah melarang menawan gajah di Mekah ini dan menyerahkan urusannya kepada rasulNya dan kaum mukminin kerana di Mekah ini tidaklah dihalalkan bagi seorang pun sebelumku dan sesungguhnya pernah dihalalkan buatku pada suatu masa di suatu hari dan juga tidak

dihalalkan bagi seseorang setelah aku. Maka tidak boleh diburu binatang buruannya, tidak boleh dipotong durinya, dan tidak boleh diambil barang temuan di sana kecuali untuk diumumkan dan dicari pemiliknya. Barangsiapa yang dibunuh maka keluarga mangsa memiliki dua pilihan apakah dia akan meminta tebusan wang atau meminta balasan dari keluarga mangsa". Maka berkatalah al-Abbas, "Kecuali pohon Idzhir, kerana pohon itu kami gunakan sebagai wangian di kuburan kami dan di rumah kami". Maka Rasulullah SAW bersabda, "Ya, kecuali pohon Idzhir". Lalu berdiri Abu Syah, seorang penduduk Yaman dan berkata, "Wahai Rasulullah, tuliskanlah buatku? Rasulullah SAW berkata, "Tuliskanlah buat Abu Syah". Berkata,al-Walid bin Muslim ; Aku bertanya kepada al-Awzai : "Apa yang dia maksudkan dengan meminta tuliskanlah buatku wahai Rasulullah?" Dia berkata, "Isi khutbah tadi yang dia dengar dari Rasulullah SAW". [HR al-Bukhari, Bab : Bagaimana Mengumumkan Barang Temuan Milik Penduduk Mekah][7].

Analisis Hadis 8:
Hadis ini menceritakan bahawa Nabi SAW telah menyampaikan khutbah yang panjang kepada para sahabat r.anhum ketika peristiwa pembukaan Kota Mekah. Setelah selesai khutbah, salah seorang sahabat Nabi SAW yang bernama Abu Syah telah meminta dituliskan untuknya semua isi kandungan khutbah tersebut dan seterusnya Nabi SAW meminta para sahabat untuk menuliskannya untuk Abu Syah. Boleh difahami disini bahawa Nabi SAW telah menggunakan dua kaedah iaitu syarahan umum kepada semua sahabat dan mengarahkan disediakan nota catatan kepada seorang sahabat yang memerlukan. Dengan merujuk kepada dimensi diri Nabi SAW yang merupakan orang yang paling fasih ucapannya walaupun tidak boleh membaca dan menulis maka sebab di sebalik pemilihan kaedah syarahan ini boleh difahami dengan jelas. Namun, timbul pertanyaan apakah rahsia di sebalik kebenaran Nabi SAW. untuk mencatat isi khutbahnya?. Ini kerana Nabi SAW pernah melarang untuk menulis hadis sebagaimana sabdanya : "*Janganlah kalian menulis dariku, barangsiapa menulis dariku selain al-Quran hendaklah dihapus, dan ceritakanlah dariku dan tidak ada dosa. Barangsiapa berdusta atas namaku dengan sengaja, maka hendaklah dia menyiapkan tempatnya dari neraka*". [HR Muslim, Bab : Hukum menulis hadis][6]. Dengan merujuk kepada jawapan al-Awzai sebagaimana diceritakan di dalam hadis, maka boleh difahami bahawa dimensi murid iaitu Abu Syah yang berasal daripada Yaman yang tidak mampu menghafal dengan baik telah menyebabkan Nabi SAW memberikan arahan khusus kepadanya. Bahkan, menurut Mohd. Muhiden larangan menulis hadis ini sebenarnya ditujukan kepada sahabat yang kuat hafalan[11]. Ini untuk menggalakkan mereka menyampaikan hadis secara lisan. Kesimpulannya, merujuk kepada dimensi guru iaitu kemampuan untuk menyampaikan khutbah dengan baik dan juga dimensi murid iaitu kemampuan untuk menghafal atau tidak isi pengajaran guru maka rahsia di sebalik kaedah syarahan yang dilaksanakan oleh Nabi SAW dapat diikuti dengan lebih tepat.

4. Kesimpulan

Kesimpulannya, ada tiga dapatan yang boleh disenaraikan di sini. Pertama, terdapat empat dimensi guru yang akan mempengaruhi pendidikan Nabi SAW iaitu pertama, sifat kemanusiaan Nabi SAW seperti lupa, marah, sayang, benci dan seumpamanya, kedua, keterbatasan ilmu mahupun fizikal yang dimiliki oleh Nabi SAW seperti tidak mengetahui tarikh kiamat, takdir dan lain-lain, ketiga, objektif pendidikan Nabi SAW yang dinilai ialah kata-kata bukan perbuatannya dan, keempat, kebolehan dan kepakaran Nabi SAW yang tersendiri seperti kemahiran berpidato, kefasihan berbahasa dan lain-lain. Seterusnya dapatan kedua ialah terdapat empat dimensi murid yang akan mempengaruhi pendidikan Nabi SAW iaitu pertama, perbezaan biodata para sahabat r.anhum merangkumi sifat, fizikal, umur dan sebagainya, kedua, perbezaan pengetahuan sedia ada dan kefahaman ilmu para sahabat ra.anhum, ketiga, perbezaan peranan, minat, kelebihan dan kelemahan diri yang wujud di kalangan para sahabat r.anhum dan, keempat, perbezaan gaya pembelajaran, kaedah belajar dan kemahiran belajar di kalangan para sahabat r.anhum seperti kemahiran menulis, membaca, mendengar dan menghafal. Seterusnya dapatan ketiga ialah terdapat lapan

kaedah pendidikan yang digunakan oleh Nabi SAW iaitu demonstrasi, debat, perbandingan, soal jawab, perbincangan, uji minda, bercerita dan syarahan. Ringkasan dapatan kajian boleh difahami sebagaimana Jadual 1.

Jadual 1 : Kaitan Kaedah Pendidikan Nabi SAW. dengan Dimensi Guru dan Murid

No.	Permasalahan Hadis	Kaedah Pendidikan	Dimensi Guru	Dimensi Murid
1	Nabi SAW mengimamkan solat jemaah di atas mimbar dan para sahabat menjadi makmum.	Kaedah Demontrasi	Guru ingin mengajar para murid tatacara solat dengan sempurna di tempat yang senang dilihat oleh semua murid.	Bilangan murid yang ramai dan kedudukan mereka yang jauh di belakang.
2	Nabi SAW tidak memarahi seorang sahabat r.a. yang meminta izin untuk berzina	Kaedah Debat	Guru cuba memahami dan membetulkan salah faham murid.	Murid kurang ilmu, salah faham konsep dan jahil tentang hakikat sesuatu perkara
3	Nabi SAW tidak menjawab secara terus pertanyaan seorang sahabat r.a. tentang hukum menggantikan nazar haji ibunya yang telah mati.	Kaedah Perbandingan	Guru ingin menggalakkan pembelajaran kendiri murid melalui perbandingan hukum	Murid memiliki ilmu asas dan ada kemampuan untuk membuat keputusan hukum bersama-sama guru
4	Nabi SAW tidak menjawab soalan seorang sahabat perihal tarikh kiamat.	Kaedah Soal Jawab	Guru telah mengabaikan soalan yang tidak bermanfaat sebaliknya ingin menumpukan kepada isu yang lebih bermanfaat	Murid telah tersilap memilih keutamaan soalan yang akan ditanya
5	Nabi SAW membuka perbincangan untuk pertanyaan seorang sahabat tentang takwil mimpi	Kaedah Perbincangan	Guru ingin memberi peluang kepada semua murid dan menggalakkan perbincangan di antara mereka tentang sesuatu isu	Murid memiliki ilmu asas dan mampu memberikan pandangan dan jawapan seterusnya ingin meminta pengesahan daripada guru
6	Nabi SAW menguji kemampuan ilmu seorang sahabat yang akan diutus menjadi guru	Kaedah Uji Minda	Guru ingin menilai, menguji dan mengiktiraf kemampuan murid serta mendoakannya	Murid telah diberikan tugas khas dan juga memiliki kedudukan yang istimewa.
7	Nabi SAW bercerita tentang kisah kebaikan seorang lelaki kepada seekor anjing	Kaedah Bercerita	Guru ingin memberikan peneguhan hati dan motivasi kepada para murid	Murid bersedia untuk memberikan respon dan bersama-sama menyelami hikmah dan hukum disebalik cerita
8	Nabi SAW menyampaikan khutbah panjang dan mengarah untuk menulis isinya kepada seorang sahabat yang meminta nota catatan	Kaedah Syarahan	Guru mempunyai kemampuan untuk memberi khutbah dan syarahan dengan baik	Murid mempunyai kemampuan untuk mendengar dan memahami isi kandungan syarahan dengan baik.

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INFLUENCE OF HEAT INPUT ON THE HARDNESS OF COLD-ROLLED STEEL JOINTS USING COLD ARC WELDING TECHNOLOGY

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ABSTRACT

This study examines the effect of heat input on the hardness of cold-rolled steel joints using Cold Arc welding technology. Cold Arc welding, a type of Gas Metal Arc Welding (GMAW), is designed for low heat input, making it ideal for welding thin materials with minimal thermal distortion. The research focuses on how varying levels of heat input influence the hardness and microstructure of the welded joints made from cold-rolled steel plates of different thicknesses. Experimental results indicate that lower heat inputs result in higher hardness, while higher heat inputs lead to reduced hardness and coarser grain structures in the weld metal and heat-affected zone (HAZ). The study involves preparing and welding specimens with different heat inputs, followed by hardness testing using a Vickers hardness tester. The data analysis reveals significant correlations between heat input and hardness values. The findings provide valuable insights into optimizing welding parameters for achieving superior mechanical properties and structural integrity in cold-rolled steel joints, enhancing their performance and reliability in industrial applications. This research contributes to the broader understanding of heat management in welding processes and its impact on material properties.

Keywords: Cold Arc Welding, Heat Input, Hardness Testing

1. Introduction

Cold Arc welding is a modern variation of Gas Metal Arc Welding (GMAW) that uses low heat input to improve the quality and productivity of welding, particularly for thin materials[1],[2]. This technique minimizes thermal distortion and reduces common welding issues such as cracking and residual stress, enhancing the overall integrity of the welds.

Automated welding, including robotic welding, is becoming increasingly important in the industry to enhance production efficiency and consistency. However, managing heat input is crucial as it directly affects the microstructure and mechanical properties of the welded joints [3],[4].

Hardness testing is a key method for evaluating the mechanical properties of welded joints. It provides insights into the material's resistance to deformation and wear. By conducting hardness tests, the effects of different heat inputs on the welded joints' hardness and identify the optimal welding parameters.

This study focuses on the Cold Arc welding process and its impact on the hardness of cold-

rolled steel joints. The primary objective is to understand how varying heat inputs influence the hardness and microstructure of the welds. By analyzing the hardness test results, we aim to optimize the welding parameters to achieve superior mechanical properties and ensure the durability and reliability of the joints in industrial applications.

2. Methodology

The Cold Arc welding machine from the EWM brand was used for this study. The welding process employed solid welding wires of grade ER70S-6 for mild steel, with a diameter of 1.0 mm. A shielding gas mixture consisting of 82% Argon and 18% CO2 was used to protect the weld area from contamination. Two different joint thicknesses were prepared: 1.0 mm and 0.8 mm, using a lap joint configuration. The choice of these thicknesses and joint types was based on data obtained from a comprehensive literature review [5]. Specific welding parameters from previous studies were referred to, ensuring consistency and reliability in the experiments [6].

The process parameters used in the experiments are presented in Table 1. These parameters were carefully selected to cover a range of heat inputs, allowing observation of their effects on the hardness and mechanical properties of the welded joints. By following these methods, the study aims to systematically investigate the impact of different heat inputs on the hardness of cold-rolled steel joints, using the Cold Arc welding technique.

Table 1. Welding Parameter

Level	Current (A)	Voltage (V)	Welding speed (mm/min)	Heat input (KJ/mm)
1	40	3.77	550	0.016
2	40	10.50	550	0.046
3	48	14.5	350	0.119

3. Result and Discussion

A schematic of the weld geometry is shown in Figure 1, highlighting three zones: base metal, Heat Affected Zone (HAZ), and fusion zone. Measurements were taken on both Plate 1 and Plate 2 during the experiment. Increasing energy input affects the mechanical and metallurgical properties of the weld metal and increases the consumption of flux, which can lead to changes in the alloying elements.

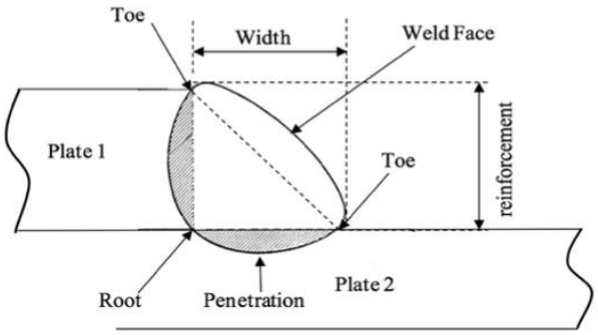


Fig. 1. Weld bead geometry characteristics

Table 2 shows the effect of hardness on the parent metal. Figure 2 illustrates that hardness results are not equal. Increasing the heat input from 0.016 KJ/mm to 0.119 KJ/mm increased the hardness of the sample test width from 179.3 HV to 239.1 HV.

Table 2. Microhardness result

Distance from the centre	Heat Input (KJ/mm)		
	0.016	0.046	0.119
	Hardness (HV)		
-5	130.1	132.1	125.8
-4	124.7	127.1	128.8
-3	145.7	150.9	170.6
-2	158.3	163.6	189.4
-1	179.3	209.1	239.1
1	135.2	139.6	174
2	127.4	128.1	144.3
3	130.3	133.2	133.6
4	118.7	113.6	122.2
5	112.8	115.3	129.4

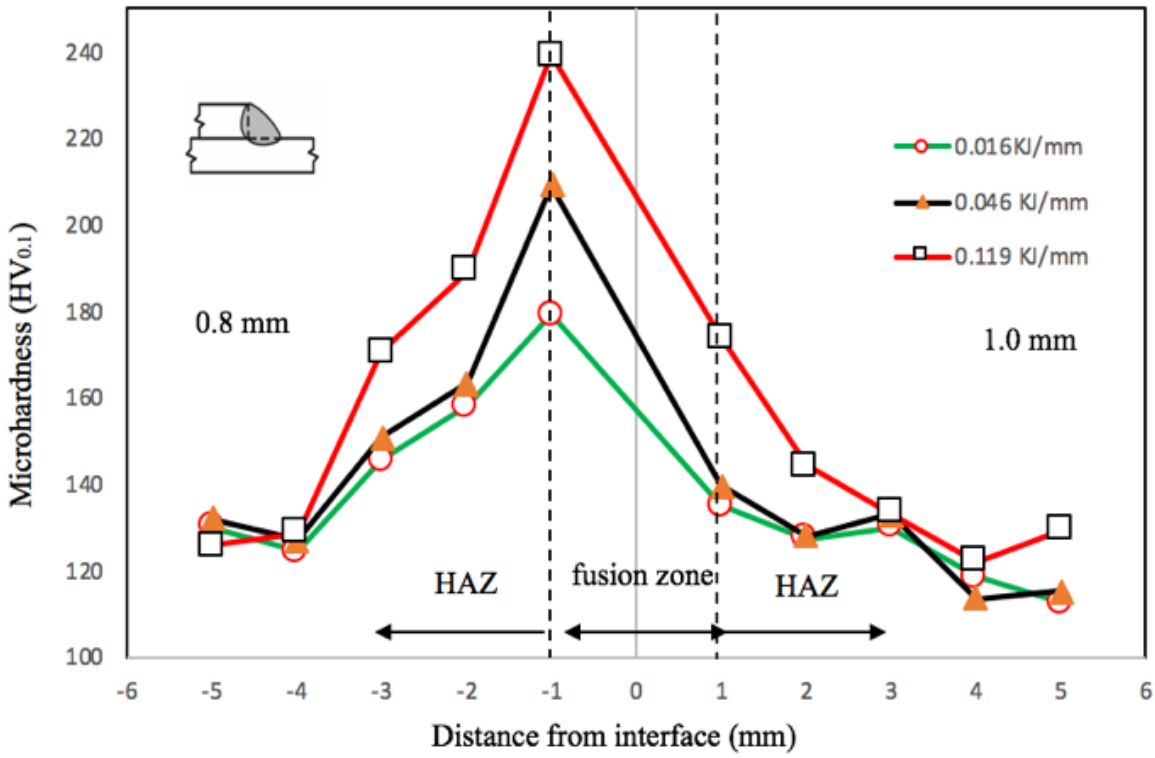
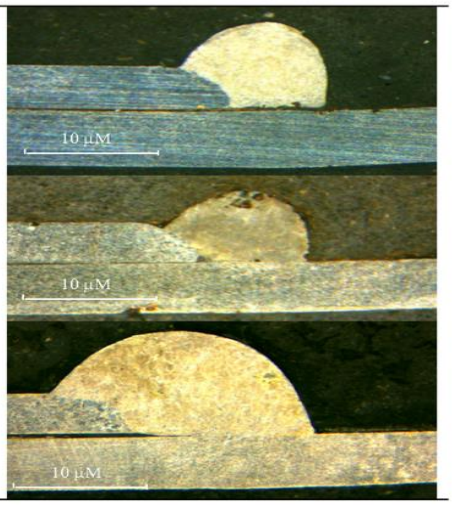


Fig. 2. Graph hardness test distance from the fusion line

As shown in Figure 2, the microhardness values are related to heat input along a distance line. The results show that higher heat input significantly affects the microhardness of the weld plate. This finding is consistent with previous research [4], [7], and [8]. The lowest hardness value is found in the base material zone, while hardness increases closer to the fusion zone. The impact is more significant in the 0.8 mm thick plate compared to the 1.0 mm thick plate. The thinner plate is more affected by heat input due to its higher temperature saturation over a small thickness, whereas for lower temperatures, the effect is greater.

Table 3 shows the macrograph of the specimen tested in this experiment. It is important to determine the right welding speed and current to minimize microstructural changes and distortions in the product. The proper welding speed and current are essential to minimize microstructural changes and distortions in the final product.

Table 3. Macrograph of the specimen test

Heat input (KJ/mm)	Macrostructure profile
0.016	
0.046	
0.119	

4. Conclusion

The study demonstrates that the heat input significantly influences the hardness of cold-rolled steel joints welded using Cold Arc technology. The following are the summaries that have been obtained.

i. Effect of Heat Input on Hardness: As the heat input increased from 0.016 KJ/mm to 0.119 KJ/mm, the hardness of the welded joints also increased. This was observed in the hardness measurements, which rose from 179.3 HV to 239.1 HV.
ii. Variation in Hardness Across Zones: The hardness was lowest in the base material zone and increased towards the fusion zone. This gradient indicates that the area closest to the weld is most affected by the heat input, experiencing the highest increase in hardness.
iii. Impact of Plate Thickness: The study found a more pronounced effect of heat input on the thinner plates (0.8 mm) compared to the thicker plates (1.0 mm). Thinner plates are more susceptible to changes in temperature due to their smaller volume and higher surface area-to-volume ratio.
iv. Consistency with Previous Research: The results align with findings from previous studies, reinforcing the understanding that higher heat inputs lead to increased hardness in the welded joints.
v. Implications for Welding Parameters: To optimize the mechanical properties of welded joints, it is crucial to carefully select welding parameters such as speed and current. Proper control of these parameters can help minimize undesirable microstructural changes and distortions in the final product.

This research underscores the importance of controlling heat input during welding to achieve desired mechanical properties and maintain the structural integrity of the weld. Proper parameter selection is essential for producing high-quality welds with consistent and reliable properties.

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HUBUNGAN PENGUASAAN TOPIK PERSAMAAN (ALGEBRA) DENGAN KEPUTUSAN PEPERIKSAAN AKHIR BAGI SUBJEK *BUSINESS MATHEMATICS*: SATU KAJIAN TERHADAP PELAJAR DIPLOMA, JABATAN PERDAGANGAN, POLITEKNIK TUANKU SULTANAH BAHIYAH

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ABSTRAK

Persamaan algebra merupakan kemahiran asas yang penting dalam kursus matematik bermula dari peringkat sekolah rendah, menengah dan seterus ke peringkat Institut Pengajian Tinggi (IPT). Kajian ini dijalankan bertujuan untuk mengkaji tahap penguasaan pelajar berkaitan topik persamaan algebra, hubungan penguasaan topik persamaan (algebra) terhadap keputusan peperiksaan akhir dan hubungan antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah. Analisis kolerasi digunakan bagi mencapai objektif kajian ini. Hasil kajian jelas menunjukkan bahawa terdapat hubungan yang signifikan di antara penguasaan pelajar berkaitan topik persamaan algebra terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics*. Ini menunjukkan bahawa pelajar yang menguasai topik persamaan (algebra) mendapat keputusan yang baik dalam peperiksaan akhir bagi subjek *Business Mathematics.* Manakala tiada hubungan yang signifikan di antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir. Ini bermaksud markah penilaian kerja kursus yang tinggi bukan jaminan untuk pelajar lulus dengan baik dalam peperiksaan akhir bagi subjek *Business Mathematics.* Diharapkan hasil kajian ini dapat membantu pensyarah subjek *Business Mathematics* untuk memberi penekanan agar pelajar dapat menguasai topik persamaan (algebra) dengan baik. Dicadangkan juga pensyarah untuk membangunkan satu inovasi pengajaran dan pembelajaran untuk membantu pelajar dalam penguasaan topik tersebut dengan lebih baik. Seterusnya dapat meningkatkan peratusan pencapaian kelulusan cemerlang pelajar dalam subjek *Business Mathematics* dan sekaligus melahirkan graduan yang celik matematik.

Kata Kunci: Persamaan Algebra, Penilaian Kerja Kursus, Penilaian Akhir, *Business Mathematics*, Diploma, Perdagangan.

1. Pengenalan

Dalam pendidikan peringkat diploma, terutamanya dalam bidang perniagaan, subjek matematik memainkan peranan yang penting. *Business Mathematics* adalah salah satu subjek teras yang memerlukan penguasaan konsep-konsep asas matematik termasuk algebra. Algebra, yang merangkumi topik-topik seperti persamaan linear, persamaan kuadratik, dan sistem persamaan, adalah asas penting yang digunakan dalam pelbagai aplikasi perniagaan seperti analisis data, penyediaan bajet, dan peramalan kewangan. Oleh itu, penguasaan dalam topik persamaan dianggap sebagai faktor penting yang boleh mempengaruhi prestasi pelajar dalam subjek ini [5].

Di peringkat diploma Politeknik Malaysia Kementerian Pendidikan Tinggi (KPT) kursus *Business Mathematics* merupakan salah satu kursus teras disiplin (Discipline Core) yang wajib diambil oleh pelajar Diploma Pengajian Perniagaan dan Diploma Perakaunan. Kursus

ini perlu dipelajari oleh para pelajar sebagai pengetahuan asas dalam menyediakan pengetahuan tentang pelbagai konsep asas matematik, masalah pengurusan dan teknik penyelidikan operasi asas berdasarkan kewangan dan kaedah kuantitatif. Kursus ini turut menekankan penggunaan konsep penyelesaian matematik dalam perniagaan dan pengurusan. Ianya secara tidak langsung memberi maklumat dan pendedahan kepada pelajar tentang amalan asas dalam dunia perniagaan dan kewangan.

2. Sorotan Kajian

Algebra adalah satu komponen penting dalam pendidikan matematik yang membentuk asas bagi banyak konsep dan aplikasi dalam pelbagai bidang termasuk perniagaan. Penguasaan dalam algebra adalah penting kerana ia membina kemampuan pelajar untuk menyelesaikan masalah, berfikir secara analitikal dan memahami hubungan antara pembolehubah dalam situasi perniagaan [3].

Beberapa kajian menunjukkan bahawa terdapat hubungan yang signifikan antara penguasaan algebra dengan prestasi akademik dalam mata pelajaran matematik peringkat tinggi. Pelajar yang mempunyai asas algebra yang kukuh cenderung untuk mencapai keputusan yang lebih baik dalam subjek matematik yang lebih kompleks [18]. Ini kerana pemahaman yang baik terhadap konsep algebra membolehkan pelajar mengaplikasikan pengetahuan tersebut dalam pelbagai konteks, termasuk dalam *Business Mathematics.*

Terdapat beberapa faktor yang mempengaruhi penguasaan pelajar dalam algebra. Kajian oleh [12] menunjukkan bahawa sikap pelajar terhadap matematik, kaedah pengajaran yang digunakan, serta sokongan yang diberikan oleh guru memainkan peranan penting dalam menentukan tahap penguasaan algebra. Pelajar yang menghadapi kesukaran dalam memahami konsep asas algebra sering kali menunjukkan prestasi yang lemah dalam subjek-subjek matematik yang memerlukan penggunaan algebra.

Kajian oleh [8] menunjukkan bahawa penguasaan algebra mempunyai korelasi yang positif dengan keputusan peperiksaan dalam subjek *Business Mathematics*. Pelajar yang menunjukkan penguasaan yang tinggi dalam topik persamaan algebra cenderung untuk mencapai keputusan yang lebih baik dalam peperiksaan akhir subjek ini. Ini menunjukkan bahawa pemahaman yang mendalam tentang algebra adalah kritikal bagi kejayaan akademik dalam *Business Mathematics.*

Selain itu, menurut [6] hubungan antara penguasaan algebra dan prestasi dalam matematik perniagaan di kalangan pelajar kolej menunjukkan bahawa pelajar yang menunjukkan penguasaan tinggi dalam topik algebra cenderung untuk mencapai prestasi yang lebih baik dalam peperiksaan matematik perniagaan.

Kajian oleh [2] pula meneliti cabaran yang dihadapi oleh pelajar diploma dalam menguasai topik algebra. Antara cabaran utama yang dikenalpasti termasuk kesukaran dalam memahami konsep asas algebra dan kurangnya sokongan akademik yang mencukupi. Kajian ini mencadangkan agar lebih banyak sokongan diberikan melalui sesi tutor dan bahan pembelajaran tambahan untuk membantu pelajar mengatasi cabaran ini.

2.1 Permasalahan kajian

Berdasarkan pencapaian akademik dalam peperiksaan akhir semester Jabatan Perdagangan di Politeknik Sultanah Bahiyah (PTSB) bagi kursus *Business Mathematics* untuk Sesi 1 2022/2023, sebanyak 4.82% pelajar telah gagal. Peratusan kegagalan ini meningkat kepada 19.15% pada Sesi 2 2022/2023. Ini bermakna bilangan pelajar yang gagal meningkat daripada 8 orang pada Sesi 1 2022/2023 kepada 27 orang pada Sesi 2 2022/2023. Peningkatan bilangan dan peratusan kegagalan yang lebih tiga kali ganda ini amat

membimbangkan pensyarah yang mengajar kursus tersebut. Pelbagai persoalan timbul bagi mendapatkan kepastian tentang permasalah ini. Mengambil kira keperluan terhadap kepentingan dan penguasaan pelajar bagi kursus *Business Mathematics*, penyelidik ingin mendapatkan kepastian apakah tahap penguasaan pelajar dalam topik persamaan algebra, mengkaji hubungan penguasaan topik persamaan (algebra) terhadap keputusan peperiksaan akhir dan hubungan antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

2.2 Persoalan Kajian

i. Apakah tahap penguasaan pelajar berkaitan topik persamaan algebra dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah?

ii. Apakah terdapat hubungan yang signifikan di antara penguasaan pelajar bagi topik persamaan algebra terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah?

iii. Apakah terdapat hubungan yang signifikan di antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah?

2.3 Objektif Kajian

i. Mengenalpasti tahap penguasaan pelajar berkaitan topik persamaan algebra dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

ii. Mengenalpasti hubungan di antara penguasaan pelajar berkaitan topik persamaan algebra terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

iii. Mengenalpasti hubungan antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

2.4 Skop Kajian

Kajian ini dijalankan ke atas 294 orang pelajar Diploma Pengajian Perniagaan dan Diploma Perakaunan di Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah yang mengambil subjek *Business Mathematics* iaitu melibatkan pelajar semester satu dan semester dua sahaja.

3. Metodologi Kajian

3. 1 Populasi dan sampel Kajian

Populasi kajian terdiri daripada keseluruhan pelajar semester satu dan semester dua program Diploma Pengajian Perniagaan dan Diploma Perakaunandi Politeknik Tuanku Sultanah Bahiyah. Jumlah populasi kajian terdiri daripada 294 orang pelajar yang mengambil kursus *Business Mathematics* bagi Sesi 1 2022/2023 dan Sesi 2 2022/2023. Manakala sampel kajian adalah kesemua 294 orang pelajar [7].

3.2 Rekabentuk Kajian

Instrumen kajian adalah menggunakan data sekunder yang diperolehi daripada Sistem Pengurusan Maklumat Politeknik (SPMP) bagi mendapatkan data berkaitan markah penilaian kerja kursus dan markah Kuiz 1 kursus *Business Mathematics* bagi Sesi 1 2022/2023 dan Sesi 2 2022/2023. Manakala keputusan rasmi Peperiksaan Akhir kursus *Business Mathematics* bagi Sesi 1 2022/2023 dan Sesi 2 2022/2023 diperolehi daripada Unit Peperiksaan Politeknik Tuanku Sultanah Bahiyah. Data sekunder yang diperolehi adalah betul dan tidak diragui bagi memperolehi keputusan kajian yang tepat.

3.3 Analisis Data

Analisis deskriptif digunakan untuk menganalisa data yang diperolehi daripada sumber sekunder. Penyelidik menggunakan perisian *Statistical Packages for Science Social* 26 (SPSS 26) untuk mendapatkan min dan sisihan piawai bagi menjawab persoalan kajian yang pertama dalam menilai tahap penguasaan pelajar berkaitan topik persamaan algebra dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah. Melalui analisis deskriptif, tahap penguasaan pelajar boleh dikategorikan kepada tahap rendah, sederhana, dan tinggi berdasarkan min skor yang diperolehi [10]. Markah penilaian kerja kursus bagi Kuiz 1 dipilih untuk menguji tahap penguasaan pelajar kerana penilaian Kuiz 1 merangkumi topik persamaan algebra.

Jadual 1. Penilaian Kerja Kursus Kuiz 1

Kerja Kursus	Topik terlibat	Markah
Kuiz 1	Persamaan (algebra)	100

Manakala analisis Korelasi Pearson dilaksanakan untuk menjawab persoalan kajian kedua dan ketiga bagi mengenalpasti hubungan di antara penguasaan pelajar bagi topik algebra terhadap keputusan peperiksaan akhir dan hubungan di antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

4. Keputusan dan Perbincangan

4.1 Analisis Deskriptif

Analisis menunjukkan nilai min yang ditafsir berdasarkan skala *Likert* Tiga Mata yang memaparkan skor min bagi menentukan tahap penguasaan pelajar dalam topik persamaan algebra bagi kursus *Business Mathematics* [8]. Data kajian yang diperolehi diinterpretasikan dengan merujuk dirujuk sebagai indikator bagi setiap nilai purata skor min yang diprolehi.

Jadual 2. Pentafsiran Skor Min (Tahap penguasaan)

Tahap penguasaan	Skor Min
Rendah	0 - 49
Sederhana	50 - 74
Tinggi	75 -100

4.2 Dapatan Kajian

Jadual 3 di bawah menunjukkan analisis latarbelakang responden dalam peratusan dan frekuensi (*f*). Daripada 294 pelajar semester 1 & 2, Program Diploma Pengajian Perniagaan (DPM) adalah 79.9%, manakala Program Diploma Akauntansi (DAT) adalah 20.1% yang menjadi responden. 83 orang adalah lelaki dan 211 orang adalah perempuan. Responden

yang tertinggi adalah berbangsa India iaitu sebanyak 48.5% dan yang terendah adalah lain-lain bangsa iaitu seramai 12%.

Jadual 3. Latar Belakang Responden

Bil	Kategori	Kumpulan	Frekuensi (*f*)	Peratus (%)
1.	Jantina	Lelaki	83	28.2
		Perempuan	211	71.8
2.	Bangsa	Melayu	105	35.8
		Cina	34	11.5
		India	143	48.5
		Lain-lain	12	4.2
3.	Program	Diploma Pengajian Perniagaan (DPM)	235	79.9
		Diploma Akauntansi(DAT)	59	20.1

Objektif 1: Mengenalpasti tahap penguasaan pelajar berkaitan topik persamaan algebra dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

Jadual 4 menunjukkan analisis deskriptif min bagi penilaian kursus Kuiz 1. Nilai skor min yang diperolehi adalah 61.25. Ini bermaksud penguasaan pelajar berkaitan topik persamaan algebra dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah berada pada tahap sederhana.

Dapatan ini disokong oleh kajian yang dijalankan [1] mendapati bahawa majoriti pelajar menunjukkan tahap penguasaan yang sederhana dalam topik persamaan algebra. Hasil analisis menunjukkan pelajar menghadapi kesukaran dalam memahami konsep asas dan aplikasi persamaan algebra dalam konteks perniagaan. Manakala kajian yang dijalankan oleh [11] turut mendapati bahawa majoriti pelajar menunjukkan tahap penguasaan yang rendah dalam topik persamaan algebra. Namun, terdapat faktor lain yang mempengaruhi penguasaan pelajar termasuk kaedah pengajaran, tahap minat pelajar, dan ketersediaan sumber pembelajaran [10].

Jadual 4. Analisis deskriptif min

Penilaian Kursus	N	Minimum	Maksimum	Min	Sisihan piawai
Kuiz 1	294	22.5	100	61.25	8.83

Objektif 2: Mengenalpasti hubungan di antara penguasaan pelajar berkaitan topik persamaan algebra terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

Jadual 5. Ujian Korelasi Pearson bagi Kuiz 1, Penilaian Kerja Kursus dan Peperiksaan Akhir

Kolerasi	Ujian Statistik	Kuiz 1	Penilaian Kerja Kursus	Peperiksaan Akhir
Kuiz 1	Pearson Correlation	1		
	Sig. (2-tiled)			
Penilaian Kerja Kursus	Pearson Correlation	0.503	1	
	Sig. (2-tiled)	0.00		
Peperiksaan Akhir	Pearson Correlation	0.003	.422	1
	Sig. (2-tiled)	0.00	0.03	

Merujuk kepada Jadual 5, hasil dapatan jelas menunjukkan bahawa terdapat hubungan yang signifikan di antara penguasaan topik persamaan algebra (Kuiz 1) dan keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah. Nilai p ialah 0.00 < 0.05, menunjukkan

bahawa hubungan ini adalah signifikan. Kuiz 1 mempunyai nilai p=0.00, manakala peperiksaan akhir mempunyai nilai p=0.003. Nilai p yang diperolehi dalam kajian ini menyokong dapatan bahawa penguasaan algebra memainkan peranan penting dalam menentukan prestasi akademik pelajar. Manakal nilai korelasi (r) yang diperoleh ialah 0.53 dan ini menunjukkan terdapat hubungan positif sederhana di antara penilaian kerja kursus (Kuiz 1) terhadap peperiksaan akhir. Ini menunjukkan bahawa pelajar yang lebih mahir dalam algebra cenderung untuk mencapai keputusan yang lebih baik dalam peperiksaan akhir subjek *Business Mathematics.*

Dapatan ini selari dengan kajian oleh [9],[14] dan [15] di mana terdapat kolerasi positif yang signifikan antara penguasaan algebra dan prestasi akademik dalam matematik perniagaan. Pelajar yang menguasai algebra dengan baik cenderung untuk mencapai keputusan yang lebih tinggi dalam peperiksaan matematik. [10] turut menyokong bahawa terdapat korelasi positif yang signifikan antara penguasaan algebra dan keputusan peperiksaan akhir. Kajian ini mendapati kedua-dua pemboleh ubah ini, menunjukkan bahawa penguasaan dalam algebra adalah penentu penting kepada kejayaan akademik dalam subjek *Business Mathematics*.

Objektif 3: Mengenalpasti hubungan antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

Berdasarkan Jadual 5, hasil dapatan menunjukkan bahawa tidak terdapat hubungan yang signifikan antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah. Ini kerana nilai $p = 0.422 > 0.05$, menunjukkan bahawa hubungan tersebut tidak signifikan.

Keputusan kajian ini turut disokong oleh [17], mendapati bahawa tiada hubungan di antara markah kerja kursus dan prestasi peperiksaan akhir. Faktor seperti tekanan peperiksaan dan kebolehan mengurus masa dalam peperiksaan akhir mungkin berbeza daripada kerja kursus yang lebih terstruktur dan dijangka. Dapatan kajian ini juga selari dengan kajian oleh [16], turut mendapati tiada hubungan signifikan antara penilaian kerja kursus terhadap keputusan peperiksaan akhir dan mencadangkan perlunya pendekatan penilaian yang lebih holistik.

5. Kesimpulan dan Cadangan

Secara keseluruhannya, kajian yang dijalankan telah berjaya mencapai objektif yang telah ditetapkan di mana hasil dapatan menunjukkan bahawa tahap penguasaan pelajar dalam topik persamaan algebra dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah adalah sederhana.

Dapatan kajian juga menunjukkan jelas bahawa terdapat hubungan signifikan di antara penguasaan topik persamaan algebra dengan keputusan peperiksaan akhir dalam subjek Business Mathematics. di antara penguasaan pelajar berkaitan topik persamaan algebra terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah. Hubungan positif ini adalah bersifat sederhana.

Selain itu, dapatan kajian jelas menunjukkan tiada hubungan yang signifikan di antara markah penilaian kerja kursus terhadap keputusan peperiksaan akhir bagi subjek *Business Mathematics* dalam kalangan pelajar Diploma, Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah.

Diharapkan hasil dapatan kajian ini dapat digunakan untuk pensyarah mempelbagaikan lagi pendekatan pengajaran interaktif seperti pembelajaran berasaskan masalah (PBL) dan penggunaan teknologi seperti perisian matematik interaktif [13]. Ini bagi meningkatkan pemahaman dan minat pelajar terhadap topik persamaan algebra. Selain itu, pensyarah digalakkan mengadakan lebih banyak sesi latihan dan tutorial yang fokus pada persamaan algebra untuk memastikan pelajar benar-benar memahami konsep asas sebelum beralih ke topik yang lebih kompleks.

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STUDENT ENGAGEMENT, INTERACTION AND ENJOYMENT ON GAME BASED LEARNING ACTIVITIES IN THE CLASSROOM: MASTERSQUARE GAME

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ABSTRACT

Non-digital game-based learning (NDGBL) uses physical or analog games like board games, card games, role-playing games, and physical activities to teach and engage students without electronic devices. Using non-digital games in the classroom creates an engaging and effective learning environment. The MasterSquare game was introduced in the Principles of Management course to enhance educational outcomes and encourage active learning. This study aims to evaluate the MasterSquare Game's effectiveness in increasing student engagement, fostering social interaction, and enhancing class enjoyment. The use of games in education has the potential to change traditional classroom settings into interactive spaces. This study, which included 65 students from Politeknik Tuanku Sultanah Bahiyah's Commerce Department, demonstrated that using NDGBL in the classroom improves students' engagement, enjoyment, and social interactions.
Keywords: NDGBL, engagement, enjoyment, students' interaction.

1. Introduction

One of the most ancient and effective pedagogical concepts that has been used throughout human history is learning through games. The goal of game-based learning environments is to increase students' motivation and engagement levels when they are learning difficult subjects [1][2]. Students can freely move around in game-based learning settings thanks to the usage of learning-promoting game components, and games have been designed to assist successful self-regulation (SRL) learning [3]. Active learning environments have replaced traditional classrooms at colleges and universities with the goal of facilitating student-centred instruction [3][4]. The development of active learning classrooms is a component of the larger educational movement that aims to get students interested and participating in their education. When students actively participate in the learning process, they are said to be engaged [5]. Participation in learning activities by students is referred to as engagement, and it usually includes behavioural, emotional, and cognitive aspects [6].

Students' behaviour and activities are referred to as the behavioural dimension. By responding to the teacher's questions and completing the required assignments, students can engage in active participation in class. The main way to affect student engagement is through the teacher's capacity to create an immediate learning environment [7]. They can also exhibit a more passive involvement by following guidelines and directives. Affective responses of students to the educational process and classroom setting are all included in the category of emotional engagement. This demonstrates how much students respect and identify with their school activities. As an example, students who are emotionally engaged will express feeling happy, proud, inquisitive, and enthusiastic about their studies and assignments. Students' invisible mental processes that encourage commitment to learning and skill mastery are known as cognitive engagement. It involves the application of self-regulated and deep-processing techniques, such as maintaining attention and concentration, avoiding distractions, and attempting to comprehend the course material and find answers to issues [6]. In general, doing, feeling, and thinking are broad definitions for these three aspects of student engagement.

Nowadays, a variety of instructional approaches to small group learning are referred to as collaborative learning. These techniques include, but are not limited to, cooperative learning, team-based learning, peer tutoring, study groups, project-based learning, problem-based learning, and learning communities [8]. The most beneficial approach to active learning is collaborative learning. Instructors use collaborative learning as a strategy to speed up learning and boost student performance [4]. Students who engage in collaborative learning manage their relationships, create content, and participate in group activities [4]. Students act as learning resources for one another in an interactive learning environment by interacting with one another, observing one another's work, exchanging ideas, and forming group decisions [4]. Students' enjoyment in the classroom is a critical component that has a big influence on their learning results and attitudes. Studies have indicated that when academic experiences are enjoyable for students, it increases their motivation, engagement, and willingness to actively participate in class activities [9]. According to Pekrun [10] enjoyment is a stimulating and positive emotion, recognized as an activity-related emotion. It occurs when an individual's personal goals meet the learning task [11], as long as the task is regarded positively, and the activity is viewed as adequately manageable [12]. Positive emotions among students, such as enjoyment in class, are a vital element of the educational experience and have an impact on learning and successful outcomes.

Fostering student involvement, improving peer connection, and making sure learning is enjoyable are all important objectives in modern education. When it comes to sustaining student attention and encouraging active engagement, traditional teaching methods frequently fall short. Educators are investigating alternative techniques in response, such as game-based learning. However, since digital games have received so much attention, there is a knowledge gap about the ability of non-digital games to produce similar educational results.

This research seeks to determine the ability of this game and answer the issue of how effective is the MasterSquare Game in improving student engagement, social interaction, and enjoyment during class sessions. Due to that, the MasterSquare game was created with the intention to be a fun and adaptable approach for lecturers and students to engage in class activities. MasterSquare Game is a non-digital game based learning game also known as NDGBL. The NDGBL method makes use of already-existing physical game models including dice, cards, board games, and inventive ideas from the lecturer. One type of reinforcement used in the NDGBL approach is task completion within the game, which allows students to develop tactics, explore options, and come up with answers on their own without teacher assistance [13] [14]. As part of regular class activities, the lecturer used the MasterSquare Game to encourage student participation, feedback, engagement, and enjoyment throughout the class. MasterSquare was developed to embody the fundamental concepts covered in the Principles of Management course. The objective of this research is to determine the students' engagement, social interactions and overall enjoyment levels with the game-based learning sessions offered by MasterSquare Game.

2. Methodology

This study employed a quantitative research design to examine the levels of student engagement, interaction, and enjoyment with game-based learning activities, specifically using the MasterSquare game, in the classroom. This study involved 65 participants among Commerce Department students of Politeknik Tuanku Sultanah Bahiyah. The sample was selected through targeting students of the Principles Management course. Data were collected using a structured questionnaire designed to assess student's engagement, interaction and enjoyment related to the MasterSquare game. Students are given a questionnaire to complete after playing the MasterSquare Game. This questionnaire is designed to gather their feedback on their experience of the game. The questionnaire consisted of four sections. The demographic information section, engagement, enjoyment and social interaction section. The

data collected uses descriptive analysis methods and is analysed by using percentage and mean.

2.1 MasterSquare Game

The MasterSquare game is designed to be a fun and flexible way for lecturers and students to adapt in class sessions. This game is a non-digital game. As part of regular class activities, the lecturer used MasterSquare Game in class to gain students participation, feedback, engagement and feel the enjoyment during class session. This game is developed based on the nature of the course, which is the Principles of Management course. This course provides information on basic functions in management which are planning, organizing, leading and controlling. This course also covers the fundamentals of strategic management which briefly introduce the real-life operation of an organisation. The lecturer developed course-related questions for the game and the questions are divided by topic in the course. The rules of the game are created straightforward and simple for the students to easily understand and follow.

3. Result and Discussion

The survey involved 65 responses from the Commerce Department of Politeknik Tuanku Sultanah Bahiyah students. 61.5% of respondents are female and the rest 38.5% are male students. All the respondents are students of the Principles of Management course which the MasterSquare Game is implemented during class sessions. Table 1.0 provides an overview of the demographic details.

Table 1. Demographic profile of respondents

Item	Description	Frequency	Percentage
Gender	Male	25	38.5%
	Female	40	61.5%
	Malay	46	70.8%
Race	Indian	16	24.6%
	Chinese	1	1.5%
	Other	2	3.1%
Age	18-21	65	100%

Most of the respondents are Malay students representing 70.8% followed by Indian students 24.6%, and the rest are represented by Chinese and other race students. The age of all respondents is between 18 to 21 years old. It shows the respondents represent Gen Z (1995 - 2015) students who were born in the year between 2003 to 2006. Besides that, 90.7% of the respondents agree that they love playing games. The most played games by respondents are online games (70.8%), board games (61.5%), educational games (60%), card games (55.4%), video games (49.2%), outdoor games (1.5%) and other games (1.5%) respectively. For this statement, respondents can choose more than one option provided. This shows, that all the participants are being exposed and knowledgeable in game concepts.

Table 2. Students' engagement result

No.	Item	Mean
1	I wanted to complete the game	4.46
2	I wanted to explore all the options available to me	4.45
3	I do not care how the game ended	4.08
4	I found the game satisfying	4.52

5	I felt absorbed in the game	3.26
6	I felt that time passed quickly	4.42
7	I felt excited during the game	4.57

Participants' responses from the survey indicated that 87.7% *(mean: 4.46)* of students wanted to complete the game and 89.2% *(mean: 4.45)* confirmed that they wanted to explore all options available to them in the game. The item *"I found excited during the game" (4.57)* scored the highest mean followed by items *"I found the game satisfying" (4.52)* and *"I wanted to complete the game" (4.46).* It reveals that the students were excited throughout the game, as seen by their highest mean of 4.57. The items are responded to on a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

Table 3. Students' enjoyment result

No.	Item	Mean
1	I feel worried when playing MasterSquare Game.	2.55
2	I feel happy when playing MasterSquare Game.	4.42
3	I feel exhausted when playing MasterSquare Game.	2.51
4	I feel miserable when playing MasterSquare Game.	2.32
5	I talk to myself when playing MasterSquare Game.	3.11
6	I make loud comments even if nobody is around when playing MasterSquare Game.	3.18

Based on the response, 58.5% of students feel happy when playing MasterSquare Game in the classroom. The highest mean score by item *"I feel happy when playing MasterSquare game" (4.42)* meanwhile the item *"I feel miserable when playing MasterSquare game"* scores the lowest mean which is 2.32. This result confirmed that students feel happy adopting MasterSquare Game during class sessions. Response options reflect a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

Table 4. Students' social interaction result

No.	Item	Mean
1	I felt cooperative toward other classmates.	4.31
2	I strongly collaborated with other classmates.	4.31
3	The cooperation in MasterSquare Game is helpful to my learning.	4.42
4	MasterSquare Game supports a social interaction between the players.	4.37
5	MasterSquare Game supports communities within the game.	4.38
6	MasterSquare Game supports communities outside the game.	4.11

In terms of social interaction, the highest mean score is 4.42 for the item *"The cooperation in MasterSquare Game is helpful to my learning"* followed by *"MasterSquare Game supports communities within the game" (4.38)* and *"MasterSquare Game supports a social interaction between the players".* The lowest score mean by item *"MasterSquare Game supports communities outside the game".* The items *"I felt cooperative toward other classmates"* and *"I strongly collaborated with other classmates"* share the same value of score mean (4.31).

The purpose of this study was to determine the ability of this game and answer the issue of how effective the MasterSquare Game is to improve student engagement, social interaction, and enjoyment during class sessions. The findings in this study indicated that MasterSquare Game encouraged student engagement, social interaction, and enjoyment during class sessions. As can be seen from the result, the game was successful in making the players feel excited and positively engaged. It means most students wanted to finish the game since it was interesting enough to keep them interested the entire time. A lively classroom is created when most students are enthusiastically engaged in the session; these students then tend to share their enthusiasm with their peers and lecturers [12][16]. Based on previous study, it is proven that game-based learning improved student engagement, motivation and academic performance [15]. In future versions or comparable instructional game designs, game

designers should use these findings by stressing components that increase excitement, contentment, and the drive to explore.

Both the high mean score for happiness and the high percentage of students who reported feeling happy show that MasterSquare Game efficiently promotes pleasant emotional experiences during class sessions. Meanwhile, the low mean score for feeling miserable suggests that the game generally does not lead to negative emotional experiences for students. This is encouraging because it suggests that the game is well-liked and does not negatively impact students' emotional well-being in the classroom. The NDGBL methodology is thought to be more beneficial and successful than the traditional learning method in improving participants' performance in the Isometric Transformation topic. The NDGBL's more enjoyable and captivating nature made it possible for players to compete in the game and its new difficulties, which improved players' understanding of isometric transformation. The NDGBL provided participants with a competitive learning environment and the chance to attempt new things in a game-like setting, which increased their motivation to participate and learn [14]. According to a previous study, college students who interact more often also report feeling stronger as part of the community. Thus, this leads to better enjoyment in the classroom [17]. Prior research in foreign language proficiency by Chengchen and Li [22] indicates that greater levels of foreign language enjoyment (FLE) suggest the longest independent predictive effect on later foreign language success, and this effect might persist for at least two months. This proves that feeling enjoyment in class will boost student achievement in education. According to this outcome, using the MasterSquare game in the classroom allowed students to have fun as they learned.

The study's findings show that players' cooperative interactions in the game were beneficial to their learning process. Cooperative gameplay likely facilitated knowledge sharing, problem-solving, and teamwork, enhancing educational outcomes. According to the results of a previous study, stronger interaction between teachers and students was linked to both lower disciplinary issues and higher academic achievement [16]. The participants also acknowledged that the MasterSquare Game promote social interaction among players, which is crucial for building relationships and enhancing engagement within class groups. Peer and educator interactions among students are essential to fostering academic success and engagement. For teaching, learning, and student engagement to be effective, there must be positive interactions between educators and learners [16]. The lower mean score for supporting communities outside the game suggests that while MasterSquare Game enhances internal social dynamics, its connection or influence on broader external communities may be perceived as less impactful. However, there may be opportunities to further integrate its impact beyond the immediate gaming environment to enhance broader community connections. Overall, these results indicate that MasterSquare Game successfully promotes positive social interactions, cooperation, and collaboration among students.

While NDGBL demonstrates numerous benefits, it is not without limitations. A primary limitation of NDGBL in this investigation would be the sample size. The participants were limited to Principles of Management course students who were experienced with the MasterSquare game. Future research that might include students from different disciplines could provide a broader perspective on the effectiveness and applicability of MasterSquare Game (NDGBL) across various courses and student demographics.

4. Conclusion

This research is significant in improving students' engagement, social interaction and enjoyment by applying NDGBL (MasterSquare game) in the classroom. In terms of emotional engagement, students seem to see MasterSquare Game favourably, especially when it comes to promoting happiness and reducing negative emotional reactions when using it in the classroom. Overall, this game portrays a positive impact among participants, highlighting

strong engagement, satisfaction and motivation to explore and complete the game. These days, students want immediate information, visual learning methods, and interactive peer learning—which replaces teacher-centric, instruction-based learning—as well as challenges and problem-solving tasks [15]. The results of the study suggest that educators should incorporate game based learning into their lessons rather than relying solely on conventional methods. Empirical studies have repeatedly shown that active learning (i.e NDGBL) promotes positive attitudes, gets students engaged in learning processes, develops critical thinking skills, leads to better retention of materials, yields better learning outcomes, and motivates students for further study. It has been proven that NDGBL approach benefits various subjects, including English grammar [18], biology [19], chemistry [20], accounting [21] as well as mathematics [14]. Future research should involve more students to gain a more comprehensive understanding of the efficacy of NDGBL and to ensure that the findings are dependable and broadly applicable. Further research also could explore the relationship between student engagement, social interaction, and enjoyment while playing MasterSquare Game to understand their combined impact on learning outcomes and overall student satisfaction.

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IoT-BASED SMART AGRICULTURE MONITORING SYSTEM

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ABSTRACT

This paper provides the development of an Internet of Things (IoT)-based smart agriculture monitoring system designed to monitor the environmental temperature, soil moisture, and crop soil humidity. The system uses a temperature and humidity sensor for temperature and humidity monitoring and a soil moisture sensor to detect the soil moisture level. These sensors are connected to an ESP32 microcontroller, the central element for data processing and communication. These sensors transmit data to a smartphone application, which gives users immediate access to essential information about their crops. When temperature and humidity levels surpass predetermined limits, a water pump will function to ensure that crops receive sufficient water at the right time. Using the Blynk App, users may remotely monitor and control the system using the mobile application, which provides real-time alerts. By utilising IoT technology, an intelligent agricultural field monitoring system that tracks temperature and humidity in the soil can be achieved. Implementing this system may eliminate water waste, save labour costs, and develop more sustainable and effective farming techniques. This technology will make farming operations more intelligent and increase agricultural output.
Keywords: Internet of Things, Agriculture, monitoring, Blynk App

1. Introduction

Growth in the agricultural sector is necessary for the development of the country's economic condition. There is a need to increase productivity, particularly in agriculture, resulting from the growing world population [1]. However, there will be times when supply and demand are out of balance. To improve agricultural productivity, managing and maintaining capital and labour remains a challenging task [2]. In this era, many farmers still use traditional farming methods, which results in low-yield crops and fruits [3]. Farmers are still using manual methods of checking their crops. In this method, the farmers manually verify all the parameters and calculate the readings [4]. Agriculture must advance along with the global trend toward new technology and applications. Sustainable and effective farming methods are becoming increasingly important due to climate change, the busiest daily life, and the loss of natural resources. Conventional farming practices frequently depend on physical labour and subjective judgment, which can result in wasteful water use, unpredictable crop yields, and higher operating expenses. Furthermore, these problems are made worse by unpredictable weather patterns, making it challenging for farmers to maintain ideal growing conditions [5].

Agriculture is the primary source of food grains and other raw materials, so it is the foundation of life for the human species and essential to the expansion of the national economy. Modern technologies can assist farmers in remotely monitoring their crops and promptly taking corrective action. They can raise crop quality and productivity without requiring much labour [6]. Figure 1 provides an overview of the elements of IoT-based smart agriculture, while Figure 2 shows smart agriculture integrated with an IoT monitoring system [3].



Fig. 1. IoT for agriculture



Fig. 2. IoT monitoring system

By incorporating the concept of IoT-based systems into agriculture, farmers can monitor environmental conditions to help them monitor useful information from farms as well as to help improve the quality and quantity of products [7]. This project aims to develop an IoT-based smart agriculture monitoring system that addresses the above critical challenges. The system consists of one main part: a detection system, including a temperature sensor and humidity, a moisture sensor, and a soil moisture detector. The second part is system control, which is monitoring and controlling using an application on a smartphone to ensure optimal water usage and enhance crop health.

This system is not only designed to monitor the soil moisture, temperature and humidity of the crop soil but also purposely tries to lower labour costs, increase the overall sustainability of agricultural techniques, and intentionally improve crop productivity and water management. This Internet of Things (IoT)-based solution offers a contemporary way to address the urgent problems facing agriculture, paving the path for a more robust and efficient farming industry by giving farmers access to real-time data and automated control.

2. Methodology

The IoT-based smart agriculture monitoring system is designed to monitor temperature and humidity levels in real time and automatically water the plant based on predefined thresholds. The system uses the temperature and humidity sensor (DHT11) for temperature and humidity monitoring and a soil moisture sensor to detect soil moisture level. These sensors connect to an ESP32 microcontroller, and the ESP32 DOIT DEVKIT V1 will serve as the central processing unit for the system. These sensors transmit data to a smartphone application, which gives users immediate access to essential information about their plant. When the temperature and humidity levels surpass predetermined limits, the water pump will function to ensure that crops receive sufficient water at the right time. All this data is displayed on the mobile application, which shows the humidity and temperature values, allowing users to set thresholds and control parameters remotely. The temperature can be adjusted to a certain level and depends on the type of plant being grown. ESP32 offers greater capabilities than

ESP8266 and has more features, such as dual-core processing, Bluetooth, and Wi-Fi connectivity, which are necessary for the Internet of Things applications [8].

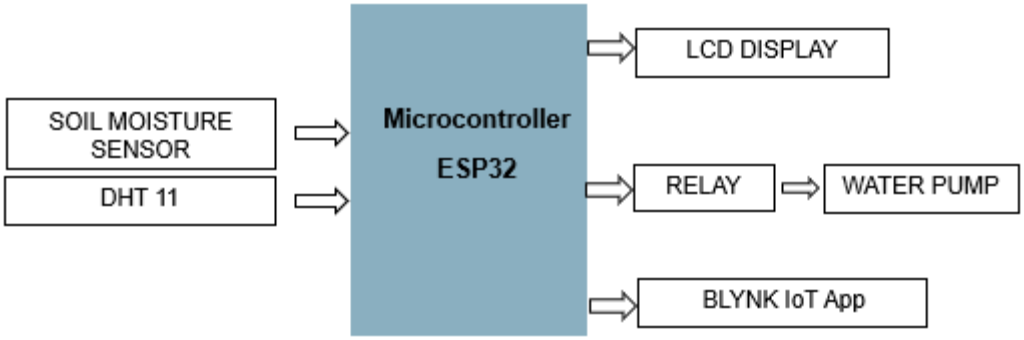


Fig. 3. The block diagram of the system

Figure 3 shows the block diagram of the IoT-based smart agriculture monitoring system, which explains the process flow of this project. ESP32 microcontroller comes with GPIOs and is ideally suited to the IoT project, which can operate with components such as sensors, actuators, and a user interface that can support various protocols [9]. The system comprises a soil moisture sensor and DHT11 (humidity and temperature sensor) as the input and is connected to the ESP32. These sensors continuously monitor the environmental conditions, such as soil moisture, humidity, and temperature, and then convey this data to the ESP32 at regular intervals. The ESP32 processes the incoming data, comparing the current readings to predefined threshold values for soil moisture, humidity and temperature. If the soil moisture level falls below a certain threshold, indicating the soil is dry, the ESP32 activates the water pump by sending a signal to the relay. This relay acts as a switch to a motor and allows water flow and irrigates the soil. Once the soil moisture sensor detects the desired moisture level has been reached, the ESP32 signals the water pump to stop, thus stopping the water flow.

Besides, the system also comes with an LCD connected to the ESP32 and indicates the status of the system and sensor readings in real-time. The Blynk IoT App is linked with the ESP32 to enable remote monitoring and control. The ESP32 can transmit data to the Blynk app over Wi-Fi, and users can monitor the system's status and the surrounding environment from their smartphones or other internet-connected devices. Additionally, users can sign up to get messages and alerts when specific conditions are fulfilled, including low soil moisture or extremely high temperatures. Users may easily establish or modify threshold values for temperature, humidity, and soil moisture using the Blynk IoT App. If necessary, they can also operate the water pump manually. This combination of automated decision-making, real-time sensor data, and simple user interfaces offers better crop health and efficient water use and can improve agricultural productivity.

2.1 Prototype Development

The process of developing a system starts from constructing a prototype model. In this method, a basic prototype is built, tested, and repeated until the desired outcomes are achieved. Users can simulate the real system experience by interacting with the prototype to better understand the needs and features of the system. The development process starts with thoroughly examining the system requirements and intended results. This involves developing a conceptual model and detailing the system's functionalities to ensure that the project objectives are achieved. The first prototype is built using this framework, and the Proteus software is used to design and simulate the circuit. Figure 4 shows the project circuit connection using Proteus software. The simulation results are then evaluated to determine

whether the functionality meets the project's goals, providing some basis for determining any required improvements or modifications.

Testing and modification are done continuously on the prototype as it is improved. The simulation input provides a guide for this improvement, enabling focused changes to improve functionality and performance. More testing and simulations are carried out to confirm these modifications and ensure the prototype progresses to fulfil the project requirements. The project proceeds to the PCB preparation stage after the prototype performs well enough in simulations. This involves constructing the real PCB board and integrating all the prototype's tested parts and configurations. The finished PCB is put through a complete testing process to ensure it satisfies all project requirements. The project's goal is to create a fully functional and dependable IoT-based smart agriculture for monitoring system through this continuous improvement method. The above process ensures that the result is tested, practical to use, and fulfils the specified requirements, finally offering an effective solution to contemporary agriculture challenges.

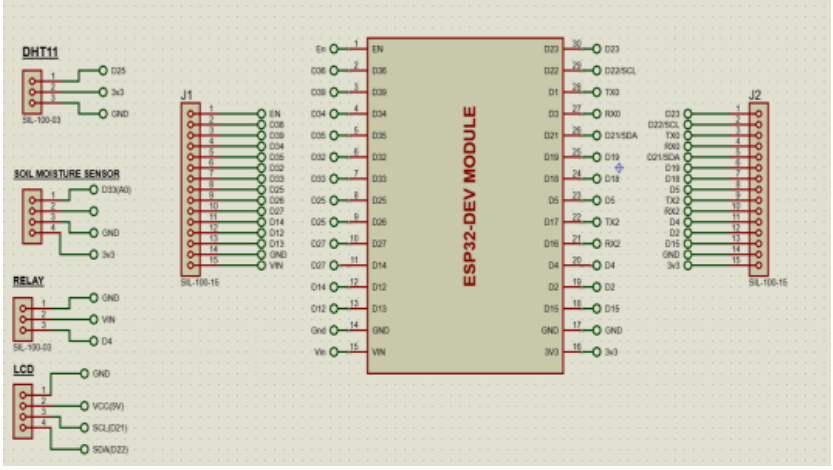


Fig. 4. Project circuit connection

Table 1. GPIO pin connection table

No	Input/ Output	ESP32 Pin #	Pin Mode	Name Pin	Note
1	DHT 11	GPIO25	INPUT	DHTPIN	Measure the temperature and humidity of ambient air.
2	Soil moisture	GPIO 33	INPUT	soil moisture	Measure the volumetric water content of the soil.
3	5V relay	GPIO 04	OUTPUT	relay pin	Control water pump on or off through relay.
4	LCD	GPIO 21 GPIO 22	OUTPUT	SDA SCL	Display data at LCD.
5	Water pump	-	OUTPUT	-	Water the plant according to the programme instructions.

Table 1 shows the ESP32 GPIO pin connections for various peripherals. GPIO25 is configured as an input to measure the temperature and humidity of ambient air (DHT11), while GPIO 33 acts as an input of a soil moisture sensor to measure the volumetric water content of the soil; GPIO 04 functions as an output for a relay to control the water pump on or off for supplying the water; LCD set to GPIOs 21 (SDA) and 22 (SCL) are arranged as outputs for communication and data display.

The Blynks app manages all the functionality for data analysis. It is an application run entirely on the Internet. It is a user-friendly interface developed to manage and simplify IoT projects.

Without requiring complex coding or technical knowledge, it enables users to construct unique mobile and web applications for controlling and monitoring IoT devices. Real-time temperature and humidity information gathered from the installed sensors is shown on the dashboard. As the data is updated continuously, farmers or users can stay updated about the condition of the environment. Besides, the dashboard's graphs and charts provide historical temperature and humidity data trends. With the aid of this app, users can examine trends over time and make wise decisions. By instantly notifying the user's mobile device when environmental parameters surpass predetermined thresholds or when there are systemic problems, the Blynk dashboard further improves communication. Users can generate reports and access thorough analytics to understand better how various elements affect crop health and watering requirements. Users may enhance their agricultural practices by using the dashboard to view important performance parameters like crop health indicators, water usage, and system efficiency. The application is easy for users to navigate and utilise because the Blynk dashboard is a relatively easy user interface.



Fig. 5. Blynk app

3. Result and Discussion

The results of the experiment and a thorough analysis of the data are presented in this section. The main goal of this study was to find out how well different sensors attached to an ESP32 microcontroller performed in terms of monitoring environmental conditions. In addition to monitoring a water pump via a 5V relay, data was gathered from soil moisture and temperature and humidity sensors (DH11). The real-time data was presented on an LCD. The data analysis highlights key findings, patterns, and abnormalities, which assesses the system's dependability and effectiveness. Table 2 shows this research data analysis for temperature, humidity, and soil moisture. This study is essential to understanding the actual implications of our findings and evaluating the system's potential application and constraints in real-world situations.

Data on soil moisture, temperature, and humidity sensors (DHT11) are collected and analysed to ensure optimal plant growth and minimise negative effects from severe weather. Warmth and high humidity can harm plants; thus, these conditions are important. This study focuses on three critical temperature phases: the flowering phase, which happens at 28°C and 40–50% humidity; the vegetative phase, which occurs at 25°C and 60–70% humidity; and additional critical temperature thresholds [10][11].

Table 2. Data analysis

Variable Name	Symbol	Function
Temperature	°C	Collect ambient temperature
Humidity	%	Collect air humidity
Soil Moisture	%	Collect soil moisture

Air temperature and humidity data are collected to avoid the adverse effects of high temperature or high humidity, which can cause the soil to dry out and plants to die or grow with poor quality. Plant growth occurs at three different temperature points: the lowest, highest, and moderate. For example, in this research, the ideal humidity and temperature for flowering plants vary depending on the specific plant species. However, many flowering plants generally develop well at temperatures between 20°C -30°C and humidity levels around 40% to 60%. Some tropical flowering plants may be more suitable at a slightly higher humidity level, in the range of 60% to 80%. It is essential to learn the specific needs of plant species because Malaysia has a variety of flora with different requirements for growth and optimal flowering [12].

Besides, data on soil temperature were gathered, which is crucial for both plant nutrient availability and the best possible functioning of soil organisms. All the data points—which included soil temperature, moisture content, air temperature, and humidity—were transferred to the Blynk cloud for additional analysis after being shown on an LCD. With real-time changes and insights into soil and plant health made possible by this extensive monitoring system, plant care may be improved and more effective.

After logging into the Blynks application, the user gains access to the system status notifications. Users also can regulate the water pumps to water the plant according to the setting. The soil moisture sensor measures soil moisture, and the DHT 11 sensor measures air temperature and humidity and transmits the data to ESP32. The ESP32 processes this signal and activates and deactivates the water pumps while simultaneously transmitting motor status and soil data to the user via the application. The actual prototype that is used for this system is represented in Figures 6, 7, and 8.

The Blynk dashboard in Figure 9 for this project provides a comprehensive overview of key environmental parameters over a one-day observation period. The dashboard's Online Status indicator shows that the system is currently online and operational, as indicated by the green status. There are two irrigation indicators. The first indicates whether irrigation is currently active, and the second shows a toggle switch for manual control. In the dashboard, irrigation appears to be off, as the grey status shows. The Automation Status toggle switch indicates whether the automatic irrigation control based on sensor thresholds is active. The switch is green, indicating that automation is enabled.

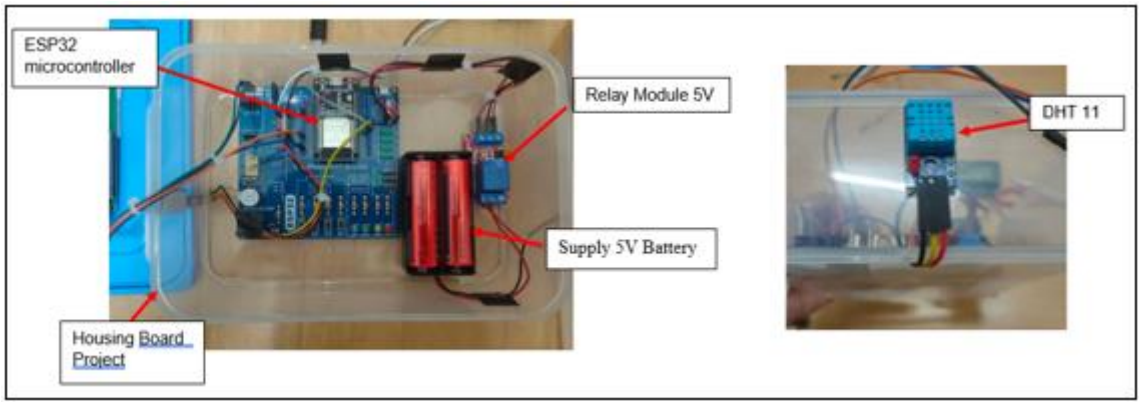


Fig. 6. Main components

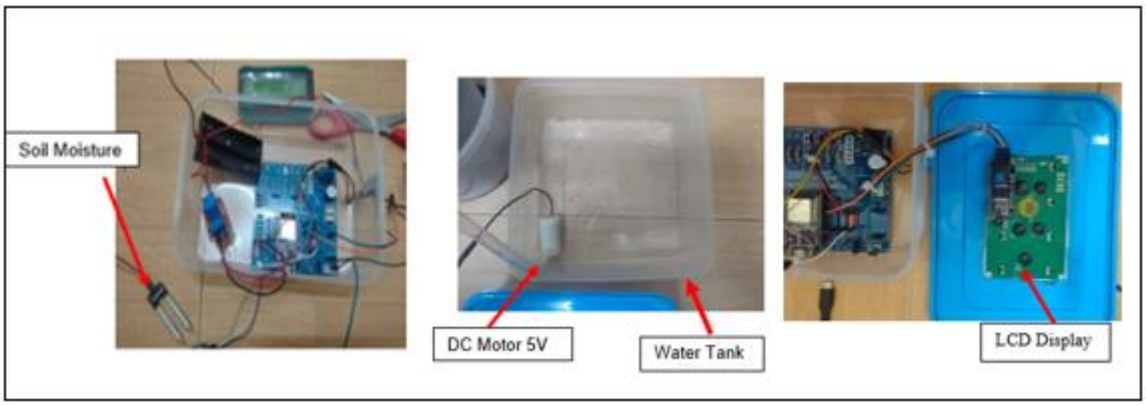


Fig. 7. Irrigation system components



Fig. 8. The entire prototype

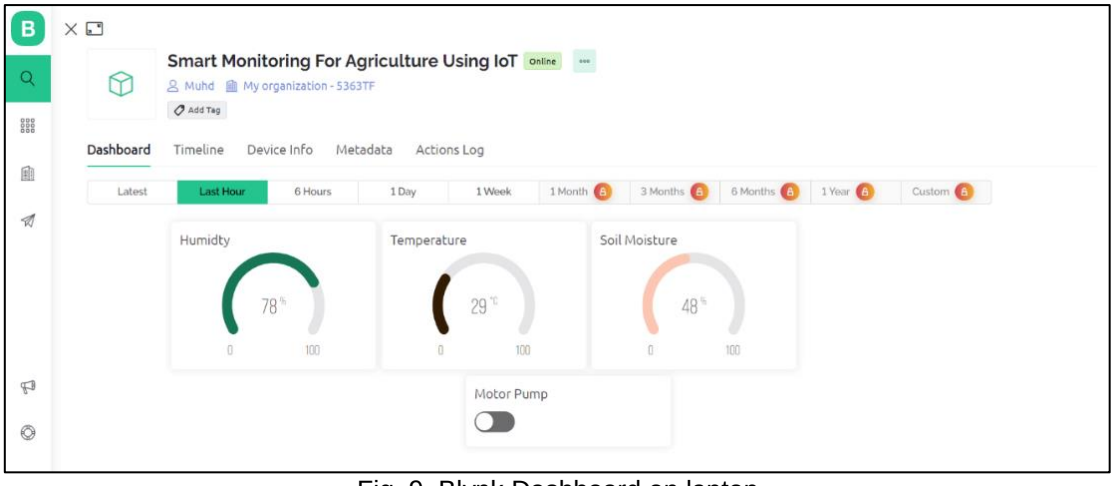


Fig. 9. Blynk Dashboard on laptop



Fig. 10. Blynk Dashboard in Mobile Phone

The Environmental Parameters are shown in the details in Table 3.

Table 3. Environmental Parameters

Variable Name	Reading	Description
Temperature	The gauge shows a current temperature of 29°C.	The overall trend of the temperature over the observation period displays slight variations between 27.5°C and 29°C.
Humidity	The gauge shows a current humidity level of 78%.	Throughout the monitoring period, there were only slight variations in the consistently high humidity levels.
Soil Moisture	The gauge indicates a soil moisture level of 52%.	The variations in soil moisture level during the observation period show a rather constant moisture level with minor declines.

According to the data in Table 3, the high humidity level indicates a relatively moist climate, which can be advantageous for some crops but also requires close observation to avoid mould growth or other moisture-related problems. Given the moderate temperature, various crops may be able to grow successfully there.

The data on soil moisture level indicates that the soil has sufficient moisture since it is greater than 50%. This ensures that crops are not under pressure from water, which is essential for plant health. However, this research was not equipped with a soil temperature sensor. It would be better if a soil temperature sensor were placed in the soil to identify the soil temperature so that the data would be greater.

The water supply is not on right now, but if the automation feature is activated, the system is set up to turn it on automatically. If necessary, users can manually adjust irrigation via the dashboard. Enabling automation ensures that the system will react to environmental changes on its own and maintain ideal growing conditions by turning on the irrigation as needed.

4. Conclusion

This project's implementation of an IoT-based smart agriculture monitoring system is an immense step forward for contemporary farming methods. Through a network of sensors, this system continuously records temperature and humidity, which are two important environmental characteristics. This is to ensure that crops receive the best care possible

through timely and accurate irrigation. Users will have a user-friendly and reliable tool for remote control, in-depth analytics, and real-time data monitoring by integrating the Blynk dashboard application. The system's capacity to automate irrigation based on preset criteria improves healthy plant growth by preserving stable soil moisture levels. The system also saves water by avoiding over-irrigation.

This research significantly can reduce human expenses and improve decision-making when integrated with the ease of remote access and personalised notifications. Generally, this research shows how IoT technology could potentially be utilised to develop more effective and sustainable agriculture methods. Such a system may result in better agricultural practices, more yields, and enhanced crop health. By implementing this technology, farmers may minimise their influence on the environment, maximise resource utilisation, and create the foundation for a more intelligent and sustainable agricultural future.

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WIRED FOR WORDS: NEURAL SECRETS OF SECOND LANGUAGE MASTERY

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ABSTRACT

This study investigates the neural correlates of second language (L2) acquisition through a comparative analysis of immersive and traditional classroom environments among students at Polytechnic Sultan Salahuddin Abdul Aziz Shah. We analyse neuroimaging studies using a systematic review methodology to identify brain regions and neural activation patterns associated with L2 learning. Our objectives include identifying key brain regions involved in L2 acquisition, comparing neural activation patterns between immersive and traditional methods, and assessing their educational efficacy. The findings reveal that immersive learning engages broader neural networks, including enhanced activation in Broca's, Wernicke's, and the prefrontal cortex, facilitating deeper syntactic processing, semantic integration, and memory consolidation. In contrast, traditional methods predominantly activate localised regions, such as Broca's area and the angular gyrus, emphasising explicit language processing and vocabulary retrieval. These results underscore the potential of immersive learning environments to optimise L2 teaching strategies, enhancing language acquisition outcomes in educational settings.

Keywords: Second language acquisition, Neural activation patterns, Immersive language learning, Traditional classroom methods, Neuroimaging studies.

1. Introduction

Second language (L2) acquisition is a complex cognitive process involving various brain regions and neural networks. Understanding the neural mechanisms underlying L2 learning can provide valuable insights for optimising teaching methods and enhancing language acquisition. This study aims to investigate the neural correlates of L2 acquisition by comparing immersive and traditional classroom environments. By examining the brain regions and neural activation patterns associated with different teaching methods, we aim to identify effective strategies for L2 instruction at Polytechnic Sultan Salahuddin Abdul Aziz Shah.

Language processing in the brain involves multiple regions, primarily located in the left hemisphere. Broca's area, situated in the left inferior frontal gyrus, is crucial for speech production and syntactic processing. Wernicke's area, located in the left superior temporal gyrus, plays a significant role in semantic processing and comprehension. Additionally, the prefrontal cortex and hippocampus are involved in memory consolidation and executive functions, which are essential for language learning.

Immersive learning environments, which provide extensive exposure to the target language, are believed to engage broader neural networks than traditional classroom methods. Studies have shown that immersive learning can enhance syntactic processing, semantic integration, and memory consolidation by increasing activation in Broca's area, Wernicke's area, and the bilateral prefrontal cortex and hippocampus. In contrast, traditional classroom methods often focus on explicit language instruction and vocabulary memorisation, primarily activating localised regions in the left hemisphere, such as Broca's area and the angular gyrus.

A systematic review of neuroimaging studies on L2 acquisition reveals that immersive learning methods consistently result in broader neural activation patterns. These patterns are associated with improved language processing and retention, suggesting that immersive environments may be more effective for L2 learning. However, traditional classroom methods are also advantageous, particularly in enhancing explicit language processing and vocabulary retrieval.

2. Methodology

This study employs a systematic review methodology to synthesise and analyse existing neuroimaging research on second language (L2) acquisition. By comparing the neural activation patterns associated with immersive and traditional classroom environments, we aim to identify effective teaching strategies for L2 instruction. The systematic review approach allows us to aggregate findings from multiple studies, providing a comprehensive understanding of the neural correlates of L2 learning.

We conducted a thorough literature search using several academic databases, including PubMed, Scopus, and Google Scholar. The search terms included combinations of keywords such as "second language acquisition," "L2 learning," "neuroimaging," "fMRI," "PET," "immersive learning," "traditional classroom," "brain activation," and "neural correlates."

To ensure the relevance and quality of the studies included in our review, we applied the following inclusion criteria:

i. Study Focus: The study must investigate L2 acquisition using neuroimaging techniques.
ii. Participants: The study must involve human participants, specifically students learning a second language.
iii. Teaching Methods: The study must compare immersive and traditional classroom environments.
iv. Outcome Measures: The study must report on brain regions and neural activation patterns associated with L2 learning.
v. Publication Type: Only peer-reviewed journal articles were included.

Studies were excluded if they:

i. Did not use neuroimaging techniques.
ii. Focused on non-human subjects.
iii. Did not provide a clear comparison between immersive and traditional classroom environments.
iv. Were not peer-reviewed.

Two independent reviewers extracted data from the included studies using a standardised data extraction form. The extracted data included:

i. Study Characteristics: Author(s), publication year, sample size, age of participants, language being learned, and teaching methods used.
ii. Neuroimaging Techniques: Type of neuroimaging technique (E.g., fMRI, PET), scanning parameters, and analysis methods.
iii. Brain Regions: Specific brain regions are reported to be activated during L2 acquisition.
iv. Neural Activation Patterns: Patterns of neural activation associated with immersive and traditional classroom environments.
v. Outcome Measures: Behavioural outcomes related to L2 learning (E.g., language proficiency, retention rates).

We assessed the quality of the included studies using a modified version of the Newcastle-Ottawa Scale (NOS) for observational studies. The quality assessment focused on three domains: selection of study groups, comparability of groups, and ascertainment of outcomes.

We performed a descriptive analysis to summarise the characteristics of the included studies. This included calculating the frequency and percentage of studies reporting activation in specific brain regions and comparing the prevalence of neural activation patterns between immersive and traditional classroom environments.

We conducted a meta-analysis using random-effects models to compare neural activation patterns between immersive and traditional classroom environments. This approach allowed us to account for heterogeneity across studies. The primary outcome measure was the difference in activation levels in key brain regions, including Broca's area, Wernicke's area, the prefrontal cortex, and the hippocampus.

We used the following statistical methods for data analysis:

i. Effect Size Calculation: We calculated the effect sizes (Cohen's d) for neural activation differences between immersive and traditional classroom environments.
ii. Heterogeneity Assessment: We assessed heterogeneity using the I^2 statistic, which measures the percentage of variation across studies that is due to heterogeneity rather than chance.
iii. Publication Bias: We evaluated publication bias using funnel plots and Egger's test.

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3. Results and Discussions

Our systematic review included 30 studies that met the inclusion criteria. The studies varied in sample size, participant age, languages being learned, and neuroimaging techniques used. Most studies used functional magnetic resonance imaging (fMRI) to investigate neural activation patterns.

The analysis revealed that immersive learning environments engage broader and more diverse neural networks compared to traditional classroom methods. Key findings include:

i. Broca's Area: Increased activation in the left inferior frontal gyrus, associated with enhanced syntactic processing.
ii. Wernicke's Area: Greater activation in the left superior temporal gyrus is linked to improved semantic integration.
iii. Prefrontal Cortex: Enhanced activation in the bilateral prefrontal cortex, which is involved in executive functions and working memory.
iv. Hippocampus: Increased activation in the bilateral hippocampus is associated with memory consolidation and retrieval.
v. Additional Brain Regions: Engagement of regions involved in multisensory integration (e.g., superior parietal lobule) and emotional processing (e.g., amygdala) may contribute to deeper language learning and retention.

In contrast, traditional classroom methods primarily resulted in localised neural activation patterns. Key findings include:

i. Broca's Area: Activation in the left inferior frontal gyrus, associated with explicit language processing and syntactic analysis.
ii. Angular Gyrus: Increased activation in the left angular gyrus, linked to vocabulary retrieval and language comprehension.

iii. Limited Engagement: Less extensive activation in the prefrontal cortex and hippocampus, suggesting limited effects on memory consolidation and executive functions.

The meta-analysis revealed significant differences in neural activation patterns between immersive and traditional classroom environments:

i. Broader Neural Networks: Immersive learning methods resulted in broader and more distributed neural activation, engaging multiple brain regions simultaneously.
ii. Enhanced Syntactic and Semantic Processing: Immersive environments showed significantly greater activation in Broca's and Wernicke's areas, indicating enhanced syntactic and semantic processing.
iii. Improved Memory Consolidation: In immersive learning, greater activation in the hippocampus and prefrontal cortex was observed, suggesting improved memory consolidation and executive functions.
iv. Localised Activation: Traditional classroom methods resulted in more localised activation patterns, primarily in regions associated with explicit language processing and vocabulary retrieval.

The findings suggest that immersive learning methods are more effective in engaging comprehensive neural networks and promote deeper language processing and retention. While beneficial for explicit language processing and vocabulary retrieval, traditional classroom methods may not engage the broader neural networks necessary for holistic L2 acquisition.

The results of this study provide a neuroscientific foundation for optimising L2 teaching strategies at Polytechnic Sultan Salahuddin Abdul Aziz Shah. Integrating immersive elements into traditional classroom instruction can enhance language learning outcomes by engaging broader neural networks and promoting deeper language processing. For example, incorporating interactive simulations, multimedia resources, and real-life language use can complement traditional explicit instruction, leading to more effective L2 acquisition.

Future research should explore the long-term effects of different teaching methods on neural activation patterns and language learning outcomes. Additionally, investigating individual differences in neural responses to various teaching methods can provide insights into personalised L2 instruction strategies. Longitudinal studies are needed to examine the sustained impact of immersive and traditional classroom methods on neural plasticity and language proficiency.

4. Conclusion

In conclusion, this study has delved into the neural correlates of second language (L2) acquisition by comparing immersive and traditional classroom environments. Through a systematic review of neuroimaging studies, we have identified distinct patterns of neural activation associated with each learning method. Immersive learning environments engaged broader neural networks, fostering enhanced syntactic processing, semantic integration, and memory consolidation in regions like Broca's, Wernicke, and the prefrontal cortex. Conversely, traditional classroom methods predominantly activated localised brain regions such as Broca's area and the angular gyrus, emphasising explicit language processing and vocabulary retrieval.

These findings underscore the efficacy of immersive learning environments in promoting comprehensive neural engagement and deeper language processing compared to traditional methods. Integrating immersive elements into traditional classroom instruction could significantly enhance language learning outcomes.

Future research should explore the longitudinal effects of these learning methods on neural plasticity and language proficiency. Additionally, investigating individual differences in neural responses to various teaching strategies could provide insights into personalised L2 instruction. By unravelling the neural secrets of second language mastery, educators can refine teaching methodologies to better meet students' diverse learning needs, ultimately advancing educational practices and improving language acquisition outcomes.

This study contributes valuable insights into the neuroscientific foundations of L2 acquisition, offering a pathway for optimising educational strategies at Polytechnic Sultan Salahuddin Abdul Aziz Shah and beyond.

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KESEDIAAN PELAJAR KOLEJ KOMUNITI SEBERANG JAYA TERHADAP PEMILIHAN KERJAYA DALAM BIDANG PENGAJIAN

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ABSTRAK

Program Sijil Teknologi Elektrik (SKE) di Kolej Komuniti Seberang Jaya (KKSJ) direkabentuk bertujuan untuk memperkenalkan pelajar kepada teknologi elektrik dan elektronik. Program SKE melahirkan graduan seramai 60 ke 80 orang setahun. Pada tahun 2023, graduan SKE di KKSJ adalah seramai 62 orang pelajar. Berdasarkan data analisis statistik Unit Pengesanan Graduan KKSJ didapati daripada 62 orang graduan tersebut hanya sebanyak 35.5% yang berjaya meyandang pekerjaan dalam bidang pengkhususan yang dipelajari di kolej. Graduan memilih kerjaya bukan hanya bergantung kepada bidang pengajian di institusi malah terdapat faktor lain yang menyebabkan graduan berarah kepada kerjaya lain. Justeru itu satu kajian telah dijalankan untuk mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek motivasi, afektif dan pengalaman. Kajian ini menggunakan analisis pendekatan kuantitatif deskriptif untuk mengenalpasti skor min secara kaedah tinjauan dan soal selidik digunakan sebagai instrumen kajian. Seramai 40 orang pelajar SKE semester 4 telah dipilih secara rawak sebagai responden. Secara keseluruhannya dapat dilihat dan dikenalpasti, purata skor min tahap kesediaan pelajar Program Sijil Teknologi Elektrik (SKE) KKSJ memilih kerjaya mengikut bidang pengajian dari aspek motivasi, afektif dan pengalaman ialah 3.51 dan berada pada tahap sederhana.Didapati purata skor min tahap kesediaan pelajar SKE meyandang pekerjaan mengikut bidang pengajian dari aspek motivasi ialah 3.37 dan dikenalpasti sebagai tahap sederhana manakala aspek pengalaman menunjukkan skor min sebanyak 3.86 dan berada pada tahap tinggi. Skor min bagi aspek afektif terhadap kesediaan pelajar juga menunjukkan tahap sederhana iaitu sebanyak 3.29. Dengan ini pihak pengurusan kolej, barisan pensyarah SKE dan Unit Pengesanan Graduan harus meneliti semula semua inisiatif yang telah dibuat untuk membentuk graduan ke arah bidang kemahiran. Pihak kolej perlu mengorak langkah yang lebih efisien seperti menganjurkan program-program yang berteraskan aspek afektif dan meningkatkan motivasi untuk menjana dan meningkatkan peratus graduan KKSJ memiliki pekerjaan mengikut bidang pengkhususan kelak. Kualiti graduan diukur berdasarkan kebolehan graduan memiliki pekerjaan mengikut bidang pengkhususan yang telah dipelajari dan kemahiran yang dimiliki.
Kata Kunci*:* Kesediaan, Pelajar SKE, Pekerjaan Dalam Bidang

1. Pengenalan

Pendidikan dan Latihan Teknikal dan Vokasional (TVET) merupakan laluan kepada lepasan Sijil Pelajaran Malaysia (SPM) untuk meneroka kerjaya dalam bidang yang mempunyai potensi besar terhadap masa depan pelajar. Antara bidang TVET yang makin mendapat sambutan tinggi untuk diceburi ialah seperti pemasangan dan penyelenggaraan elektrik, pembaikan dan servis, rekaan grafik, terapi kecantikan, penyediaan dan pembuatan makanan [1]. Graduan TVET mempunyai peluang pekerjaan yang tinggi, di mana ia menjadi pilihan utama majikan disebabkan oleh kemahiran *hands on* yang dimiliki oleh lepasan pengajian TVET. Bidang TVET sentiasa bergerak seiring dengan permintaan industri dan perkembangan teknologi semasa [2]. Ini terbukti dengan peratusan kebolehpasaran graduan TVET Kementerian Pendidikan Tinggi pada tahun 2023 iaitu sebanyak 94.5% dan telah menunjukkan peningkatan sebanyak 1.2 % berbanding tahun 2022 (*Graduate References Hub for Employment and Training* (GREat), 2023). Namun begitu, kebolehpasaran graduan tidak menunjukkan graduan memiliki pekerjaan dalam bidang pengajian yang telah dipelajari di institusi. Kualiti graduan tidak boleh diukur hanya berdasarkan kebolehan graduan menempatkan diri dalam sesuatu sektor

pekerjaan malah perlu dikenalpasti graduan mampu memiliki pekerjaan mengikut bidang pengkhususan yang telah dipelajari dan kemahiran yang dimiliki. Menurut kajian didapati bukan semua graduan yang berjaya mendapat kerjaya yang diiginkan berdasarkan kemahiran dan kelulusan yang dimiliki [3]. Selain itu kajian lain turut membuktikan tidak terdapat hubungan yang signifikan di antara ciri-ciri pendidikan dan ciri-ciri pekerjaan untuk mengenalpasti kesepadanan antara bidang pengajian yang dipelajari dengan peluang pekerjaan yang diceburi [4].

1.1 Latar Belakang Kajian

Kolej Komuniti Seberang Jaya (KKSJ) merupakan salah satu institusi peneraju TVET yang menawarkan Program Sijil Teknologi Elekrik (SKE), Sijil Sistem Komputer dan Rangkaian (SSK) dan Program Diploma In Electronics Instrumentation Technology (DEI) dan berlokasi di Negeri Pualu Pinang. Program Sijil Teknologi Elektrik (SKE) khususnya telah direkabentuk bertujuan untuk memperkenalkan pelajar kepada teknologi elektrik dan elektronik pada masa kini. Pelajar berkemampuan untuk mengendalikan kerja-kerja pemasangan dan perkhidmatan berkaitan industri elektrik dan elektronik. Program ini menawarkan kursus teras seperti Teknologi Elektrik, Alat Pengukuran, Peranti Semikonduktor, Sistem Digit, Lukisan Terbantu Komputer, Motor Elektrik Satu Fasa, Pemasangan Elektrik Satu Fasa, Motor Elektrik Tiga Fasa, Pemasangan Elektrik Tiga Fasa, Pengawal Logik Bolehaturcara, Sistem Automasi Bangunan, serta kursus Sistem Keselamatan dan Penyamanan Udara. Melalui program ini, pelajar bukan sahaja berpeluang mendapatkan pengetahuan secara teori malah dilengkapi dengan latihan amali untuk meningkatkan kemahiran *hands on* selaras dengan keperluan industri semasa. Di samping itu pelajar-pelajar juga didedahkan dengan aktiviti penyelidikan, aktiviti *Corporate Social Responsibility* (CSR) dan pembangunan projek-projek inovasi yang berteraskan elektrik dan elektronik dengan menggunakan teknologi baru yang selaras dengan perkemabangan terkini IR4.0.

1.2 Pernyataan Masalah

Secara umumnya, Program Sijil Teknologi Elektrik (SKE) melahirkan graduan seramai 60 ke 80 orang setahun. Pada tahun 2023, graduan SKE Kolej Komuniti Seberang Jaya adalah seramai 62 orang pelajar (Unit Peperiksaan & Penilaian, KKSJ). Berdasarkan data analisis statistik unit pengesanan graduan KKSJ didapati daripada 62 orang graduan tersebut hanya sebanyak 35.5% yang berjaya meyandang pekerjaan dalam bidang pengkhususan yang dipelajari di kolej. Manakala 64.5% graduan lagi bekerja dalam bidang pekerjaan yang tidak sepadan dengan bidang pengajian teknologi elektrik. Selain itu, dapatan kajian juga menunjukkan bukan semua pelajar dalam bidang kejuruteraan mampu berjaya memiliki jawatan sebagai jurutera selepas pengajian [5]. Ini menjelaskan bahawa graduan memilih kerjaya bukan hanya bergantung kepada bidang pengajian di institusi malah terdapat faktor lain yang menyebabkan graduan berarah kepada kerjaya lain. Antara faktor-faktor yang telah dikenalpasti berdasarkan kajian-kajian lepas ialah pengalaman yang diperoleh daripada persekitaran pelajar itu sendiri, motivasi dan bimbingan serta sifat afektif yang dimiliki oleh pelajar seperti minat, serta kesukaan [6]. Justeru itu satu kajian telah dijalankan untuk meninjau sejauh manakah faktor-faktor tersebut menjadi penyumbang kepada tahap kesediaan pelajar KKSJ memilih kerjaya dalam bidang pengajian. Dengan memahami faktor-faktor yang mempengaruhi kesediaan pelajar maka persekitaran yang lebih kondusif boleh diwujudkan untuk pembangunan profesional dan kemajuan dalam bidang elektrik. Kesediaan kerjaya pelajar mengikut bidang pengajian juga ditinjau supaya dapat membantu mereka berhadapan dengan cabaran dunia profesional hari ini [5].

1.3 Objektif Kajian

Secara khususnya, kajian ini telah dijalankan untuk:

i. Mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek motivasi.
ii. Meninjau tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek afektif.
iii. Mengukur tahap kesediaan pelajar KKSJ terhadap pemilihan kerjaya mengikut bidang pengajian dari aspek pengalaman.

1.4 Kepentingan Kajian

Kajian ini penting untuk memberi panduan dan gambaran situasi semasa kepada pihak tertentu seperti di bawah supaya dapat mengorak inisiatif yang lebih progresif untuk menjana dan meningkatkan peratus graduan TVET Kolej Komuniti Seberang Jaya dalam sektor pekerjaan mengikut bidang pengkhususan kelak.

Kajian ini penting bagi pihak pengurusan KKSJ untuk mengenalpasti strategi yang sewajarnya dalam melatih individu yang berkemahiran tinggi dalam bidang elektrik. Di samping itu dengan adanya dapatan kajian ini para pensyarah Unit Teknologi Elektrik juga dapat memberi dorongan, dan motivasi yang lebih proaktif kepada bakal pelajar yang menjejak ke alam pekerjaan bagi mewujudkan semangat bekerja mengikut bidag pengkhususan dan kemahiran yang dimiliki. Dengan ini ia dapat membantu memenuhi kesepadanan tenaga kerja mengikut bidang pengajian dan mencapai kualiti kebolehpasaran graduan yang unggul.

Kajian ini juga akan menjadi panduan kepada pelajar bidang elektrik untuk menangani cabaran dalam membina kerjaya mengikut bidang pengajian yang dipelajari di kolej. Kajian ini seharusnya dapat meningkatkan motivasi, dan minat pelajar dalam bidang kejuruteraan elektrik. Justeru itu ia dapat meningkatkan jumlah bilangan graduan yang menerajui pekerjaan kelak berdasarkan bidang pengkhususan di kolej.

Dengan adanya dapatan kajian ini maka Unit Pengesanan Graduan dapat meneliti semula aspek-aspek yang mendorong kepada kemerosotan peratus graduan bekerja dalam bidang pengkhususan pengajian dan seterusnya mengambil langkah yang drastik untuk meningkatkan pencapaian peratus kesepdanan pekerjaan graduan pada masa akan datang.

1.5 Batasan Kajian

Kajian ini hanya menumpukan kepada tahap kesediaan pelajar Program Sijil Teknologi Elektrik (SKE) KKSJ terhadap pemilihan kerjaya mengikut bidang pengajian berdasarkan aspek motivasi, afektif dan pengalaman. Responden yang dikaji terdiri daripada pelajar SKE semester 4 yang telah menjalani latihan industri pada Sesi II 2023/2024.

2. Tinjauan Literatur

2.1 Afektif

Afektif adalah domain yang berkait emosi seperti nada, rasa, penerimaan dan penolakkan. Selain itu, perasaan, nilai, penghargaan, minat dan sikap turut dimasukkan dalam domain afektif. Penerapan domain afektif dalam aktiviti pembelajaran di kelas dan melalui program – program yang dijalankan di institusi membantu pelajar menanamkan ciri – ciri afektif dalam diri pelajar [7]. Dalam pada itu, YB Dato' Haji Muhammad Yusof Apdal Timbalan Menteri Pendidikan Tinggi (2023), dalam ucapannya menyebut pencapaian yang cemerlang dalam bidang akademik mahupun kemahiran insaniah menyumbang kepada kebolehpasaran graduan yang tinggi. Kemahiran insaniah pelajar amat berkait rapat dengan kesediaan pelajar dalam pemilihan kerjaya pada masa mendatang. Keterlibatan pelajar dalam program – program serta aktivti yang dijalankan di institusi akan membentuk kemenjadian diri pelajar sebagai persiapan menghadapi dunia pekerjaan sebenar. Manakala mengikut dapatan kajian

yang dijalankan terhadap 104 orang pelajar dari Universiti Kebangsaan Malaysia (UKM) didapati faktor minat yang berpunca dari diri pelajar itu sendiri mempengaruhi dalam pemilihan kerjaya [8]. Selain itu, kajian lain juga menyatakan bahawa pelajar lebih cenderung memilih pekerjaan yang diminati dan kehendak ibu bapa tidak mempengaruhi pemilihan kerjaya pelajar [9]. Di samping itu, didapati antara faktor-faktor penting yang perlu diberi perhatian apabila seseorang individu menentukan hala tuju kehidupan, dan memilih pekerjaan ialah sama ada pekerjaan yang dipilih bersesuaian dengan kelulusan, minat dan personality [10]. Berdasrkan kajian yang dijalankan turut menyatakan bahawa faktor minat mempengaruhi pelajar Jabatan Kejuruteraan Elektrik (JKE) di Politeknik Sultan Mizan Zainal Abidin untuk kesediaan pembentukan dan perancangan kerjaya [5]. Minat dan kecenderungan memainkan peranan untuk mempengaruhi pencapaian pembelajaran seseorang [11].

2.2 Motivasi

Mengikut definisi Kamus Dewan Bahasa dan Pustaka motivasi ialah semangat kuat yang ada pada diri sendiri yang mendorong berusaha untuk mencapai sesuatu kejayaan. Motivasi merupakan satu dorongan bagi seseorang individu ke arah mencapai sesuatu matlamat [12]. Walaupun seseorang individu mempunyai bakat dan kebolehan yang hebat namun sekiranya tidak dipenuhi dengan tahap motivasi yang tinggi. maka kejayaan sukar dicapai. Justeru itu, faktor motivasi dilihat sebagai faktor yang sangat penting dalam pemilihan sesuatu kerjaya. Aspek motivasi dan bimbingan dari pengurusan politeknik memainkan peranan penting dalam menentukan tahap kesediaan pelajar dalam pembentukan kerjaya [5]. Projek pelajar dan aktiviti CSR yang dijalankan di institusi juga memberi motivasi untuk pelajar memilih kerjaya yang diingini [13][14]. Selain itu, kajian yang dijalankan mendapati bahawa kebolehan pelajar melakukan latihan praktikal di institusi dapat memberi motivasi dan membina keyakinan pelajar terhadap kemampuan mereka berjaya dalam bidang kerjaya yang sedang diikuti [13].

2.3 Pengalaman

Salah satu aspek yang penting dalam menentukan kerjaya seseorang ialah faktor pengalaman. Kamus Dewan menyatakan bahawa pengalaman ialah pengalaman yang telah dialami atau dilalui oleh sesorang individu. Pelajar biasanya belajar melalui pengalaman mereka sendiri. Kajian oleh mendapati pelajar yang lebih banyak pengalaman kerja dalam pelbagai bidang maka mereka lebih bersedia untuk membentuk kerjaya di masa depan [15]. Latihan industri penting dalam persediaan pelajar untuk menghadapi dunia pekerjaan. Latihan industri akan menambah pengalaman kerja mengikut kemahiran yang ada pada diri. Pelajar juga menganggap pengalaman dari latihan industri, penglibatan dalam program kesukarelawan, projek akhir pelajar dan aktiviti ilmiah yang lain dljalankan di politeknik banyak memberi pengalaman untuk mereka menghadapi alam pekerjaan [5].

3. Metodologi Kajian

Metodologi bagi kajian ini akan membincangkan mengenai reka bentuk kajian, cara pengumpulan data, populasi dan kaedah persampelan serta tujuan kajian rintis yang telah dijalankan bagi menguji kesahan borang soal selidik.

3.1 Reka Bentuk Kajian & Pengumpulan Data

Kajian ini menggunakan analisis pendekatan kuantitatif deskriptif untuk mengenalpasti skor min secara kaedah tinjauan dan soal selidik digunakan sebagai instrumen kajian. Item dalam borang soal selidik telah diadaptasi daripada kajian [5]. Borang soal selidik telah diedar melalui *google form* dan pelajar dikehendaki memilih jawapan berdasarkan skala likert yang diberi. Aras skala likert tersebut adalah seperti di Jadual 1**.** Data yang dikumpul telah dianalisis menggunakan IBM Statistical Package for Social Science (SPSS) versi 26.0.

Jadual 1. Skala Likert Yang Digunakan Dalam Borang Soal Selidik

Aras	Tahap
1	Sangat Tidak Setuju (STS)
2	Tidak Setuju (TS)
3	Kurang Setuju (KS)
4	Setuju (S)
5	Sangat Setuju (SS)

Item dalam soal selidik (*google form*) dibahagikan kepada dua bahagian seperti berikut:

Bahagian A: Demografi Responden - 3 Soalan
Bahagian B: Tahap kesediaan pelajar Program Sijil Teknologi Elektrik Kolej Komuniti Seberang Jaya bagi pemilihan kerjaya dalam bidang - 25 Soalan

3.2 Populasi dan Kaedah Persampelan

Populasi kajian ialah seramai 45 orang pelajar semester 4 Program Sijil Teknologi Elektrik (SKE) Kolej Komuniti Seberang Jaya bagi Sesi II: 2023/2024. Teknik persampelan rawak mudah telah digunakan. Berdasarkan pengiraan sampel Krejcie dan Morgan (1970) sampel kajian ini adalah seramai 40 orang responden.

3.3 Kajian Rintis

Kajian rintis telah dijalankan terhadap 15 orang pelajar. Saiz sampel seramai 10-30 orang memadai untuk memenuhi tujuan perbincangan awal yang berkesan tentang ujian [16]. Nilai kebolehpercayaan (Cronbach's Alpha) yang diperoleh ialah 0.975 dan menunjukkan soal selidik yang dibina adalah pada aras kesahan tinggi.

4. Analisis Dapatan Kajian dan Perbincangan

Skor min telah dianalisis untuk menginterpretasi tahap kesediaan bagi setiap objektif yang dikaji. Jadual 2 ialah interpretasi skor min bagi skala Likert 5 mata bagi menilai tahap kesediaan pelajar SKE memilih kerjaya dalam bidang pengajian dari aspek motivasi.

Jadual 2. Jadual Interpretasi Skor Min

Min Skor	Tahap
1.00-2.33	Rendah
2.34-3.66	Sederhana
3.67-5.00	Tinggi

4.1 Analisis Objektif 1: Mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek motivasi.

Jadual 3 menunjukkan analisis skor min bagi setiap pernyataan tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek motivasi. Tahap kesediaan pelajar telah diklasifikasi mengikut Jadual 2.

Mengikut analisis di Jadual 3, didapati purata keseluruhan skor min tahap kesediaan pelajar SKE KKSJ memilih kerjaya mengikut bidang pengajian elektrik dari aspek motivasi ialah 3.37 dan berada pada tahap sederhana. Sebanyak 53.4% pelajar telah menyatakan bahawa mereka kurang bermotivasi untuk menyandang pekerjaan dalam bidang pengajian. Pelajar SKE masih kurang bermotivasi untuk mengejar kerjaya dalam bidang elektrik selepas menamatkan program di Kolej Komuniti Seberang Jaya. Ini dapat disokong dengan pernyataan no.1 iaitu “Saya bermotivasi untuk mengejar kerjaya dalam bidang elektrik

selepas menamatkan program di Kolej Komuniti Seberang Jaya." yang telah mencatat skor min yang rendah dan merupakan bacaan skor min yang paling rendah daripada 9 item pernyataan yang diajukan kepada pelajar bagi aspek motivasi.

Jadual 3. Analisis Objektif Pertama
(Mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek motivasi)

Bil	Item	Skor Min	Tahap
1	Saya bermotivasi untuk mengejar kerjaya dalam bidang elektrik selepas menamatkan program di Kolej Komuniti Seberang Jaya.	2.33	Rendah
2	Saya mendapat tunjuk ajar dan sokongan yang mencukupi dari kolej dalam memahami peluang kerjaya dalam bidang elektrik.	3.13	Sederhana
3	Saya mendapat bimbingan dari pensyarah kolej terhadap pemahaman mengenai pilihan kerjaya dalam bidang elektrik.	3.50	Sederhana
4	Saya memperoleh input yang jelas daripada taklimat unit latihan industri kolej terhadap pemahaman mengenai pilihan kerjaya dalam bidang elektrik.	3.95	Tinggi
5	Kolej komuniti membantu saya dalam merancang perkembangan diri dan kerjaya dalam bidang elektrik selepas tamat pengajian.	3.10	Sederhana
6	Aktiviti ilmiah (sebarang program yang dianjurkan) di kolej membantu meningkatkan pengetahuan saya tentang peluang kerjaya dalam bidang elektrik.	3.48	Sederhana
7	Aktiviti CSR memberikan motivasi kepada saya dalam pemilihan kerjaya dalam bidang elektrik.	3.85	Tinggi
8	Pentaksiran dan penilaian projek meningkatkan kemahiran saya dan seterusnya memberi motivasi kepada saya untuk memilih kerjaya dalam bidang elektrik.	3.98	Tinggi
9	Saya bersemangat untuk bekerja dalam bidang elektrik kerana Latihan Industri memberi pengetahuan yang jelas berkaitan pekerjaan dalam bidang.	3.13	Sederhana
	Purata Skor Min	3.37	Sederhana

Selain itu hanya sebanyak 62% daripada 40 orang pelajar menyatakan kolej komuniti membantu dalam merancang perkembangan diri dan kerjaya dalam bidang elektrik selepas tamat pengajian. Skor min bagi pernyataan tersebut ialah 3.10 dan menunjukkan tahap sederhana. Namun begitu, pelajar SKE berpendapat bahawa pentaksiran dan penilaian projek meningkatkan kemahiran pelajar dan seterusnya memberi motivasi kepada mereka untuk memilih kerjaya dalam bidang elektrik. Ia dapat disokong dengan pernyataan no.8 yang mencatat skor min paling tinggi iaitu 3.98 daripada 9 item soalan aspek motivasi dan menunjukkan tahap tinggi bagi pernyataan tersebut. Mereka turut menyatakan input yang jelas diperoleh daripada taklimat unit latihan industri kolej terhadap pemahaman mengenai pilihan kerjaya dalam bidang elektrik di mana skor min yang dicatat ialah 3.95 dan berada pada tahap tinggi. Aktiviti CSR turut memberikan motivasi kepada pelajar SKE dalam pemilihan kerjaya dalam bidang elektrik di mana skor min yang dicatat bagi pernyataan tersebut ialah 3.85 dan menunjukkan tahap tinggi. Walaupun motivasi dan bimbingan dari pengurusan kolej memainkan peranan penting dalam menentukan tahap kesediaan pelajar dalam pembentukan kerjaya namun sekiranya tidak dipenuhi dengan tahap motivasi yang tinggi dari dalam diri sendiri maka kejayaan sukar dicapai [12]. Pelajar yang bermotivasi tinggi dan baik dalam sebarang bidang biasanya akan berpencapaian tinggi dalam bidang tersebut [17][18]. Berdasarkan dapatan objektif pertama dapat dilihat bahawa Unit Latihan Industri KKSJ mahupun pihak pengurusan Kolej Komuniti Seberang Jaya telah mengorak banyak inisiatif secara berterusan dalam memastikan pelajar meneroka kerjaya dalam bidang selepas tamat pengajian namun motivasi pelajar itu sendiri yang menjadi kekangan bagi pelajar untuk mereka bekerja dalam bidang pengajian elektrik. Pelajar SKE perlu lebih bersemangat untuk

bekerja dalam bidang elektrik kerana Latihan Industri memberi pengetahuan yang jelas berkaitan pekerjaan dalam bidang.

4.2 Analisis Objektif 2: Mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek afektif.

Jadual 4 menunjukkan analisis skor min bagi setiap pernyataan tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek afektif. Tahap kesediaan pelajar telah diklasifikasi mengikut Jadual 2.

Jadual 4. Analisis Objektif Kedua
(Mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek afektif)

Bil	Item	Skor Min	Tahap
1	Saya berasa seronok untuk memilih kerjaya dalam bidang elektrik.	2.98	Sederhana
2	Minat saya tinggi untuk bekerja dalam bidang elektrik.	2.96	Sederhana
3	Saya berinspirasi dari pensyarah di kolej yang memainkan peranan dalam pembentukan minat dan kesediaan saya untuk memilih kerjaya dalam bidang elektrik..	3.30	Sederhana
4	Saya bersifat positif terhadap untuk memilih kerjaya dalam bidang elektrik.	3.15	Sederhana
5	Saya tidak akan putus asa untuk memilih kerjaya dalam bidang elektrik.	3.05	Sederhana
6	Hobi atau aktiviti luar kelas memberikan impak terhadap minat saya dalam pemilihan kerjaya dalam bidang elektrik.	3.95	Tinggi
7	Keinginan saya untuk bekerja dalam bidang elektrik tinggi kerana saya dapat mengaplikasi kemahiran yang saya pelajari dalam kelas.	3.10	Sederhana
8	Saya berasa kagum untuk bekerja dalam bidang pengkhususan yang dipelajari di kolej.	3.00	Sederhana
9	Saya bertekad untuk bekerja dalam bidang elektrik walaupun ditawarkan dengan gaji yang rendah.	2.90	Sederhana
	Purata Skor Min	3.29	Sederhana

Mengikut analisis di Jadual 4, didapati purata keseluruhan skor min tahap kesediaan pelajar SKE KKSJ memilih kerjaya mengikut bidang pengajian elektrik dari aspek afektif ialah 3.29 dan ia berada pada tahap sederhana. Secara keseluruhannya hanya sebanyak 58% pelajar yang bertekad untuk bekerja dalam bidang elektrik walaupun ditawarkan dengan gaji yang rendah. Ini dapat dilihat melalui pernyataan no 9 iaitu “Saya bertekad untuk bekerja dalam bidang elektrik walaupun ditawarkan dengan gaji yang rendah” mencatat skor min yang paling rendah daripada 9 item soalan yang dikaji untuk aspek afektif. Seterusnya dapat dilihat bahawa sebanyak 79% pelajar berpendapat bahawa hobi atau aktiviti luar kelas memberi impak terhadap minat pelajar dalam pemilihan kerjaya dalam bidang elektrik di mana pernyataan tersebut mencatat skor min yang paling tinggi daripada kesemua item yang dikaji bagi aspek afektif. Selain itu, hanya sebanyak 59.6% pelajar SKE menzahirkan perasaan seronok memilih kerjaya dalam bidang elektrik. Di samping itu, jumlah peratusan pelajar yang menyatakan bahawa minat mereka tinggi untuk bekerja dalam bidang elektrik adalah sebanyak 59.2%. Manakala hanya 60% pelajar menyatakan perasaan kagum untuk bekerja dalam bidang pengkhususan yang dipelajari di kolej. Ini dapat disokong dengan pernyataan no.8 iaitu “Saya berasa kagum untuk bekerja dalam bidang pengkhususan yang dipelajari di kolej” yang mencatat skor min pada tahap sederhana iaitu 3.00. Penerapan domain afektif dalam aktiviti pembelajaran di kelas dan melalui program – program yang dijalankan di institusi membantu pelajar menanamkan ciri – ciri afektif seperti minat, suka, yakin dalam diri pelajar [7]. Elemen afektif juga memainkan peranan untuk mempengaruhi pencapaian seseorang [11]. Pelajar SKE perlu yakin dan tidak putus asa untuk memilih kerjaya dalam bidang elektrik di mana hanya sebanyak 61% pelajar menyatakan rasa tidak putus asa dalam pemilihan

kerjaya dalam bidang pengajian elektrik. Ini kerana kejuruteraan elektrik merupakan bidang TVET yang makin mendapat sambutan tinggi untuk diceburi. Dapatan kajian bagi objektif kedua menunjukkan faktor afektif mempengaruhi pelajar Program Sijil Teknologi Elektrik (SKE) di Kolej Komuniti Seberang Jaya untuk kesediaan pembentukan dan perancangan kerjaya. Tahap kesediaan terhadap pemilihan kerjaya dalam bidang dari aspek elemen afektif masih sederhana dalam kalangan pelajar SKE.

4.3 Analisis Objektif 3: Mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek pengalaman.

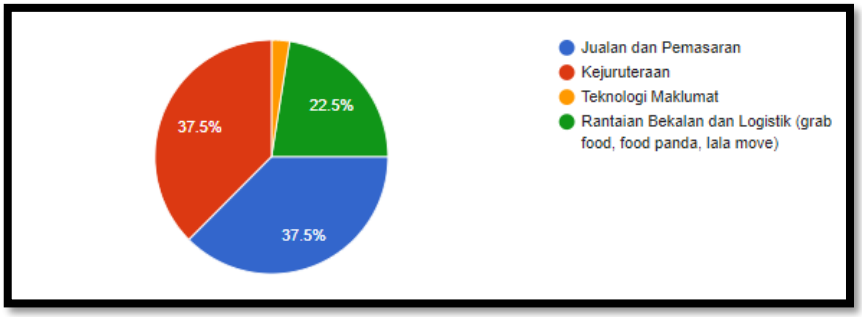
Jadual 5 menunjukkan analisis skor min bagi setiap pernyataan tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek pengalaman. Tahap kesediaan pelajar telah diklasifikasi mengikut Jadual 2.

Jadual 5. Analisis Objektif Ketiga
(Mengenalpasti tahap kesediaan pelajar KKSJ memilih kerjaya mengikut bidang pengajian dari aspek pengalaman)

Bil	Item	Skor Min	Tahap
1	Saya memperoleh pengalaman dari latihan industri terhadap kesediaan saya untuk memasuki dunia pekerjaan dalam bidang elektrik.	3.93	Tinggi
2	Pengalaman latihan industri memberikan gambaran yang lebih jelas tentang realiti dunia kerja kepada saya.	3.68	Tinggi
3	Interaksi dengan rakan sebaya di kolej memainkan peranan dalam membentuk persepsi dan kesediaan saya terhadap kerjaya dalam bidang.	3.60	Sederhana
4	Penglibatan dalam aktiviti sukarela (CSR) mempengaruhi pengalaman dan kesediaan saya untuk melibatkan diri dalam bidang pekerjaan yang saya pelajari di kolej.	3.83	Tinggi
5	Penglibatan dalam projek pelajar di kolej mempengaruhi kesediaan saya dalam memilih dan membentuk kerjaya dalam bidang pengajian yang saya pelajari. .	3.50	Sederhana
6	Penglibatan dalam aktiviti ilmiah di kolej mempengaruhi kesediaan saya dalam memilih dan membentuk kerjaya dalam bidang pengajian yang saya pelajari di kolej.	3.80	Tinggi
7	Penglibatan dalam pekerjaan sambilan mempengaruhi pengalaman dan kesediaan saya untuk melibatkan diri dalam bidang pekerjaan elektrik.	3.50	Sederhana
	Purata Skor Min	3.86	Tinggi

Mengikut analisis di Jadual 5, didapati purata keseluruhan skor min tahap kesediaan pelajar SKE KKSJ memilih kerjaya mengikut bidang pengajian elektrik dari aspek pengalaman ialah 3.86 dan ia berada pada tahap tinggi. Sebanyak 78.6% pelajar menyatakan bahawa mereka memperoleh pengalaman dari latihan industri terhadap kesediaan untuk memasuki dunia pekerjaan dalam bidang elektrik. Ini dapat disokong dengan pernyataan no.1 iaitu “Saya memperoleh pengalaman dari latihan industri terhadap kesediaan saya untuk memasuki dunia pekerjaan dalam bidang elektrik” menunjukkan skor min yang paling tinggi daripada 7 item yang dikaji bagi aspek pengalaman. Manakala sebanyak 76.6% pelajar bersetuju bahawa penglibatan dalam aktiviti sukarela (CSR) mempengaruhi pengalaman dan kesediaan mereka untuk melibatkan diri dalam bidang pekerjaan yang dipelajari di kolej. Ini dapat disokong dengan pernyataan no.4 iaitu “Penglibatan dalam aktiviti sukarela (CSR) mempengaruhi pengalaman dan kesediaan saya untuk melibatkan diri dalam bidang pekerjaan yang saya pelajari di kolej” yang juga mencatat skor min tinggi. Di samping itu, sebanyak 70% pelajar turut setuju bahawa penglibatan dalam projek pelajar di kolej mempengaruhi kesediaan mereka dalam memilih dan membentuk kerjaya dalam bidang pengajian dipelajari. Pelajar berasa pengalaman dari latihan industri, penglibatan dalam

program kesukarelawan, projek akhir pelajar dan aktiviti ilmiah yang lain dljalankan di institusi banyak memberi pengalaman untuk mereka menghadapi alam pekerjaan [5]. Selain itu, sebanyak 70% pelajar setuju bahawa penglibatan dalam pekerjaan sambilan mempengaruhi pengalaman dan kesediaan mereka untuk melibatkan diri dalam bidang pekerjaan elektrik. Ini dapat dilihat pada dapatan data yang telah dianalisis untuk pekerjaan sambilan pelajar SKE di Rajah 1. Berdasarkan Rajah 1 di bawah, hanya sebanyak 37.5% pelajar yang melibatkan diri dalam pekerjaan sambilan bidang elektrik dan 60% lagi bekerja dalam bidang jualan pemasaran dan rantaian bekalan dan logistik seperti *grab food, food panda* dan *lala move*. Apabila pelajar SKE selesa dengan pekerjaan sambilan di luar bidang maka mereka tidak mahu berusaha untuk mencari pekerjaan lain khususnya dalam bidang elektrik meskipun kejuruteraan elektrik bergerak seiring dengan permintaan industri dan perkembangan teknologi semasa. Pelajar biasanya belajar melalui pengalaman mereka sendiri. Pelajar SKE lazimnya bekerja secara sambilan dalam bidang selain bidang kejuruteraan dengan gaji yang tinggi maka mereka tidak bertekad dan bersemangat untuk bekerja dalam bidang kejuruteraan. Pernyataan ini disokong dengan analisis dapatan kajian yang diperoleh bagi aspek afektif di mana “Saya bertekad untuk bekerja dalam bidang elektrik walaupun ditawarkan dengan gaji yang rendah” menampilkan catatan skor min yang sangat sederhana iaitu 2.90 dan menunjukkan pelajar SKE kurang setuju dengan pernyataan tersebut. Graduan kejuruteraan lebih berminat menceburi bidang lain selepas tamat pengajian kerana merasakan pekerjaan sambilan mereka memberi gaji yang lebih tinggi manakala bidang jurutera hanya mampu menawarkan gaji antara RM2000-RM3000 dan ia tidak setimpal dengan beban kerja [19]. Secara keseluruhannya dapatan bagi tahap kesediaan terhadap pemilihan kerjaya dalam bidang dari aspek pengalaman mencatatkan skor min yang tinggi dan jelas membuktikan bahawa pengalaman terdahulu yang dimiliki oleh pelajar sebelum memasuki alam pekerjaan sebenar memberi kesan yang besar untuk memilih kerjaya mengikut bidang.



Rajah 1: Bidang Pekerjaan Sambilan Yang Diceburi Oleh Pelajar SKE

5. Kesimpulan dan Cadangan

Bab 5 ini akan membincangkan mengenai kesimpulan daripada analisis dapatan kajian yang telah dijalankan dan cadangan untuk manambahbaik faktor-faktor yang memberi kesan terhadap tahap kesediaan pelajar SKE KKSJ memilih kerjaya mengikut bidang pengajian serta menyatakan cadangan kajian lanjutan.

5.1 Kesimpulan

Secara keseluruhannya dapat dilihat dan dikenalpasti, purata skor min tahap kesediaan pelajar Program Sijil Teknologi Elektrik (SKE) KKSJ memilih kerjaya mengikut bidang pengajian dari aspek motivasi, afektif dan pengalaman ialah 3.51 dan berada pada tahap kesediaan yang sederhana. Justeru itu, pihak pengurusan kolej, pensyarah SKE dan Unit Pengesanan Graduan KKSJ perlu mengembleng tenaga untuk mengambil inisiatif yang lebih efisien dan positif secara beterusan dengan memberi dorongan yang lebih tinggi, dan motivasi yang lebih proaktif kepada bakal pelajar yang menjejak ke alam pekerjaan bagi mewujudkan semangat bekerja mengikut bidang pengkhususan dan kemahiran yang dimiliki. Ini dapat ditinjau daripada dapatan kajian dimana sebanyak 53.4% pelajar berpendapat bahawa mereka kurang bermotivasi untuk mengejar kerjaya dalam bidang elektrik selepas

menamatkan program di Kolej Komuniti Seberang Jaya. Ini kerana aspek afektif yang rendah dalam diri pelajar menyebabkan mereka tidak cenderung untuk memiliki kerjaya dalam bidang. Hanya sebanyak 58% pelajar yang bertekad untuk bekerja dalam bidang elektrik walaupun ditawarkan dengan gaji yang rendah. Graduan kejuruteraan lebih berminat menceburi bidang lain selepas tamat pengajian kerana merasakan bidang jurutera hanya mampu menawarkan gaji antara RM2000-RM3000 dan ia tidak setimpal dengan beban kerja [19]. Dengan ini para graduan beralih ke prospek kerjaya yang menguntungkan mereka. Begitu juga dengan pelajar SKE Kolej Komuniti Seberang Jaya yang lebih cenderung menceburi bidang pekerjaan sambilan mereka berbanding bidang kejuruteraan. Berdasarkan dapatan kajian juga didapati sebanyak 60% pelajar bekerja secara sambilan dalam bidang jualan pemasaran dan rantaian bekalan dan logistik seperti *grab food, food panda* dan *lala move.* Apabila pelajar selesa dengan pekerjaan sambilan mereka maka mereka tidak berusaha untuk mencari pekerjaan lain khususnya dalam bidang elektrik. Di samping itu generasi muda tidak lagi berminat untuk memilih kerjaya dalam bidang kejuruteraan disebabkan pengaruh media sosial [20].

5.2 Cadangan

Berikut adalah cadangan untuk meningkatkan tahap kesediaan pelajar SKE KKSJ memilih kerjaya mengikut bidang pengajian:

i. Penganjuran Program Berkonsepkan Penerapan Aspek Afektif

   Barisan pengurusan kolej dan pensyarah Program Sijil Teknologi Elektrik khususnya perlu meningkatkan lagi peganjuran program berkonsepkan penerapan aspek afektif dalam diri pelajar supaya pelajar lebih minat, seronok, bersemangat untuk memilih kerjaya dalam bidang kemahiran yang dimiliki. Mereka perlu bertekad untuk bekerja dalam bidang walaupun dipengaruhi oleh banyak faktor luaran lain. Peganjuran program yang berkonsepkan menyemai sikap afektif dalam diri pelajar juga dapat meningkatkan motivasi dan keyakinan pelajar untuk berusaha secara berterusan supaya dapat menyandang pekerjaan dalam bidang.

ii. Pelaksanaan Program Secara Kolaborasi Bersama Persatuan Jurutera Malaysia

   Unit Latihan Industri KKSJ perlu mengambil inisiatif yang lebih proaktif untuk melaksanakan program secara kolaborasi bersama Persatuan Jurutera Malaysia dalam menangani masalah pelajar bekerja di luar bidang pengajian. Dengan adanya peglibatan wakil Persatuan Jurutera Malaysia maka pelajar akan lebih membuka minda dan memahami akan kepentingan bidang kejuruteraan untuk masa depan dan pelajar tidak hanya bergantung kepada pekerjaan sambilan. In kerana kualiti graduan tidak boleh diukur hanya berdasarkan kebolehan graduan menempatkan diri dalam sesuatu sektor pekerjaan malah perlu dikenalpasti graduan mampu memiliki pekerjaan mengikut bidang pengkhususan yang telah dipelajari dan kemahiran yang dimiliki [3].

5.3 Cadangan Kajian Lanjutan

Kajian ini boleh ditambahbaik dengan menggunakan reka bentuk kajian jenis kualitatif dengan menemuramah pelajar dan staf berkaitan persepsi masing-masing terhadap tahap kesediaan pelajar memikul pekerjaan mengikut bidang pengajian. Selain itu, kajian lanjutan juga boleh dibuat terhadap tahap kesediaan pelajar menyandang pekerjaan mengikut bidang pengajian dan kemahiran yang dimiliki di peringkat Kolej Komuniti Zon Utara dan tidak menumpu pada Program Sijil Teknologi Elektrik sahaja.

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APPLICATION OF DRONE TECHNOLOGY IN DEMARCATION SURVEYING

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ABSTRACT

The word 'DRONE' (Dynamic Remotely Operated Navigation Equipment), also known as Unmanned Aircraft System (UAS), has become increasingly popular among the public because they are easy to operate. The use of drone technology is also used in land surveying. The Department of Survey and Mapping Malaysia (DSMM) also issued the Geomatics Survey Regulation 2021 which provides guidelines on Topographic Surveying using Photogrammetry Airborne Survey method or UAS method. Drone technology in land surveying activities requires high accuracy. Licensed Land Surveyors have started using these high-resolution images to assist in the presentation of Cadastral Survey data, namely Certified Plan by JUPEM. However, based on the latest survey regulations, namely the Geomatics Survey Regulations 2021, mapping using drones is not specifically linked to Cadastral Surveying. The Geomatic Survey Regulations 2021 place drone mapping under the heading of Topographic Surveying. Therefore, this study aims to identify the potential of drones in Cadastral Survey activities through a Demarcation survey. This research contains 3 objectives which is, to explore the use of drone in Demarcation survey, to test the accuracy of coordinate from the drone and to evaluate the potential of using drone technology in Demarcation Surveying by identifying the advantages of using drone. The study area is a land lot numbered PT3530 in Kulim District, Kedah. This land lot is a cemetery waqf land under the supervision of At-Taufiq Mosque, Ayer Merah Kulim, Kedah. This land lot covers an area of 1.5 hectares. This study used the onion model by Saunders with positivism philosophy. This is a mono method experimental study. Therefore, two methods of Demarcation survey were carried out to find the coordinate of boundary mark of land lot PT3530, which is the Demarcation survey supported by GNSS and the Demarcation survey supported by drone. Afterwards, their result compared with the original coordinate. The coordinates of the drone at the boundary mark Pvc2 show a difference of -0.022m North and -0.021m East. While the coordinates of the boundary mark PpbL5 show a difference of -0.020m North and -0.021m East. Regardless of some of the disadvantages of drones, the use of drones in Demarcation surveys gives more advantages to the results of Cadastral Surveying. The drone is able to provide coordinate accuracy up to 0.030 meters horizontally. In this study, the use of drones in demarcation survey is not too complicated. The result of this study proves that drones can provide very accurate and high-resolution products in accordance with the needs of Demarcation Surveys in Cadastral Surveying by identifying the advantages of using this technology. Thus, with more advantages and benefits from drone technology, this study proposes the wide use of this technology in Cadastral Surveying especially in Demarcation surveys.

Keywords: cadastral surveying; orthophoto; demarcation survey; drone; aerial mapping

1. Introduction

The word 'DRONE' (Dynamic Remotely Operated Navigation Equipment), also known as Unmanned Aircraft System (UAS), has become increasingly popular among the public because they are easy to operate [1][2]. This is due to drone technology that has become more and more sophisticated and has been used in various fields, especially in the field of photography, agricultural filming, military, national border surveillance, search and rescue operations [1]. In the field of education, drone technology has also been used in the field of teaching and learning from primary school to university level [1]. The Malaysian Government

supports the development of drones by placing drone technology on the third strategy and Strategy 12 in the Fourth Industrial Revolution (4IR) National Policy. Strategy 3 is to support equipping the workforce with IR4.0 technology while strategy 12 explains that the Government is ready to update its regulatory approach and review the rules that hinder the development of IR4.0 technology [3]. Through the 2024 budget, the Government has allocated RM10 million for drone technology activities under MRANTI [4]. The use of drone technology is also used in land surveying. The Department of Survey and Mapping Malaysia (DSMM) also issued the Geomatic Survey Regulation 2021 which provides guidance on topographic mapping using drones. Drone technology in land surveying activities requires high accuracy. The accuracy of mapping using drones depends on several factors such as flight altitude, weather, the number of control stations and the specifications of the drone used [5]. Drones with high specifications can produce a high-resolution image and provide accurate coordinates, that are complete enough to be used for topographic mapping and Cadastral Survey. Licensed Land Surveyors have started using these high-resolution images to assist in the presentation of Cadastral Survey data, namely Certified Plan by JUPEM. However, based on the latest survey regulations, namely the Geomatics Survey Regulations 2021, mapping using drones is not specifically linked to Cadastral Surveying. The Geomatics Survey Regulations 2021 describe about drone mapping under the heading of Topographic Surveying. The difference between North and East coordinates from drone and GNSS measurements is +0.380cm and +0.351cm [6]. This difference is within the tolerance of the Cadastral Surveying coordinate data which is 3.0cm Northing and 3cm Easting set by the Department of Survey and Mapping Malaysia (DSMM) [5]. The result from the previous study shows that the high-resolution images from drones are suitable for cadastral surveying [6]. Therefore, this study aims to identify the potential of drones in Cadastral Survey activities through a Demarcation survey. This research contains 3 objectives which is, to explore the use of drone in demarcation survey, to test the accuracy of coordinate from drone and to evaluate the potential of using drone technology in Demarcation Surveying by identifying the advantages of using this technology. This study used the onion model by Saunders with positivism philosophy [7]. This is a mono method experimental study. Therefore, two methods of Demarcation survey were carried out to find the coordinate of boundary mark of land lot PT3530, which is the Demarcation survey supported by GNSS and the Demarcation survey supported by drone.

2. Methodology

To have a better understanding of the components of this research, the research onion has been chosen for describing the overall research design in achieving the aim and objectives of the research. This is because the research onion elaborates in more detail how the objectives of this research could be achieved in each of the layers. Also, the layers are arranged systematically to make them easier to follow and understand. This study used a mono quantitative experimental approach and used paired sample T-test analysis to prove the accuracy of coordinate by drone. Therefore, two methods of Demarcation survey were carried out to find the coordinate of boundary mark of land lot PT3530, which is the Demarcation survey supported by GNSS and Demarcation survey supported by drone. A demarcation survey is the activity of identifying the location of boundary marks by determining the coordinates of each boundary mark of the land lot. In order to achieve the first objective and third objectives, a study area is set at land lot numbered PT3530 in Kulim District, Kedah. This land lot is a cemetery waqf land under the supervision of At-Taufiq Mosque, Ayer Merah Kulim, Kedah. This land lot covers an area of 1.5 hectares. The DJI Phantom 4 Pro V2.0 is used to explore the procedure to produce an accurate orthophoto with guidance by Geomatics Surveying Regulation 2021. For achieving the second objective, Demarcation survey with GNSS is set at the same area which is land lot PT3530. The result from both activities are compare with the original coordinates stated in Certified Plan by JUPEM. From both activities from this study can obtain third objectives which is to evaluates the potential of drone in Demarcation Survey.

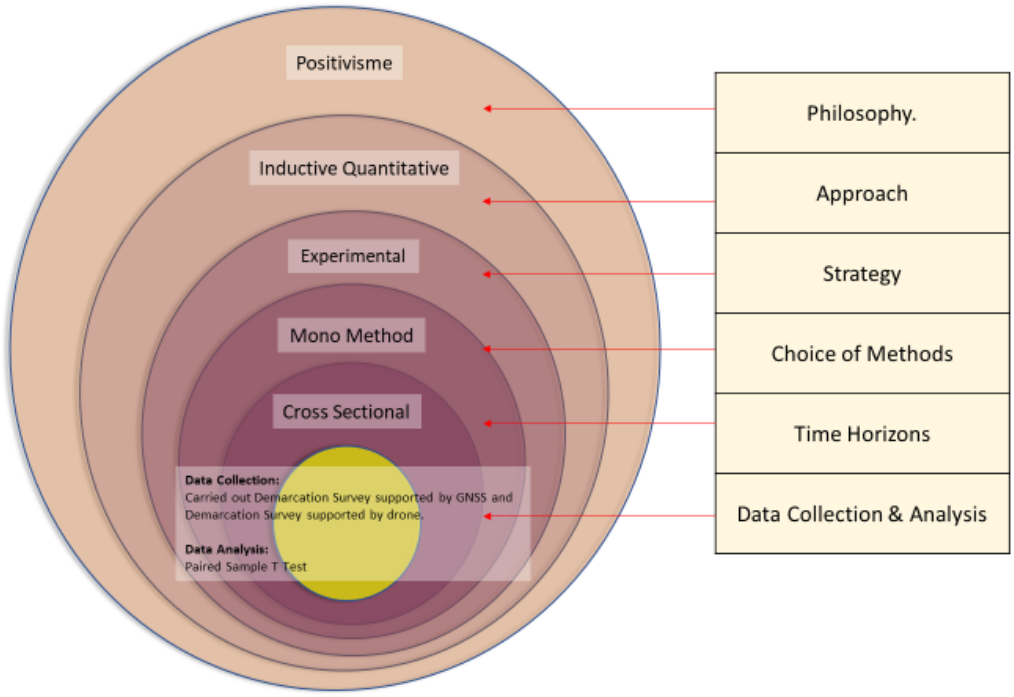


Fig. 1. Research Model

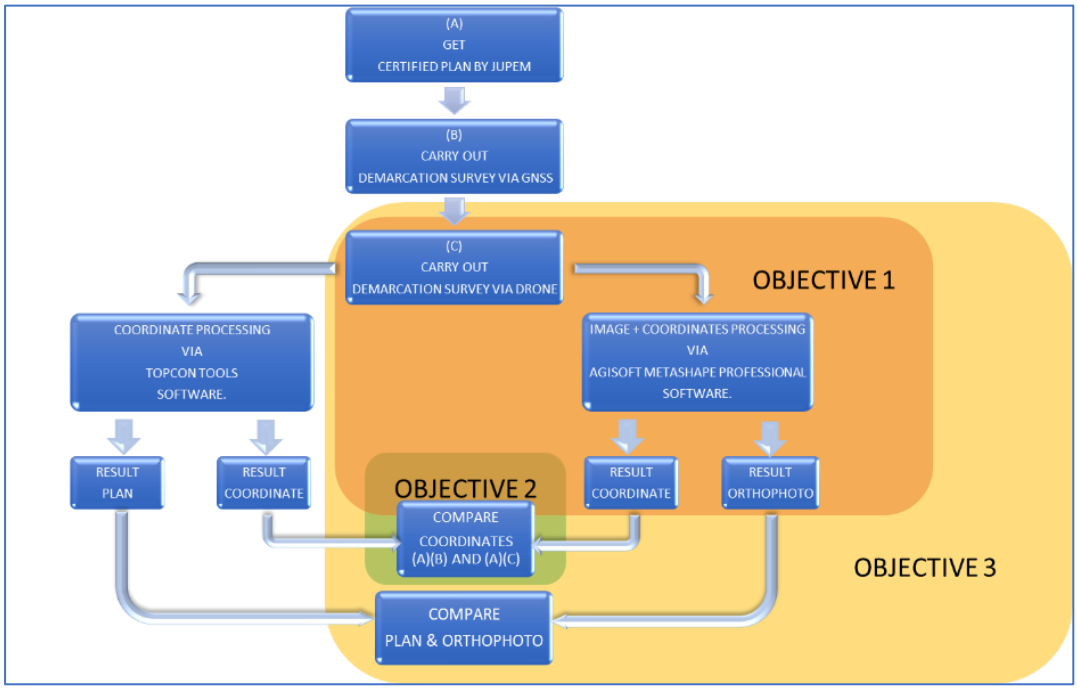


Fig. 2. Method of Research

2.1 Data Acquisition.

a. Demarcation Supported by Global Navigation Satellite System (GNSS) Method.

In order to achieve the second objective, a Demarcation survey supported by GNSS is carried out. A Cadastral Reference Mark (CRM) was established in the land Lot PT3530. Static GNSS

observations were carried out for 1 hour by using CRM 1 station in the Tuanku Sultanah Bahiyah Polytechnic area (PTSB) as a reference station (Base Station). While the boundary mark 1 of Lot PT3530 (BKL 1) has been used as CRM 1 for this study. Figure 12 shows, the location of CRM 1 (Base Station) in Tuanku Sultanah Bahiyah Polytechnic (PTSB) and (CRM 1) in Lot PT3530 area for this study. The MyLot application provided by JUPEM is used to roughly identify the position of boundary mark. All five (5) boundary marks for land lot PT3530 were found by using this application. Henceforth, the rapid static observation method was carried out on each boundary mark using the Topcon Hiper V GNSS tool. This activity was carried out to obtain the coordinate value of the boundary mark to determine the position of the boundary stone in the conditions as in the Certified Plan. The observations were carried out for 30 minutes for each station. For this observation, CRM 1 is used as a reference station (base station). Figure 4 shows the position of the boundary mark for the land lot PT3530, CRM 1 (BKL1), station 2 (Pvc2), station 3 (PbL3), station 4 (PpBL4) and station 5 (PpBL5). Topcon Tools software was used for data processing purposes. The observation coordinate was compared with the original coordinate which was stated in the Certified Plan by JUPEM. The difference tolerance of coordinate is ±0.030m for horizontal component which is Northing and Easting and ±0.060m for vertical component which is the height of the point.



Fig. 3. GNSS Observation for CRM



Fig. 4. GNSS Observation for CRM.



Fig. 5. Location of Boundary Mark

b. Demarcation Survey Supported by Drone Method.

This activity aims to achieve objectives 1 and 3. A Demarcation Survey, supported by the DJI Phantom 4 Pro V2.0 drone, was conducted to explore drone technology's application in Demarcation surveying and to assess its advantages in this field. To ensure precise image coordinates, five (5) Ground Control Points (GCP) were established across the entire PT3530 land lot using the Topcon Hiper V GNSS tool. The GNSS coordinate observation method remains consistent with previous GNSS procedures. The study utilized the DJI Phantom 4 Pro

V2.0 drone, with flight planning conducted via Pix4Dcapture Pro software. Flight parameters were set at an 80% front overlap and 60% side overlap, adhering to the Geomatics Survey Regulations 2021, at a flight height of 80 meters at medium speed. This planning resulted in the capture of 200 images. Agisoft Metashape Professional software was employed to process these images into an orthophoto.



Fig. 6. GCP Location



Fig. 7. Type of Drone used: DJI Phantom 4 Pro V2.0

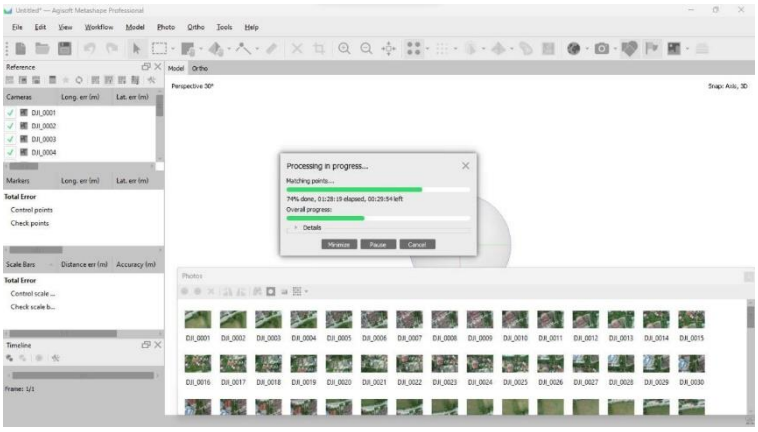


Fig. 8. Drone Image Processing Using Agisoft Metashape Professional Software

3. Result and Discussion

The reference coordinates of all boundary marks for land lot PT3530 are detailed in Table 1, sourced from the Certified Plan issued by JUPEM. According to the Circular from the Director General of Survey and Mapping No.5 Year 2021, the allowable deviation between the coordinates of the actual boundary marks (as per the Certified Plan) and those obtained through GNSS observation must not exceed specified tolerances. These tolerances are set at a maximum of 0.030 meters for both Northing and Easting coordinates, while the allowable deviation in height should not exceed 0.060 meters [8].

Table 1. Coordinate of Boundary Mark from Certified Plan by JUPEM

Point	N / S(m)	E / W(m)
CRM 1 (BKL 1)	-67871.634	-11186.278
Pvc2	-67767.006	-11202.989
PbL3	-67724.862	-11043.043
PpbL4	-67837.821	-11086.466
Ppbl 5	-67869.373	-11091.492

The results of GNSS observations using the Topcon Hiper V are presented in Table 2, displaying the coordinates of the five boundary marks for land lot PT3530. These coordinates were processed using Topcon Tools software. Table 3 shows the coordinates of five Ground Control Points (GCPs) captured by the DJI Phantom 4 Pro V2.0 drone, processed using Agisoft Metashape Professional software. To meet Objective 2, both sets of observations share two common points: Pvc2 and PpBL5.

Table 2. Coordinate Boundary Mark from GNSS Data Observation

Point	N / S(m)	E / W(m)
CRM 1 (BKL 1)	-67871.612	-11186.278
Pvc 2	-67767.016	-11202.989
PbL3	-67724.891	-11043.043
PpBL4	-67837.853	-11086.479
PpBL 5	-67869.384	-11091.471

Table 3. Coordinate of Boundary Mark from GNSS of Drone

Point	N / S	E / W
GCP 1/Pvc 2	-67767.038	-11203.010
GCP 2	-67748.746	-11022.378
GCP 3/PpbL5	-67869.404	-11091.492
GCP 4	-68001.109	-11079.914
GCP 5	-67911.401	-11161.418

The comparison of GNSS observation data from the Topcon Hiper V with coordinates from the Certified Plan demonstrates high accuracy, as shown in Table 4. At Station 1 (CRM 1), the observed data is within tolerance, with a deviation of -0.022 meters in the North and 0.000 meters in the East. Similarly, the coordinates of the second boundary mark (Pvc2) are within tolerance, with a deviation of -0.010 meters in the North and 0.000 meters in the East. However, for the third boundary mark (PbL3), there is a deviation of -0.031 meters in the North, while the East remains within tolerance at 0.000 meters. The fourth boundary mark (PpBL4) shows deviations of -0.032 meters in the North and -0.013 meters in the East. The fifth boundary mark (PpBL5) exhibits deviations of -0.011 meters in the North and 0.021 meters in the East. The larger errors observed for the third and fourth boundary marks are attributed to their positions under the tree and roof. These findings highlight the limitations of the GNSS method.

Table 4. Differences of Coordinate of Boundary Mark from Cadastral Plan and GNSS Observations

Point	N / S (m)	E / W(m)
CRM 1 (BKL 1)	-0.022	0
Pvc2	-0.010	0
PbL3	-0.031	0
PpbL4	-0.032	-0.013
PpbL5	-0.011	+0.021

The comparison from the drone also shows the height accuracy value. A comparison of coordinates focusing on 2 boundary marks. At boundary mark Pvc2, the drone's coordinates show a deviation of -0.022 meters in Northing and -0.021 meters in Easting. Similarly, at boundary mark PpbL5, there is a deviation of -0.020 meters in Northing and -0.021 meters in Easting.

Table 5. Differences of Coordinate of Boundary Mark from Certified Plan and GNSS of Drone

Point	N / S (m)	E / W(m)
GCP 1/Pvc 2	-0.022	-0.021
GCP 3/PpbL5	-0.020	-0.021

The coordinates accuracy from both observation methods shows minimal error compared to the actual values. Results from both methods remain within specified tolerances, with differences typically ranging between 2 to 3 centimetres. The Topcon Hiper V tools can achieve 0.000m difference compared to coordinates from the drone with a minimum 0.020m. However, Topcon Hiper V also recorded the biggest difference which is 0.032m compared to DJI Phantom 4 Pro V2.0. The drone shows the consistence accuracy with a maximum difference recorded by 0.022m. This is a good result, considering the drone method in this study uses minimal conditions. This study demonstrates that drone technology, under optimal conditions, can achieve height accuracy within the tolerances outlined in Circular No. 5 of 2021 from the Director General of Survey and Mapping. This finding confirms the fulfilment of the study's second objective.



Fig. 9. A Comparison image between (a) Orthophoto and (b) Google Maps

The entire process of demarcation survey using the DJI Phantom 4 V2.0, from flight planning to the final orthophoto product, serves to achieve the first and third objectives. In exploring the drone's application in demarcation surveys, the guidelines outlined in the Geomatics Survey Regulation 2021, specifically for Topographic Surveying using Photogrammetry Airborne Survey methods, are followed. The findings related to the first objective are detailed in the methodology.

Compared to GNSS methods, drone-assisted Demarcation survey offer significant advantages in terms of ease of use and personnel requirements. The flight application (Pix4Dcapture Pro) and processing software (Agisoft Metashape Pro) are straightforward and user-friendly. This approach enables rapid and cost-effective data collection over large areas, reducing reliance on extensive ground-based surveys and cutting operational costs. Moreover, it minimizes fieldwork time and enhances accessibility to challenging terrains or boundary marks obscured by trees or roof, areas where GNSS methods may struggle.Table 4 shows the large error of GNSS data for boundary mark PbL3 and PpbL4 which are located under the tree and roof.

The drone's capability to capture high-resolution imagery and precise geospatial data ensures accurate boundary delineation and comprehensive mapping of land parcels. For example, Figure 9(a) depicts an orthophoto image, contrasted with Figure 9(b) showing a Google Maps image, highlighting the orthophoto's ability to faithfully represent the actual surroundings of the land lot. Figure 10 displays the Demarcation survey result using the GNSS method (Proposed Certified plan), whereas Figure 11 overlays the Proposed Certified plan with the orthophoto from the drone method, offering a clearer depiction of boundary marks and surrounding conditions.

Moreover, drone technology supports efficient monitoring and updating of cadastral maps by offering real-time data, facilitating prompt detection of changes or encroachments on land lot. Based on the experience and data collected in this study, drone methods significantly enhance the efficiency, accuracy, and safety of demarcation surveying processes. Indirectly, these advancements underscore the crucial role of drone technology in making cadastral surveying indispensable for modern land management and administration

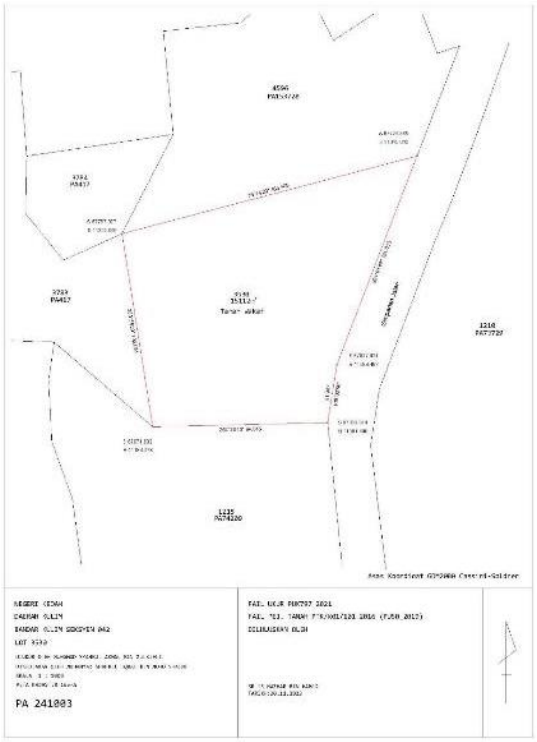


Fig. 10. Cadastral Plan Using GNSS Method

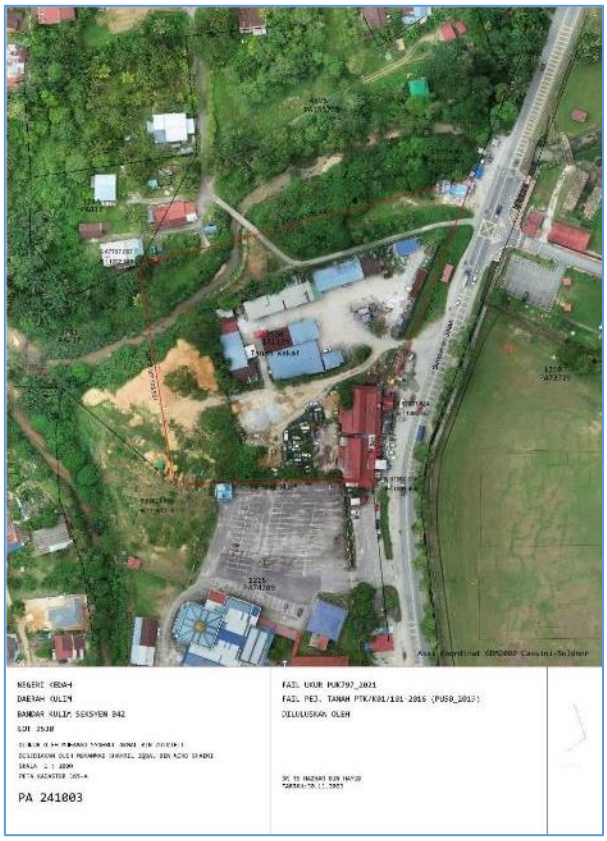


Fig. 11. Cadastral Plan with aid of Orthophoto from Drone

Table 6. Advantages of Drone in Demarcation Survey

Aided with Drone Method	Aided with GNSS Method
User Friendly: Their flight software and processing are friendly used.	User Friendly: Their software needs a time to explore because it has many usage and method.
Man Power: Required minimum 2 person to handle the tools	Man Power: Required minimum 3 person to handle the tools.
Time: Can collect more data over a large area in minimum time.	Time: Take a long time for a small area.
Cost effective: Can collect more data over a large area with minimum man power.	Cost effective: Take a long time for a small area with more man power.
Accessibility: Can apply at difficult or remote terrain or boundary mark under the tree, facilitating surveying in areas that are challenging or unsafe for GNSS methods.	Accessibility: Cannot apply at a dangerous and risk area (mountains and hills)
Result: Plan can present in 3D.	Result: Plan can present in 2D.
Result: Attractive data presentation	Result: Poor data presentation
Result: Plan with spatial data.	Result: Only a plan.
Result: Data can used for various analysis.	Result: Only for one purpose.

4. Conclusion

In this study, a drone of the DJI Phantom 4 Pro V2.0 type was used in Demarcation Surveying. The demarcation survey supported by drone technology was carried out in accordance with the guidelines outlined in the Geomatics Survey Regulation 2021, specifically for Topographic Surveying using Photogrammetry Airborne Survey methods. Despite some inherent limitations, the utilization of drones in Demarcation surveying offers significant advantages over usual methods in Cadastral Surveying. The study reveals that drone technology is straightforward to implement for supporting Demarcation surveys. The preparation of tools and procedures is uncomplicated, with user-friendly flight planning and image processing software that are easy to install and operate. These software solutions feature organized toolbars and intuitive interfaces. The drone can reduce time on the field work. For this study, it takes only 20 minutes for a drone to capture image of that area. This activity also required minimum manpower and reduce the cost of the work. In terms of accuracy, the drone-generated orthophoto accurately depicts the positions of boundary marks within specified tolerances. The drone achieves horizontal coordinate accuracy up to 0.030 meters, presenting Demarcation survey data in a visually compelling and comprehensible manner. The presentation of Demarcation survey data aided with drone orthophoto is more interesting and easier to understand. The drone is able to capture high-resolution imagery and precise geospatial data to ensure accurate boundary delineation and comprehensive mapping of land lot. It helps the landlord to find the position of the boundary marks of his land. The study proves that drones can provide very accurate and high-resolution products in accordance with the needs of Demarcation Surveys in Cadastral Surveying. This finding, it proves that drones are able to provide image results that have high coordinate data accuracy and can be used in Demarcation survey. In conclusion, this study found that drones have revolutionized cadastral surveying by offering unparalleled advantages in efficiency, accuracy, and accessibility. Their ability to swiftly collect high-resolution data over large areas at a reduced cost has supported surveying practices. By improving the accuracy of boundary delineation, facilitating surveying in challenging terrains, and enabling real-time monitoring of land parcel changes, drones will

become an essential tool for Cadastral Surveying. Drone technology in cadastral surveying not only enhances productivity and precision but also ensures sustainable and effective land mapping for future development and resource management strategies. Thus, with more advantages and benefits from drone technology, this study proposes the wide use of this technology in Cadastral Surveying especially in Demarcation surveys.

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UNEARTHING GREEN BUILDING INDEX (GBI): AN EXPLORATORY STUDY IN MALAYSIAN CONSTRUCTION INDUSTRY

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ABSTRACT

Sustainable construction has recently been highlighted to reduce the greenhouse effect around the world. Green building is a chance to use our resources more efficiently while creating an ecologically balanced and healthier building. Green Building Index (GBI) is a tool that provides an opportunity for developers and building owners to design and construct green, sustainable buildings that can provide energy savings, water savings, a healthier indoor environment, better connectivity to public transport and the adoption of recycling and greenery for their projects and reduce our impact on the environment. However, there are challenges in adapting to the GBI. Minorly local construction organisations are still reluctant to employ such green building index tools, although it is promoted to benefit their project. Therefore, this research aims to identify the Green Building Index (GBI) Critical Success Factor (CSFs) in green building construction. This research was implemented using a qualitative method, in which data was collected through in-depth virtual interviews. Five professional participants from green building and sustainable construction in Malaysia were involved in this study. This research employed Yin's 5-step analytics procedures to analyse the data. Based on the findings, this research has witnessed several factors and themes that influence the implementation of the Green Building Index (GBI). The researcher found that the criteria factor in the GBI instrument is complicated and requires certain costs to meet the requirements of green buildings. Apart from that, other emerging themes are the challenges to educating clients and developers on the GBI specifications. This study provides valuable information and practical guidance for construction industry players (project managers, contractors and designers), thereby reducing the setbacks for implementing green building projects.
Keywords: Green Building, Green Building Index (GBI), Critical Success Factor (CSFs)

1. Introduction

Green building is an outcome of a design that focuses on increasing the efficiency of resource use energy, water, and materials while reducing building impacts on human health and the environment during the building's lifecycle through better siting, design, construction, operation, maintenance and removal. Green buildings have been well-received by governments worldwide as a strategy for improving the sustainability of the construction industry in terms of the environment, economy, and society [1]. However, the adoption rate of green buildings in the Malaysian construction industry is lower than in other developed countries [2].

The main obstacle to the adoption of green buildings in the country is to create a paradigm shift in environmental issues for all Malaysians, especially those in the construction industry. Lack of awareness, even from the architects, consultants, and clients, has been repeatedly mentioned in existing research as the key issue to the slow progress and reluctance to get involved in green buildings [3]. Most of the constructed buildings in Malaysia are not categorised as green-responsive buildings because Malaysia is very much lacking in the application and development of green buildings compared to other countries such as Australia, Japan, and Singapore [4].

According to [5], various green building rating tools have been developed and are being followed in different parts of the world. Since the last decade, many assessment tools have been developed, and new ones are still being developed around the world. Malaysia has developed its rating tools, namely the Green Building Index (GBI), which is specifically suitable for its tropical weather and community needs to nurture green building development in Malaysia (GBI, 2020). It is the first Malaysian rating system for evaluating green design and performance of buildings in Malaysia. GBI is a green rating system for buildings and was developed by the Malaysian Institute of Architects (MIA) and the Association of Consulting Engineers Malaysia (ACEM) [6].

GBI is a comprehensive rating system for evaluating the environmental design and performance of buildings based on the following six main criteria: Energy Efficiency (EE), Indoor Environmental Quality (IEQ), Sustainable Site Planning & Management (SM), Material & Resources (MR), Water Efficiency (WE) and Innovation (IN) (GBI, 2010). According to [7], the GBI assessment process is divided into three stages: first, Application and Registration; second, Design Assessment (DA); and third, stage Completion and Verification Assessment (CVA). Four (4) classifications are given for the GBI certifications to show the gradient of compliance with the requirements on a 100-point scale, namely Platinum (86 – 100 points), Gold (76 – 85 points), Silver (66 – 75 points) and Certified (50 – 65 point) [8].

[9] argued that more research should be carried out on Malaysia's green design development and methodologies. This is because green building design in Malaysia is measured below the accepted average. [10] in their research state that more research and development (R&D) processes should be encouraged to accelerate the implementation of the GBI certificate in the construction industry. Therefore, this research aims to identify the Green Building Index (GBI) Critical Success Factors (CSFs) in green building construction.

There has been very little prior research and references pertaining to the themes that emerged from the Green Building Index (GBI) rating tool. Therefore, this research expands the methodology used to analyse the variables in this study. Since much existing research has been conducted using the quantitative method, this study will use qualitative methods to analyse the collected data. To distinguish this research from others, this research uses Yin's 5-step analytics procedures to analyse the collected data. This will consider the methodological gap for this research. Hence, this research narrows down to identifying the Critical Success Factors (CSFs) of the GBI rating tool in developing green buildings. In analysing the collected data, this research develops themes related to this field. This might be a contributor to the upcoming research.

2. Methodology

This research paper was implemented using a qualitative method, in which data was collected through in-depth virtual interviews. The participant was chosen based on their working experience in green building development. According to [11], qualitative research allows for a more in-depth examination and comprehension of how groups or individuals view social or human problems. A structured interview directly follows the researcher's word usage, phrases, and meaning. In contrast, a qualitative interview seeks to understand participants "on their own terms and how they make sense of their own lives, experiences, and cognitive process". Therefore, this paper encourages participants to use their own terms rather than the researcher's choice.

This research interviewed five participants who are professionals in the green building and sustainable construction industry in Malaysia. Participant 1 is working at Pulau Langkawi, Kedah, while Participant 2 is from Batu Kawan, Pulau Pinang; Participant 3 is from Sungai

Petani, Kedah; Participant 4 is from Putrajaya, and Participant 5 is from Kuala Lumpur. Table 2.1 shows the details of the participants.

Table 1. Respondent Profile

Participant Name (P)	Company Name	Position	Department	Working Experience
P1	Company A	Resident Architect	Architect and Interior Consultant	25 Years
P2	Company B	Project Manager	Project Management Department	25 Years
P3	Company C	GBI Facilitator	GBI Services	10 Years
P4	Company D	Sr. Technical executive	Highway Concessionaires	8 Years
P5	Company E	GBI Facilitator	GBISB	23 Years

Interviewees were contacted by email and phone call to arrange an appointment. The researchers gave a brief explanation of the study's purpose, the study's procedures, and the study participants' expectations, including how long it takes to participate. This is essential to ensure that participants can be assured that they are participating in the study voluntarily and can withdraw from the study at any time. Participants are also convinced that their privacy is protected and that they know the risks and benefits expected from the study. The interview was conducted via Google Meet and phone call. Each interview that was conducted was recorded. The recorded attachment was used to transcribe the participant's opinion. Manual coding was employed to conduct the data analysis procedure. Figure 1 demonstrates the interview process for this study.

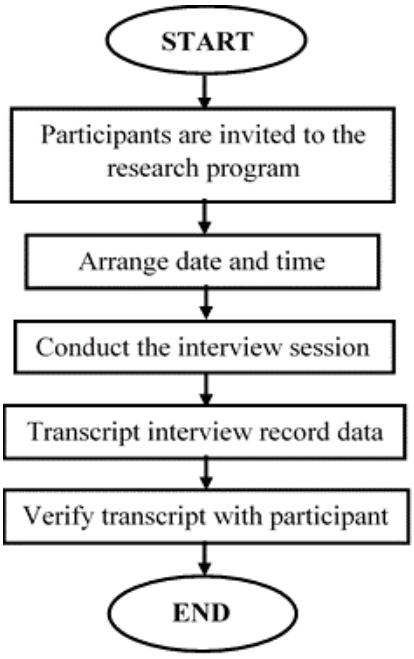


Fig. 1. Procedures for Interview Session

The role of a researcher is to observe, collect, and deliver accurate textual data or information regarding the nature of the study. While the interview session is conducted, take advantage of unexpected chances to collect data by responding to participant replies and providing questions that will elicit answers that will improve the study's findings. The five-step data analysis procedure, presented in Figure 2 below, was proposed by [12].

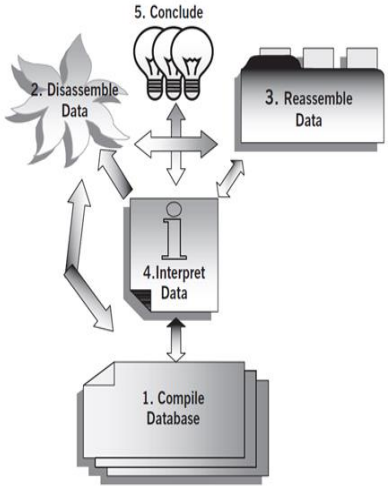


Fig. 2. Five steps procedure for qualitative data analysis

The first stage of the five-stage analytic approach proposed by [12] is compiling data. The primary goal of this step is to categorise and arrange the notes or data obtained from the field activities. Based on this research, the data was collected from the recorded interviews. Based on the recorded video, each content and opinion that the participant answered will be scripted. Once the transcript is done, it is necessary to read it several times to become acquainted with each participant's responses.

The second stage is data disassembling. The goal at this stage is to split down the gathered data into smaller fragments by identifying them with codes or levelling them. The researcher must identify particular actions, opinions, and explanations and assign them level 1 or code them to categorise the fragment. If there is a comparable item, the researcher must classify it as level 2 and place it in a bigger category. As part of the trial-and-error process, this step might be recursive between the prior two stages. Based on this research, once the script is familiarised, the next stage is breaking down the phrase into smaller fragments or keywords.

Reassembling is the third stage. The goal of this stage is to find a believable and fascinating pattern. The researcher must include the usage of substantive themes to arrange the fragment items and build a new formation that is distinct from the original text. To accomplish this, [12] introduced three arrays, which use hierarchy, create matrices, and cooperate with other arrays. Next, to see whether there is any difference in participant reaction and experience, [12] advocated three procedures: constant comparison, negative instant, and rival thinking. In addition, there is a necessity in the search procedure to mix and match the code question till it satisfies the researcher.

The fourth stage is interpreting data. This step aims to analyse the data derived from reassembling data and data arrays following the research question and purpose. The reassembled data must be scanned from a larger viewpoint until it is narrowed down to a close agreement with the study's purpose. The final stage is the conclusion. In this stage, the researcher wraps up the whole study process.

3. Result and Discussion

This part draws upon the main themes and presents findings that arose from the interview process and following data analysis. Participants are asked two parts of the questions. Part A involves participant demographics, and Part B focuses on factors influencing Critical Success Factors (CSFs) in implementing GBI as rating tools in green buildings. All the participant

answers were transcribed, followed by summarising the overall transcript and determining key themes that emerged from the following data analysis.

3.1 Critical Success Factors (CSFs) in the Implementation of the Green Building Index (GBI)

The research findings have been analysed in three elements to discuss the Critical Success Factors (CSFs): embracing the Green Building Index (GBI) as a rating tool, challenges in implementing GBI as a rating tool, and improving GBI criteria.

3.1.1 Embracing Green Building Index (GBI) as a Rating Tool

For this part, the researcher wants to assess the level of knowledge of participants who are professionals in green building and sustainable construction regarding the rating tool used to evaluate green buildings in Malaysia. Based on the responses from *Participant 1* and *Participant 2*, they stated that the Green Building Index (GBI) is a rating tool that evaluates green buildings that prioritise energy-saving consumption. Many definitions have been made to ascertain what a green or sustainable building is, but they all revolve around the same functions. For a building to be green, it is vital for the environmental impact of all its constituent parts and design decisions to be evaluated. Apart from that, based on responses from *Participant 3*, *Participant 4* and *Participant 5*, they informed that GBI is a rating tool to evaluate green buildings and measure the degree of greenness of a building. All participants agreed that GBI is a rating tool suitable for Malaysia's environment. This point is supported by existing studies from [13], who stated that GBI was developed to suit Malaysia's local climate, integrating the local building codes and making it uniquely appropriate in the Malaysian context.

Most participants responded that their company used GBI as their rating tool for green buildings. However, some companies have just considered GBI, and others have been using GBI for a long time as a green building rating tool. *Participant 2* and *Participant 4* responded that they had just used GBI as a rating tool for green buildings. *Participant 1, Participant 3,* and *Participant 5* already have more experience in GBI because most of their projects involve GBI. To answer the question, how do they implement GBI? *Participant 1* states that it is very challenging to consider GBI because it is complicated to comply with every set procedure and very hard to achieve the desired target. While *Participant 2* also claims that not all projects use GBI as a rating tool because it depends on the budget and nature of the project. If the financial allocation is sufficient and the top management allows GBI, they will implement it in their project. He mentions that, currently, two of their project use GBI as a rating tool. However, *Participant 4* responded that they used consultant services to guide them regarding processes, procedures, and other related GBI. The GBI consultant guides developers or clients to build and meet green building criteria.

According to empirical findings, most participants responded that their company used GBI as a green building rating tool because it suits a building project and follows Malaysian Standards. *Participant 1* stated that, for all private projects, their company used the GBI rating tool rather than the MyCREST rating tool for government green building projects. There are dissimilarities in terms of criteria for green building between both rating tools. While *Participant 2* outlines that GBI is more popular and known other than rating systems. It is well known because GBI was developed by the Malaysian Institute of Architects (MIA) and the Association of Consulting Engineers Malaysia (ACEM). In addition, various bodies and agencies, such as the Construction Industry Development Board (CIDB) and the Institute of Engineers Malaysia (IEM), regularly hold campaigns about GBI. Moreover, *Participant 3* claims that GBI is recognised by the World Green Building Council (WGBC). WGBC only recognise one green rating tool for each country, and it is professional to use GBI as Malaysia's green building rating tool. It is similar to *Participant 4* and *Participant 5,* which they agreed that the GBI rating system is the best practice in measuring how greener the building is.

In terms of measuring the success of their company using GBI as a rating tool, most participants stated that it depends on the target and intentions of the clients. If they want to reach the platinum level in the green building rating, then the challenges are more complex than if they only want to reach the certified level. Therefore, *Participant 3* mentioned that a GBI Certified level is relatively easy to achieve, and almost all green buildings in Malaysia fulfil the certified requirements in line with UBBL and MS1525. However, *Participant 1* outlined that achievable indices must meet all established criteria to meet the requirement. Every project that wishes to use GBI must conduct an audit every month to determine whether components were employed to meet the GBI criteria. This is because, under GBI, each selected index level must match the criteria and specifications given by the GBI authority to be recognised.

Participant 2 and *Participant 4* stated that their company has considered GBI as a rating tool to be included in the implementation of each project, so they have no single project that fully implements GBI from the beginning to the end of the project. Nevertheless, most of their new projects implement GBI as a rating tool. Even so, they agreed that the success of implementing GBI as a rating tool depends on the client's objective and the required allocation cost. While *Participant 4* also responded that client requirements and requests are more important in achieving the level of green building in terms of the GBI rating tool. Apart from that, *Participant 5* claimed that most projects that engage in sustainability right from the project's inception would likely have a better GBI implementation success rate than those that engaged in the middle of the project. Being engaged from the inception creates a better understanding and will be able to have more cost/budget control over the design.

3.1.2 Challenges in Implementing the Green Building Index (GBI)

In this part, the researcher will discuss the challenges in implementing the GBI rating tool. According to [13], the main obstacle to adopting green buildings in the country is creating a paradigm shift in environmental issues for all Malaysians, especially those in the construction industry. They also state that low investments and participation from the government and private companies in the green building movement also create a challenge for building practitioners in designing and building more efficiently. [3] state that the cost required for green building alternatives is always higher than the conventional building method. This is because the cost of a building is often measured in terms of the construction cost instead of the life cycle cost. Based on the responses from *Participant 2, Participant 3* and *Participant 4*, they agreed that cost is the most crucial challenge in implementing GBI as a green building rating tool. *Participant 2* mentioned that there would be an increase in the cost of assessing the number of points and ratings to be obtained. Each point has its own characteristics, so to get more points, then more cost is required. According to [14], a green building rating system is being developed to provide rating points to identify whether the building meets the criteria needs and to demonstrate whether the building design is green.

Apart from that, *Participant 1* claims that there are challenges to fulfilling the need for GBI Specifications because it is too complicated. He describes the difficulties faced in implementing GBI in previous green building construction compared to MyCREST rating tools, which are easier to comply with. Like *Participant 2* and *Participant 4,* they agreed on the challenges and difficulties of complying with GBI specifications. Based on the responses from *Participant 3* and *Participant 5,* another challenge in implementing GBI is to educate clients, consultants and contractors on the understanding of green requirements and intentions because most people know about the GBI but do not go into detail about the intention of the criteria. According to [3], lack of awareness, even from the architects, consultants, and clients, has been repeatedly mentioned in this research as the critical issue of slow progress and reluctance to get involved in green buildings.

3.1.3 Improvement Green Building Index (GBI) Criteria

In this section, the researcher will go a step further than challenges and look forward to improving GBI rating tools in terms of implementation. This means that the researcher will examine the possible instrument of GBI so that it can be implemented more easily in green buildings. [15] state that a green building rating system is being developed to provide rating points to identify whether the building meets the criteria needs and to demonstrate whether the building design is green or not. The green building rating system provides a framework for assessing environmental performance in green buildings.

However, according to [14], the green building rating system should be improved in technical services and resources to determine how friendly the building is environmentally based on the proper green building rating system that suits Malaysia's local context so implementation is more manageable. *Participant 1* mentioned that the GBI instrument needs to be upgraded due to numerous scopes and processes that must be followed, and it depends on the degree of the requirement. This issue will cause difficulties in achieving the required level of green building. While *Participant 2* also responded that the Green Building Index Sdn. Bhd. (GBISB) needs to review current GBI criteria or provide advice for other methods to comply with the u-value (roof criteria) requirement with minimal cost or to remove its mandatory. He claims that, even if only one point is achieved, it is mandatory to obtain and involves high costs due to the whole roofing area needing to add insulation materials thickness to achieve the required U value.

However, *Participant 3* mentioned that the characteristics required in a green building do not need to be improved as long as the construction of a green building considers the costs involved and benefits the occupants during operation. *Participant 4* suggested that the GBI category needs to be tailored to the form for Rest and Service Areas according to their current project. Apart from that, she mentions that there are too many criteria that need to be complied with in GBI, and some improvement in the priority of the criteria needs to be made. This is because focusing on certain criteria will make achieving the level of green building easier. However, *Participant 5* outlined that the sustainable consultant should be engaged at the beginning of the project rather than in the middle to know the actual project cost.

Most participants agreed that the GBI criteria meet the green building characteristics and suit Malaysia's tropical climate and culture. However, too many things need to change to fulfil the characteristics, creating an extra cost. Therefore, *Participant 1* mentions that the level of importance in the sub-criteria needs to be evaluated, and the priority should be based on the high or less important level of importance. This statement is supported by [16], which defines new weighting criteria of GBI with respect to the local conditions in Malaysia. *Participant 2* responded that some projects only target certification or at least a silver rating, which is the low rating for green building because the cost allocated is low and the criteria that need to be followed are simple. He mentions that it still achieves the characteristics of a green building according to the GBI index, even at the lowest rating. This is why achieving Gold and Platinum ratings is not easy; it involves many costs and, at the same time, needs to fulfil complex criteria. When it comes to complex criteria that need to be followed, the possibility of green building construction will decrease, and sustainable building construction cannot be achieved as planned by some countries, including Malaysia.

Therefore, *Participant 4* mentioned that, by omitting certain criteria, GBI could give another criterion to achieve the same total point to fulfil the related requirements. Apart from that, *Participant 5* also claimed that the desired criteria and the whole process in GBI must be understood by the whole team, including the clients, and not solely depend on a sustainability consultant so that it is easy to achieve green building. However, the existing criteria need to be reviewed regarding the importance of achieving green building so that it is easy for all parties to understand and apply them in their work.

4. Conclusion

The significant conclusion that this research can draw is, based on the findings, the researcher is sure that the current Malaysian construction industry is actively developing green buildings and using the Green Building Index (GBI) as a measurement tool. The researcher found that several factors influence the success of implementing GBI. First, it depends on the client's intention and target to achieve a certain level of green buildings. Second, it depends on the cost involved, where, if the cost is allocated, then GBI can be implemented in green buildings. On the other hand, this research has also witnessed the challenges the construction industry faces in implementing GBI. Based on the findings, the researcher found that cost is the most critical challenge that clients or developers should consider to achieve green buildings. Other challenges are convincing clients and developers to implement GBI in green buildings. The researcher found that the third challenge is to fulfil the need for GBI specifications because they are very complicated and difficult to comply with. Each project that uses GBI must meet the criteria because each selected index must match the criteria and specifications that have been set by the GBI authorities to be recognised.

In a nutshell, to fulfil this research purpose, the researcher will highlight the emerging themes of implementing GBI in green buildings. Firstly, the theme that emerged is that all the companies comply with the GBI in developing green building construction because it suits Malaysia's tropical culture. Secondly, the themes that emerged under challenges in implementing GBI are high cost and complicated specifications in GBI. Apart from that, other challenges include educating clients and developers on the GBI specifications. Thirdly, the theme that emerged under the GBI instruments is the improvement in terms of GBI criteria. The existing criteria need to be improved, and the features need to be maintained so that it is easier to achieve a green building rating.

The findings of this study are valuable to contractors, developers, and clients who desire to understand the Green Building Index (GBI) Critical Success Factor (CSFs) in sustainability building construction. The reader will gain an understanding of the implementation of the GBI in green building for each company and how they implement it in their project. Moreover, managers who consider GBI as a rating tool can use this research as a guide to achieve a certain level of green building. This is because this research also explains the factors influencing the success of implementing GBI as a rating tool. The empirical data gives a clear view of how each company implements GBI successfully.

For future studies, there are several suggestions for improvement, such as the scope of the study needs to be more extensive, involving all types of green sustainability development. In addition, the selected participant must be suitable for a specific company so that the feedback given is more accurate and valuable to all players in the construction industry. Moreover, further studies are recommended for those who wish to offer qualitative methods studies. This research contribution would be a guideline for upcoming studies. Overall, this research paper has only achieved the first-level themes due to respondent constraints. Upcoming research is recommended to fulfil up to the third-level themes. This will be considered the fulfilment of qualitative research proposed by [12], and at the same time, it will help to fulfil credibility and validity

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DIGITALIZATION FOR DATA COLLECTION IN PHASE CHANGE EXPERIMENT

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ABSTRACT

Digitalisation for data acquisition in phase change experiments is an innovation project for temperature detectors. This teaching tool is an innovative method used by higher educational institutions in Malaysia for teaching and learning purposes. The advantage of this tool is that it can help lecturers and students in theoretical demonstrations and practical work. The tool helps to determine phase change and temperature using Arduino and ESP32 programming software to obtain experimental data. This tool is an alternative to existing GPFDS experiments conducted manually, requiring much time, complicated handling, and inaccurate data. With the advancement of digital technology, the tool is designed for more efficient experiments and innovations, making the experimental process faster and easier. The material used for the development of this Phase Change Thermodynamic Training Kit is the ESP32, which has many advantages. Among other things, the ESP32 is a low-power chip (SoC). Also, the temperature detector uses Arduino software to display the temperature value on the LCD screen. In addition, more modern devices are also used, namely electric heat coolers to maintain low temperatures and electric fan heaters to maintain high temperatures above room temperature. To summarise, the developed training kit can solve the problems of lecturers and students by using the Arduino system. This discovery proves that this training kit can save experiment time and expand knowledge of more advanced technologies.
Keywords: Phase Change, Arduino, Teaching aid, Digitalization

1. Introduction

The expression "phase change" describes a shift in the material state. It begins with a shift in the initial temperature and continues until the phase changes. Heating solid ice cubes to a temperature at which energy is added to increase the kinetic energy of the molecules and cause them to change to the liquid phase of water is a process known as melting. The added heat energy increases the kinetic energy of the molecules until the bonds break and they undergo a phase shift. This experiment is designed to allow students to learn hands-on " by experiencing how quickly a substance undergoes a phase shift.

The term phase change refers to the transition of matter from a solid state, known as melting or freezing, to a liquid state, known as the gas, liquid, and solid phases. The transition from a liquid to a gas is known as condensation or vaporisation. There is also a direct transition from the gaseous to the solid state, known as sublimation. The process by which matter changes from one state (solid, liquid, or gas) to another is known as a phase change. These changes require the application of appropriate pressure to match the conditions or pressure changes in the environment. They are brought about by external forces acting on the cargo. The sticky energy generated by this change is also known as latent heat. The experiments of this tool are related to those of Newton.

This experiment shows that temperature fluctuations always lead to phase shifts, but temperature has a different effect. For example, thermal energy is used to increase the amount of kinetic energy that grows when objects are heated. As long as the process of phase shift is not completed, there is no temperature change in the state of matter passing from one phase

to another. There is no temperature shift because the current energy source is only used to separate the molecules and not to increase their kinetic energy.

The digitalisation of data acquisition for phase change experiments is a teaching tool designed to support the hands-on learning and teaching process. To facilitate the acquisition of experimental data, an Arduino ESP 32 is used. A 'temperature' sensor is used in an Arduino system to control the temperature of the metal. The Arduino system with IR 4.0 technology measures the temperature and uses current to generate heating and cooling power simultaneously at a certain voltage. A fan heater made of nichrome wire is used instead of a Bunsen burner to achieve a temperature of more than 10 °C above room temperature. First, the device is easy to use due to its digital and methodical style. Figure 1.0 below is a digitalisation tool for data acquisition for phase change experiments.

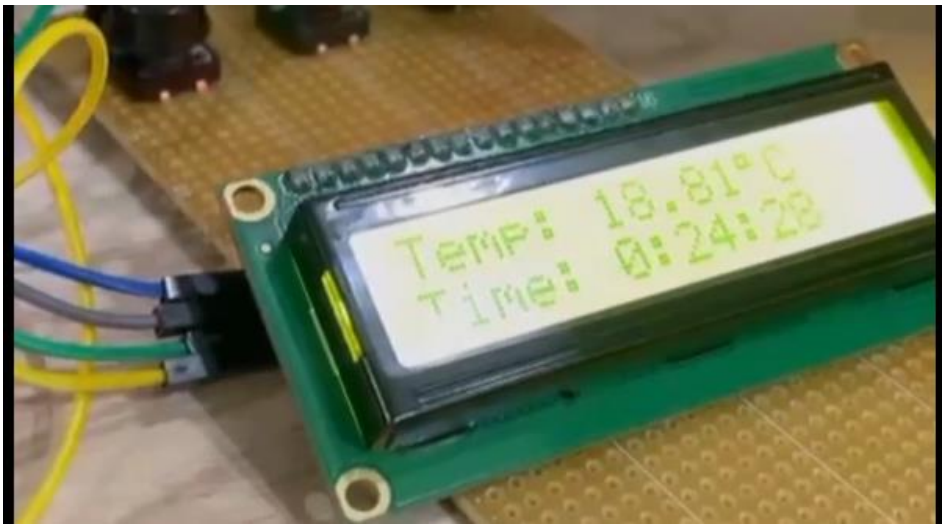


Fig. 1. Data Collection Digitalization Tool for Phase Change Experiments

An educational tool called Digitalization Data Collection for Phase Change Experiments was developed to facilitate hands-on learning and teaching methods. It uses an Arduino ESP 32 to facilitate the collection of experimental data. A temperature sensor is used in the Arduino system for temperature control. The Arduino system with IR 4.0 technology measures the temperature and uses current to generate heating and cooling power simultaneously at a certain voltage. To achieve temperatures of more than 10 °C above room temperature, a fan heater made of nichrome wire is used as a heating element instead of a Bunsen burner. It is a user-friendly operation of the experiments with a digital method to record the data. The test results are more accurate, and the management of the experiments is easier, faster, and safer.

With the old method, the device is not computerised and is difficult to use. In addition, the data obtained has not been analysed in detail. Hardware and reading errors are two types of problems that can occur. The device is shown in Figure 1.2 (left). This kit has been developed with its user-friendly interface and precise temperature control by the Arduino system. To create a "phase change training kit", the ESP 32 system was used to record temperature and time measurements.

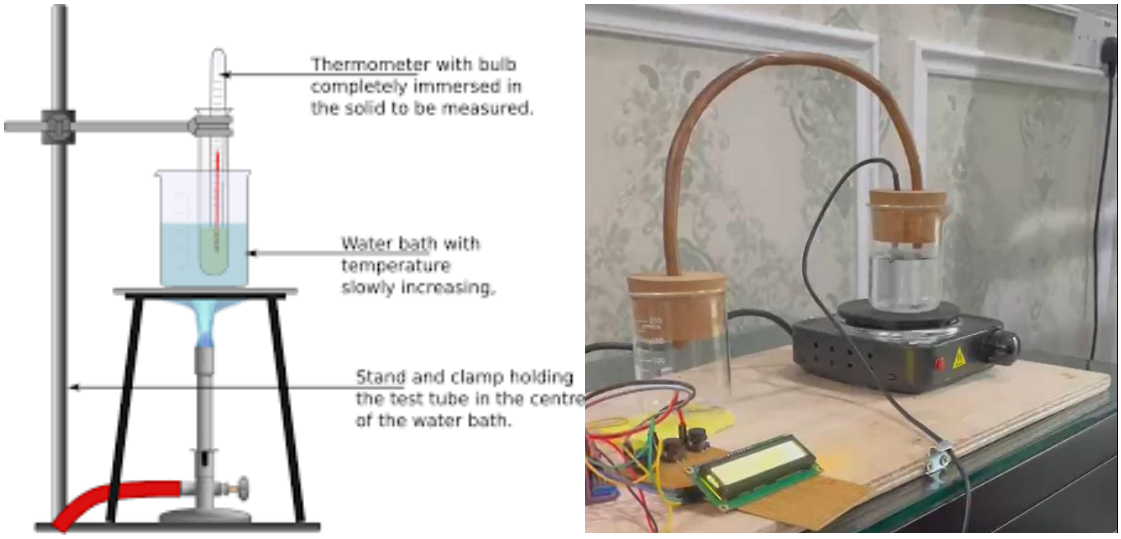


Fig. 2. Phase Change Experiment

2. Literature Review

In modern times, people have developed more complex technologies. Therefore, when developing a technology, we must conduct a thorough investigation to ensure no difficulties for future use. Various references and guidelines are used to ensure that the kit is developed to meet the technical criteria and user requirements. This study was conducted to meet the acceptable criteria. The planning should be done at the beginning of the research to ensure that the design form meets the criteria. The idea of project scope is adapted to the planned project. Various attempts have been made to find reference materials to ensure a successful project. As a result, some information about the experimental project phase change is made with Arduino. The collected results are reviewed and used to improve the project.

It was found that the Thermodynamics experimental design utilizes several new technology systems to assist in the experimentation process for various purposes. The technologies used include the Arduino system, which is used for data collection and interpretation of data screens, computers, and cell phones [1][3][4][5]. The method of previous data collection is still used in [2], although the number of those using it is decreasing.

3. Methodology

Components involve:

i. Arduino Uno
ii. LCD screen
iii. Connecting cable
iv. Temperature sensor
v. Breadboard

Arduino Uno is a microcontroller system that uses the ATnega328P. It has 14 pins for input and output purposes, 6 of which are used for output purposes that use PWM. 6 analog inputs, ceramic resonator 16MHz (CSTCE16M0V53). USB connector, ICSP power supply socket, and reset button. The Arduino is the part that acts as the brain that controls all data changes and interprets them. If the Arduino is connected to the computer via USB, a program can be created on the Arduino to control its actions.



Fig. 3. Arduino Uno and LCD Panel

With this USB connection cable, the Arduino can be connected to a computer and programmed without using an external medium. The Arduino used by Luca_McLoughlin [1], Andrei Florian [3], P Chanthamanee, P Jinda, M Mani, and Prasitpong [4], and ThothLoki [5] must be programmed with this cable to connect the Arduino and the computer.



Fig. 4. USB Cable, Temperature Sensor, and Breadboard

DHT11 is a sensor that measures the ambient temperature and humidity in the environment. This type of sensor is used by Luca_McLoughlin [1], Andrei Florian [3], P Chanthamanee, P Jinda, M Mani, and Prasitpong [4], and ThothLoki [5]. This sensor works by sending an output that is sent to the Arduino. This type of sensor is suitable for use with Arduino boards and even Raspberry Pi boards, which are also ideal for use at the 5V and 3.3V voltage levels.

Breadboard allows the creation of simple temporary electronic circuits and quickly conducts experiments with circuit design. The breadboard allows designers to connect components or wires easily thanks to the rows and columns of spring terminals mounted inside under the perforated plastic housing. This breadboard was used by Luca_McLoughlin [1], Andrei Florian [3], P Chanthamanee, P Jinda, M Mani and Prasitpong [4], and ThothLoki [5] to create a circuit for connecting the temperature sensor to Arduino Uno and Arduino Mega systems.

4. Result and Discussion

The following are the data values obtained from testing the material's melting and boiling points. Table 1 shows the average data for the water boiling point test, while Table 2 shows the average data for the water melting point test. The study's objective is to compare the validity of the use value of new and old tools.

Table 1. Experiment 1 (Average Boiling Points)

Time (min)	Digital (℃)	Manual (℃)
0	26	26
2	31	31
4	46	48
6	64	68
8	78	74
10	88	79

12	92	98
14	96	100
16	98	100
18	99	100
20	100	100

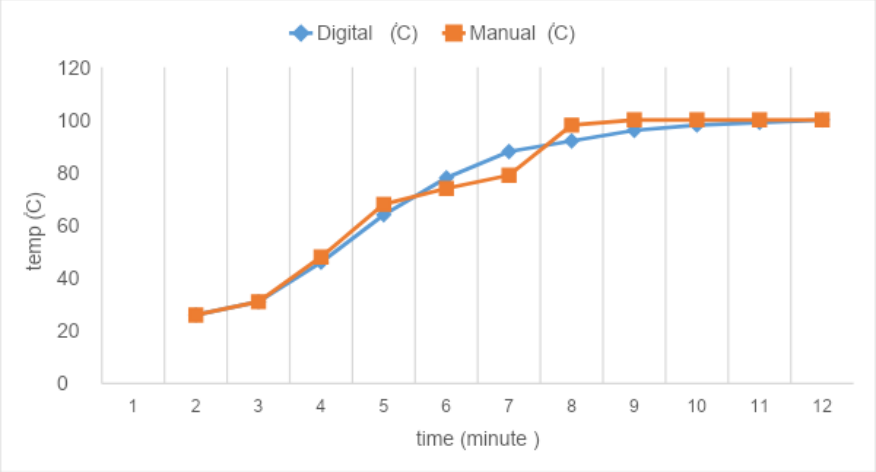


Fig. 5. Graph temperature vs time for Digital and Manual Tools (Boiling)

Figure 5. is a graph of the temperature change from the liquid phase to the gas phase to determine the boiling point of a substance. The findings of the experimental data prove that the value obtained from digital data has a higher validity than manual data. Digital data values are more accurate with a uniform distribution of graph points. Table 2 shows the average water boiling point value for the two data sets.

Table 2. Experiment 2 (Average Melting Point)

Time (min)	Digital (C)	Manual (C)
0	-0.8	1
2	13	6
4	15	6
6	50	30
8	81	58
10	100	74
12	101	83
14	101	97
16	101	100
18	101	101
20	-0.8	101

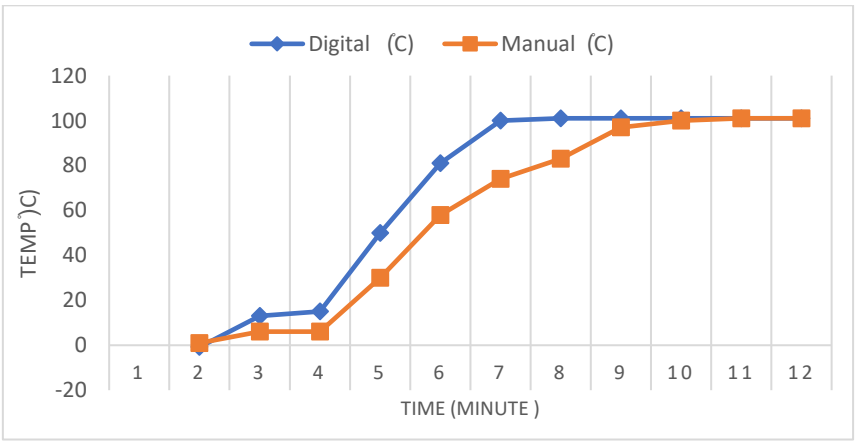


Fig. 6. Graph temperature vs time for Digital and Manual Tools (Melting)

Figure 6 shows the temperature change from the solid to the liquid phase to determine a substance's melting point. The data findings show that the value obtained from digital data is highly valid compared to the use of manual data. Digital data values are more accurate than experimental data. Table 2 shows the average water boiling point value for the two data used.

5. Conclusion

Digitisation for Data Collection in Phase Change Experiments is a tool developed to improve upon existing tools. This tool was developed using new technology to determine the temperature value of the material. Users who conduct experiments using this tool can speed up the use of time compared to using conventional methods. This is because this experiment is equipped with a DS18B20 sensor to determine the zero temperature of the material. The safety factor is also considered when this device uses a nichrome wire heating fan as a heater compared to the conventional Bunsen burner method. Phase changes can be seen from the values and graphs produced.

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APPENDIX 1

Table 3 to Table 6 shows the experimental data obtained from the Two-Phase Change Experiment.

Table 3. Manual Eksperimen (Average Melting Point)

Time (min)	1	2	3	4	5	Average
0	1	1	1	1	1	1
2	6	6	6	6	6	6
4	7	6	6	6	6	6
6	30	31	30	29	28	30
8	60	60	58	56	58	58
10	73	73	75	74	73	74
12	83	84	83	83	84	83
14	96	98	97	96	96	97
16	100	100	100	100	100	100
18	101	101	101	101	101	101
20	101	101	101	101	101	101

Table 4. Digitalization Eksperimen (Average Melting Point)

Time (min)	1	2	3	4	5	Average
0	-0.8	-0.8	-0.8	-0.8	-0.8	-0.8
2	13	14	13	12	13	13
4	15	15	14	15	14	15
6	50	51	50	51	50	50
8	82	82	81	80	80	81
10	100	100	100	100	100	100
12	101	101	101	101	101	101
14	101	101	101	101	101	101
16	101	101	101	101	101	101
20	101	101	101	101	101	101

Table 5. Manual Eksperimen (Average Boiling Points)

Time (min)	1	2	3	4	5	Average
0	26	25	27	26	26	26
2	30	31	31	31	30	31
4	47	49	49	49	49	48
6	66	68	70	70	68	68
8	74	82	86	74	74	74
10	81	90	96	79	79	79
12	97	98	100	99	99	98
14	100	100	100	100	100	100
16	100	100	100	100	100	100
18	100	100	100	100	100	100
20	100	100	100	100	100	100

Table 6. Digitalization Eksperimen (Average Boiling Points)

Time (min)	1	2	3	4	5	Average
0	26	26	26	26	26	26
2	31	30	31	31	30	31
4	46	47	45	46	46	46
6	64	64	63	63	64	64
8	78	78	78	78	77	78
10	88	87	87	88	88	88
12	92	91	92	91	92	92
14	96	95	96	96	96	96
16	98	98	98	98	98	98
18	99	99	99	99	99	99
20	100	100	100	100	100	100

VIRTUAL REALITY TECHNOLOGY FOR IMPROVING MATHEMATICS ABILITIES IN CHILDREN WITH AUTISM SPECTRUM DISORDER: A PROPOSED STUDY FRAMEWORK

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ABSTRACT

The education of mathematics for children with autism spectrum disorder (ASD) has received limited attention despite evidence that many ASD children exhibit proficiency, with some displaying exceptional abilities. Educators face the challenge of developing and implementing effective teaching strategies tailored to their needs. Word problems offer opportunities for applying computational procedures and analytical reasoning in real-world contexts, which is essential for creating accurate situational models from narrative texts. Given the diverse academic and cognitive challenges faced by children with ASD, achieving competence in word problem comprehension remains a significant hurdle. This study explores a conceptual framework for enhancing word problem comprehension among autistic children using virtual reality (VR) learning technology. VR environments may benefit children by addressing cognitive and perceptual difficulties, enhancing planning, problem-solving, and behaviour regulation, and offering effective communication tools for those with limited expressive language. The research focuses on understanding these children's learning capacities in solving word problems through VR integration, aiming to enhance their comprehension of problem situations. Ultimately, this project aims to expand the application of technology in special education contexts.

Keywords: Autism Spectrum Disorder (ASD); Mathematics Education; Word Problem Comprehension; Virtual Reality (VR), Special Education Technology

1. Introduction

Every child in Malaysia has the right to an education that allows them to achieve their particular potential according to the Malaysia Education Plan 2013-2025 (PPPM 2013-2025). Malaysia signed and ratified the Convention on the Rights of Persons with Disabilities in 2008, which calls for improving education quality and recognises the right of persons with disabilities to education. Hence, teachers need to overcome students' limitations, regardless of physical, mental, or social limitations. The People with Disabilities (OKU) Registration Form identifies six sub-categories of learning disabilities: global developmental delay (GDD), down syndrome, attention deficit hyperactivity disorder (ADHD), autism spectrum disorder (ASD), intellectual, and specific learning disorders including dyscalculia, dysgraphia, and dyslexia.

In Malaysia, it is estimated that 9,000 new-borns with autism each year (National Autism Society of Malaysia) [1]. Autism spectrum disorder (ASD) is a lifelong developmental disability that starts in early childhood and affects daily functioning. According to the Ministry of Health (MOH), the number of ASD diagnoses in Malaysia has increased over the past decade. Research [2] also shows that autism rates worldwide have risen from 62 in 10,000 a decade ago to 100 in 10,000 (or 1 in 100) today. The increase is attributed to better awareness, improved diagnosis, and more community support. As one of the fastest-growing neurological disorders, there is a need to adapt educational technologies to help children with autism.

Children with ASD often face additional challenges, such as comorbid disorders, that can make the traditional school environment difficult for them. Research on the developmental challenges associated with ASD has identified specific ways in which these disorders impact cognitive, linguistic, communicative, and social skills—critical areas for academic success. For instance, children with ASD tend to struggle with attention, information processing, and social cognition, which are essential for effective learning and classroom participation, as highlighted by researchers like Baron-Cohen et al. [3] and Frith and Happé [4]. Moreover, difficulties with information integration, abstract reasoning, and cognitive flexibility, crucial for problem-solving and critical thinking, are common among individuals with ASD [5]. These challenges underscore the importance of adapting educational approaches to better support the academic learning needs of children with ASD.

In recent years, there has been a surge of interest in assisting ASD children to succeed in their mathematical studies. Mathematics is an essential academic skill for all students, especially ASD children, to achieve self-sufficiency in life functioning and social community. [6]–[8]. Word problem (WP) solving is one area of focus for the researchers. WP is a math exercise that ties solutions to real-world problems to mathematical ideas that are explained in words instead of mathematical notation. WP is significant because it includes abilities and functions that are required in daily life [7]; [9]-[12]. WP, on the other hand, is a difficult activity for children with ASD since it demands not just mathematical skills but also reading comprehension, memory organisation, and real-world reasoning [13]. Hence, the use of virtual reality in the mathematics problem-solving education of children with autism is vital, as it can facilitate the effective learning of mathematics problem-solving by children with ASD.

For over twenty years, there has been a growing body of research into the potential of VR as a teaching aid for students with autism spectrum disorder (ASD). Scientists have found that virtual reality (VR) can help students with autism spectrum disorder (ASD) learn [14]. This is because virtual reality (VR) allows us to replicate any event and learning context, tailoring it to the student's sensory preferences [9] and allowing us to define, control, and manipulate the degree and number of characteristics of spoken and nonverbal communication [10]. In addition, the visual-spatial learning style that has been found to be a particular strength and primary channel for children with ASD is well-suited to the immersive environment provided by VR technology [15]. In addition, it enables the frequent and methodical execution of activities in environments that are highly analogous to those in the real world but without the associated risks [16].

The importance of mathematics in helping children with autism spectrum disorder acquire the abilities necessary to keep up with today's fast-paced technological world cannot be overstated. According to the National Council of Teachers of Mathematics [17], "mathematical competency" and "the ability to employ mathematical skills in everyday life and the future" are essential for all students. Thus, the students in the Malaysian Special Education Curriculum (Learning Issue) are expected to employ information and communication technology abilities that are up-to-date with technological advances.

There is minimal research on WP solving abilities in children with ASD. Early detection of WP issues allows for early treatment of the student. As a result, effective teaching strategies using VR can aid students with learning difficulties, which will positively impact their future lives and careers. Thus, this research aims to provide a proposed conceptual framework for teaching autistic children word problem comprehension using VR technology. This study has two specific objectives: (a) to determine the word problem-solving abilities of autistic children and (b) to determine how a virtual reality environment can be integrated into the word problem-solving learning process.

2. Literature Review

2.1 Word problem Ability among ASD children

According to [18], teaching word problem-solving to children with ASD may be twice as challenging as educating children without a learning disability. This is because the majority of students with ASD have been diagnosed with intellectual difficulties and a low IQ [18]. The criteria for School Mathematics stress a solid grasp of mathematical concepts, a high level of procedural knowledge, and the capacity to apply mathematical knowledge to solve real-world issues [16]. Hence, regardless of whether a student has a disability, all students are expected to develop the requisite math skills. Despite the fact that it involves several academic and cognitive components, assisting children with ASD in developing competency in WP solving has proven to be difficult [9], [20].

WP is a mathematical exercise in which the problem's information is delivered as text rather than mathematical notation. WP requires students to use semantic and arithmetic skills to solve problems [13]. Students with autism, on the other hand, have difficulty understanding the problem situation or semantic information due to their linguistic impediment [23]-[25]. In addition, WP frequently incorporates contextual information or knowledge that children may encounter in their daily lives [9], [20], [26], [27]. Due to their little experience with the community and their limited and repetitive patterns of behaviour, interest, or interests, [28] the children will find the learning process tough.

This result is comparable with the findings of previous studies [29] – [31], which revealed that autistic children with an IQ of 80 or higher or High Functioning Autism (HFA) may have difficulties with WP solving and mathematical reasoning. Mathematical reasoning is related to cognitive ability since it assists in comprehending a problem circumstance and adjusting to solve a new problem. Thus, when solving WP, students will apply their understanding to real-world problem scenarios in which mathematical reasoning has become an essential tool for making tangible judgements.

2.2 Potential of Virtual Reality Learning Environment

VR is a digital and physical learning platform that has been successfully designed to aid learning [32]-[34]. The mechanism is a 3D simulation of a real-world learning environment capable of constructing everything imaginable, with users able to interact with the environment [34]. The VR learning environment is experiential and intuitive, providing a novel learning experience comparable to hands-on learning, simulation, and concept visualisation.
In recent years, VR-based interactions between human-computer interaction activities have been recommended for ASD intervention. VR has various advantages, including its immersive, controllable, replicable, sensory, stimulating, and intervention and enhancement of individualised learning methodologies. The three main advantages of using VR are as follows: (1) Virtual reality enables users to experience real-world scenarios [34]. (2) Virtual reality can foster a sense of social presence in cyberspace [35]; [36] (3) Embodiment in VR will provide the user with a sense of self and presence, which can encourage a higher level of cognitive interaction with other avatars [37]. Moreover, the qualities of active, authentic, and contextualised learning processes obtained from VR's interactive experiences can facilitate the user's comprehension of abstract concepts [38].

2.3 VR and word problem comprehension

The virtual reality (VR) learning environment capitalises on the visual learning ability of ASD individuals. To support the development of mental representations, the virtual reality (VR) learning environment presents the learner with a virtual scenario representing the textual information in the word problem. The controlled visual content can be explored and interacted

with repeatedly. Children with ASD are able to actively participate in interactive and immersive simulated circumstances by experiencing a virtual environment as if they were physically present. In a virtual scenario, including the buying and selling of goods or services, for instance, will provide pupils with a sensation of presence in the buying and selling activities. As the learner engages in math activities, they gain everyday experience with mathematical knowledge and develop their contextual knowledge. Moreover, visuals and sound were efficient in teaching abstract concepts to autistic children [39].

Interactivity in a VR system, which comes from the user's interactions with an avatar, can help create and visualise mental models. VR stimulates imagination by triggering memories of images or concepts related to objects, events, and situations, blending past experiences with new ones. This continuous interaction between text and world knowledge is helpful in forming problem scenarios. VR, therefore, has the potential to assist learners in building accurate mental representations, or situation models, when understanding word problems. Visualising and constructing these models helps students better grasp the meaning of the text, improving their comprehension of word problems.

This approach aligns with constructivist principles [40]– [48]. It suggests that incorporating VR technology into learning processes can enhance how children with ASD understand the context of mathematical problems in real-world situations. The VR experience is expected to help create problem scenarios that will aid these children in solving word problems more effectively.

2.4 Problem situation

To understand a narrative text, a reader must create a mental image or situation model [40]. A situation model is the mental picture of the events in the text that the reader forms as they read. As the reader builds this model, text information and world knowledge interact constantly [40]. Readers often rely on their past experiences and vocabulary to make sense of what they are reading. Without an accurate mental image, their comprehension will be limited. Additionally, research shows that mental imagery enhances reading comprehension [44], [45]. Relevant background information must be readily available in students' memory to accurately represent a scenario in a text [46]. Studies have found that children with limited prior knowledge tend to score lower on comprehension tests [47]. Norman's [48] finding that children's mental representations are shaped by their prior knowledge and experiences supports this. Therefore, it can be argued that children are more likely to understand a problem in a text if they have a strong foundation of prior knowledge.

3. Methodology

The study framework consists of three key components: (i) an investigation into the word problem-solving (WP) abilities of autistic children, (ii) an intervention using a virtual reality (VR) learning environment to enhance WP skills, and (iii) the use of VR to support the formation of mental schemas. Figure 1 illustrates the conceptual framework guiding this research. The study proposes a VR-based learning approach for teaching word problems to autistic children as its primary conclusion.

To develop the research objectives, a preliminary investigation will be conducted to identify challenges and refine the study's focus. This initial phase aims to gain a comprehensive understanding of how autistic children currently acquire and process mathematical word problems. Building on these insights, the study will explore how the integration of VR technology can improve problem-solving skills by facilitating the formation of problem schemas among these children.

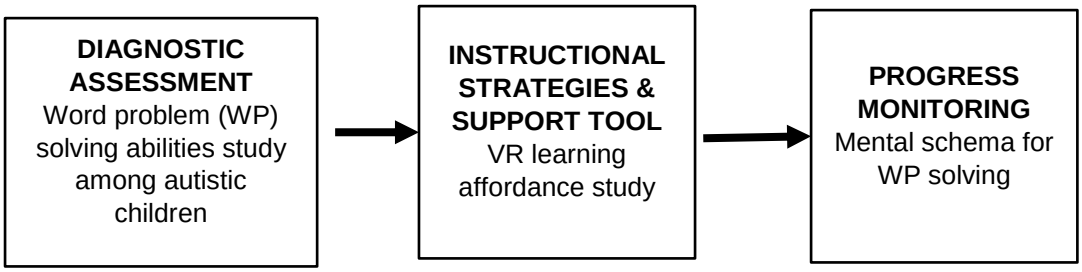


Fig. 1. Study Framework

This framework represents a systematic approach designed to enhance word problem-solving abilities in autistic children. The process begins with a Diagnostic Assessment, where the specific difficulties these children face when tackling word problems are carefully evaluated. This assessment phase involves detailed studies that might include standardised tests and observational methods to identify the unique challenges and needs of the children. Based on these findings, the next phase focuses on Instructional Strategies & Support Tools, particularly through Virtual Reality (VR) learning environments. By leveraging VR's immersive and interactive capabilities, tailored instructional strategies are developed to address the identified difficulties, providing individualised support and reinforcing learning in a controlled setting. The final phase, Progress Monitoring, involves continuous tracking of the children's development to assess the effectiveness of the interventions. This phase looks at changes in the children's mental schemas for solving word problems, ensuring that the strategies have the desired impact. Adjustments to the instructional approaches may be made based on the ongoing assessment of progress, creating a dynamic and responsive learning process.

3.1 Diagnostic Assessment

To thoroughly understand the word problem-solving abilities of autistic children, a multifaceted approach to data collection is essential. This approach begins with observational studies, where researchers directly observe children as they engage with word problems. This method allows for identifying specific behavioural patterns and difficulties that might not be captured through other assessment forms. Complementing this, standardised tests are utilised to evaluate cognitive abilities, comprehension, and problem-solving skills in a structured environment. These tests are designed for autistic children, ensuring the assessments are tailored to their unique learning needs. Additionally, interviews and questionnaires with educators, therapists, and parents provide valuable insights into the children's abilities and challenges from those who know them best. This combination of methods creates a comprehensive and nuanced understanding of each child's problem-solving capabilities, facilitating targeted and effective support strategies.

3.2 Instructional Strategies & Support

VR-based instructional sessions are implemented, and detailed data is collected on the children's performance during these sessions. This allows for direct observation of how the instructional strategies influence their problem-solving abilities in a controlled environment. Additionally, user interaction logs are utilised to record and analyse how children engage with the VR tool. These logs provide insights into the specific actions, behaviours, and patterns of interaction within the virtual environment, offering valuable information on how effectively the tool supports learning. Pre- and post-tests are administered to further assess the impact of VR interventions. These tests are designed to measure the children's problem-solving abilities before and after the VR sessions, allowing for a clear comparison and quantification of improvements. By combining these data collection methods, the study aims to

comprehensively evaluate the effectiveness of VR-based instructional strategies in enhancing the word problem-solving skills of autistic children.

3.3 Progress Monitoring

Progress monitoring is central to this process, where regular assessments are conducted to observe and document shifts in how children approach and solve word problems. These assessments include a series of structured tasks that reflect different types of word problems, designed to gauge the child's understanding and problem-solving strategies at various stages.

By administering pre- and post-intervention assessments, researchers can compare the children's performance before and after applying VR-based instructional strategies. This comparison allows for identifying specific changes in mental schemas, such as improved problem categorisation, better recognition of relevant information, or more efficient application of problem-solving strategies.

Qualitative data collection techniques, such as think-aloud protocols or interviews, can also be employed to gain deeper insights into the children's thought processes. During these sessions, children might be asked to verbalise their reasoning while solving problems, providing a window into their cognitive strategies and how these evolve over time.

The study will involve a sample of autistic children aged 8 to 12, selected from specialised educational settings. The participants will be chosen based on their difficulties with mathematical word problems (WP) and their potential to benefit from a virtual reality (VR) intervention. Parental consent and ethical approval will be obtained before the study.

3.4 AssessmentTools

3.4.1 Baseline Assessment

To establish a starting point, a standardised Word Problem-Solving Test will be administered to evaluate the initial abilities of the participants. This test will measure their comprehension, problem-solving strategies, and accuracy in solving mathematical word problems. In addition to this, cognitive and behavioural assessments, such as the Autism Diagnostic Observation Schedule (ADOS), will be conducted to gather relevant data on the participants' cognitive abilities and problem-solving processes. These assessments will also help in understanding the current state of mental schema formation related to word problem-solving.

3.4.2 Post-Intervention Assessment

Following the intervention, the same standardised Word Problem-Solving Test used in the baseline assessment will be administered to evaluate any improvements in the participants' abilities. To further assess the effectiveness of the intervention, data on the participants' interaction with the VR environment will be collected, including metrics such as time spent, engagement levels, and task completion rates. Additionally, further cognitive tests and observational data will be utilised to examine any changes in mental schema formation that may have resulted from the intervention, providing a comprehensive view of the impact on word problem-solving skills.

4. Result and Discussions

The study systematically explored the challenges and interventions related to word problem-solving in autistic children, yielding significant insights into their cognitive processes and the effectiveness of targeted instructional strategies. Table 1 below concisely presents the

findings from the framework phase, highlighting the key results and insights gained from the study.

Table 1. Research Findings

Components	Key Findings
Diagnostic Assessment	Observational Studies Identified behavioural patterns such as difficulty grasping abstract concepts and focusing on irrelevant details in word problems. Standardised Tests Revealed struggles with cognitive processing, comprehension, and problem-solving strategies, confirming the need for tailored support. Interviews & Questionnaires Provided qualitative insights from educators, therapists, and parents, complementing the observational and test data and offering a comprehensive understanding of each child's problem-solving abilities.
Instructional Strategies & Support	VR-Based Instructional Sessions Enhanced children'schildren's engagement and problem-solving abilities in a controlled environment. User Interaction Logs Revealed improved focus and better problem-solving steps and highlighted areas needing further support, particularly in transferring skills to non-VR contexts.
Progress Monitoring	Pre and Post-Tests Demonstrated measurable improvements in problem-solving abilities, especially in problem categorisation and application of logical steps. Structured Assessments Showed significant progress in problem categorisation, information recognition, and strategy application. Pre- and Post-Intervention Assessments Provided evidence of positive impact from VR-based strategies, with clear improvements in problem-solving abilities. QualitativeTechniques Think-aloud protocols and interviews revealed more systematic approaches and clearer articulation of reasoning, indicating the development of more effective problem-solving strategies over time.

The Diagnostic Assessment phase identified several challenges that autistic children face when solving word problems. Through observational studies, researchers noticed that these children often struggle with abstract concepts and tend to focus on irrelevant details. Standardised tests confirmed these difficulties, highlighting issues with cognitive processing and comprehension. Additionally, interviews with educators, therapists, and parents provided valuable insights into the children's strengths and weaknesses. This led to a comprehensive understanding of their problem-solving abilities and the need for targeted instructional strategies.

VR-based instructional sessions were implemented during the Instructional Strategies & Support phase to enhance the children's problem-solving skills. The controlled VR environment significantly improved their engagement and focus. User interaction logs revealed better navigation and more consistent application of strategies. Pre- and post-tests showed measurable improvements in problem categorisation and logical reasoning, indicating that the VR-based approach effectively addressed the children's specific needs.

The Progress Monitoring phase tracked changes in the children's problem-solving strategies over time. Regular assessments showed significant improvements in their ability to categorise problems, recognise relevant information, and apply strategies efficiently. Pre- and post-intervention comparisons confirmed the positive impact of the VR sessions, while qualitative techniques like think-aloud protocols provided deeper insights into the children's evolving cognitive processes. These findings underscore the importance of continuous monitoring and

adaptive strategies in supporting the sustained development of problem-solving skills in autistic children.

5. Conclusion

In conclusion, the study effectively identified and addressed the unique challenges autistic children face in solving word problems through a comprehensive and multi-phase approach. The Diagnostic Assessment phase provided critical insights into these children's specific difficulties, emphasising the need for tailored instructional strategies. The implementation of VR-based instructional sessions in the Instructional Strategies & Support phase demonstrated significant improvements in the children's engagement and problem-solving abilities, with measurable progress in key areas such as problem categorisation and logical reasoning. Finally, the Progress Monitoring phase confirmed the long-term benefits of these interventions, highlighting the importance of continuous assessment and adaptive strategies in fostering sustained improvements. Overall, this approach shows great promise in enhancing the problem-solving skills of autistic children, offering a robust framework for future educational interventions.

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COMPARATIVE CASE ABOUT 4G VERSUS 5G IN TERMS OF ANTENNA LOCATION IN THE BUILDING

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ABSTRACT

Wireless communication systems depend mainly on the effective use of wireless antennas to provide users with a smooth wireless connection [1]. As a result, antenna technology is vital in determining the overall performance of a wireless communication system [2]. The implementation of near-user wireless communication systems is required for reliable communication in indoor environments. However, building such a system is difficult due to variables such as interference and multipath propagation [3]. The scope of this study includes a comparison of antenna placement tactics used in 4G and 5G networks. The number and type of antennas, their placement and orientation, and the factors that affect their position are all examined in this analysis. Furthermore, this research investigates the effect of antenna sites on network performance indicators such as signal strength, coverage and performance comparisons between 4G and 5G networks in various scenarios. The findings of this study are likely to help develop network design and optimisation for indoor settings. Research findings can help enterprises choose the best antenna technology for specific indoor areas depending on their individual needs. Finally, this research aims to improve the reliability and performance of indoor wireless communication systems in 4G and 5G networks.
Keywords: Indoor antenna placement, signal quality, network performance, 4G, 5G, building materials, wireless communication systems.

1. Introduction

This study addresses the importance of antenna location within buildings and its impact on signal quality, coverage area, and overall network performance in 4G and 5G networks. With wireless communication systems heavily reliant on wireless antennas to ensure seamless connectivity, the significance of antenna technology in determining system performance cannot be overstated. However, deploying wireless communication systems in indoor environments poses unique challenges due to interference and multipath propagation.

To shed light on this topic, a comparative case study was conducted to examine the effects of indoor antenna placement on 4G and 5G networks. The study also investigated how building materials and other environmental factors influence signal quality and network performance. The research findings offer valuable insights into the benefits and limitations of each technology within indoor settings, empowering network designers and optimisers to make informed decisions concerning indoor coverage.

Furthermore, the study's outcomes have significant practical implications for industries seeking to select the most suitable antenna technology for indoor locations based on their requirements. By carefully considering the research findings, businesses can ensure optimal wireless connectivity and enhanced user experience.

This research encompasses a comprehensive analysis of antenna placement strategies utilised in 4G and 5G networks, including the types and quantities of antennas, their locations and orientations, and the key factors influencing their placement decisions. Additionally, the study examines the influence of antenna location on crucial network performance metrics such

as signal strength, coverage, data rates, and latency. Comparative evaluations of 4G and 5G networks are conducted across various scenarios to provide a comprehensive understanding of their performance differences.

By addressing these essential aspects of indoor antenna placement, this research contributes to the existing body of knowledge in wireless communication systems. The findings can guide the design and optimisation of indoor networks and support the future development of more efficient and reliable wireless communication systems.

2. Literature Reviews

The fourth generation of wireless networks, or 4G, have replaced the third generation. 4G technology offers its clients better speed and IP-based multimedia services. With 4G, customers can receive voice, data, and streamed multimedia "Anytime, anywhere" thanks to an integrated, worldwide network that can offer a complete IP solution. The demand for implementing such technologies that combine all these systems into a single unified system is considerable. Now, we have several technologies, each capable of fulfilling activities like supporting phone traffic utilising voice over IP (VoIP), broadband data access in mobile situations, and others. Since 4G is about smoothly integrating terminals, networks, and applications, it offers a solution to this issue [5].

Data communication between base and mobile stations is the foundation of LTE. A radio access node and the nodes of the reference station make up the network. The standard uses frequency and time duplexes to facilitate bidirectional data transfer between the base and mobile stations [6]. The network uses CS-FDMA in the uplink and OFDMA in the downlink to offer multiple access. Unlike the 256 OFDM technology used in the Wi-Fi standard, OFDMA is based on a set of modular orthogonal carriers up to 2048. While OFDM requires the device to broadcast data utilising every subcarrier, OFDMA allows for data transmission on subcarriers of a user-dedicated channel, increasing the transmitter's power. The primary benefits of OFDMA technology include the decrease of mutual interference in equipment with omnidirectional antennas and increased adaptability of equipment with various antenna types [6].

Fifth Generation (5G) Technology is a more recent generation of mobile networks in wireless communication. The fifth generation (5G) mobile networks offer everyone access to high-speed internet whenever and wherever they are. Because of its innovative characteristics, such as linking people and controlling gadgets, machines, and objects, 5G is a little different. Different degrees of performance and capabilities offered by the 5G mobile technology will create new user experiences and link new businesses [8].

5G provides high speed, low latency, and highly scalable connectivity between multiple devices and IoT worldwide. 5G provides a flexible model to develop a modern generation of applications and industry goals [9, 10]. There are many services offered by 5G network architecture stated below: Massive machine-to-machine communications: 5G offers novel, massive machine-to-machine communications [11], also known as the IoT [12], that provide connectivity between lots of machines without any involvement of humans. This service enhances the applications of 5G and provides connectivity between agriculture, construction, and industries [13].

2.1 Type of antenna installed for In-Building Coverage (IBC)

This case study focuses on clients inside buildings to enjoy cellular facilities and the placement of antennas on structures. Inside antenna range is influenced by three things:

i. Physical impediments
ii. The energy produced by the booster

iii. The outer antenna receives and broadcasts the outside signal

There are two different indoor antenna types available:
Omnidirectional dome antennas transmit and catch signals coming from all directions. They have 360-degree coverage and are intended for central places. Their power is determined by their "gain," just like external antennas. For maximum effects, they must be attached to the ceiling. They provide coverage for a single floor but are not appropriate for many floors.



Fig. 1. Omnidirectional dome Antenna

Panel antennas have a specific orientation, enabling the best reception in the desired locations. A directional antenna is more powerful than an omnidirectional one, like an outdoor one; therefore, the signals it broadcasts are stronger. It can be fixed to the ceiling or the wall. The panel antenna can be mounted on the wall in some areas (such as a long, narrow hallway), but we often advise installing it on the ceiling for the best coverage.



Fig. 2. Panel antenna

2.2 Antenna Positioning

Inside antenna installation depends on the layout of the building and the area that wants to be covered. The optimal location for the antenna is in the centre of the ceiling if the architect just wants to enhance the signal in a single space, such as a home office. A directional antenna is required if the designer wants a stronger signal to reach numerous stories. Once more, the middle of the space the designer wants to cover is the best position for the antenna to be affixed to a ceiling. As indicated above, in some unique room layouts, it may be better to position a directional panel antenna, for example, on the wall of a long, narrow corridor [14]. One indoor antenna could not provide enough coverage, depending on the signal intensity at the time and the construction materials of the building's walls and floors. To extend coverage to numerous stories and huge buildings, the designer might wish to install extra antennas and splitters.

The location of the indoor antenna for IBC (In-Building Coverage) should be chosen depending on several variables, including the building's structure, the frequency bands being used, the

signal's strength, and external interference. The reasons listed below should be considered when choosing the indoor antenna position for IBC:

i. Building Structure: The position of the indoor antenna for IBC is significantly influenced by the building structure. The antenna should be positioned so the signal can pass through building walls and reach all necessary locations. If the building has concrete walls, the antenna might need to be placed near a window or on a higher floor for improved signal penetration.

ii. Frequency Bands: Various frequency bands have varied signal propagation characteristics. For instance, compared to low-frequency bands like 700 MHz, high-frequency bands like 5G have shorter wavelengths and demand more antennas to cover the same region. Hence, it is crucial to choose the placement of the antenna to guarantee that all necessary frequency bands are well covered.

iii. Signal Strength: Choosing an antenna site that will give everyone in every area the best possible signal is essential. The signal strength of the closest cell tower should be considered, as it will help establish the necessary antenna gain and placement. A signal strength survey can help find the ideal location for the antenna.

iv. Interference: Choosing the indoor antenna's placement is important to reduce interference from outside sources. For instance, positioning the antenna close to electrical or other wireless equipment could result in interference and worse signal quality. As a result, it is important to choose the antenna site in a way that minimises interference from other sources.

Planning the indoor building coverage of a building or office structure can help to increase the signal strength, Reference Signal Received Power (RSRP) value, and Signal Interference Noise Ratio (SINR) [15].

3. Methodology

To conduct testing and analysis, the location of the antenna close to it must first be examined to determine whether the area is a top priority. The Penang International Airport (PIA) was chosen as the study's sample location since it is the first in the northern region to have 5G indoor coverage installed and maintained by G2R Sdn. Bhd.

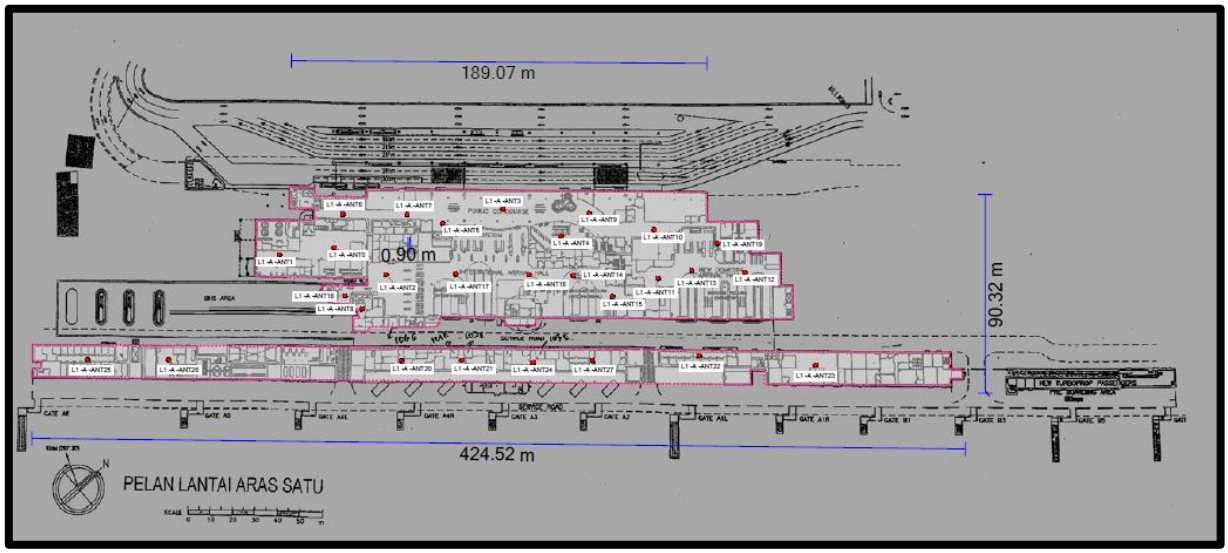


Fig. 3. Antenna location for Level Penang International Airport for 4G LTE system. Source: Technical Proposal Penang Airport Rev5

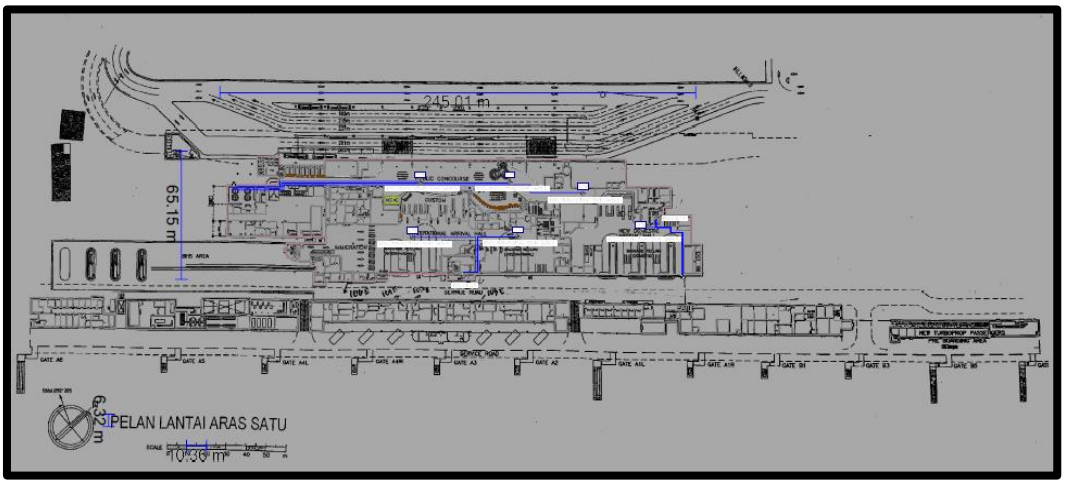


Fig. 4. Antenna location for Level 1 Penang International Airport for 5G system. Source: Technical Proposal PIA Rev4

When comparing the antenna positions for the 4G and 5G systems, there is a difference where part of the 5G antenna is not positioned. This assumes that the region inside the yellow box will be served by 5G coverage from the outside (outside coverage).

Estimating the radiated power from each antenna is essential to ensure accurate and effective field tests. Radiated power refers to the amount of electromagnetic energy transmitted by an antenna into the surrounding space. By calculating the approximate value of the radiated power, researchers can anticipate the signal strength and coverage area that can be expected during the field tests.

Calculating the radiated power involves considering various factors, such as the characteristics of the antennas, their gain, the transmitted power from the base station, and any losses that may occur in the transmission process. These calculations consider the specific technical specifications of the antennas, including their frequency range, radiation patterns, and polarisation.

Several methods exist for calculating the output of each antenna that will be installed or utilised in prediction software. One of them is to use Microsoft Excel. All values for cables, antennas, jumpers, and other components will be entered here, and some formulas will be used to get the final value for each antenna.

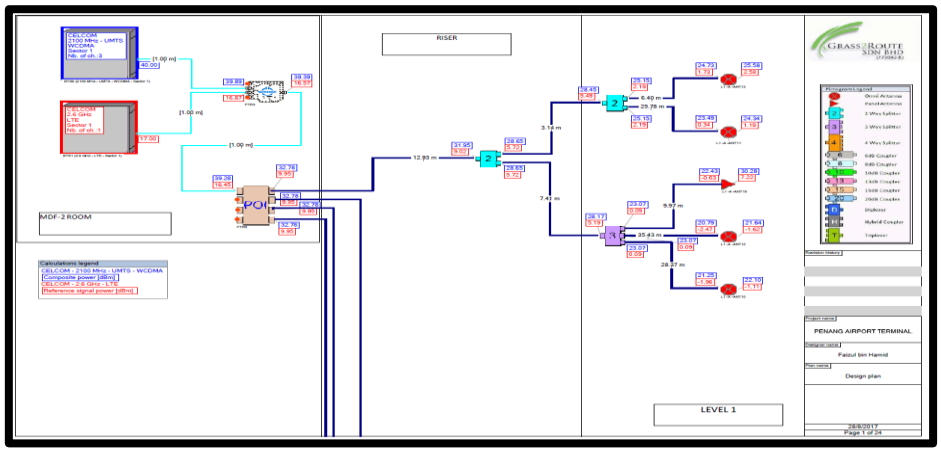


Fig. 5. Calculations that resulting the final value at each antenna. Source: Technical Proposal Penang Airport Rev5

Table 1. Prediction EIRP calculation (4G LTE) for the antenna at Level 1 PIA.

Antenna Label	Reference Signal Power (dBm)	
L1 – A – ANT0	-2.71	Sector 3
L1 – A – ANT1	-2.71	
L1 – A – ANT2	-18.46	
L1 – A – ANT3	-4.89	
L1 – A – ANT4	-4.39	
L1 – A – ANT5	-6.72	Sector 3
L1 – A – ANT6	-2.70	
L1 – A – ANT7	-6.32	
L1 – A – ANT8	-6.44	
L1 – A – ANT9	-3.79	Sector 1
L1 – A – ANT10	-1.11	
L1 – A – ANT11	1.19	
L1 – A – ANT12	-1.62	
L1 – A – ANT13	2.58	
L1 – A – ANT14	-4.22	
L1 – A – ANT15	-5.32	
L1 – A – ANT16	-2.67	
L1 – A – ANT17	-3.43	
L1 – A – ANT18	-6.34	
L1 – A – ANT19	7.22	

Table 2. Prediction EIRP calculation (5G) for the antenna at Level 1 PIA.

Antenna Label	Reference Signal (dBm)
ECO-L1-N2RU-1(3.5GHz-5GNR-Sector2)	-8.15
ECO-L1-N2RU-2(3.5GHz-5GNR-Sector2)	-8.15
ECO-L1-N2RU-3(3.5GHz-5GNR-Sector2)	-8.15
ECO-L1-N2RU-4(3.5GHz-5GNR-Sector1)	-8.15
ECO-L1-N2RU-5(3.5GHz-5GNR-Sector1)	-8.15
ECO-L1-N2RU-6(3.5GHz-5GNR-Sector1)	-8.15

4. Result and Discussion

The 4G system operates at a frequency of 2600 MHz (megahertz), while the 5G system operates at a higher frequency of 3.5 GHz (gigahertz). Generally, higher frequencies used by the 5G system provide higher data rates and lower latency, enabling faster and more responsive communication. However, higher-frequency signals have shorter wavelengths and are more prone to attenuation and signal loss when encountering obstacles like buildings, trees, or other physical barriers.

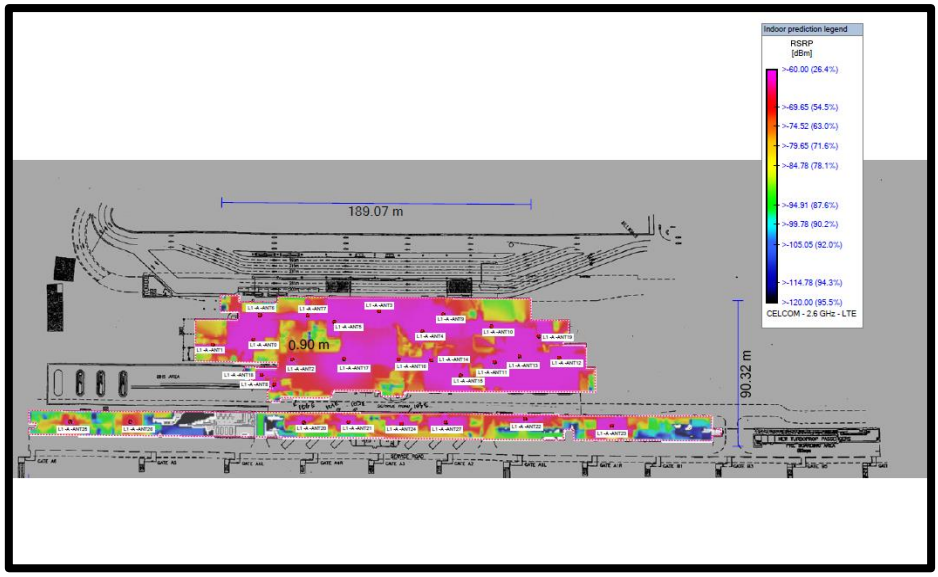


Fig. 6. Coverage prediction for 4G LTE 2.6GHz. Source: Technical Proposal Penang Airport Rev5

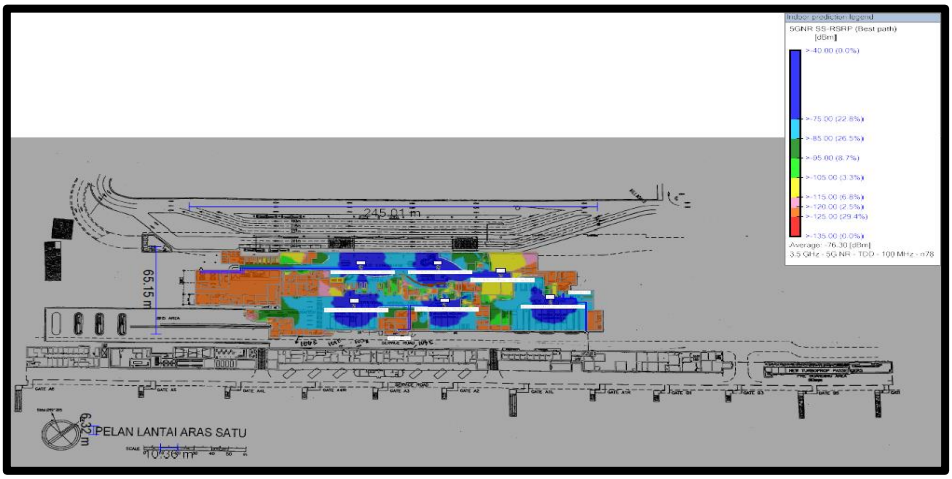


Fig. 7. Coverage prediction for 5G 3.5GHz, TDD 100MHz. Source: Technical Proposal PIA Rev4

Due to the 5G system's higher frequency, its signals experience more significant attenuation and signal degradation than the lower-frequency 4G system. This attenuation reduces the 5G system's coverage range, resulting in a smaller coverage area than the 4G system.

On the other hand, the 4G system's lower frequency enables it to penetrate obstacles and propagate over longer distances with reduced signal loss. Consequently, the 4G system provides a larger coverage area as its signals can reach farther and provide connectivity to a broader geographical region.
It's important to note that while the 5G system may have a smaller coverage area, it compensates for this limitation by offering higher data rates and improved network performance in areas with sufficient signal strength. The focus of the 5G system is on delivering faster and more reliable connections in high-density urban areas and specific target zones rather than providing extensive coverage over large geographic areas.

A small cell is a low-powered cellular base station that runs on many frequencies. Small cells can have transmission ranges from 40 feet (12 meters) to a few kilometres, depending on their size. These mini-base stations communicate with the main cellular network by Ethernet, fibre, or wireless.

Small cells can be used both indoors and outdoors. Indoor varieties are typically small—about the size of a pizza box. These mega- or gigabit-speed devices can be mounted on the ceilings and walls of homes and buildings to improve cellular network signal strength. This trend is expected to continue as more 5G small cells are installed in offices, residences, and apartments [16].

The antenna coverage prediction for 5G discloses that the location of the antenna is referred to as a "hotspot." A hotspot is an area with concentrated and robust network coverage in the context of wireless communication. The hotspot area has been recognised as the key focus for PIA visitors in this scenario. This area's categorisation as a hotspot suggests that it has a high density of network activity and is projected to draw many visits or users. The placement of the 5G antenna in this hotspot suggests that it is properly positioned to meet the communication needs and demands of PIA visitors.

Furthermore, as a hotspot, this location will likely provide consistent and dependable 5G service, allowing visitors to enjoy seamless and uninterrupted access. The emphasis on ensuring solid network performance in the hotspot corresponds with giving visitors a great digital experience, increasing their overall satisfaction and engagement during their visit to PIA.

Overall, the designation of the antenna placement region as a hotspot in the context of the 5G coverage forecast emphasises the importance of responding to tourists' communication needs at PIA. It represents a concerted effort to provide a superior network experience, allowing visitors to access high-speed data services and remain connected throughout their visit to the institution or destination.

With the coming of 5G, the importance of inside coverage has grown, and increasing indoor coverage and the user experience has emerged as an industry-wide consensus. According to industry projections, interior scenarios will enable operators to achieve higher 5G business success, emphasising the importance of high-quality indoor coverage networks [17].

The fact that 70% of data traffic is generated indoors explains the importance placed on indoor coverage. Another problem is the arrival of a high-band spectrum, which makes it difficult to rely on outdoor networks to provide complete inside coverage. According to a 4G network study, for specific office buildings with the same outdoor macro coverage, the area with worse coverage on the 3.5 GHz band is doubled compared to the 1.8 GHz band. This demonstrates that enhancing indoor coverage has been a top focus for operators in the 5G era.

The most visible difference between 5G and 4G is that 5G can enable industrial transformation and speed industrial digitalisation. Outdoor-only industrial firms, such as unmanned aerial vehicles (UAVs) and self-driving cars, comprise a small percentage of all industrial businesses. Most industrial clients and businesses are located indoors.

Furthermore, common users and industrial clients are present in any indoor setting. Airlines, for example, are industrial customers in airports, tales in shopping malls, physicians and nurses in hospitals, and manufacturing workshops in factories, all of which have many ordinary consumers. Operators are getting close to serving industrial customers when providing 5G indoor coverage to regular users. As a result, 5G interior coverage serves as a bridge connecting operators to valuable new services.

The new digital indoor system is a recent idea that has garnered broad attention from major equipment makers who have suggested their solutions. The digital indoor system's core design consists of a baseband processing unit (BBU), a remote CPRI data aggregation unit, and a remote radio unit (picoRRU). The BBU may be shared with macrocells. The remote CPRI data aggregation unit oversees branching and aggregating the pRRU's CPRI data. The

pRRU then processes the radio signal. The BBU is linked to the remote CPRI data aggregation unit through optical fibre, and the aggregation unit is linked to the pRRU via ethernet cable or optical fibre [18].

The new digital indoor distribution system has the following advantages over the previous indoor distribution system [18]:

a) The network architecture is straightforward, and there is no need to calculate connection budgets.
b) Sharing the same network O&S with macro cells for easier control
c) When compared to active DAS, it eliminates the problem of uplink ground noise lift.
d) Simple capacity control and flexible cell merging and splitting.
e) One net both inside and outside to promote macro-micro cooperation.
f) Allow for multiple-stream transmission.

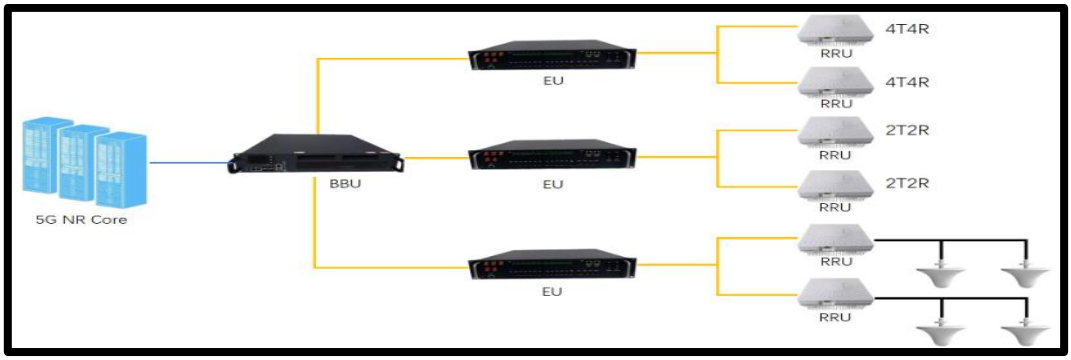


Fig. 8. Digital 5G Indoor Architecture system

The digital indoor system is the future interior coverage solution trend. It could be used in various settings, but the cost is relatively significant. It is typically used in high-traffic areas, huge shopping malls, upscale office buildings, and other locations with heavy traffic and a demanding user experience [18].

The choice of construction materials and the placement of the antenna play a crucial role in determining the strength of cellular coverage within a building. It is essential to recognise that various materials, including clear glass, plasterboard, plaster, brick, stone, concrete, and cement, can all hinder cellular radio frequencies.

For instance, clear glass directly threatens mobile signals by significantly reducing signal transmission. Plasterboard, plaster, and brick also contribute to signal loss in different ways. The high water content of the plasterboard creates a shielding effect that restricts the strength of the signal. Similarly, plaster and brick exhibit substantial attenuation rates, weakening and diminishing signal propagation.

Even though concrete and cement are commonly used in construction, their high density and thickness can adversely affect mobile signals. These materials can absorb and block radio waves, negatively impacting signal penetration.

Considering these factors becomes paramount during the construction of any structure, particularly when reliable cellular service is essential. By carefully selecting construction materials and taking the appropriate steps to optimise antenna placement, we can substantially enhance the overall quality of cellular coverage throughout the entire building. This will result in a more reliable and robust cellular experience for all users within the premises.

Table 3. Source: https://www.wilsonamplifiers.com/blog/11-major-building-materials-that-kill-your-cell-phone-reception

Material	Loss value
Clear Glass	-4dB
Drywall & Sheetrock	-2dB
Plaster	-8dB to -16dB
Brick & stone	-8dB to -28dB
Concrete & Cement	-10dB to -20dB

Because they can weaken and impede radio frequency waves, building materials substantially impact cellular coverage within a building [18]. Building materials can weaken or even erase radio frequency waves utilised for cellular service, including 4G, LTE, and 5G. Concrete, metal, and tinted low-E glass are among the primary causes of poor cell phone coverage indoors, even in regions where cellular towers are nearby [19, 20]. When constructing and erecting a facility that requires optimal cellular coverage, it is critical to consider building materials.

5. Conclusion

Ultimately, the classification of the antenna placement zone as a hotspot within the PIA 5G coverage projection emphasises the need to meet tourists' communication needs. This strategic strategy demonstrates the company's dedication to providing an extraordinary network experience, allowing guests to take advantage of high-speed data services and seamless connectivity throughout their visit.

The introduction of 5G technology has increased the relevance of interior coverage, and there is growing industry consensus on the importance of improving indoor coverage and the user experience. Higher business success in the 5G era is inextricably linked to the availability of high-quality indoor coverage networks. This industry-wide accolade emphasises the importance of meeting users' connectivity needs in indoor environments.

PIA demonstrates its grasp of tourists' developing demands in an increasingly connected world by designating the antenna deployment location as a hotspot and prioritising inside coverage. This approach improves the visitor experience and helps the institution or destination remain competitive and relevant in a technology-driven market.

As 5G evolves and expands, the emphasis on interior coverage networks will remain, driven by the need to meet customers' increased expectations for seamless connectivity and high-quality services in various indoor environments. Organisations can position themselves for success in the 5G era by recognising the importance of inside coverage and prioritising the user experience, ensuring that visitors and customers benefit from the full potential of this revolutionary technology.

In conclusion, PIA's classification of the antenna placement zone as a hotspot indicates a forward-thinking approach to meeting travellers' communication demands. The emphasis on interior coverage and user experience corresponds with industry estimates and the consensus regarding the importance of high-quality indoor coverage networks in attaining success in the 5G scenario. By embracing these trends, businesses may use 5G technology to provide superior connections and improve the overall experience of visitors and consumers.

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A PROOF-OF-CONCEPT STUDY OF TULEVA CUBE SYSTEM AS A SOLUTION FOR RURAL AREA COVERAGE SYSTEM

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ABSTRACT

Rural communities continue to face major barriers to accessing high-speed broadband internet, hampering remote job and distance education prospects. The difficulties of deploying infrastructure like fibre-optic cables and cell towers in sparsely inhabited and rocky areas exacerbate the problem. Cellular convergence, which integrates several cellular network architectures and technologies, such as 2G, 3G, 4G, and 5G, intends to improve communication experiences by allowing seamless network switching and access to various services. It also refers to the combination of wireless technologies, such as Wi-Fi and cellular networks, to provide a unified communication experience. This study assesses the effectiveness of the Tuleva Cube, a multi-carrier convergent system built by Grass2Route in Malaysia that provides an alternative to traditional infrastructure by allowing data transmission over a single channel and many carrier frequencies. The study confirms the Tuleva Cube's capacity to provide steady and dependable internet connectivity in previously inaccessible areas by evaluating signal output from strategically placed access points in rural areas. This novel technology shortens deployment schedules and lowers costs compared to traditional telecom towers, making it a promising solution for expanding broadband connectivity in remote areas.

Keywords: Multi-carrier, convergence, internet, data, voice, cellular communication

1. Introduction

Ookla's data reveals that Malaysia's median mobile internet connection speed increased by 11.79 Mbps (+48.0 percent) in the twelve months at the start of 2023. Meanwhile, Ookla's data shows that fixed internet connection speeds in Malaysia increased by 16.66 Mbps (+22.3 percent) during the same period [1]. However, a sizable discrepancy in rural internet access still needs serious improvement. This disparity highlights the significant variations among urban, suburban, and rural areas in terms of accessibility and calibre of Internet connectivity. Many people who live in rural areas have trouble getting high-speed broadband internet, which makes it difficult for them to do remote work or take distance education courses. Building communication towers in sparsely populated areas might not be financially feasible. Several variables cause the challenges of delivering internet connectivity in rural locations. First, because they frequently have lower population densities than cities, rural areas make it harder and more expensive to build and maintain the infrastructure needed for high-speed Internet, such as Fiber-optic cables or cell towers. Infrastructure development and upkeep are further complicated by the characteristics of rural areas, which include rugged terrain and dispersed population centres.

Grass2Route (G2R) proposed the Tuleva Cube (Made in Malaysia by G2R), a multi-carrier convergent system that is faster to deploy and more affordable than building a new telco tower. A multi-carrier convergent system is a type of telecommunication that transmits data over a single communication channel while utilising various carrier frequencies. Broadband internet access, digital television broadcasting, and wireless communication are only a few examples of the uses of this kind of technology.

Wireless and optical convergent technologies might enable mobile network systems to meet wireless systems' needs, such as high throughput, latency, coverage, and geographical positioning. For example, Enhance Mobile Broadband (eMBB) can benefit from the wide bandwidth available from optical communication, followed by millimetre waves (mm Waves) and terahertz (THz) communications [2].

A multi-carrier convergent system divides the available communication bandwidth into a few sub-channels or sub-carriers, each capable of transmitting data simultaneously. A multi-carrier convergent system can use the available bandwidth more effectively and produce better data speeds than a single-carrier system by utilising numerous sub-carriers.

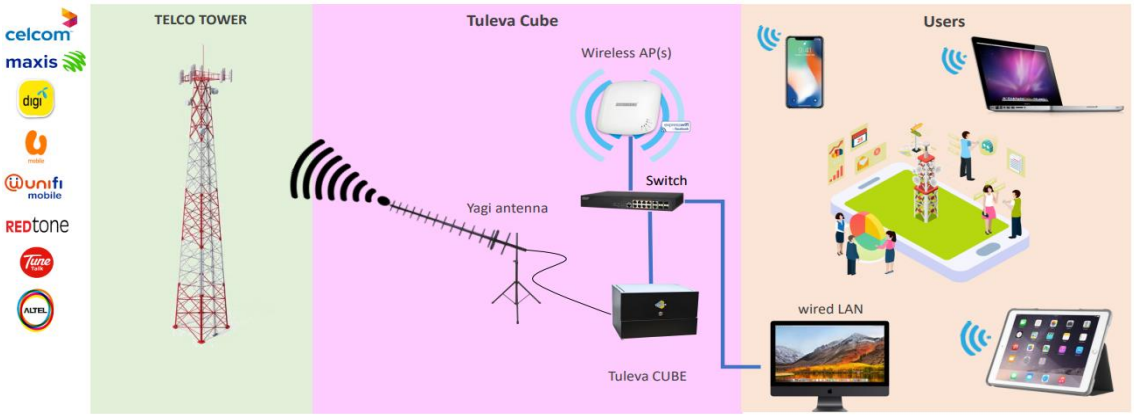


Fig. 1. Shows how the Tuleva Cube system works

Data is typically modulated into each subcarrier in a multi-carrier convergent system using a method like Orthogonal Frequency Division Multiplexing (OFDM) or its variations. Even when they are sent simultaneously over the same channel, these modulation algorithms ensure that the subcarriers are orthogonal and do not interfere with one another.

Long-Term Evolution (LTE), a wireless broadband communication standard that uses OFDM to transmit data over many sub-carriers, is one example of a multi-carrier convergent system [3]. Another illustration is Digital Video Broadcasting - Terrestrial (DVB-T), a digital television broadcasting standard that also employs OFDM to transfer data over several sub-carriers.
The hybrid Tuleva Cube device combines and amplifies numerous wired and wireless internet gateways for both uplink and downlink traffic. Using Tuleva CUBE's configuration options, five wired and wireless service provider inputs can be combined.

Table 1. Tuleva CUBE wireless AP utilises TWO (2) frequencies

Radio	2.4GHz	5.0GHz
Operating frequency (MHz)	2400-2500	5150 - 5350
Wireless protocol	802.11 n	802.11 ac

Wired LAN cables can also be used to connect to a Tuleva Cube. For crucial services like those provided by the government, banks, and security, Tuleva Cube meets the Service Level Agreement (SLA) at 99.9%. The contract includes an SLA that details, among other things [4]:

a. The kind of service that a service provider offers,
b. The quality of the service, and
c. What happens if the service provider does not meet the requirements?

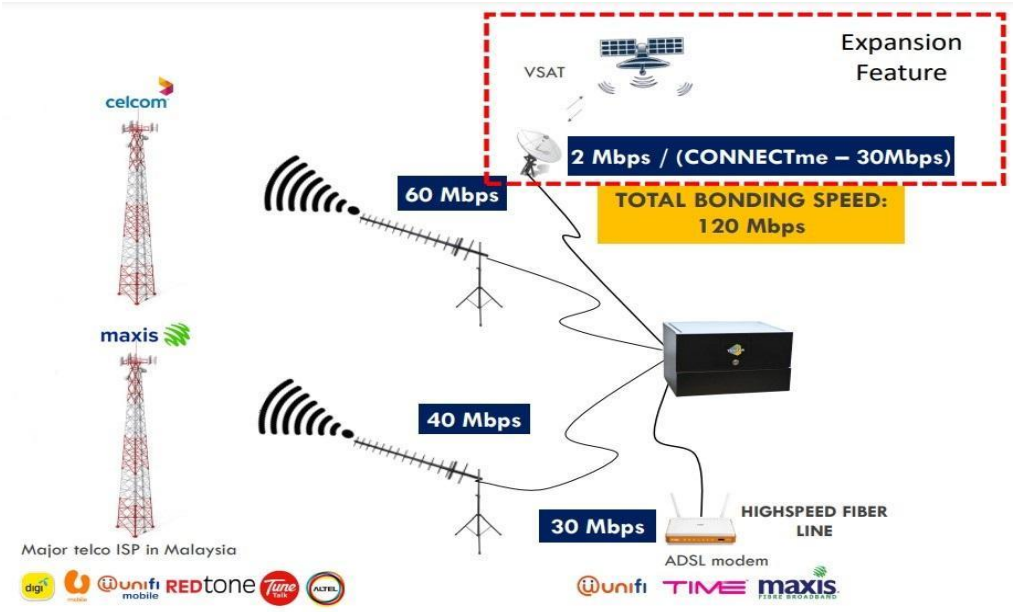


Fig. 2. Connections between the telco tower and the TC system

Networking devices, known as wired and wireless internet gateways, connect a local area network (LAN) to the Internet. An Ethernet cable is often used to connect a wired internet gateway to the local area network (LAN) and offer access to the Internet via a modem or broadband service. The gateway may also have additional functions, such as a router that allows several LAN devices to share an internet connection and a firewall that protects data by preventing unauthorised access to the LAN.

On the other hand, a wireless internet gateway offers communication to the LAN and the Internet via wireless signals rather than a physical cable. This kind of gateway often includes a wireless router that enables numerous devices to connect to the Internet through Wi-Fi, along with a modem or broadband service to provide the connection to the Internet. Additionally, wireless internet gateways may offer features like parental controls, which let users limit access to websites or categories of content, and guest access, which enables network guests to connect to the Internet without using the private LAN.

Both wired and wireless internet gateways are frequently used in homes and companies to provide internet access and allow networked devices to communicate with one another and the Internet. The decision between a wired and wireless gateway is frequently influenced by the design of the building, the quantity and distribution of network-connected devices, and the speed and dependability of the internet connection.

2. Methodology

Cellular convergence applies to the integration of diverse cellular network architectures and communication technologies to provide customers with a seamless communication experience. It intends to provide seamless network switching and access to phone, data, and multimedia services across several networks, including 2G, 3G, 4G, and 5G. The use of shared network infrastructure to support multiple cellular technologies is a vital component of cellular convergence, as many cellular network providers are introducing 4G and 5G networks that share a basic network architecture, allowing consumers to switch between the two without having to reconnect or change the settings on their device.
Cellular convergence integrates different wireless communication technologies, such as Wi-Fi and cellular, to produce a unified communication experience. Smartphones and other

mobile devices can switch between Wi-Fi and cellular networks based on which network provides the best connection.

Cellular convergence seeks to deliver a consistent user experience across different cellular networks and technologies, allowing users to get services regardless of their location or network resources. As cellular networks evolve and new technologies are introduced, it is a critical area of interest for network operators and technology providers.

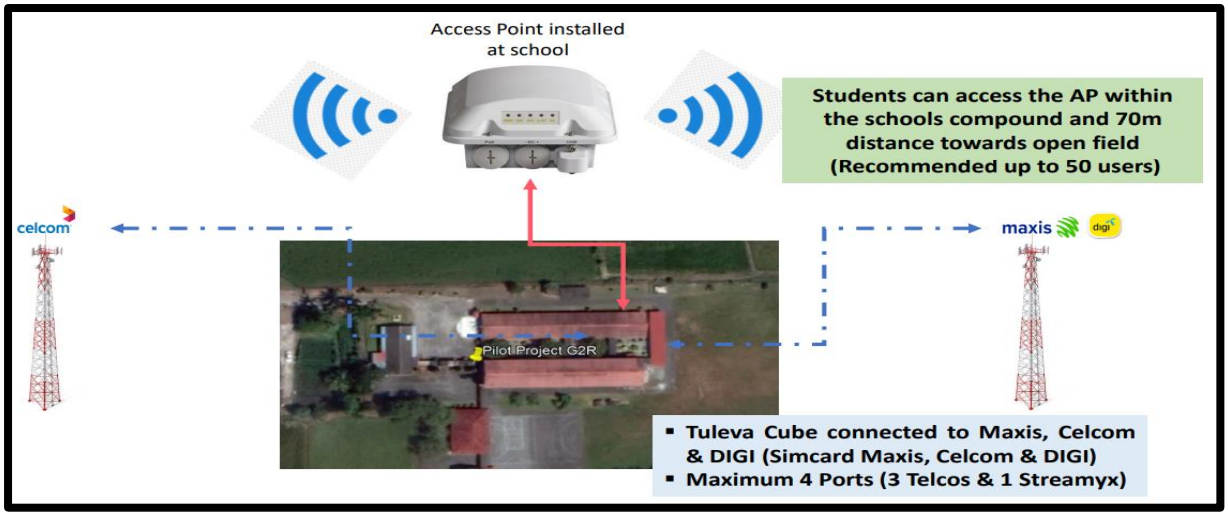


Fig. 3. Location of the Access Point (AP) placed in the school and how the AP spreads the coverage signal

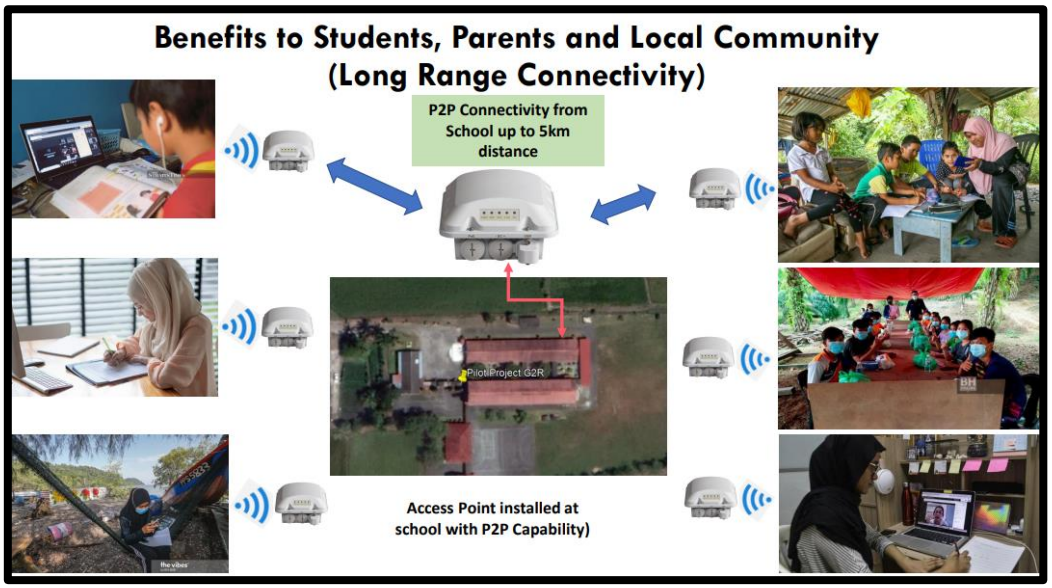


Fig. 4. P2P connectivity from the school to the area

Peer-to-peer (P2P) communication is a type of connection in which two or more devices or nodes connect to a network and interact directly without using a central server or other intermediaries. This type of connection is known as peer-to-peer (P2P) communication because it allows users to share files over the Internet without a central file server. P2P file sharing is an example of a P2P connection since it allows users to share files over the Internet without needing a central file server.

Another example of a P2P connection is peer-to-peer messaging software, which allows users to connect directly with one another rather than through a central messaging server. When

the benefits of a decentralised network exceed the problems and complexity of the deployment, P2P networking is frequently used in applications.



Fig. 5. Another project that involves G2R

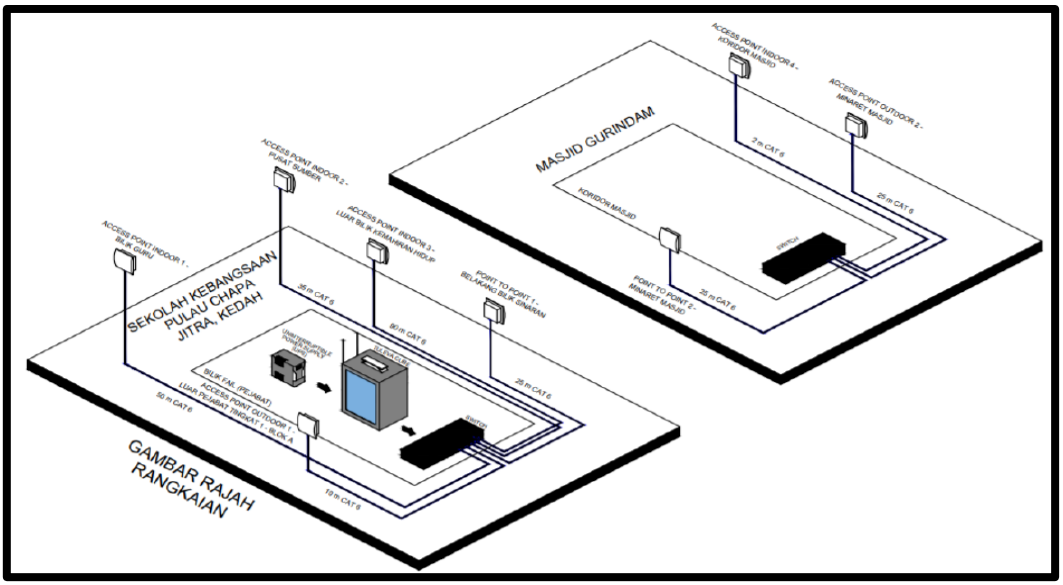


Fig. 6. Network Block Diagram of the TC system



Fig. 7. Installation of Hardware devices at the school

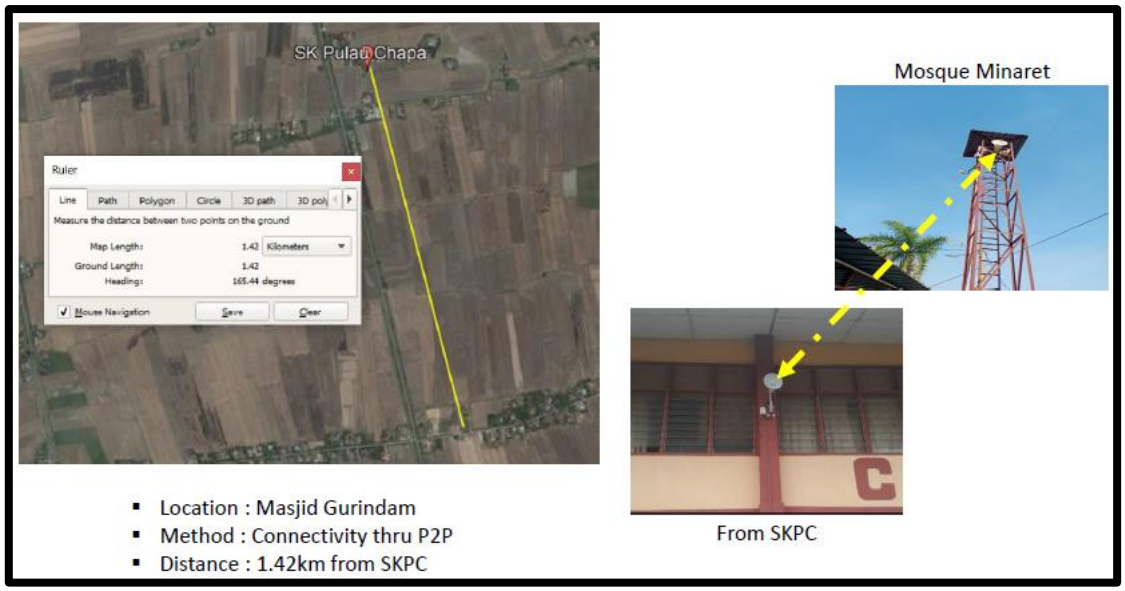


Fig. 8. Connectivity link between SK Pulau Chapa and Masjid Gurindam

3. Result and Discussion

3.1 Internet Speed Tests

One of these quick and simple methods to assess the speed of the Internet is to run a speed test. It offers the ideal means of determining whether the Internet is operating more slowly than it ought to or whether there are any other underlying problems on the other end. When performing a speed test, two elements are considered. There are many alternatives for what performance element to assess for free, like the Okla speed test. When running an internet speed test, the download and upload speeds are undoubtedly the elements a person needs to look at first. A download speed is measured by how quickly content can be downloaded from a specific internet server to a device [5].

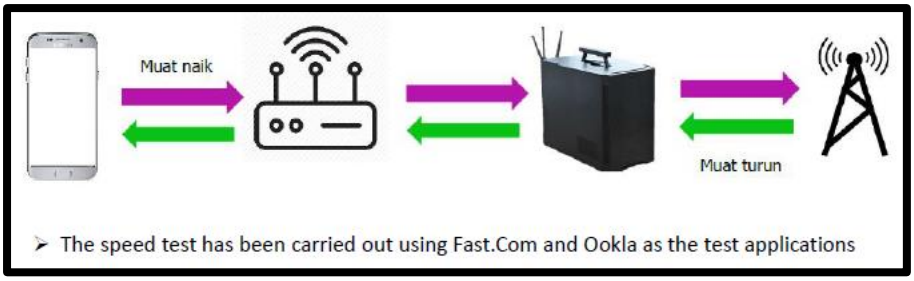


Fig. 9. Internet Speed Test

The network speed test serves a vital purpose that extends beyond evaluating network speed. Its primary objective is to delve into the intricate dynamics of upload and download times, offering valuable insights into each network's sustained performance. This rigorous assessment focused on two prominent telco providers, Maxis and Celcom, putting their capabilities to the test under real-world conditions.

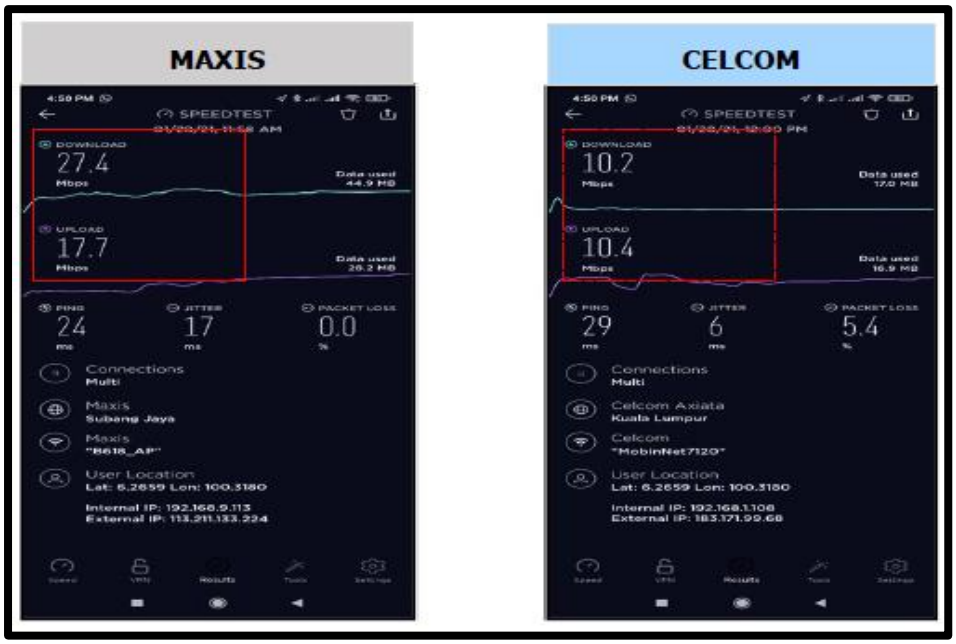


Fig. 10. Speed test at the SKPC (Smartphone test result)

By examining the two networks comprehensively (Figure 9), the researchers gain a comprehensive understanding of their resilience and stability. Beyond mere speed measurements, the researchers uncover each network's ability to consistently deliver optimal upload and download speeds over an extended period. This meticulous evaluation equips us with the knowledge needed to make informed decisions regarding network selection, enabling us to navigate the digital landscape with confidence and reliability.

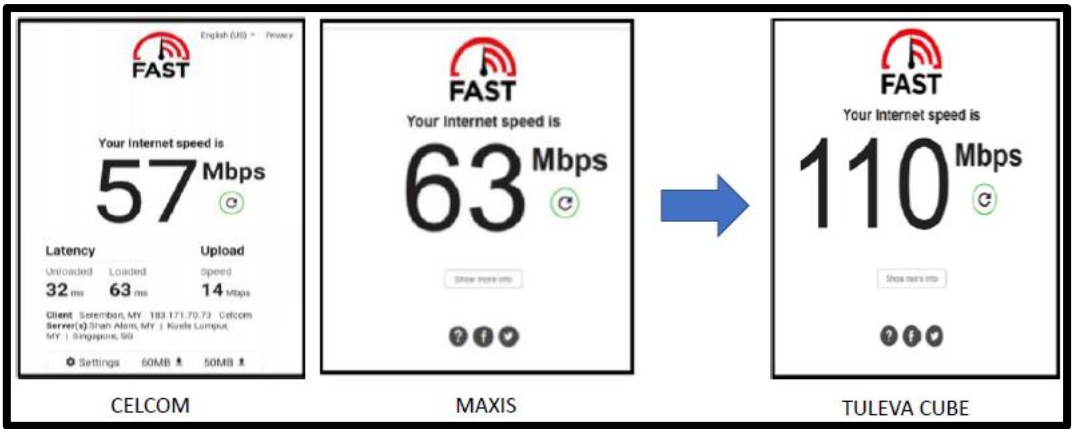


Fig. 11. Speed test at the SKPC (Test Using SIM card Injector by TC)

The compelling graphic presented above (Figure 10) vividly portrays the remarkable result of G2R company's ambitious pursuit of network convergence. Through the harmonious integration of numerous communication networks into a cohesive and unified infrastructure, G2R aims to revolutionise the way we connect. This convergence empowers a singular network capable of seamlessly catering to myriad services, encompassing traditional phone and data and video and multimedia applications.

The driving forces behind these transformative endeavours are multi-faceted and compelling. The demand for an unparalleled communication experience, fuelled by the ever-evolving technological landscape, necessitates a convergence of networks. By consolidating resources, leveraging innovative advancements, and maximising network efficiency and utilisation, G2R company is poised to reshape communication, unlocking new possibilities and propelling us toward a more integrated and interconnected future.

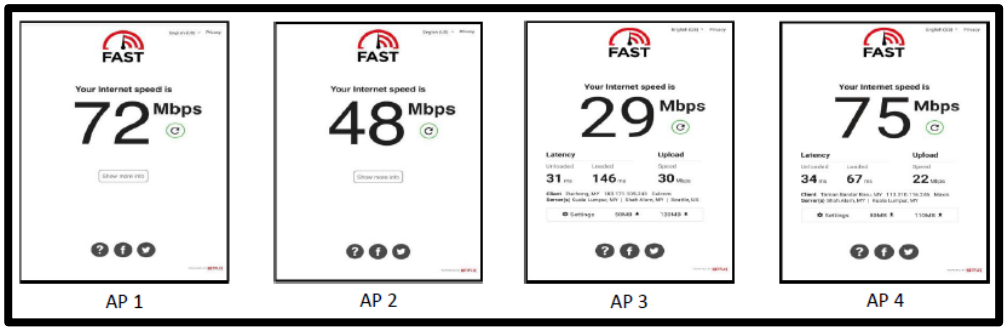


Fig. 12. Signal output results for four APs installed at SKPC

By analysing the signal output results obtained from the four Access Points (APs) meticulously deployed at SK Pulau Chapa and two APs at Masjid Gurindam, it becomes evident that this meticulously engineered system possesses the remarkable ability to deliver uninterrupted internet services to remote regions that were once challenging to reach. These results prove the system's efficacy in ensuring a consistently stable and reliable internet connection, even in the most inaccessible areas.

3.2 Network Management System (NMS)

A network management system, or NMS, is a program or collection of programs that perform a few essential tasks and enables network engineers to manage a network-independent component inside a larger network management framework.

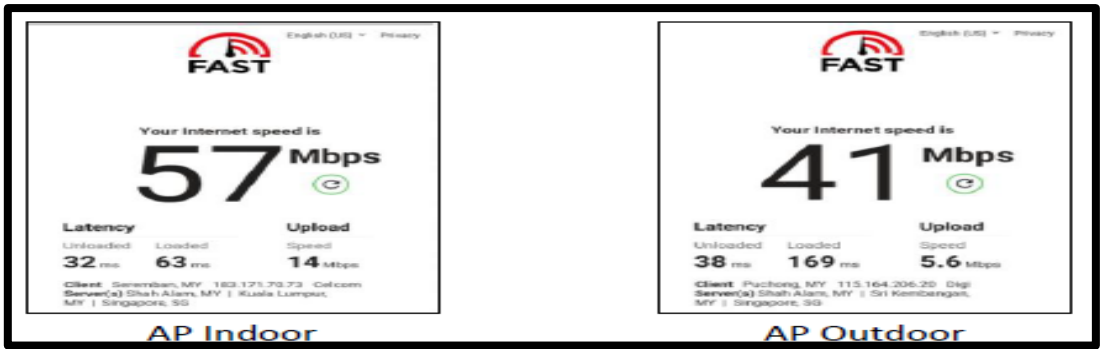


Fig. 13. Signal output results for TWO (2) APs installed at Masjid Gurindam

An NMS identifies, sets up, monitors, updates, and troubleshoots an enterprise network's wired and wireless network devices. The performance data gathered from each network component is then shown in a system management control application, enabling network engineers to make the necessary changes.

An efficient monitoring process is essential to ensuring smooth and trouble-free operation of all installed devices. G2R has advanced proactive system oversight by creating a cutting-edge Network Management System (NMS) in recognition of the importance of doing so. With the help of this cutting-edge system, any possible difficulties that might develop with the network infrastructure can be quickly identified and fixed.

The NMS monitors the functionality and overall condition of the installed systems, acting as a watchful keeper to prevent disturbances. It enables G2R to swiftly discover and address anomalies or vulnerabilities by continuously gathering and analysing crucial data points, maintaining optimal system performance at all times.

With this state-of-the-art NMS, G2R is dedicated to providing flawless reliability and reducing downtime. Incorporating such a potent monitoring system improves operational effectiveness and brings peace of mind. It enables proactive maintenance and prompt problem-solving, eventually leading to a seamless and uninterrupted user experience.

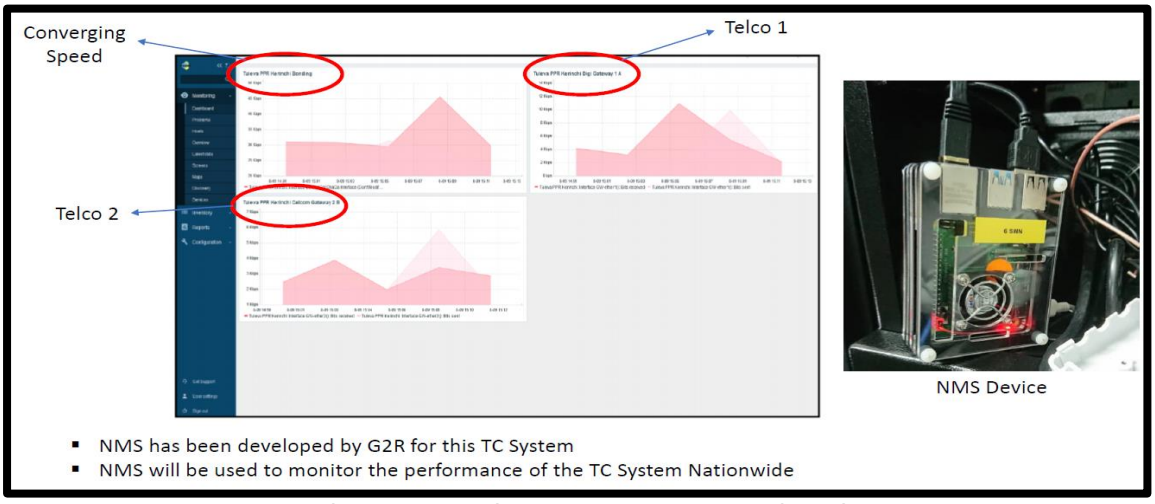


Fig. 13. Network Management System (NMS)

4. Conclusion

The Tuleva Cube, manufactured in Malaysia by Grass2Route, is a multi-carrier convergent system that offers shorter deployment timelines and lower costs as an alternative to traditional telco tower building. This novel method allows data transmission over a single channel and many carrier frequencies, enabling applications such as wireless communication, digital television, and broadband internet access. Organisations can cut lead times and costs associated with traditional infrastructure development by using the Tuleva Cube, leading to improved connection and telecommunication services.

While G2R's ambitions of developing a 'Voice Call and Data for Advanced TC System' to show their commitment to improving their capabilities, they must prioritise the technology itself and the user experience and safety considerations. G2R can ensure that its innovative TC system genuinely revolutionises transportation in a user-friendly and secure manner by focusing on seamless integration, intuitive interfaces, and solid security measures.

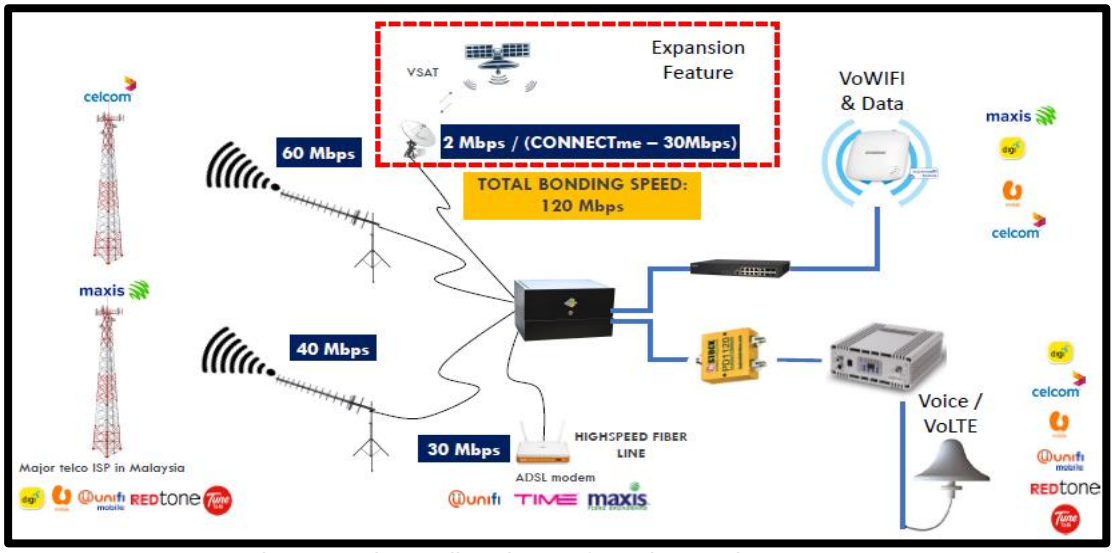


Fig. 14. Voice Call and Data for Advanced TC System

Table 2. Comparison between the Tuleva Cube and Telco Tower systems.

Characteristics	Tuleva Cube	Telco Tower
Deployment time	5 days	1 - 2 years
Site acquisition time	3 days	3 to 12 months
Cost variation	Minimal	Very high and depends on the steel market price
Cost impact for 2915 sites	RM 550mil (approximately)	RM 2.9bil (approximately)
Participation of Bumiputera Contractors	Can grow as many as sites awarded	Minimal. Only a few towers erectors in Malaysia
Creation of Job Opportunities	There are numerous opportunities. Only basic CME and IT installation skills are required.	Minimal because tower erection requires specialized knowledge.

According to the Table 2, this TC system not only demonstrates the company's capacity to create a system that facilitates, but it can also provide employment possibilities and lower the cost of construction and installation. Furthermore, facilities for obtaining internet access are easily accessible, and individuals in rural areas can enjoy the same amenities as those in cities. This initiative is greatly appreciated, especially by Bumiputera businesses like G2R Sdn.

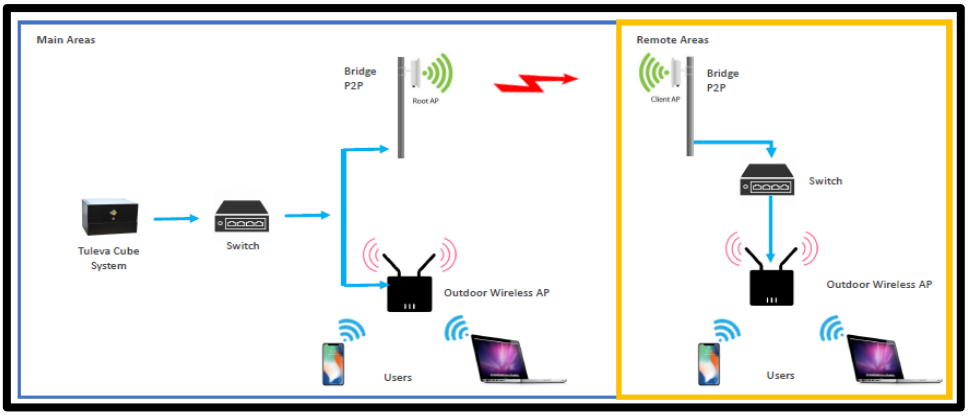


Fig. 15. Advanced TC System with Outdoor AP System.

Acknowledgements

We want to offer our heartfelt appreciation and gratitude to everyone who helped complete this technical study, "A Proof-of-Concept Study of Tuleva Cube System as a Solution for Rural Area Coverage System". Special thanks to Mr. Hezri (CEO), Mr. Nazmi (HOU TC), Mr. Irham (HOD PMO), Mr. Zainal (Field Engineering Expert), and Mr. Jamil (NMS) from G2R Sdn. Bhd. for their invaluable support, advice, and mentorship, which greatly guided this research. Author's Contribution

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HUBUNGAN ANTARA KESEDARAN ALAM SEKITAR DENGAN TINGKAH LAKU MESRA ALAM SEKITAR DI KALANGAN PELAJAR POLITEKNIK TUANKU SULTANAH BAHIYAH

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ABSTRAK

Di Malaysia masalah alam sekitar seperti pencemaran air, pencemaran udara, pengurusan sisa pepejal, penebangan hutan dan penggunaan tenaga memberi kesan kepada sumber asli dan ekosistem negara. Kesedaran alam sekitar dalam komuniti adalah penting untuk memupuk amalan mampan, meningkatkan kesihatan awam, dan membina masyarakat yang berdaya tahan dan bersatu padu. Pelajar IPT juga tidak terkecuali untuk memupuk amalan tersebut dalam menangani cabaran alam sekitar secara kolektif pada masa kini. Kajian ini bertujuan untuk melihat hubungan antara kesedaran alam sekitar dengan tingkah laku mesra alam sekitar dalam kalangan pelajar Politeknik Tuanku Sultanah Bahiyah (PTSB). Respondan seramai 60 orang pelajar Semester 1 Sesi II:2023/2024. Data kajian dikumpul menggunakan borang soal selidik dan dianalisis menggunakan *Statistical Package for Sosial Science* (SPSS). Daripada analisis nilai p yang diperolehi adalah 0.006 iaitu kecil daripada 0.05. Ia menunjukkan hubungan yang signifikan antara kesedaran alam sekitar dengan tingkah laku mesra alam sekitar dalam kalangan pelajar PTSB. Namun begitu, hubungan tersebut adalah lemah kerana nilai r yang adalah 0.348. Berdasarkan analisis ini terbukti tahap kesedaran yang tinggi dalam kalangan pelajar Semester 1 (Kejuruteraan) tidak memberi impak terhadap tingkah laku mesra alam sekitar yang diamalkan oleh mereka. Kajian ini juga mencadangkan agar program kesedaran alam sekitar diperkukuhkan dan dipertingkatkan bagi mencapai impak yang lebih besar terhadap tingkah laku pelajar.

Kata Kunci: Perhubungan, Kesedaran, Tingkah laku, Alam Sekitar

1. Pengenalan

Dalam beberapa tahun kebelakangan ini, kebimbangan yang semakin meningkat tentang kemerosotan alam sekitar dan perubahan iklim telah menekankan keperluan untuk amalan mampan. Institusi pendidikan memainkan peranan penting dalam memupuk kesedaran alam sekitar dan menggalakkan tingkah laku mesra alam dalam kalangan pelajar. Pelbagai program dan kempen telah dijalankan untuk memberi kesedaran pendidikan terhadap alam sekitar bagi melahirkan generasi yang terdidik dan mempunyai kesedaran tinggi terhadap aspek pemeliharaan alam sekitar. Ia telah dimulai seawal diperingkat prasekolah sehingga di peringkat pengajian tinggi. Justeru itu, pendidikan alam sekitar merupakan usaha terancang yang dilakukan untuk mendidik masyarakat sama ada secara formal atau sebaliknya supaya kesejahteraan dan kelangsungan hidup masyarakat dan alam sekitar terpelihara dan berkekalan. Menurut [1] pendidikan alam sekitar adalah suatu proses pendidikan tentang alam sekitar, melalui alam sekitar dan untuk alam sekitar. Agensi-agensi Kerajaan Negeri dan Persekutuan serta komuniti korporat dan Badan-badan Bukan Kerajaan (NGO) turut bekerjasama dengan Jabatan Alam Sekitar dalam melaksanakan aktiviti-aktiviti kesedaran alam sekitar secara berterusan untuk menjana kesedaran masyarakat terhadap mencintai alam sekitar.

Kesedaran alam sekitar dalam kalangan pelajar adalah penting untuk memupuk rasa tanggungjawab dan menggalakkan amalan lestari dari usia muda. Ia mendidik pelajar tentang

isu alam sekitar, seperti perubahan iklim, pencemaran, kekurangan sumber dan pemuliharaan. Kesedaran ini boleh disepadukan melalui aktiviti kokurikulum seperti kelab alam sekitar dan projek khidmat masyarakat. Pendidikan tentang alam sekitar sering kali dimulakan pada peringkat awal kanak-kanak. Jika pelajar tidak menerima pendedahan awal yang mencukupi, mereka mungkin tidak mengembangkan kesedaran dan kepentingan terhadap alam sekitar ketika memasuki institusi Pendidikan tinggi. Dengan melibatkan pelajar dalam aktiviti praktikal, seperti kitar semula, penanaman pokok dan inisiatif penjimatan tenaga, institusi pendidikan boleh membantu pelajar memahami kesan tindakan mereka terhadap alam sekitar. Selain itu, memupuk budaya penjagaan alam sekitar boleh memberi inspirasi kepada pelajar untuk menjadi penyokong kelestarian dalam komuniti mereka. Keperluan untuk meningkatkan pemahaman pelajar tentang tenaga boleh diperbaharui dan pembangunan mampan melalui pendekatan inovatif seperti modul pembelajaran mudah alih telah menunjukkan peningkatan tahap pengetahuan dan minat dengan ketara dalam mengejar kerjaya dalam sektor tenaga boleh diperbaharui [2]. Jika masyarakat di sekeliling mereka tidak menunjukkan kepedulian terhadap alam sekitar, pelajar mungkin merasa bahawa isu ini tidak penting.

Institusi pendidikan seperti Politeknik Tuanku Sultanah Bahiyah (PTSB) turut memberikan sokongan positif kepada kerajaan dengan mempertingkatkan kesedaran budaya hijau. Ini terbukti apabila Unit Pengurusan Teknologi Hijau (UPTH) ditubuhkan bagi bertujuan untuk mewujudkan budaya hijau dan melestarikan politeknik bagi melahirkan modal insan yang mengamalkan budaya hijau dalam industri dan masyarakat di luar sana seiring dengan hasrat Jabatan Pendidikan Politeknik Kolej Komuniti (JPPKK) untuk mengorientasi semula institusi Technical and Vocational Education and Training (TVET) ke arah menjadi institusi TVET Hijau [3]. Usaha ini juga terkandung dalam Hala Tuju Transformasi Politeknik (2010-2020). Antara program yang sering dianjurkan di PTSB adalah seperti Kempen *Zero Plastic, Kempen Waste to Wealth*, Pengumpulan Pakaian Terpakai, Kesedaran Hari Bumi, Kempen Tree For Life dan Gotong–Royong Perdana. Tambahan pula, menangani isu alam sekitar seperti pengurusan sisa pepejal adalah penting untuk pembangunan mampan, menonjolkan kepentingan meningkatkan kesedaran di kalangan pelajar dan masyarakat secara amnya [4].

Melalui aktiviti kesedaran alam sekitar seperti ini, pelajar berperanan sebagai pengurus alam sekitar, memerlukan pendekatan holistik yang merangkumi pemahaman tanggungjawab, nilai, kemahiran praktikal, dan strategi tingkah laku [5]. Namun begitu, kemerosotan penjagaan alam sekitar adalah disebabkan pembudayaan tingkah laku mesra alam masih di tahap yang rendah. Tingkah laku mesra terhadap alam sekitar contohnya seperti kitar semula, penjimatan elektrik dan air serta mengurangkan penggunaan kertas adalah antara tindakan kecil yang memberi signifikan dalam kelestarian alam sekitar. Melalui pembudayaan seperti ini membolehkan amalan hijau diterapkan dalam diri seseorang.

Banyak kajian telah dilakukan bagi melihat tahap pengetahuan dan tindakan kesedaran tingkah laku dalam menjaga alam sekitar. Pelbagai langkah telah diambil bagi meningkatkan pengetahuan dan kesedaran terhadap kepentingan menjaga alam sekitar. Program Pendidikan Alam Sekitar diperingkat sekolah telah memberi pengetahuan asas dan membina kemahiran serta sikap positif ke atas alam sekitar [6]. Namun tindakan tingkah laku dan kebijaksanaan dalam memastikan kelestarian alam sekitar masih lagi ditahap yang kurang memuaskan. Melalui kajian [7] tahap pengetahuan alam sekitar pelajar sekolah menengah di Langkawi adalah baik namun begitu tingkah laku mereka terhadap pendidikan alam sekitar adalah ditahap sederhana. Kebanyakkan pelajar peka terhadap masalah yang berlaku dan isu-isu yang berkaitan dengan alam sekitar tetapi kesedaran untuk mengatasi isu tersebut adalah pada tahap yang minimum [8]. Hal ini turut dilihat apabila dapatan [9] menunjukkan tidak terdapat hubungan yang signifikan antara tahap kesedaran pelajar dengan amalan baik dalam pendidikan alam sekitar di kalangan pelajar sekolah menengah sekitar Hulu Selangor. Kajian-kajian ini menunjukkan tiada amalan pembudayaan terhadap alam sekitar di kalangan pelajar walaupun mempunyai tahap kesedaran tentang kepentingan alam sekitar.

Kebanyakan pelajar sedar akan peranannya dalam menjaga alam sekitar namun secara individunya mereka sendiri tidak mengamalkan amalan pembudayaan alam sekitar tersebut [10].

Demi kelestarian alam sekitar, setiap individu perlu bertanggungjawab untuk memastikan alam ini terjaga demi kepentingan generasi akan datang. Namun sejauh mana kesedaran amalan hijau ini menjadi tingkah laku secara sukarela sebagai amalan harian dalam membantu alam sekitar kita? Justeru itu, objektif kajian ini untuk mengenal pasti hubungan antara kesedaran alam sekitar dengan tingkah laku mesra alam sekitar di kalangan pelajar PTSB. Kajian ini diharap dapat menunjukkan hubungan tahap kesedaran yang tinggi di kalangan pelajar seiring dengan tingkah laku mesra alam untuk kelestarian alam sekitar. Selain itu, kajian ini juga menyumbang kepada wacana yang lebih luas mengenai pendidikan alam sekitar dan peranan kritikalnya dalam membentuk masa depan yang lebih mampan.

2. Metodologi

2.1 Reka Bentuk Kajian

Kajian ini bertujuan untuk mengenal pasti hubungan antara kesedaran alam sekitar dengan tingkah laku mesra alam sekitar di kalangan pelajar. Lokasi kajian adalah PTSB, Kulim Kedah kerana ia mempunyai pelbagai ciri yang boleh mencapai objektif kajian.

2.2 Populasi dan Sampel Kajian

Kajian ini memberi fokus kepada pelajar semester 1 Sesi II 2023/2024 yang mengambil kursus Kejuruteraan di PTSB, Kulim Kedah. Responden yang telah menjawab soal selidik adalah 60 orang pelajar. Berdasarkan Krejcie dan Morgan (1970) saiz sampel yang sesuai untuk kajian ini ialah 52 respondan.

2.3 Instrumen Kajian

2.3.1 Soal Selidik

Dalam kajian ini, soal selidik telah dipilih sebagai instrumen kajian bagi mendapat maklumat yang dikehendaki. Ini kerana menurut [11] soal selidik adalah amat bersesuaian dengan kajian tinjauan kerana kaedah ini berupaya menghasilkan item-item yang boleh dipercayai. Kajian ini telah menggunakan soal selidik yang diubah suai daripada kajian [12]. Maklum balas diukur dengan 10 item soalan untuk Bahagian B iaitu kesedaran respondan terhadap alam sekitar dan Bahagian C berkenaan sikap respondan terhadap alam sekitar. Maklum balas bagi bahagian ini menggunakan 5 pilihan skala Likert: (1=Sangat Tidak Setuju, 2=Tidak Setuju, , 3=Setuju dan 4=Sangat Setuju).

2.3.2 Kebolehpercayaan

Kajian rintis ialah penyelidikan awal yang dijalankan dalam kelompok kecil sebelum kajian utama dijalankan. Objektif utama kajian rintis adalah untuk menguji kesahan dan kebolehpercayaan instrumen kajian sebelum digunakan dalam penyelidikan utama [17]. Kebolehpercayaan instrumen kajian diuji menggunakan pekali Alpha Cronbach. Kajian rintis sangat penting untuk memastikan kualiti penyelidikan primer. Dalam penyelidikan kuantitatif, kajian rintis bertujuan untuk mengesahkan kebolehpercayaan instrumen. Menurut [16] jumlah responden untuk ujian rintis adalah sepuluh peratus daripada sampel sebenar. Dalam kajian ini, seramai 10 orang responden telah menjawab kajian rintis dan pekali kebolehpercayaan Alpha Cronbach bagi Bahagian B yang diperolehi 0.861 manakala bagi Bahagian C adalah 0.793. Ini jelas menunjukkan nilai ini memberi maksud instrumen yang digunakan berada dalam keadaan sangat baik dan boleh digunakan dalam penyelidikan yang sebenar.

2.3.3 Kesahan

Soal selidik ini telah disemak oleh beberapa pakar dalam bidang berkenaan untuk memastikan kesahan kandungan. Ini untuk memastikan soal selidik adalah jelas dan mudah difahami oleh responden. Dengan tahap kebolehpercayaan dan kesahan yang tinggi, soal selidik ini adalah instrumen yang sesuai untuk mengukur kesedaran dan sikap responden terhadap alam sekitar.

2.4 Teknik Pengumpulan

Data yang diperolehi daripada soal selidik dianalisis menggunakan perisian SPSS versi 21.0 dengan menggunakan kaedah Kolerasi Pearson bagi mengetahui kesignifikanan hubungan di antara kesedaran respondan terhadap alam sekitar dan sikap respondan terhadap alam sekitar. Jadual 2.1 menunjukkan Nilai Indek Korelasi yang digunakan dalam kajian ini.

Jadual 1. Kolerasi Pearson

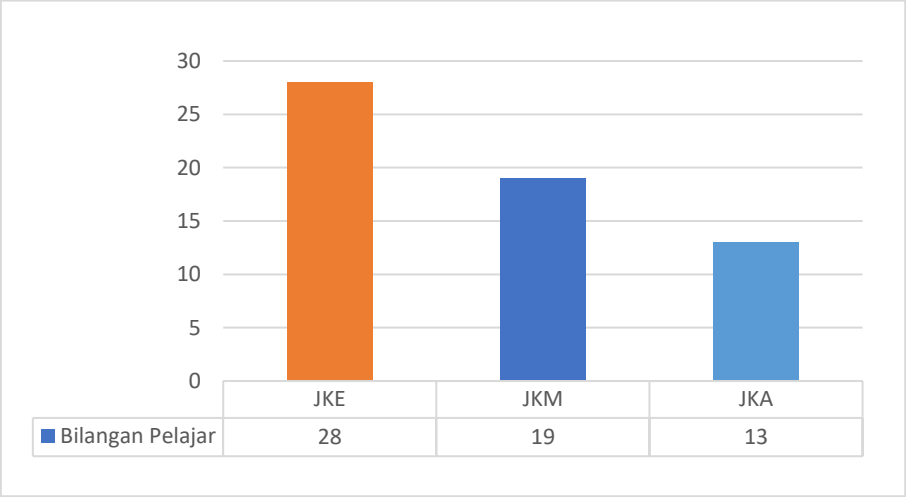
Nilai Indek Kolerasi	Interpretasi Hubungan
0.00-0.20	Sangat Lemah
0.21-0.40	Lemah
0.41-0.70	Sederhana
0.71-0.90	Kuat
0.91-1.00	Sangat Kuat

Sumber: [13]

3. Keputusan dan Perbincangan

3.1 Demografi

Jumlah kesemua sampel yang terlibat dalam kajian ini adalah seramai 60 orang pelajar seperti rajah 1



Rajah 1. Bilangan respondan mengikut jabatan

Berdasarkan Rajah 3.1 bilangan pelajar daripada JKE merupakan kumpulan responden terbesar dalam kajian ini, hampir separuh daripada jumlah keseluruhan. Diikuti dengan JKM yang merupakan kumpulan kedua terbesar, mewakili hampir satu pertiga daripada keseluruhan responden. Kumpulan terkecil dalam kajian ini ialah JKA di mana bilangan pelajar adalah 13 tetapi masih mewakili lebih daripada satu perlima daripada jumlah keseluruhan. Pembahagian ini memberikan gambaran mengenai komposisi responden mengikut jabatan. Ketidakseimbangan ini penting untuk dianalisis kerana ia boleh mempengaruhi hasil kajian,

terutama jika terdapat perbezaan ketara dalam kesedaran dan tingkah laku mesra alam antara jabatan-jabatan yang dikaji.

3.2 Ujian Pekali Kolerasi

Pelajar memahami peranan mereka dalam menjaga alam sekitar, tetapi ini tidak selalu diterjemahkan ke dalam tindakan yang berkesan [12]. Pengetahuan tentang isu-isu seperti kitar semula dan pencemaran adalah penting, namun pelaksanaan amalan mesra alam sering kali terhad disebabkan faktor-faktor seperti kemudahan yang tidak mencukupi dan kurangnya insentif untuk terlibat dalam aktiviti kitar semula [15]. Oleh itu, kajian ini bertujuan untuk menyelidik hubungan antara tahap kesedaran alam sekitar dan amalan tingkah laku mesra alam sekitar di kalangan pelajar. Dalam melihatat hubungan antara tahap kesedaran alam sekitar dan amalan tingkah laku mesra alam sekitar ujian pekali kolerasi dilaksanakan.

Ujian ini membantu mengukur kekuatan dan arah hubungan antara dua pemboleh ubah. Ini boleh menunjukkan sama ada terdapat hubungan positif, negatif, atau tiada hubungan antara pemboleh ubah yang dikaji. Dalam kajian ini, ujian pekali korelasi membantu menentukan sejauh mana kesedaran alam sekitar mempengaruhi tingkah laku mesra alam. Nilai p yang diperolehi daripada kajian ini adalah 0.006 iaitu kurang daipada 0.05. Ini menunjukkan hubungan yang signifikan antara tahap kesedaran dengan tingkah laku terhadap alam sekitar. Hubungan positif lemah yang diperhatikan antara kesedaran dan tingkah laku menunjukkan bahawa peningkatan dalam kesedaran terhadap alam sekitar cenderung untuk disertai dengan peningkatan dalam tingkah laku mesra alam.

Manakala nilai r yang diperolehi adalah 0.348 iaitu berada pada tahap lemah. Walaupun korelasi ini tidak terlalu kuat, ia masih menunjukkan bahawa kesedaran memainkan peranan penting dalam mendorong tingkah laku yang mesra alam. Nilai r mencerminkan bahawa hanya sebahagian daripada varians dalam tingkah laku yang dapat dijelaskan oleh kesedaran. Ini menunjukkan bahawa terdapat faktor-faktor lain yang juga mempengaruhi tingkah laku mesra alam, yang mungkin merangkumi faktor-faktor seperti norma sosial, pendidikan, kemudahan, dan sikap peribadi. Dalam konteks kajian ini, adalah penting untuk memberi tumpuan kepada pendekatan pendidikan dan kempen kesedaran yang dapat meningkatkan pengetahuan serta pemahaman pelajar tentang kepentingan tingkah laku mesra alam. Memandangkan hasil kajian ini, adalah wajar untuk mencadangkan agar program kesedaran alam sekitar diperkukuhkan dan dipertingkatkan bagi mencapai impak yang lebih besar terhadap tingkah laku pelajar.

Jadual 2. Ujian Korelasi Pearson antara kesedaran dan tingkah laku

		Kesedaran	Tingkah Laku
Kesedaran	Pearson Correlation, R	1	.348**
	Sig.(2-Tailed)		0.006
	N	60	60
Tingkah Laku	Pearson Correlation, R	.348**	1
	Sig.(2-Tailed)	0.006	
	N	60	60

Tambahan pula, kajian ini membuka ruang untuk penyelidikan lanjut untuk meneroka faktor-faktor lain yang mungkin menyumbang kepada tingkah laku mesra alam. Penyelidikan lanjutan boleh mengkaji aspek seperti peranan media, pengaruh rakan sebaya, dan dasar-dasar institusi dalam membentuk tingkah laku mesra alam. Secara keseluruhannya, hasil kajian ini menunjukkan bahawa usaha untuk meningkatkan kesedaran alam sekitar adalah penting dan berpotensi untuk meningkatkan tingkah laku mesra alam di kalangan pelajar, namun usaha tersebut perlu disokong dengan pendekatan yang komprehensif dan pelbagai dimensi. Justeru itu mengintegrasikan pendidikan alam sekitar ke dalam kurikulum pendidikan

tinggi adalah penting untuk memupuk budaya kelestarian dan menyediakan pelajar untuk menyumbang kepada matlamat alam sekitar dan sasaran pembangunan mampan Malaysia [14].

4. Kesimpulan dan Cadangan

Berdasarkan dapatan yang diperolehi beberapa kesimpulan utama dapat diambil iaitu terdapat hubungan positif lemah antara kesedaran alam sekitar dan tingkah laku mesra alam ($r = 0.348$, $p = 0.006$). Ini menunjukkan tahap kesedaran yang tinggi dalam kalangan pelajar Semester 1 (Kejuruteraan) tidak memberi impak terhadap tingkah laku mesra alam sekitar yang diamalkan oleh mereka. Kedua, responden terdiri daripada pelajar dari tiga jabatan: JKE (46.7%), JKM (31.7%), dan JKA (21.7%). Pengagihan yang tidak seimbang ini menunjukkan bahawa hasil kajian lebih banyak mencerminkan pandangan dan tingkah laku pelajar dari JKE. Walaupun terdapat hubungan yang signifikan antara kesedaran dan tingkah laku mesra alam, korelasi lemah ini menunjukkan bahawa faktor-faktor lain juga memainkan peranan penting dalam mempengaruhi tingkah laku mesra alam.

Responden daripada aliran pengajian berlainan boleh menyebabkan kepada faktor korelasi lemah. Beberapa kajian lepas telah mengenalpasti bahawa faktor perbezaan aliran pengajian di institut pengajian tinggi (IPT) boleh menyumbang kepada tingkah laku terhadap alam sekitar. Ia disokong oleh kajian [18] yang mendapati bahawa tahap komitmen terhadap alam sekitar dipengaruhi oleh aliran pengajian pelajar di universiti melalui ketidakseragaman kurikulum serta strategi pengajaran dan pembelajaran mata pelajaran yang berkaitan dengan pendidikan alam sekitar yang berbeza. Menurut kajian [19], kursus-kursus aliran pengajian Sains Tulen banyak menghabiskan aktiviti di luar dan ini menyumbang kepada perubahan sikap pelajar terhadap alam sekitar. Oleh yang demikian aliran pengajian seperti ini akan mempunyai tingkah laku yang tinggi terhadap alam sekitar. Ia turut disokong oleh dapatan kajian [20] terhadap pelajar aliran Pengurusan Pelancongan, Politeknik Sultan Idris Shah di mana telah menunjukkan tahap kesedaran alam sekitar yang tinggi serta tahap tingkah laku mesra alam sekitar yang tinggi disebabkan para pelajar didedahkan dan mengikuti pembelajaran alam sekitar secara langsung seperti terkandung dalam struktur pengajian kurikulum program Diploma Pengurusan Pelancongan Politeknik.

Berdasarkan kesimpulan ini, beberapa cadangan dapat diberikan untuk meningkatkan kesedaran dan tingkah laku mesra alam di kalangan pelajar. Pertama, meningkatkan program pendidikan alam sekitar yang lebih komprehensif dan berkesan di semua aliran pengajian dengan memasukkannya ke dalam struktur pengajian kurikulum khususnya di politeknik. Program ini harus menekankan kepentingan tingkah laku mesra alam dan memberikan maklumat praktikal tentang bagaimana pelajar dapat mengamalkannya dalam kehidupan harian mereka. Kedua, memandangkan pengagihan responden yang tidak seimbang, adalah penting untuk menyesuaikan pendekatan kesedaran alam sekitar mengikut keperluan dan konteks setiap jabatan. Ketiga, melakukan penyelidikan lanjut untuk mengenal pasti faktor-faktor lain yang mempengaruhi tingkah laku mesra alam. Ini boleh termasuk kajian tentang peranan norma sosial, pengaruh rakan sebaya, dan sokongan institusi dalam membentuk tingkah laku mesra alam di kalangan pelajar. Keempat, mengadakan kempen kesedaran alam sekitar secara berkala yang melibatkan aktiviti praktikal dan interaktif untuk menggalakkan pelajar mengamalkan tingkah laku mesra alam. Aktiviti seperti gotong-royong, kitar semula, dan penggunaan sumber tenaga yang boleh diperbaharui boleh membantu meningkatkan kesedaran dan tingkah laku mesra alam. Dengan melaksanakan cadangan-cadangan ini, diharapkan kesedaran dan tingkah laku mesra alam di kalangan pelajar dapat ditingkatkan, seterusnya menyumbang kepada usaha pemeliharaan dan pemuliharaan alam sekitar yang lebih berkesan. Meningkatkan kesedaran alam sekitar dalam kalangan pelajar adalah kunci untuk memupuk tingkah laku mampan dan menggalakkan penglibatan aktif dalam usaha perlindungan alam sekitar.

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ECO-FRIENDLY SOUNDPROOF WALLS: HARNESSING MUDCAKE WASTE

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ABSTRAK

Panel dinding memainkan peranan penting dalam pembinaan bangunan, tidak hanya sebagai struktur tetapi juga sebagai penyelesaian untuk mengurangkan gangguan bunyi dari persekitaran luar. Dalam usaha untuk memperkenalkan inovasi mesra alam, kajian ini mengkaji penghasilan panel dinding kedap bunyi dengan menggunakan bahan buangan gula (*mudcake*), serat buluh, simen, dan pasir. Penyelidikan ini menggunakan aplikasi Sound Level Meter untuk mengukur nilai decibel bunyi yang diserap oleh panel-panel ini. Tiga sampel panel dengan dimensi 300 mm × 300 mm dan ketebalan 30 mm telah diuji, setiap satu dengan peratusan campuran bahan yang berbeza. Keputusan awal menunjukkan bahawa campuran dengan 30% simen, 30% pasir, 40% *mudcake* dan 10% serat buluh memberikan hasil penyerapan bunyi terbaik, dengan pengurangan sehingga 61.5 dB. Berdasarkan data ini, panel dinding berskala lebih besar dengan ukuran 600 mm × 800 mm, dengan ketebalan 100 mm dihasilkan menggunakan formulasi yang sama. Ujian lanjut dan analisis menunjukkan bahawa panel ini mampu mengurangkan bunyi bising dengan efektif, walaupun faktor persekitaran seperti aliran udara dan bunyi latar mempengaruhi keberkesanan penyerapan bunyi. Kajian ini membuktikan bahawa penggunaan bahan buangan gula (*mudcake*) dalam penghasilan panel dinding kedap bunyi bukan sahaja berkesan tetapi juga menawarkan penyelesaian yang lebih murah dan mesra alam berbanding produk komersial sedia ada. Dengan demikian, panel ini bukan sahaja meningkatkan kualiti hidup penghuni bangunan tetapi juga menyumbang kepada kelestarian alam sekitar.
Kata Kunci: *Mudcake*, serat buluh, *sound level meter* (dB), mesra alam

1. Pengenalan

Di era kini, perkembangan sektor pembinaan begitu maju dan pesat. Inovasi-inovasi terkini dalam produk binaan kini banyak yang menggunakan bahan kitar semula, yang memberikan banyak keuntungan serta kemudahan kepada pengguna. Salah satu inovasi menarik adalah panel dinding.

Panel dinding adalah produk pembinaan yang terdiri daripada pelbagai bahan yang digunakan untuk keperluan pemasangan dinding di sesebuah bangunan. Panel dinding ini hadir dalam pelbagai ukuran yang disesuaikan dengan saiz bangunan. Kini, panel dinding pasang siap yang menggunakan Sistem Bangunan Berindustri (IBS) semakin digalakkan dalam pembinaan bangunan moden.

Sistem Bangunan Berindustri (IBS) bukan hanya popular dalam industri pembinaan, tetapi juga diaplikasikan dalam sektor lain seperti automotif, di mana komponen-komponen kenderaan dihasilkan dari pelbagai bahagian kilang dan disatukan pada pemasangan akhir.

Panel dinding yang bersifat akustik ditakrifkan oleh ciri-ciri yang menentukan kualiti bunyi berkaitan dengan pendengaran. Kajian oleh [2] menyatakan bahawa data tentang sifat penebat akustik produk bangunan mesti disahkan melalui ujian makmal di bilik akustik khas. Keberkesanan penyerapan akustik ini berkaitan dengan ketebalan bahan yang digunakan [3].

Penghasilan panel dinding dari bahan kitar semula dapat mengurangkan kuantiti pembuangan bahan kitar semula. Hal ini menghasilkan produk mesra alam dengan kualiti yang baik mengikut standard panel dinding. Bahan-bahan seperti pelepah pisang, serat pokok kenaf dan serat kelapa telah digunakan untuk menghasilkan panel dinding ini.

Panel dinding ini memiliki banyak kelebihan seperti kedap bunyi, kalis air, dan meredam kebakaran. Ujian mampatan dan kedap bunyi dilakukan untuk memastikan panel dinding ini memenuhi standard yang diperlukan.

Bahan kitar semula seperti pelepah pisang, serat pokok kenaf, dan serat kelapa memberikan ciri-ciri seperti sifat fizikal yang baik, penebat haba, dan sifat akustik yang unggul. Menurut kajian oleh Jabatan Geografi dan Alam Sekitar, kebisingan trafik memberi kesan kepada tingkah laku pelajar serta mengganggu proses pengajaran dan pembelajaran di sekolah.

Kajian ini bertujuan menghasilkan panel dinding IBS yang mengandungi *mudcake*, abu terbang, dan serat buluh serta menentukan sifat akustik panel dinding kedap bunyi yang direkabentuk menggunakan bahan-bahan ini melalui ujikaji dan simulasi.

Bahan utama dalam penghasilan panel dinding ini adalah *mudcake* sebagai komponen utama agen pengikat, dengan serat buluh dan abu terbang sebagai gentian tambahan. Ujian akustik dilakukan mengikuti ISO 354 dan dianalisis untuk menentukan kesan kepelbagaian kandungan bahan dan hubungan sifat akustik dengan ketumpatan serta kekuatan mampatan panel dinding.

Penggunaan bahan kitar semula dalam penghasilan panel dinding ini juga menyumbang kepada pemuliharaan sumber asli dan perlindungan alam sekitar. Kecekapan tenaga merupakan aspek penting dalam pembangunan bangunan hijau di negara kita. Selain menjimatkan kos pembinaan, panel dinding mesra alam ini memberikan produk berkualiti tinggi dengan harga berpatutan, menjadikan ia pilihan bijak bagi pengguna. Dengan menggunakan bahan kitar semula, industri pembinaan dapat menetapkan rintangan haba yang baik serta penyerapan bunyi yang berkesan. Kajian ini membuka peluang penggunaan *mudcake* yang lebih meluas dalam pembinaan di Malaysia, menjanjikan masa depan yang lebih lestari dan inovatif.

2. Metodologi

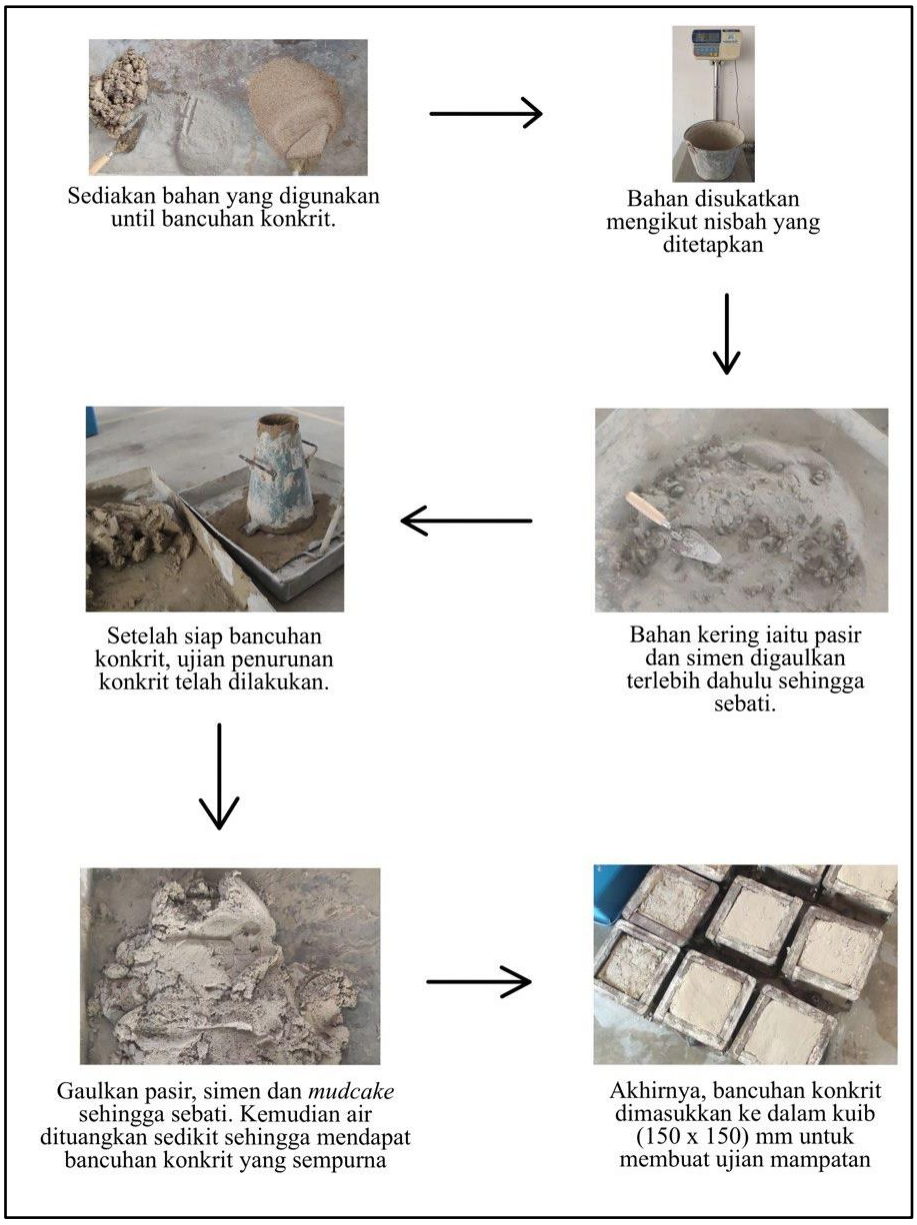
Reka bentuk kajian adalah pelan tindakan yang terperinci yang menggariskan langkah-langkah sistematik untuk melaksanakan kajian. Ia berfungsi sebagai panduan kritikal bagi penyelidik dalam mengumpul data, menganalisis hasil, dan membuat interpretasi. Dengan reka bentuk kajian yang baik, penyelidik dapat membuat inferens yang tepat tentang pemboleh ubah yang dikaji.

Penggunaan konkrit ringan dalam sektor pembinaan telah bermula sejak lama. Meskipun dahulu jarang digunakan, kini ia semakin popular dalam industri pembinaan kerana sifatnya yang serbaguna dan efisien. Menurut [9], penggunaan kerikil sebagai agregat ringan dalam

konkrit bermula di negara-negara maju seperti Amerika Syarikat, England, dan beberapa negara Eropah lain sejak akhir abad ke-19. Salah satu inovasi utama dalam konkrit ringan adalah konkrit berbuih yang memiliki ketumpatan lebih rendah daripada konkrit biasa. "Draf Kod Model Standard Antarabangsa untuk Pembinaan Konkrit" oleh Konvensyen Komuniti Eropah mengenai Konkrit mengkategorikan konkrit ringan berdasarkan ketumpatannya, menonjolkan kelebihan konkrit berbuih dalam pembinaan moden.

[6] mendefinisikan aditif sebagai bahan yang ditambahkan ke dalam campuran konkrit biasa untuk mengawal dan mengubah sifat konkrit mengikut keperluan struktur tertentu. Contohnya, aditif digunakan untuk meningkatkan kualiti konkrit tanpa mengurangkan jumlah air dalam campuran, serta untuk melambatkan atau mempercepat pengerasan konkrit. Aditif juga meningkatkan kebolehkerjaan konkrit, terutama dalam projek yang memerlukan penggunaan mesin pam. Dengan reka bentuk kajian yang baik dan penggunaan konkrit ringan serta aditif yang tepat, sektor pembinaan dapat mencapai hasil yang luar biasa. Memahami sejarah dan evolusi konkrit ringan, serta peranan penting aditif, adalah kunci untuk mencipta struktur yang tahan lama dan berkualiti tinggi. Industri pembinaan kini berada di ambang transformasi besar dengan inovasi konkrit ringan yang terus berkembang.

Untuk ujian mampatan konkrit, bahan yang digunakan terdiri daripada simen, pasir, *mudcake* dan air dengan nisbah yang telah ditetapkan [1]. Tiga nisbah campuran yang digunakan ialah 1:3:6, 2:3:5, dan 3:3:4, merujuk kepada jumlah simen, pasir dan *mudcake* berdasarkan standard *lightweight* LC20/22 mengikut BS EN 13055-1:2020. Setiap bahan ditimbang mengikut nisbah yang ditetapkan dan digaul sehingga sebati seperti yang ditunjukkan dalam Rajah 1. Campuran ini diuji menggunakan ujian penurunan konkrit untuk memastikan kebolehkerjaan campuran tersebut. Selepas ujian penurunan, campuran dimasukkan ke dalam acuan berbentuk kiub bersaiz 150mm x 150mm x 150mm. Kiub konkrit ini akan diuji kekuatan mampatannya selepas 7, 14, dan 28 hari. Setiap tempoh ujian melibatkan tiga sampel kiub untuk setiap nisbah campuran. Hasil ujian mampatan dicatat dan dianalisis untuk menentukan kekuatan konkrit berdasarkan nisbah campuran dan tempoh pengawetan yang berbeza, membantu menentukan campuran terbaik yang memenuhi keperluan kekuatan konkrit yang dikehendaki.



Rajah 1. Langkah-langkah menyediakan konkrit untuk ujian mampatan

2.1 Ujian Mampatan

Jadual 1 menunjukkan nisbah campuran bahan yang dibancuh mengikut nisbah yang ditetapkan itu. Nisbah 1:3:6 diambil daripada LC20/22 merujuk pada BS EN 13055-1:2002 dengan nisbahnya mempunyai campuran seperti yang berikut di mana 1 ialah simen, 3 ialah pasir dan 6 ialah *mudcake*/agregat ringan [1].

Semua bancuhan yang dibacuh adalah mengikut pada nisbah bancuhan yang ditetapkan supaya analisis keputusan dapat dilaksanakan dengan tetap dan kesalahan pada campuran dapat dikesan.

Jadual 1. Bancuhan nisbah bahan bagi sampel (150 × 150 x 150) mm

NO	Sampel	Tempoh (Hari)	Nisbah Campuran	Bahan (kg)			
				Simen	Pasir	*Mudcake*	Air
A	A1	7	1:3:6	0.500	1.500	3.000	0.300
		7		0.500	1.500	3.000	0.300
		7		0.500	1.500	3.000	0.300
	A2	14		0.500	1.500	3.000	0.300
		14		0.500	1.500	3.000	0.300
		14		0.500	1.500	3.000	0.300
	A3	28		0.500	1.500	3.000	0.300
		28		0.500	1.500	3.000	0.300
		28		0.500	1.500	3.000	0.300
B	B1	7	2:3:5	1.000	1.500	2.500	0.3500
		7		1.000	1.500	2.500	0.3500
		7		1.000	1.500	2.500	0.3500
	B2	14		1.000	1.500	2.500	0.3500
		14		1.000	1.500	2.500	0.3500
		14		1.000	1.500	2.500	0.3500
	B3	28		1.000	1.500	2.500	0.3500
		28		1.000	1.500	2.500	0.3500
		28		1.000	1.500	2.500	0.3500
C	C1	7	3:3:4	1.500	1.500	2.000	0.900
		7		1.500	1.500	2.000	0.900
		7		1.500	1.500	2.000	0.900
	C2	14		1.500	1.500	2.000	0.900
		14		1.500	1.500	2.000	0.900
		14		1.500	1.500	2.000	0.900
	C3	28		1.500	1.500	2.000	0.900
		28		1.500	1.500	2.000	0.900
		28		1.500	1.500	2.000	0.900

2.2 Ujian Akuatik (*Sound Meter*)

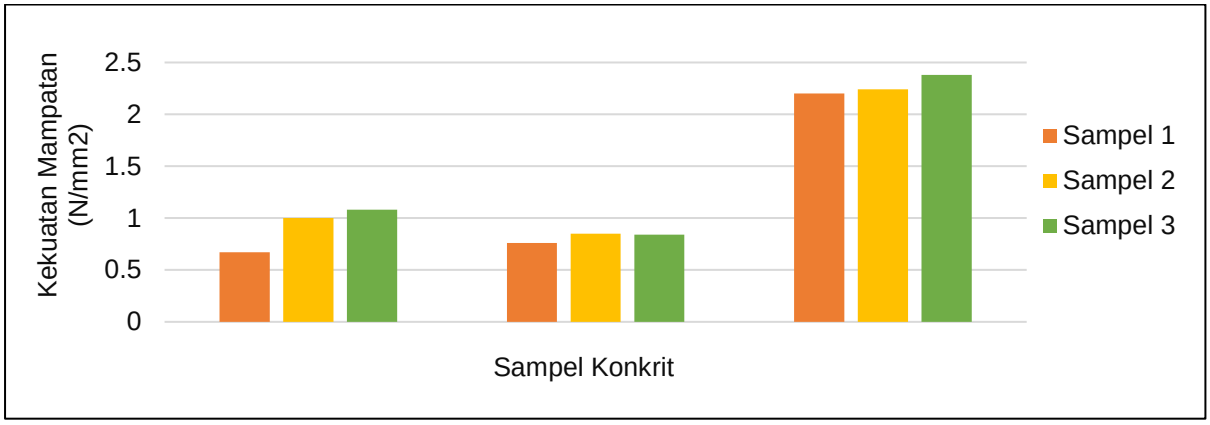
Jadual 2 menunjukkan nisbah bahan untuk bacuhan panel dinding untuk membuat ujian akuatik bagi mendapatkan nilai desibel (dB) dengan mengukur menggunakan aplikasi '*Sound Meter'.* Nilai terbaik yang direkodkan akan dijadikan rujukan untuk mendirikan panel dinding yang sebenar. Bagi sampel A menggunakan nisbah 3:3:4 daripada 10 kg dalam menghasil panel dinding [2][4]. Seterusnya, sampel B menggunakan 3 kg simen, 3 kg pasir dan 4 kg *mudcake* daripada 10 kg dan 5% daripada jumlah tersebut ialah serat buluh. Akhirnya, sampel C menggunakan nisbah 3:3:4 daripada 10 kg juga dan 10% daripada jumlah itu ialah serat buluh. Jadi, ketiga-tiga sampel yang telah siap akan diuji dengan menggunakan aplikasi '*Sound Meter*' dalam unit dB meter untuk menentukan nilai decibel.

Jadual 2. Nisbah bancuhan panel dinding

Sampel	Bahan (kg)			
	Simen	Pasir	Mudcake	Serat Buluh
A	3	3	4	0%
B	3	3	4	5%
C	3	3	4	10%

3. Keputusan dan Perbincangan

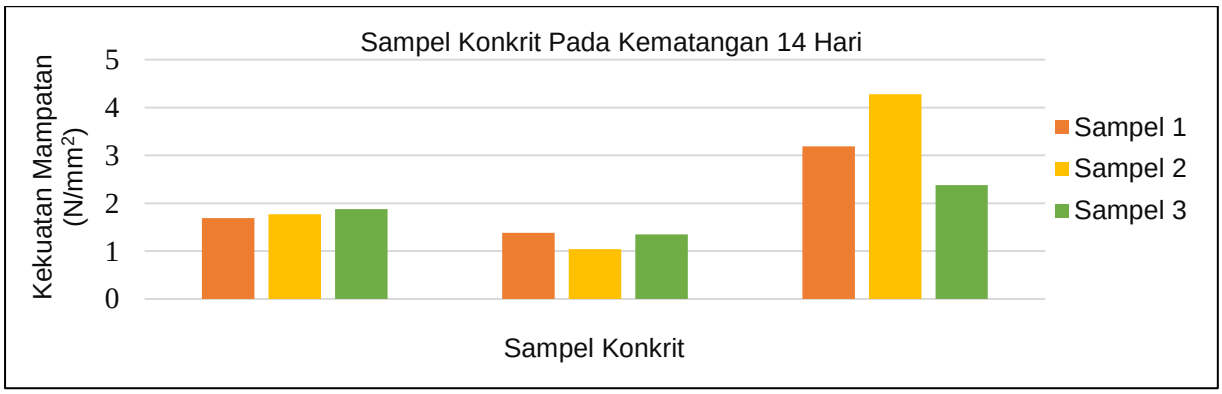
Ujian telah dilakukan ke atas sampel kiub pada tempoh 7 hari, 14 hari dan 28 hari. Dalam ujian ini, beban telah dikenakan pada sampel secara berterusan sehingga mencapai beban maksimum dan kegagalan pada struktur konkrit. Berdasarkan pada ujian yang telah dilakukan, data bagi beban maksimum dan kekuatan telah direkodkan. Formula di bawah menunjukkan pengiraan kekuatan mampatan yang dapat dikira dengan menggunakan beban maksimum(N) bahagi dengan keluasan sampel (mm^2).



Rajah 2. Sampel Konkrit pada Kematangan 7 Hari

$$\text{Kekuatan Mampatan} = \frac{\text{Beban Maksimum (N)}}{\text{Keluasan Sampel } (mm^2)}$$

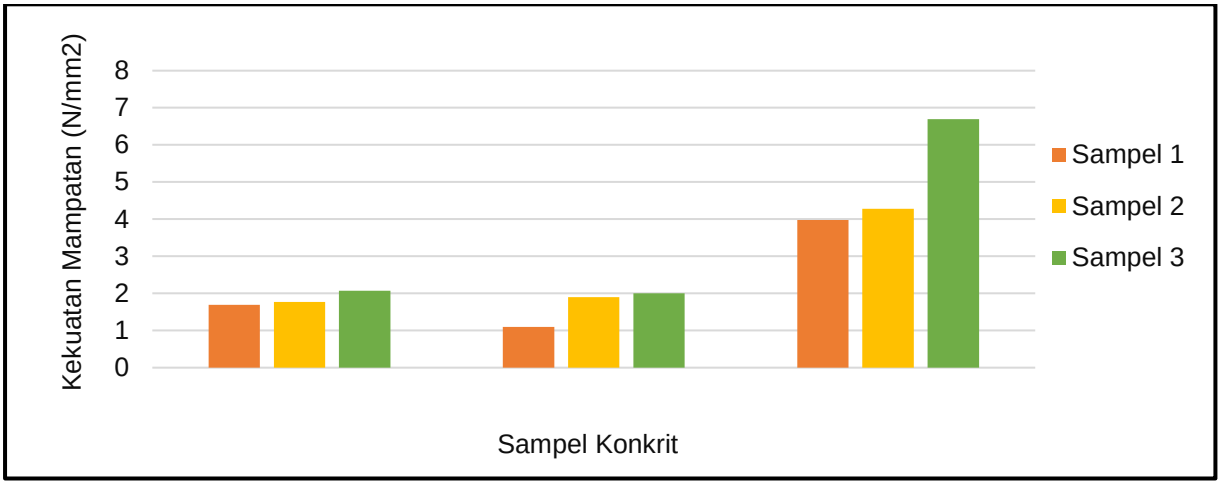
Dalam Rajah 2, telah menunjukkan keputusan bagi Ujian Kekuatan Mampatan pada usia kematangan konkrit pada hari ke-7. Berdasarkan keputusan diatas, purata kekuatan mampatan yang paling tinggi ialah pada nisbah 3:4:4 iaitu 2.27 N/mm^2, diikuti dengan nisbah 1:3:6 bagi purata nilai kekuatan mampatan ialah 0.92 N/mm^2 dan yang paling kecil ialah nisbah 2:3:5 iaitu 0.82 N/mm^2 bagi purata kekuatan mampatan. Oleh itu, nisbah sampel yang mencapai purata nilai kekuatan mampatan yang paling sesuai ialah nisbah 3:3:4. Nisbah 2:3:5 mempunyai purata kekuatan yang paling kecil mungkin disebabkan oleh bancuhan bahan yang tidak sempurna atau berlaku kesilapan semasa menjalankan ujian kekuatan mampatan.



Rajah 3. Graf Sampel Konkrit pada Kematangan 14 Hari

Dalam Rajah 3, menunjukkan keputusan Ujian Kekuatan Mampatan pada usia kematangan konkrit pada hari ke-14. Ketiga-ketiga nisbah mempunyai kekuatan mampatan yang berbeza. Nisbah yang mempunyai purata kekuatan mampatan yang paling tinggi ialah nisbah 3:3:4 iaitu 3.87 N/mm^2. Seterusnya, purata nilai kekuatan mampatan yang sederhana ialah nisbah

1:3:6 iaitu 1.78 N/mm^2 dan yang paling kecil ialah nisbah 2:3:5 iaitu 1.26 N/mm^2. Jadinya, nilai yang didapati adalah berbeza sikit antara satu sama lain.

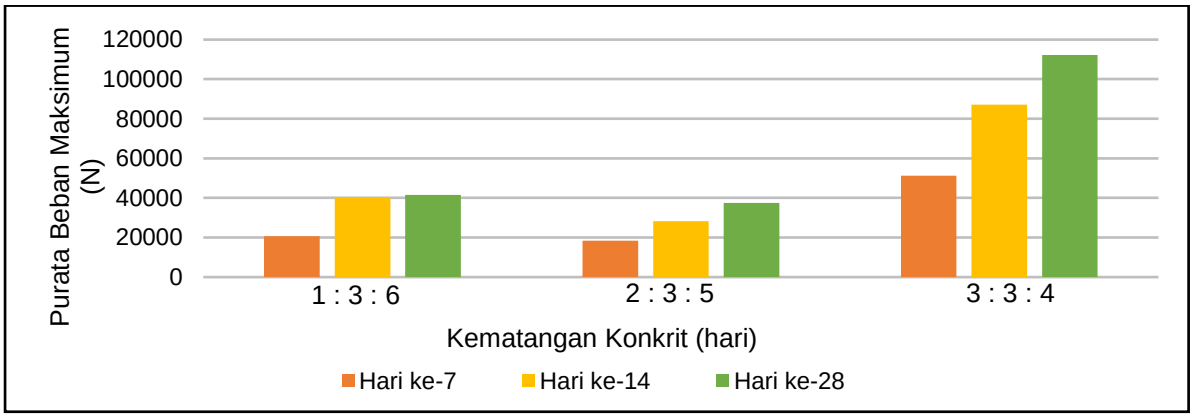


Rajah 3. Graf Sampel Konkrit pada Kematangan 28 Hari

Dalam Rajah 4, menunjukkan keputusan Ujian Kekuatan Mampatan pada usia kematangan konkrit pada hari ke-28. Ketiga-ketiga nisbah mempunyai kekuatan mampatan yang berbeza. Nisbah yang mempunyai purata kekuatan mampatan yang paling tinggi ialah nisbah 3:3:4 iaitu 4.98 N/mm^2. Seterusnya, purata kekuatan mampatan yang sederhana ialah nisbah 1:3:6 iaitu 1.83 N/mm^2 dan yang paling kecil ialah nisbah 2:3:5 iaitu 1.67 N/mm^2. Oleh kerana nisbah 3:3:4 mempunyai purata kekuatan mampatan yang paling tinggi, jadi nisbah ini adalah paling sesuai untuk bancuhan bagi panel dinding yang akan dihasilkan.

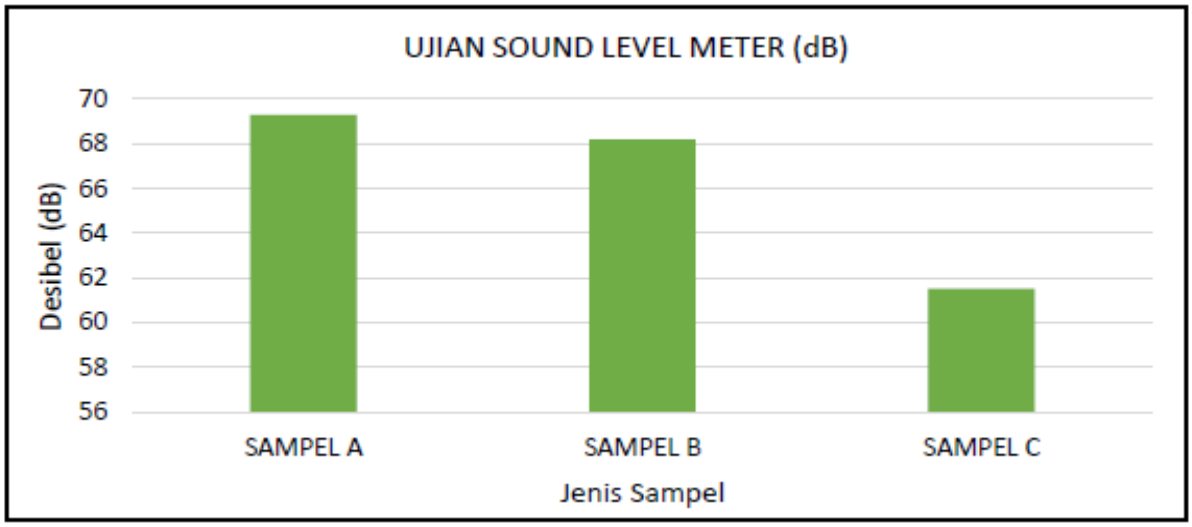
3.1 Ujian Kekuatan Konkrit

Berdasarkan ujian yang telah dilakukan keatas sampel konkrit panel dinding yang mempunyai nisbah bancuhan yang berbeza iaitu 1:3:6, 2:3:5 dan 3:3:4, nilai beban maksimum yang telah tercapai adalah berbeza. Hal ini, menyebabkan nilai kekuatan konkrit itu adalah berbeza mengikut nisbah yang berlainan. Hal ini, dapat diringkaskan bahawa keputusan Ujian Kekuatan Mampatan yang telah direkodkan akan diambil kira untuk nisbah bancuhan yang paling sesuai untuk menghasilkan panel dinding yang sempurna. Nisbah ini mewakili pasir, simen, *mudcake* dan air yang telah digaulkan sekata dan dimasukkan ke dalam kiub untuk di curing sehingga mendapat kematangan konkrit yang telah ditetapkan. Penghasilan panel dinding yang sempurna bergantung kepada keputusan Ujian Kekuatan Mampatan yang paling sesuai.



Rajah 4. Graf Ujian Kekuatan Konkrit

Berdasarkan ujian yang telah dijalankan keatas sampel panel dinding yang mempunyai nisbah bancuhan bahan tambah yang berbeza iaitu 3:3:4, dimana 30% ialah simen, 30% ialah pasir dan 40% ialah *mudcake* dan bahan tambah iaitu serat buluh dimasukkan ke dalam bacuhan iaitu 5% dan 10%. Oleh itu, nilai-nilai daripada ujian *sound level meter* dibandingkan antara sampel-sampel yang telah membuat ujian. Di mana nilai purata bagi semua sampel diambil kira untuk membuat analisis keputusan. Hal ini, keputusan ujian yang diperolehi akan dibanding dan memilih nisbah bancuhan yang paling sesuai untuk memdirikan panel dinding. Proses menganalisis data kajian ditunjukkan dalam bentuk graf , jadual . Penganalisian panel dinding kedap bunyi ini adalah merangkumi kuantiti kekedapan yang terhasil di dalam ruang serat buluh yang diletakkan [7]. Hasil daripada keputusan analisis data yang diperolehi akan dipersembahkan dalam bentuk graf. Dalam kaedah cerapan yang dijalankan , bahan serat buluh yang digunakan ialah dengan berlainan nisbah.



Rajah 5. Graf Ujian Sound Level Meter (dB)

4. Kesimpulan

Secara kesimpulannya, objektif bagi kajian ini telah dicapai seperti dinyatakan di dalam rekabentuk yang mempengaruhi konkrit panel dinding kedap bunyi pada ruang bilik bangunan adalah rekabentuk. Berikutnya untuk penghasilan panel akustik nilai ini telah mencapai kesemua objektif. Hasil daripada data ujian konkrit panel dinding menunjukkan secara keseluruhannya dengan mengunakan nilai nisbah yang ditetapkan ialah nilai yang digunakan 3:3:4 secara keseluruhan pada konkrit panel dinding dan menambahkan lagi bahan campuran seperti serat buluh bahan ini menghasilkan serapan bunyi yang kurang baik, kerana menurut kajian yang perolehi daripada konkrit panel dinding adalah semasa ujian dilakukan, frekuensi pada panel dinding tersebut adalah bunyi masih menebusi daripada dalam dan di luar permukaan. Hasil daripada kajian ini, penggunaan bahan yang tidak mencukupi dan kemungkinan dari segi nilai bancuhan seperti mudcake dan serat buluh yang mempunyai nilai bancuhan yang berbeza dengan nilai bancuhan konkrit yang biasa. Namun begitu, tembusan bunyi yang didapati pada panel dinding frekuensi tersebut yang dikaji itidak menebusi seperti ruang bilik biasa [9][10] yang tiada had kedapan bunyi. Walau bagaimanapun, panel dinding ini tidak terus masuk secara langsung kerana masih ada penebat bunyi masih menghalang bunyi tersebut

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TREN KETERLIBATAN BEKERJA SENDIRI DI KALANGAN GRADUAN POLITEKNIK TUANKU SULTANAH BAHIYAH: ANALISIS LIMA TAHUN

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ABSTRAK

Kajian ini meneroka peranan graduan dalam meneraju pertumbuhan ekonomi Malaysia dalam konteks global yang tidak stabil. Graduan siswazah diharapkan memberikan sumbangan positif melalui pembangunan kemahiran yang relevan dengan pasaran. Walau bagaimanapun, jumlah graduan yang semakin meningkat setiap tahun telah mewujudkan persekitaran pekerjaan yang sangat kompetitif, di mana sekadar bergraduat dari institusi tidak lagi menjamin kedudukan pekerjaan. Kajian ini juga menganalisis trend keterlibatan graduan dalam pekerjaan sendiri sepanjang lima tahun terakhir, menggunakan data dari soal selidik yang dijawab oleh graduan Politeknik Tuanku Sultanah Bahiyah (PTSB) dari tahun 2018 hingga 2022. Dapatan menunjukkan peningkatan signifikan dalam jumlah graduan yang bekerja sendiri, dipengaruhi oleh faktor seperti keinginan untuk fleksibiliti kerja, peluang mengekspresikan kreativiti, dan cabaran global yang mengubah lanskap kerja tradisional. Hasil kajian ini menyediakan wawasan berharga untuk institusi pengajian tinggi, majikan, dan pemegang dasar dalam membentuk landskap kerja masa depan.
Kata Kunci: Bekerja sendiri, kebolehpasaran graduan

1. Pengenalan

Dalam mendepani senario global yang tidak stabil, graduan mempunyai peranan penting dan juga perlu menyesuaikan diri dengan halangan yang mungkin timbul dalam meneraju pertumbuhan ekonomi Malaysia. Dengan adanya graduan siswazah yang diberikan fokus berterusan melalui pembangunan kemahiran yang diperlukan di pasaran, dijangkakan ianya memberikan peranan positif dalam memacu pertumbuhan ekonomi negara sedia ada. Selain itu, graduan ini juga dapat meneraju inovasi dalam banyak sektor industri dan sebagainya. Namun begitu, bergraduat daripada institusi sahaja tidak akan menjamin kedudukan pekerjaan seseorang. Bilangan pelajar bergraduat semakin meningkat tahun ke tahun dan ini mewujudkan persekitaran yang sangat kompetitif [1].

Dalam era permintaan pasaran buruh yang tinggi ini, adalah sukar bagi pelajar untuk mendapat pekerjaan sebaik sahaja bergraduasi. Dan ini akan menyebabkan siswazah akan menganggur. Terdapat beberapa faktor yang menyebabkan ini perkara ini berlaku. Antaranya adalah permintaan dan keperluan majikan yang sukar untuk dipenuhi dan gred akademik yang tinggi serta pelajar perlu dilengkapi dengan keperluan dan kemahiran dalam persekitaran kerja. Calon yang berpontensi juga harus mempunyai kelebihan tambahan seperti kemahiran kebolehkerjaan untuk membolehkan mereka bersaing. Dinyatakan oleh [2] graduan ini perlulah bersemangat untuk belajar dan boleh mengembangkan atau membangunkan sifat sedia ada, teknik dan pengalaman kerja.

Pendapatan campuran untuk golongan yang bekerja sendiri atau menjadi usahawan bebas dijangka mengalami pertumbuhan yang ketara sebanyak 13.2% disebabkan oleh prestasi ekonomi yang kukuh dan penormalan cara kerja fleksibel yang mampu mencipta lebih banyak peluang pekerjaan dan potensi pendapatan bagi kumpulan ini [3]. Ekonomi gig, semakin popular terutama di kalangan generasi muda juga adalah merupakan pekerjaan sendiri, merangkumi segmen pasaran buruh kontrak jangka pendek atau pekerjaan bebas yang

dilakukan oleh individu, yang mana ianya didorong oleh teknologi digital [4]. Walaupun bekerja sendiri bukanlah kerjaya masa hadapan, namun ianya boleh menjadi pilihan sementara ataupun menjadi alternatif bagi mengurangkan pengangguran. Walaupun pekerjaan sendiri bukanlah penyelesaian sepenuhnya, ia boleh meningkatkan dan memberikan manfaat kebajikan sosial dengan bukan sahaja menyumbang kepada pertumbuhan dan inovasi, tetapi juga meningkatkan kesihatan secara keseluruhan [5].

Kesimpulannya, untuk menghadapi situasi semasa, strategi pekerjaan perlu diubah, dan kajian tentang graduan yang bekerja sendiri menjadi sangat penting. Graduan harus bersifat kreatif dalam mencipta pekerjaan atau perniagaan, bukan hanya bergantung kepada pekerjaan tradisional. Pendedahan dan bimbingan awal dalam bidang keusahawanan sebelum tamat pengajian dapat membantu mereka bersedia menghadapi risiko masa depan. Pengetahuan tentang perniagaan akan menjadikan graduan lebih kreatif dan inovatif dalam mencipta peluang pekerjaan. Dengan fokus kepada keusahawanan, graduan tidak hanya mencari pekerjaan, tetapi juga menjadi pencipta peluang pekerjaan. Ini menekankan pentingnya kajian mengenai kerja sendiri dalam memahami dan mendukung trend ini, serta dalam menyediakan strategi pendidikan dan sokongan yang sesuai untuk mempersiapkan graduan menghadapi masa depan yang lebih fleksibel dan berorientasikan keusahawanan.

1.1 Pemasalahan Kajian

Status rakyat Malaysia yang bekerja sendiri mengalami peningkatan yang signifikan dalam beberapa tahun terakhir. Semakin banyak individu, termasuk graduan, memilih untuk menjalankan perniagaan sendiri atau menjadi usahawan bebas. Faktor-faktor seperti keinginan untuk mempunyai fleksibiliti kerja, mencapai kemerdekaan kewangan, dan mengambil alih arah kerjaya mereka sendiri telah mendorong pertumbuhan dalam keterlibatan kerja sendiri di kalangan rakyat Malaysia.

Bagi sesetengah individu, bekerja sendiri memberikan peluang untuk mengejar minat mereka yang unik, mengembangkan kebolehan keusahawanan, dan memanfaatkan peluang-peluang pasaran yang ada. Selain itu, persekitaran ekonomi yang berubah dengan cepat, termasuk kemajuan dalam teknologi digital, juga telah memudahkan individu untuk memulakan perniagaan mereka sendiri dengan kos yang rendah.

Secara keseluruhan, status rakyat Malaysia yang bekerja sendiri mencerminkan perubahan dalam landskap pekerjaan negara dan memberikan sumbangan yang penting kepada pertumbuhan ekonomi serta pembangunan masyarakat secara keseluruhannya. Dengan sokongan yang tepat, keterlibatan kerja sendiri dapat terus menjadi pilihan kerjaya yang menarik dan berpotensi untuk rakyat Malaysia di masa depan.

Graduan di Malaysia menghadapi cabaran besar dalam mencari pekerjaan tetap selepas tamat pengajian, yang mendorong ramai di antara mereka untuk memilih bekerja sendiri atau menjadi usahawan. Walaupun terdapat banyak faktor yang mempengaruhi keputusan ini, seperti keinginan untuk fleksibiliti kerja, kebebasan kewangan, dan peluang untuk mengekspresikan kreativiti, sokongan organisasi dan persekitaran ekonomi juga memainkan peranan penting. Kajian ini bertujuan untuk mengenal pasti dan menganalisis trend bekerja sendiri di kalangan graduan dan faktor-faktor ini mempengaruhi graduan untuk memilih laluan kerjaya bekerja sendiri.

1.2 Objektif

Untuk merealisasikan kajian ini, terdapat dua objektif utama, iaitu:

1. Mengenalpasti trend keterlibatan bekerja sendiri di kalangan graduan bagi sepanjang tahun 2018 sehingga 2022
2. Menganalisa hasil dapatan keterlibatan ini dan mengenalpasti faktor- faktor yang mempengaruhi keterlibatan bekerja sendiri ini.

2. Kajian Literatur

Perdebatan mengenai keboleh pasaran graduan adalah topik yang mendapat perhatian di seluruh dunia [6,7]. Masalah pengangguran merupakan isu yang memerlukan perhatian serius dan menjadi satu cabaran yang signifikan bagi graduan masa kini. Setiap tahun, beribu-ribu graduan dari pelbagai bidang dilahirkan, yang menyebabkan persaingan yang sengit dan memaksa ramai daripada mereka untuk menerima pelbagai pekerjaan demi mengekalkan kehidupan. Menurut [8] dan [1], angka ini meningkat setiap tahun. Permintaan pasaran dan faktor ekonomi sering menyebabkan ketidakseimbangan dalam pasaran tenaga kerja bagi graduan. Graduan dengan kelulusan tinggi mungkin menghadapi krisis di mana mereka terpaksa menganggur untuk tempoh yang lebih panjang [9].

Peningkatan dalam akses kepada pendidikan tinggi telah mengakibatkan pertambahan jumlah graduan yang memasuki pasaran buruh. Walaupun ini dianggap sebagai perkembangan positif yang diharapkan dapat memenuhi permintaan pekerja mahir dalam industri, realitinya adalah bahawa ramai graduan yang berkelayakan menghadapi kesulitan untuk mendapatkan pekerjaan yang sepadan dengan bidang pengajian mereka [10]. Ini menunjukkan adanya ketidaksepadanan antara kelayakan akademik dan peluang pekerjaan yang tersedia [11]. Kadar pengangguran dalam kalangan graduan berumur bawah 25 tahun pada 2017 hingga 2020 adalah empat kali lebih tinggi berbanding kumpulan umur yang lain [12]. Oleh itu, apabila seseorang pelajar gagal bekerja dalam pekerjaan yang tidak sepadan dengan pembelajarannya, ini mengakibatkan kehilangan modal insan tertentu. Atas pemasalahan ini, melibatkan graduan menerima pekerjaan di luar bidang pengajian yang mereka pelajari.

Namun begitu, bilangan penganggur terus mencatatkan penurunan pada Dec 2023 dengan merekodkan 567,30 orang dengan kadar pengangguran berada pada 3.3 peratus dan berlaku trend peningkatan kepada peratusan penduduk berkerja sendiri sebanyak 0.3 peratus kepada 3.01 juta orang daripada 3 juta orang bulan sebelumnya [13]. Diperincikan lagi, kadar pengangguran bagi belia berumur 15 hingga 24 tahun ialah sebanyak 10.6 peratus dengan jumlah seramai 306800 orang penganggur belia. [14] melaporkan bilangan siswazah yang bekerja meningkat sebanyak 6.3 peratus berbanding 2022 kepada 4.87 juta orang (2021: 4.59 juta orang). Daripada peratusan yang berkerja ini, 4.6 peratus (223.5 ribu orang) adalah siswazah bekerja dalam kategori bekerja sendiri. Dan ini dapat dilihat Jadual 1 adalah pecahan siswazah bekerja mengikut taraf pekerjaaan, Malaysia,2021-2022. Pemilihan bekerja sendiri memberi kesan kepada pengangguran, hubungan songsang dikenal pasti iaitu semakin meningkatnya bekerja sendiri, pengangguran akan berkurang [15].

Jadual 1. Siswazah bekerja mengikut taraf pekerjaan, Malaysia, 2021-2022

Taraf Pekerjaan	Majikan		Pekerja		Bekerja Sendiri		Pekerja keluarga tanpa gaji	
Tahun	Ribu Orang	%	Ribu Orang	%	Ribu Orang	%	Ribu Orang	%
2022	228.5	4.7	4368.2	89.7	223.5	4.6	47.2	1.0
2021	199.9	4.4	4145.9	90.3	194.9	4.2	49.6	1.1

Sumber: (Jabatan Perangkaan Malaysia., 2023)

Sejak kebelakangan ini, semakin ramai graduan baru berubah arah kepada pekerjaan di luar pasaran pekerjaan standard. Pekerjaan seperti kontrak, separuh masa, dan bekerja sendiri semakin popular, namun mereka sering menghadapi cabaran sebaliknya. Dinyatakan oleh [16] kini ramai golongan muda lebih terdedah untuk bekerja sendiri berikutan mempunyai pilihan untuk menetapkan jadual, dan bebas untuk membuat sebarang keputusan berbanding bekerja makan gaji. Bekerja sendiri boleh ditakrifkan sebagai status bekerja untuk diri sendiri melalui memiliki dan menjalankan perniagaan sendiri dan bukannya berkhidmat kepada orang lain/organisasi untuk mendapat upah atau gaji. Bagi graduan baru, bekerja sendiri adalah dianggap sebagai pilihan sementara sebelum mereka melangkah ke arah pekerjaan standard yang lebih konvensional.

Diantara contoh aktiviti perniagaan individu yang bekerja sendiri adalah perniagaan runcit, jualan langsung, pertanian dan perladangan, gerai, klinik, firma guaman atau pengamal profesional lain, penulis dan pelakon, remisier, dan pengendali penghantaran dalam talian / e-hailing [17]. Manakala [18] pula menyatakan bekerja akaun sendiri boleh didapati dalam pelbagai jenis peranan, mulai daripada pekerjaan bersendirian tradisional; seperti menjadi petani, nelayan, penoreh getah atau tukang tempatan, hingga menjadi pemandu teksi, sub-kontraktor yang bekerja dalam perdagangan atau sektor pembinaan dan profesional berkemahiran tinggi seperti pekerja penjagaan kesihatan (doktor, doktor gigi, ahli fisioterapi, dan sebagainya) dan penyedia perkhidmatan (peguam, akauntan, jurutera bebas dan perunding pengurusan dan lain-lain). Nyatanya lagi, ini termasuk peniaga harian yang bermain pasaran saham dari rumah dan pekerja bebas yang bekerja dipanggil "ekonomi gig" (seperti pemandu Grab atau Uber). Walau bagaimanapun, pekerja gig sering menghadapi cabaran yang berkaitan dengan kestabilan dan keselamatan pekerjaan [19]. Pertumbuhan itu mungkin berlaku disebabkan oleh sektor perkhidmatan yang berkembang, peningkatan ekonomi gig, peluang untuk "menjadi bos anda sendiri," dan sikap komuniti yang lebih positif terhadap keusahawanan [20]. Pada Bajet 2022, kerajaan Malaysia memperuntukkan RM50 juta untuk meningkatkan ekonomi gig. Pembiayaan ini akan digunakan untuk menyediakan program latihan dan peningkatan kemahiran untuk pekerja gig, serta untuk membangunkan rangka kerja pengawalseliaan yang menyeluruh untuk ekonomi gig [21].

Indikasi dari hasil kajian oleh [22] menunjukkan bahawa graduan semakin mempertimbangkan bekerja sendiri sebagai usahawan dan menjadikannya pilihan kerjaya, dengan menjadikan keusahawanan sebagai matlamat dan membina semangat yang tinggi untuk menyertai bidang ini. Selain daripada itu graduan ini mempunyai kawalan peribadi, harga diri, dan inovasi iaitu perkara yang signifikan dan positif dengan niat untuk bekerja sendiri. Hasil kajian oleh [23] menunjukkan bahawa responden mempunyai minat yang positif untuk bekerja sendiri dengan tahap yang sangat signifikan melalui dapatan kajian di mana kebanyakan bakal graduan berminat untuk menjadi usahawan bebas. Tambahan pula, kerajaan Malaysia terus menyediakan bantuan bersasar bagi yang bekerja sendiri melalui pelaksanaan inisiatif peningkatan kemahiran dan latihan semula selari dengan keutamaan nasional seperti ekonomi digital, Pendidikan dan Latihan Teknikal dan Vokasional (TVET) dan Revolusi Perindustrian Keempat (4IR) [22].

3. Metodologi Kajian

Kajian ini melibatkan penggunaan kaedah kuntitatif dengan menggunakan soal selidik instrument kajian yang dibina oleh pihak kementerian untuk menganalisis tahap trend sektor pekerjaan terutamanya. Data dikumpul melalui soal selidik yang diselesaikan oleh graduan Politeknik Tuanku Sultanah Bahiyah (PTSB) dari tahun 2018 hingga 2022. Proses pengumpulan data dilaksanakan melalui borang soal selidik secara atas talian melalui system kajian pengesanan graduan (SKPG) TVET kebangsaan. Data graduan ini melibatkan 5043 graduan yang telah memberi respon (menjawab soal selidik) daripada 5469 keseluruhan graduan yang sepatutnya terlibat pada tahun tersebut. Dapatan kajian merangkumi soalan-soalan yang telah difokuskan jawapannya, dianalisis menggunakan perisian IBM Statistical

Package for Social Science (SPSS) Versi 23. Maklumat dapatan data yang diperolehi ditunjukkan dalam bentuk deskriptif (peratus).

Dalam kajian ini, pengkaji menggunakan instrument penyelidikan yang berbentuk soal selidik yang disediakan oleh pihak kementerian. Soal selidik ini terdiri daripada enam bahagian, iaitu Bahagian A hingga Bahagian F. Bahagian A mengumpul maklumat tentang latar belakang graduan, sementara Bahagian B mengandungi maklumat terkini tentang status pekerjaan graduan. Bahagian C menanyakan tentang tahap pekerjaan graduan, sektor pekerjaan, dan pendapatan mereka. Sementara itu, Bahagian D mengumpul maklumat tentang sebab-sebab mengapa graduan belum atau tidak bekerja, manakala Bahagian E berkaitan dengan tawaran lanjutan pengajian. Bahagian F pula mengumpulkan maklumat tentang pengalaman graduan semasa mengikuti pengajian di institusi. Semua data dari soal selidik yang berjaya diisi dianalisis menggunakan perisian Statistical Package for Social Science (SPSS).

4. Keputusan

4.1 Maklumat demografi responden

Sebanyak 5,469 orang graduan PTSB telah menamatkan pengajian mereka dalam bidang Diploma dari tahun 2018 hingga 2022, dan telah memasuki pasaran kerja. Kesemua graduan ini melibatkan 12 program pengajian yang ada di tawarkan pengajian di PTSB. Daripada jumlah tersebut, sebanyak 5043 graduan telah memberikan respons kepada soal selidik ini. Jadual 2 ini merupakan jadual yang memperlihatkan analisis taburan maklumat demografi responden dalam kajian ini. Majoriti responden terdiri merupakan perempuan dengan 52.7% (n=2653) manakala responden lelaki adalah sebanyak 47.3% (n=2384). Daripada keseluruhan responden sebagaimana Jadual 2, didapati kebolehpasaran graduan (GE) PTSB berada pada tahap yang baik dan boleh dilihat seperti di Jadual 3.

Jadual 2. Taburan responden mengikut jantina tahun 2018 hingga 2022 (n=5037)

Jantina	Tahun Konvokesyen											
	2018		2019		2020		2021		2022		Jumlah	
	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%
Lelaki	450	47.4	416	43.4	479	48	505	46.5	534	51.1	2,384	47.3
Perempuan	500	52.6	543	56.6	519	52	581	53.5	510	48.9	2,653	52.7

4.2 GE PTSB Tahun 2018-2022

Sepanjang tempoh 5 tahun tersebut, didapati berlaku penurunan GE berlaku daripada tahun 2018 kepada tahun 2019 sebanyak 1.1%, iaitu daripada 95.5% kepada 94.4%. Penurunan ini turut berlaku sehingga ke tahun 2021, di mana dapat di lihat bahawa penurunan bermula pada 4.6% daripada tahun 2019(94.4%) kepada tahun 2020(89.8%). Peratusan pada tahun 2020 pula menurun sebanyak 2% kepada tahun 2021(87.8%). Pada tahun 2020 dan 2021 ini, dijangkakan penurunan ini berlaku adalah disebabkan pandemik covid-19. Pandemik ini telah merosakkan ekonomi Malaysia dengan menyebabkan penurunan nilai mata wang negara serta pengurangan dalam keluaran dalam negara kasar (KDNK), sementara juga mencetuskan kesan sosial yang signifikan dalam masyarakat Malaysia. Namun begitu, GE kembali kepada tahap yang lebih positif pada tahun 2022, dan telah dicatatkan peningkatan sebanyak 7.6% menjadikan GE pada tahun tersebut menjadi 95.4%. Secara keseluruhannya, peratusan GE selama 5 tahun adalah 92.5%. Peratusan ini telah mencapai KPI yang telah ditetapkan oleh pihak kementerian.

Jadual 3. GE PTSB bermula tahun 2018 hingga 2022 (n=4657)

Kebolehpasaran Graduan (GE)	Tahun Konvokesyen											
	2018		2019		2020		2021		2022		Jumlah	
	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%
Kebolehpasaran Graduan (GE)	907	95.5	905	94.4	896	89.8	953	87.8	996	95.4	4,657	92.5

4.3 Analisa graduan bekerja sendiri

GE PTSB diambil kira kepada 4 status pekerjaan iaitu bekerja, melanjutkan pengajian, meningkatkan kemahiran, dan menunggu penempatan pekerjaan. Peratusan ini dapat dilihat sebagaimana tercatat pada Jadual 4, iaitu peratusan ini dikelaskan mengikut tahun dan status masing-masing. Dapatan data ini menunjukkan bahawa 3661 responden adalah dalam kategori bekerja mewakili 72.7% daripada keseluruhan responden. 946 responden merupakan peratusan kedua tertinggi iaitu 18.8 yang mewakili responden yang melanjutkan pengajian. Manakala meningkatkan kemahiran dan menunggu penempatan mencapai 0.2% dan 0.8% masing-masing yang mewakili 9 dan 41 responden. Data responden daripada status pekerjaan yang bekerja daripada tahun 2018 sehingga tahun 2022 yang akan terlibat untuk kajian ini. Namun begitu dapat dilihat daripada data tersebut, didapati bahawa berlaku peningkatan peratusan responden yang bekerja saban tahun dan ini menjelaskan satu perkembangan yang positif terhadap GE PTSB.

Jadual 4. Status pekerjaan graduan tahun 2018 hingga 2022 (n=5037)

Status Pekerjaan	Tahun Konvokesyen											
	2018		2019		2020		2021		2022		Jumlah	
	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%
Bekerja	719	75.7	723	75.4	628	62.9	787	72.5	804	77	3,661	72.7
Melanjutkan pengajian	179	18.8	176	18.4	258	25.9	145	13.4	188	18	946	18.8
Meningkatkan kemahiran	1	0.1	1	0.1	1	0.1	5	0.5	1	0.1	9	0.2
Menunggu penempatan pekerjaan	8	0.8	5	0.5	9	0.9	16	1.5	3	0.3	41	0.8
Belum bekerja	43	4.5	54	5.6	102	10.2	133	12.2	48	4.6	380	7.5

Setelah mana data responden yang bekerja ini diperolehi, dilakukan pula tapisan data untuk mendapatkan data yang berkerja sendiri. Data responden yang berkeja sendiri ini dijadualkan seperti Jadual 5. Daripada Jadual 5 ini dapat dilihat bahawa untuk data berkerja sendiri ini di kategorikan atau dikelaskan kepada dua taraf pekerjaan, iaitu bekerja sendiri/ freelance dan juga menjadi majikan. Pada tahun 2018 responden yang bekerja sendiri adalah sebanyak 41 responden dan bekerja sendiri/freelace adalah 31(75.6%) berbanding berkerja sendiri sebagai majikan 10 (24.4%). Tren ini didapati meningkat pada tahun 2019 daripada 41 kepada 64 dimana 62.5% bagi bekerja sendiri/freelance dan 37.5% adalah sebagai majikan. Peratusan ini menurun kepada 56 responden pada tahun 2020 dan berlaku peningkatan semula pada tahun 2021 kepada 66 responden. Pada tahun 2020 ini, mencatatkan rekod 35 (62.5%) dan 21(37.5%) masing-masing untuk bekerja sendiri/freelace dan majikan. Manakala pecahan untuk 2021 pula adalah 50(75.8%) bagi bekerja sendiri/freelance dan 16 (24.2%) bagi majikan. Namun begitu, berlaku penurunan pada tahun 2022 sebanyak 11 orang. Penurunan

ini menjadikan jumlah bagi tahun 2022, 33 (60%) bekerja sendiri dan menjadi majikan adalah sebanyak 22 (40%).

Jadual 5. Taraf pekerjaan responden yang bekerja sendiri tahun 2018 hingga 2022

Tahun Konvokesyen												
	2018		2019		2020		2021		2022		Jumlah	
Bekerja: Taraf Pekerjaan	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%
Bekerja sendiri/Freelance	31	75.6	40	62.5	35	62.5	50	75.8	33	60	189	67
Majikan	10	24.4	24	37.5	21	37.5	16	24.2	22	40	93	33
Jumlah	41	100	64	100	56	100	66	100	55	100	282	100

Jadual 6. Responden bekerja sendiri mengikut program pengajian

Bekerja sendiri												
Tahun Konvokesyen												
Program Pengajian (Diploma)	2018		2019		2020		2021		2022		Jumlah	
	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%	Bil.	%
Akauntansi	4	9.76	7	10.77	11	20.00	5	7.58	3	5.45	31	10.99
Geomatik	3	7.32	7	10.77	5	9.09	1	1.52	6	10.91	21	7.45
Kejuruteraan Awam	10	24.39	7	10.77	13	23.64	10	15.15	9	16.36	49	17.38
Kejuruteraan Elektrik	1	2.44	6	9.23	6	10.91	7	10.61	5	9.09	25	8.87
Kejuruteraan Elektrik Dan Elektronik	1	2.44	1	1.54	5	9.09	2	3.03	3	5.45	12	4.26
Kejuruteraan Elektronik (Komputer)	3	7.32	4	6.15	2	3.64	3	4.55	3	5.45	15	5.32
Kejuruteraan Elektronik (Komunikasi)	2	4.88	2	3.08		0.00	2	3.03		0.00	6	2.13
Kejuruteraan Mekanikal	4	9.76	3	4.62	4	7.27	7	10.61	8	14.55	26	9.22
Kejuruteraan Mekanikal (Pembuatan)		0.00	2	3.08		0.00	2	3.03	1	1.82	5	1.77
Kejuruteraan Mekatronik	2	4.88	5	7.69	3	5.45	5	7.58	5	9.09	20	7.09
Pemasaran	4	9.76	10	15.38	3	5.45	9	13.64	5	9.09	31	10.99
Pengajian Perniagaan	6	14.63	10	15.38	4	7.27	13	19.70	7	12.73	40	14.18
Ukur Tanah	1	2.44		0.00		0.00		0.00		0.00	1	0.35
Jumlah	41	100.0	64	100	56	100	66	100	55	100	282	100

Keseluruhan data bagi pengesanan graduan bekerja sendiri ini juga dianalisa mengikut kepada 12 program (Diploma) yang terlibat/ditawarkan di PTSB. Bagi seseluruhan tahun 2018, dapat dilihat daripada 41 graduan, didapati bahawa Program Pengajian Kejuruteraan Awam adalah merupakan program tertinggi yang bekerja sendiri mewakili 24.39% (10). Manakala bagi tahun 2019, terdapat 2 Program Pengajian iaitu Program Pengajian Pemasaran dan Pengajian Perniagaan yang masing-masingnya sebanyak 10 (15.38%) orang graduan bekerja sendiri. Ketika tahun 2020, Program Pengajian Kejuruteraan Awam kembali mengungguli bilangan tertinggi, iaitu sebanyak 23.64% mewakili 13 graduan, daripada 56 graduan yang bekerja sendiri bagi tahun tersebut. Begitu juga pada tahun 2022, di mana

program ini turut mempunyai peratusan bekerja sendiri tertinggi, iaitu 9 (16.36%). Untuk tahun 2021 pula, peratusan tertinggi adalah 19.70% mewakili 13 graduan daripada 66 graduan keseluruhan bekerja sendiri di perolehi oleh Program Pengajian Perniagaan.

Dan daripada keseluruhan 5 tahun yang mewakili 282 graduan yang bekerja sendiri, dapat dilihat bahawa Program Pengajian Kejuruteraan Awam (17.38%) iaitu seramai 49 graduan adalah merupakan program yang teramai graduannya telah memilih untuk bekerja sendiri. Manakala Program Pengajian Ukur Tanah adalah merupakan bilangan graduan yang terendah sekali dalam memilih untuk bekerja sendiri sepanjang 5 tahun bermula 2018 sehingga 2022.

5. Perbincangan

Analisis data menunjukkan peningkatan signifikan dalam jumlah graduan yang terlibat dalam kerja sendiri sepanjang lima tahun yang lalu. Kajian ini meneroka trend keterlibatan kerja sendiri di kalangan graduan selama lima tahun terakhir. Dengan peningkatan keinginan untuk mencipta peluang kerja sendiri, semakin banyak graduan yang memilih untuk meneroka jalan kemandirian dalam kerjaya mereka. Kajian ini mengumpul data dari sumber soal selidik SKPG untuk mengenal pasti tren dan faktor yang mempengaruhi keterlibatan kerja sendiri di kalangan graduan. Dapat dilihat faktor-faktor seperti keinginan untuk mempunyai fleksibiliti kerja, peluang untuk mengekspresikan kreativiti, dan cabaran global yang mengubah landskap kerja tradisional telah menjadi pendorong utama di sebalik keputusan graduan untuk memilih kerja sendiri. Selain itu, faktor-faktor seperti sokongan organisasi, dan kepuasan kerja juga memainkan peranan penting dalam menentukan kejayaan keterlibatan kerja sendiri di kalangan graduan.

Faktor-faktor yang mempengaruhi keterlibatan kerja sendiri di kalangan graduan, boleh dilihat daripada beberapa kajian dan laporan yang telah dibuat oleh pelbagai organisasi dan penyelidik dalam perkara berkaitan ini. Daripada faktor fleksibiliti kerja, banyak kajian menunjukkan bahawa ianya adalah salah satu pendorong utama untuk bekerja sendiri. Fleksibiliti ini membolehkan individu menyeimbangkan kehidupan kerja dan peribadi dengan lebih baik. Kajian oleh [25] yang menyatakan bahawa secara umumnya pekerja bukan sahaja lebih bermotivasi, malah mereka lebih produktif dan komited terhadap kerjanya. Perkara ini turut disokong oleh [26] menekankan bahawa fleksibiliti kerja tidak hanya meningkatkan keseimbangan kerja-kehidupan tetapi juga mengurangkan tekanan dan meningkatkan motivasi di tempat kerja. Fleksibiliti ini membuatkan ramai pekerja mempertimbangkan kerja sendiri sebagai alternatif yang lebih baik.

Dengan bekerja sendiri membolehkan individu menggunakan kreativiti mereka dengan cara yang tidak mungkin dalam pekerjaan tradisional. Kajian oleh [27] menyimpulkan individu dengan keyakinan yang tinggi terhadap kemampuan kreatif mereka cenderung memilih pekerjaan sendiri. Dengan lebih banyak kebebasan untuk merancang dan menyesuaikan pekerjaan mereka, individu dalam pekerjaan sendiri dapat mencapai tahap kreativiti yang lebih tinggi [28]. Selain itu , graduan mula beralih kepada pekerjaan sendiri sebagai cara untuk menyesuaikan diri dengan keadaan ekonomi yang tidak menentu. Tambahan lagi adalah ketika mana berlaku pandemic COVID-19 telah mempercepatkan peralihan kepada bekerja sendiri dan inilah yang menjadikan cabaran global yang mengubah lanskap kerja tradisional. Dengan banyak pekerjaan tradisional yang terjejas, bekerja sendiri muncul sebagai pilihan utama bagi ramai individu yang mencari fleksibiliti dan kestabilan ekonomi [29]. Manakala, teknologi memungkinkan individu menawarkan perkhidmatan secara bebas tanpa perlu bergantung kepada majikan tradisional [30].

Sokongan dari organisasi seperti inkubator perniagaan, program mentor, dan jaringan usahawan dapat membantu graduan dalam memulakan kerja sendiri. Bantuan dari organisasi membantu mengatasi cabaran dan meningkatkan kebarangkalian kejayaan perusahaan kecil

yang bekerja sendiri ini. Dan ini adalah merupakan faktor yang menyumbang kepada salah satu faktor pencapaian graduan bekerja sendiri ini. Sebagaimana yang dijalankan oleh [31] yang menggerakkan Pelan Strategik Program Peningkatan Kemahiran yang memberi manfaat kepada graduan termasuk perancangan kerjaya di peringkat tempatan atau global, pengiktirafan dan pemanfaatan optimum kemahiran.

Dapat dilihat daripada trend yang berlaku sepanjang 5 tahun bermula tahun 2018 sehingga tahun 2022 kepada graduan PTSB ini, pemilihan bekerja sendiri dipengaruhi juga oleh faktor kepuasan kerja. Telah diperakui dalam beberapa kajian bahawa faktor-faktor yang berkaitan dengan pekerjaan seperti gaji, jam kerja, peluang kenaikan pangkat, dan keselamatan kerja memiliki pengaruh terhadap kepuasan kerja [32,33]. Selain itu, sama ada pekerjaan tersebut melibatkan kebebasan yang terhad, peluang untuk mempelajari kemahiran baru, serta memberikan ruang kepada individu untuk mencapai pencapaian diri dan mendapatkan kepuasan dari hasil kerja mereka juga turut memainkan peranan penting [34]. Kepuasan kerja merujuk kepada perasaan memuaskan yang diperoleh seseorang apabila menyelesaikan atau menjalankan tugasan yang dinikmati, dilaksanakan dengan baik, dan diberi ganjaran yang memadai. Selain itu, ia juga melibatkan perasaan kegembiraan dan kepuasan terhadap pekerjaan yang dilakukan [35].

Satu kajian yang dilakukan oleh [36] mendapati bahawa individu yang bekerja sendiri cenderung lebih menghargai cabaran yang dihadapi semasa bekerja dan memiliki lebih banyak perasaan sayang terhadap kerjaya mereka berbanding dengan mereka yang bekerja di bawah majikan dalam organisasi. Ini menunjukkan tahap kepuasan kerja yang lebih tinggi di kalangan individu ini dan menemukan bahawa mereka adalah yang paling gembira. Menurut [37] dalam penulisannya menyatakan kajian yang dilakukan oleh LinkedIn dan dilaporkan dalam HR Asia, bilangan 40 tentang faktor-faktor yang membuatkan pekerja gembira di tempat kerja, hasil maklum balas menunjukkan bahawa faktor utama adalah pekerjaan itu sendiri dan peluang untuk belajar dan meningkatkan potensi diri.

Namun begitu, graduan ini perlu bersedia daripada segi persediaan kemahiran kewangan. Ini adalah kerana kemahiran kewangan yang kukuh dalam kalangan graduan yang bekerja sendiri dianggap penting. Oleh itu, kesedaran kewangan perlu dipupuk sejak awal bagi membantu graduan menguruskan perniagaan mereka sendiri dengan lebih cekap, seperti yang dicadangkan oleh beberapa kajian [38,39]. Kajian yang dijalankan menunjukkan keperluan dan kepentingan untuk menyiasat celik kewangan dalam kalangan belia yang bekerja sendiri yang dianggap sebagai konteks baru dalam kajian celik kewangan Malaysia [40]. Di Malaysia, Bank Negara Malaysia (BNM) telah menubuhkan Agensi Kaunseling dan Pengurusan Kewangan, biasanya dikenali sebagai AKPK, pada tahun 2006 (Agensi Kaunseling dan Pengurusan Kewangan [41]. Misi AKPK adalah untuk memperkasakan komuniti (isi rumah dan perusahaan kecil dan sederhana) untuk berdaya tahan kewangan melalui pendidikan kewangan, khidmat nasihat dan pengurusan hutang yang mengutamakan pelanggan. Sejajar dengan ini, adalah baik bagi graduan berkerja sendiri dilengkapkan dengan keupayaan asas mengurus wang dengan bijak dan mencapai matlamat kewangan yang baik. Kemahiran kewangan yang mantap membolehkan mereka mengendalikan aktiviti ekonomi dengan berkesan, serta memberi sumbangan kepada pembangunan negara dan ekonomi Malaysia.

6. Kesimpulan Dan Cadangan

Kesimpulannya, kajian ini memberikan pandangan yang mendalam tentang evolusi keterlibatan kerja sendiri di kalangan graduan dan menyediakan wawasan yang berharga kepada pelbagai pihak termasuk institusi pengajian tinggi, majikan, dan pemegang dasar dalam membentuk landskap kerja masa depan. Data yang dianalisis menunjukkan perkembangan positif dalam keterlibatan graduan dalam pekerjaan sendiri dari tahun 2018 hingga 2022 di PTSB. Terdapat peningkatan dalam jumlah graduan yang memilih untuk

bekerja sendiri atau sebagai freelancer, yang mencerminkan minat dan kecenderungan untuk mengambil alih kendali peribadi dalam kerjaya mereka. Program-program daripada Diploma Kejuruteraan Awam dan Diploma Perniagaan menonjol sebagai pilihan utama di kalangan graduan yang memilih bekerja sendiri sebagai pilihan. Namun begitu, terdapat variasi dalam trend dari tahun ke tahun, dengan beberapa program mengalami peningkatan dan penurunan dalam jumlah graduan yang memilih bekerja sendiri. Analisis ini memberikan pandangan yang mendalam tentang corak pemilihan kerjaya graduan dan implikasinya terhadap landskap kerja dalam pendidikan tinggi. Namun begitu, persediaan kemahiran kewangan perlu diperhalusi oleh para graduan yang bekerja sendiri ini untuk membolehkan mereka merancang perbelanjaan dengan baik. Mengetahui cara menguruskan pendapatan, mengawal belanja, dan merancang bagi perbelanjaan peribadi dan perniagaan adalah penting untuk kestabilan kewangan jangka panjang. Memahami tren kerja sendiri di kalangan graduan dapat memberikan pandangan yang lebih jelas tentang arah pasaran kerja masa depan. Ini penting untuk membantu institusi pendidikan dalam merancang kurikulum yang relevan dan menyediakan graduan dengan kemahiran yang diperlukan dalam ekonomi yang semakin berubah. Dengan pemahaman yang lebih baik tentang faktor-faktor yang mempengaruhi keterlibatan kerja sendiri, langkah-langkah boleh diambil untuk menyokong perkembangan positif dalam kemandirian kerja graduan. Seterusnya dapat menjadi pemangkin kepada ekonomi dan pembangunan negara. Graduan yang berjaya dalam bekerja sendiri boleh menjadi pemacu pertumbuhan ekonomi, terutamanya dalam mencipta pekerjaan baru dan memperkenalkan inovasi dalam pelbagai sektor. Kajian ini menekankan kepentingan menyokong perkembangan ini untuk manfaat ekonomi negara. Adalah dicadangkan kajian lanjut boleh melihat kepada kesan jangka panjang terhadap kerjaya graduan yang memilih untuk bekerja sendiri, termasuk kebolehan mereka untuk bertahan dan berkembang dalam persekitaran ekonomi yang berubah. Serta dicadangkan juga membuat kajian perbandingan antara sektor ekonomi yang berbeza untuk melihat bagaimana faktor-faktor seperti sokongan organisasi, akses kepada teknologi, dan perubahan dalam pasaran mempengaruhi keputusan graduan untuk bekerja sendiri.

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KAJIAN AWAL TERHADAP KEMAHIRAN GENERIK DI DALAM PEMBELAJARAN DAN PENGAJARAN

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ABSTRAK

Pengetahuan dan kemahiran dipindahkan melalui proses pembelajaran dan pengajaran (PdP). Ini akan meningkat tahap kefahaman pelajar dan mungkin memperolehi keputusan yang baik. Di samping itu PdP turut meningkat kemahiran generik seperti kerja berpasukan, komunikasi serta pemikiran kritis melalui ativiti di dalam bilik kuliah seperti tugasan dalam kumpulan, ujian, pembentangan tugasan dan lain-lain. Objektif kajian adalah untuk mengenalpasti aspek kemahiran generik melalui proses PdP yang dijalankan di Politeknik Tuanku Sultanah Bahiyah. Hasil dapatan menunjukkan bahawa kemahiran generik yang diperolehi melalui PdP adalaj tinggi seperti komunikasi, penyelesaian masalah, kerja berpasukan dan pengurusan masa. Dapatan turut menunjukan bahawa kemahiran generik seperti komunikasi mempunyai hubungan yang positif dan signifikan dengan kemahiran penyelesaian masalah, kerja berpasukan dan pengurusan masa bagi tugasan yang diberikan dalam PdP.

Kata Kunci: PdP, komuniksi, penyelesaian masalah, kerja berpasukan, pengurusan masa

1. Pengenalan

Pembelajaran dan pengajaran yang baik dikaitkan dengan tahap kefahaman pelajar berkaitan dengan subjek yang diajar. Jika pelajar faham maka kemanpuan untuk menjawab peperiksaan adalah baik. Maka dengan ini PdP memainkan peranan yang penting dalam menyediakan pelajar yang mempunyai pengetahuan yang baik. Pensyarah akan memastikan PdP berjalan dengan lancar untuk memastikan pelajar akan memperolehi keputusan yang baik. Implikasi daripada PdP yang baik itu turut meningkat kemahiran generik yang diperlukan oleh majikan.

Menurut [1] perkembangan dan kemajuan dalam pendidikan tanah air mendesak guru menguasai pelbagai kemahiran generik, seperti, kemahiran berfikir, kemahiran mengguna dan mengintegrasikan teknologi maklumat dan komunikasi dalam pengajaran dan pembelajaran, kemahiran belajar, dan kebolehan menguasai bahasa kedua dengan baik .

Kemahiran generik amat diperlukan samada melancarkan proses PdP dan alam pekerjaan pada masa depan. Kemahiran yang mungkin terbentuk daripada PdP berkait dengan aktiviti di dalam kuliah seperti kemahiran komunikasi melalui aktiviti pembentangan tugasan, peningkatan kemahiran komunikasi ini memudahkan pelajar untuk berkongsi jawapan atau memberi tindak balas terhadap soalan yang ditanya oleh pensyarah. Ini menunjukkan bahawa PdP meningkat kemahiran teknikal dan generik.

Tanpa penglibatan pelajar di dalam PdP memberi kesan terhadap kemahiran generik. PdP adalah proses pemindahan pengetahuan serta kemahiran dalam meningkat kemahiran teknikal dalam bidang yang diambil. Kemahiran generik amat penting tertamanya di dalam pekerjaan. Majikan berminat mengambil pekerja yang mempunyai kemahiran generik yang

baik. PdP bukan sahaja meningkat pengetahuan dan kemahiran tetapi turut meningkat kemahiran generik dengan penglibatan pelajar dalam PdP. Menurut [2] mendapati penerapan kemahiran komunikasi, kemahiran kerja dalam kumpulan dan kemahiran kepimpinan dalam kokurikulum berada pada tahap sederhana.

Objektif kajian adalah untuk mengenalpasti aspek kemahiran generik terhasil daripada proses pembelajaran dan pengajaran di dalam bilik kuliah. Sampel kajian adalah terdiri daripada pelajar-pelajar jabatan kejuruteraan dan perdagangan Politeknik Tuanku Sultanah Bahiyah. Hasil dapatan menunjukkan bahawa kemahiran gemerik wujud dalam proses PdP yang membantu pelajar dalam memasuki alam pekerjaan. Seterusnya Dapatan turut menunjukkan bahawa PdP membentuk kemahiran generik seperti komunikasi, penyelesaian masalah, kerja berpasukan dan pengurusan masa.

2. Soroton Kajian

Kemahiran generik seringkali dikaitkan dengan aktiviti luar bilik kuliah seperti bersukan, beruniform serta berpersatuan. Penekanan yang kurang dikaitkan dengan kemahiran generik dengan PdP yang berlaku didalam bilik kuliah. PdP yang berkesan akan meningkat tahap kemahiran generik dalam kalangan pelajar. Rajah 1 menunjukkan hubungan antara pembelajaran dan pengajaran terhadap kemahiran generik.



Rajah 1. Kerangka kerja konseptual

Aktiviti-aktiviti di dalam bilik kuliah yang direka bentuk bertujuan terlibat di dalam PdP untuk meningkat kefahaman. Latihan kendiri, tugasan di dalam kumpulan, pembentangan, tutorial, dan seminar adalah untuk meningkat kefahaman dan prestasi akademik. Kefahaman kurang berkesan hanya melalui penyampaian satu hala sahaja. Penglibatan pelajar dalam aktiviti memberi pengalaman serta dapat mengingati dengan baik.

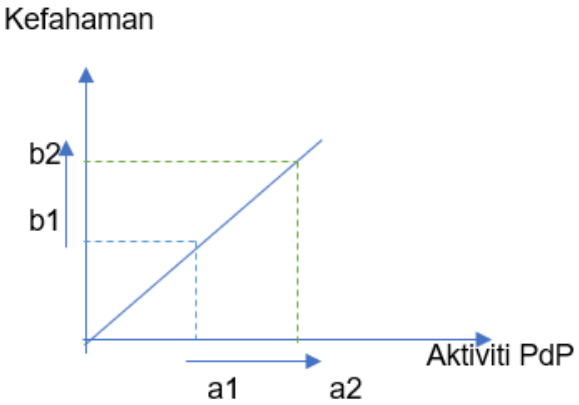
Kegagalan pelajar memahami topik dengan baik memberi kesan terhadap prestasi akademik. Kesukaran menjawab peperiksaan memberi mereka prestasi akademik yang kurang baik. Sukar untuk menghafal atau mengingati apa yang dipelajari hanya melalui penyampaian tanpa penglibatan di dalam aktiviti bilik kuliah. Maka aktiviti-aktiviti di dalam bilik kuliah berkait dengan prestasi akademik.

PdP adalah proses pemindahan pengetahuan dan kemahiran kepada pelajar melalui penyampaian. Namum demikian, PdP melalui kaedah penyampaian semata-mata kurang berkesan. Maka dengan ini aktiviti-aktiviti di dalam bilik kuliah diperlukan untuk memastikan keberkesanan PdP. Peningkatan kefahaman pelajar dikaitkan dengan peningkatan pengetahuan dan kemahiran yang dijalankan di dalam PdP.

Kenapa kefahaman itu penting? Apa akan terjadi apabila pelajar kurang faham? Adakah aktiviti-aktiviti di dalam bilik kuliah dapat meningkat kefahaman? Bagaimana? Persoalan ini harus dijawab dalam memastikan bahawa PdP yang dijalankan melalui pelbagai aktiviti dapat meningkat prestasi akademik. Adakah hanya pengetahuan dan kemahiran yang meningkat melalui aktiviti-aktiviti bilik kuliah dalam PdP? Adakah kemahiran generik juga akan meningkat melalui proses PdP?

Prestasi akademik menjadi keutamaan dan dikaitkan dengan kerjaya pada masa hadapan. Maka PdP memainkan peranan yang penting. PdP bukan sahaja mampu meningkat prestasi akademik malahan kemahiran generik turut ditingkatkan. Ini menunjukkan bahawa PdP

membantu pelajar meningkat kemahiran generik seterusnya meningkat kefahaman pelajar untuk memperolehi prestasi akademik. Di samping itu majikan turut berminat pada pelajar yang mempunyai kemahiran generik yang baik.



Rajah 6. Graf aktiviti PdP dan kefahaman di dalam kelas

Kemahiran generik diperolehi melalui aktiviti luar bilik kuliah seperti sukan, badan beruniform seperti kerja berpasukan, kepimpinan serta pengurusan masa. Namum demikian, perkara tersebut juga didapati melalui aktiviti bilik kuliah seperti tugasan berkumpulan atau individu, sesi soal jawab, pembentangan dan lain-lain. Ini menunjukkan bahwa kemahiran generik boleh didapati melalui proses PdP. Menurut [3], kemahiran berfikir, kemahiran pemudahcaraan, kemahiran belajar, kemahiran menilai dan menaksir dalam pengajaran, adalah penting untuk mempelajari Bahasa Melayu.

Jelas menunjukkan bahawa betapa pentingnya kemahiran di samping kemahiran teknikal dimana ko-kurikulum dam akademik selari dalam meningkat kemahiran generik. Selari dengan kehendak industri yang memerlukan kemahiran generik dalam perniagaannya. Persediaan bekerjaya tidak hanya fokus kepada penganugerahan akademik semata, malah kepentingan pengalaman praktikal dan kemahiran generik menjadi perkara keutamaan yang perlu dititikberatkan [4].

Kemahiran generik diperlukan oleh majikan dalam memastikan operasi berjalan dengan lancar. Institusi pengajian tinggi merupakan tempat untuk meningkat kemahiran generik ini melalui aktiviti di dalam bilik kuliah. Terdapat banyak kajian berkaitan keperluan kemahiran generik oleh majikan. Menurut [5], kemahiran generik hendaklah berfaedah dan penting untuk para pelajar menceburi semua bidang kerjaya dan memenuhi keperluan majikan masa kini.

Aktiviti di dalam bilik kuliah ketika PdP akan memberi ruang kepada pelajar untuk meningkat kemahiran generik. Contoh, latihan di dalam kuliah yang dijalankan ketika PdP akan meningkat kemahiran penyelesaian masalah. Tujuan adalah memastikan bahawa pelajar faham topik berkenaan. Namun demikian, aktiviti tersebut mampu meningkat pemikiran kritis yang amat diperlukan ketika bekerja. Ini menujukkan bahawa kemahiran generik diperolehi melalui aktiviti PdP.

PdP yang berkesan dapat meningkat kefahaman pelajar di samping mempengaruhi prestasi akademik. Maka aktiviti pembentangan tugasan dalam kumpulan atau individu adalah meningkat kefahaman. Kesan daripada aktiviti ini turut dikaitkan dengan kemahiran komunikasi yang amat penting. Pembentangan dapat meningkat kemahiran komunikasi termasuk sesi soal jawab. Ini menunjukkan bahawa aktiviti PdP mampu meningkat kemahiran generik melalui aktiviti pembentangan. Di samping itu ramai guru menyedari kepentingan komunikasi terhadap pelajar-pelajar. Di samping itu kepentingan komunikasi dikaitkan dengan keperluan industri. Tenaga pengajar turut menyedari bahawa kemahiran komunikasi

merupakan kemahiran asas yang diperlukan dalam apa-apa jua bidang pekerjaan yang diceburi [6].

Terdapat aktiviti tugasan berkumpulan samada seminar, kajian kes atau projek akhir yang diperlukan dalam meningkat kefahaman atau proses penilaian berterusan. Perbincangan antara ahli mampu meningkat pengetahuan ahli berkaitan perkara yang dibincangkan dan memberi kesan terhadap prestasi akademik. Seterusnya kesan terhadap kemahiran generik dikaitkan dengan kemahiran kerjasama berpasukan. Saling membantu dalam memastikan tugsan dapat disiapkan. Ini menunjukkan bahawa kemahiran berpasukan dikaitkan dengan aktiviti bilik kuliah yang diperlukan oleh industri.

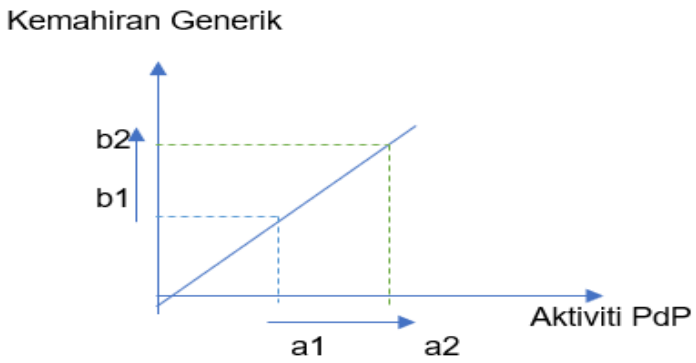
Menurut [7] menyatakan bahawa kemahiran kerja berpasukan akan meningkat apabila mempunyai pemimpin yang baik, anggota bertanggungjawab, hubungan yang erat antara ketua pasukan dan ahli-ahlinya, bersama-sama menyelesaikan masalah dan mempunyai matlamat yang jelas. Disokong oleh hasil kajian [8] yang mendapati guru kejuruteraan menerapkan kemahiran bekerja di dalam kumpulan dengan kerap. Pelajar yakin bekerja dalam berpasukan akan membantu mereka dalam menyempurnakan tugas yang diberikan. Ini akan meningkat kefahaman terhadap subjek tersebut.

Aktiviti PdP melibatkan tugasan serta latihan yang memerlukan tempoh masa untuk disiapkan. Aspek pengurusan masa penting dalam memastikan tidak berlaku kelewatan. Implikasi daripada itu dapat meningkat kemahiran berkaitan mengurus masa. Majikan berminat pada pekerja yang sering menepati masa. Ini berkaitan dengan kos yang dapat dijimatkan. Ini menunjukkan bahawa aktiviti dalam kuliah penting dalam meningkat pengurusan masa.

Etika bekerja adalah penting yang dipupuk ketika kuliah dijalankan. Tugasan yang diberikan perlu diselesaikan dengan penuh amanah dan jujur maka ini mengelakkan daripada meniru dan menipu. Ketika kuiz atau ujian dijalankan, jawapan perlu diberikan berdasarkan kefahaman pelajar yang diperolehi ketika Pd P atas ketika mengulangkaji. Ini menunjukkan bahawa PdP akan meningkat kemahiran generik.

Aktiviti PdP yang dijalankan memberi kesan ke atas kenahiran pemikiran kritikal. Pelajar perlu memikirkan untuk menyelesaikan tugasan yang diberikan. Penyelesaian masalah dalam aktiviti Pdp sangat penting dalam meningkat kefahaman. Malahan pihak industri memerlukan kemahiran pemikiran kritikal dalam menyelesaikan masalah dalam mengurangkan kos dan meningkat produktiviti dan keuntungan. Ini menunjukkan bahawa PdP memberi manfaat kepada pelajar dalam meningkat kemahiran generik.

Rajah 3 menunjukkan hubungan aktiviti PdP di dalam bilik kuliah akan meningkat kemahiran generik. Semakin banyak aktiviti PdP yang dijalankan semakin tinggi kemahiran generik yang akan diperolehi.



Rajah 7. Graf aktiviti PdP dan kemahiran generik

3. Metodologi Kajian

3.1 Sampel dan Instrumen

Sampel kajian adalah terdiri dari pelajar-pelajar pelbagai program di Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah (PTSB). Pelajar-pelajar terdiri dari program Diploma Kejuteraan Awam (DKA) serta Akauntasi (DAT), Pengajian Perniagaan (DPM) dan Pemasaran (DPR) dari semester 4 dan 5.

Kajian rintis dijalankan bagi menguji soalseidik yang dibina melalui soroton kajian. Sampel kajian yang digunakan dalam kajian rintis adalah terdiri daripada pelajar semester 3. Melalui percubaan ini maka isu-isu yang berkaitan dengan soal selidik dapat dikenalpasti. Jika soalan yang dikemukan tidak jelas dan kurang difahami akan diperbetulkan. Penting bagi responden disediakan soalan yang betul untuk mendapat jawpaan yang tepat. Bagi mencapai objektif kajian, maka soal selidik dibahagi kepada 2 bahagian. Bahagian pertama tertumpu pada demografi. Bahagian ini menumpu pada jantina, bangsa, jabatan, program serta semester. Manakala bahagian kedua menumpu kepada kemahiran generik dan PdP. Kemahiran generik terdiri daripada komunikasi, kepimpinan, kerja berpasukan, kreativiti, pengurusan masa, penyelesaian masalah, etika bekerja dan pemikiran kritikal.

Seterusnya elemen dalam PdP menumpu pada latihan individu, tugasan dalam Kumpulan, ujian, kuiz, pembentangan dan tutorial. Bahagian ini berdasarkan kepada penilaian sendiri berasas pada 4 skala Likert seperti berikut iaitu 1 adalah Sangat Tidak Setuju, 2 adalah Tidak Setuju manakala 3 ialah Setuju dan 4 adalah Sangat Setuju

Manakala Bahagian B berkaitan kemahiran generik dan PdP mengandungi 20 soalan mengandungi 4 aspek kemahiran generik dalam PdP iaitu soalan 1 hingga 6 berkaitan kemahiran generik Seterusnya soalan 7 hingga 11 berkaitan penyelesaian masalah. Manakala soalan 12 hingga 15 berkaitan dengan kerja berpasukan dan soalan 16 hingga 20 adalah kemahiran Pembentangan

3.2 Prosedur kutipan data

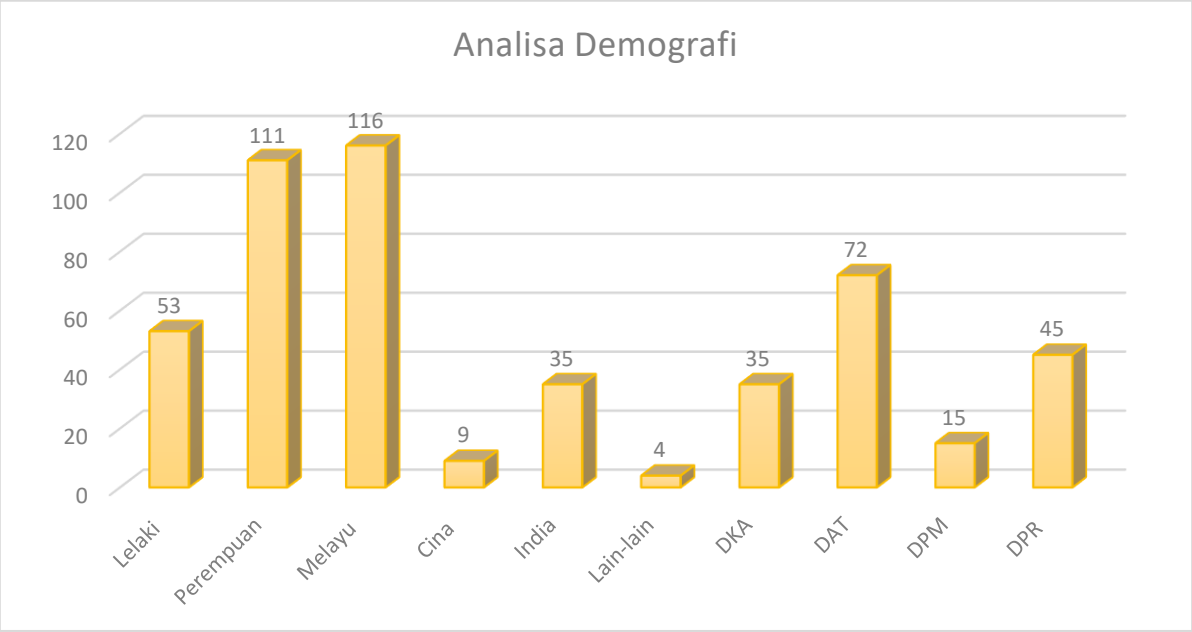
Sebelum data diagihkan kepada responden yang terdiri daripada pelajar pelbagai program di Jabatan Perdagangan PTSB, kebenaran dari Ketua Jabatan diperlukan. Kebenaran ini amat penting dalam memastikan responden memberi komitmen ketika soal selidk diagihkan untuk mendapat maklum balas dari responden. Semua maklumat berkaitan responden termasuk maklum balas adalah dikawal dan rahsia. Seterusnya, kajian ini tidak akan mendedahkan nama responden dalam memastikan responden berasa selamat. Soal selidik ini dijalankan secara bersemuka iaitu antara penyelidik dan pelajar. Data-data yang diperolehi akan dianalisa dengan menggunakan software SPSS version 28.0

4. Hasil Dapatan dan Perbincangan

4.1 Bahagian A: Analisa Demografi

Bahagian ini menumpu pada hasil dapatan dan perbincangan yang diperolehi daripada soal selidik yang dianalisa melalui SPSS. Bahagian satu berkaitan dengan demografi.

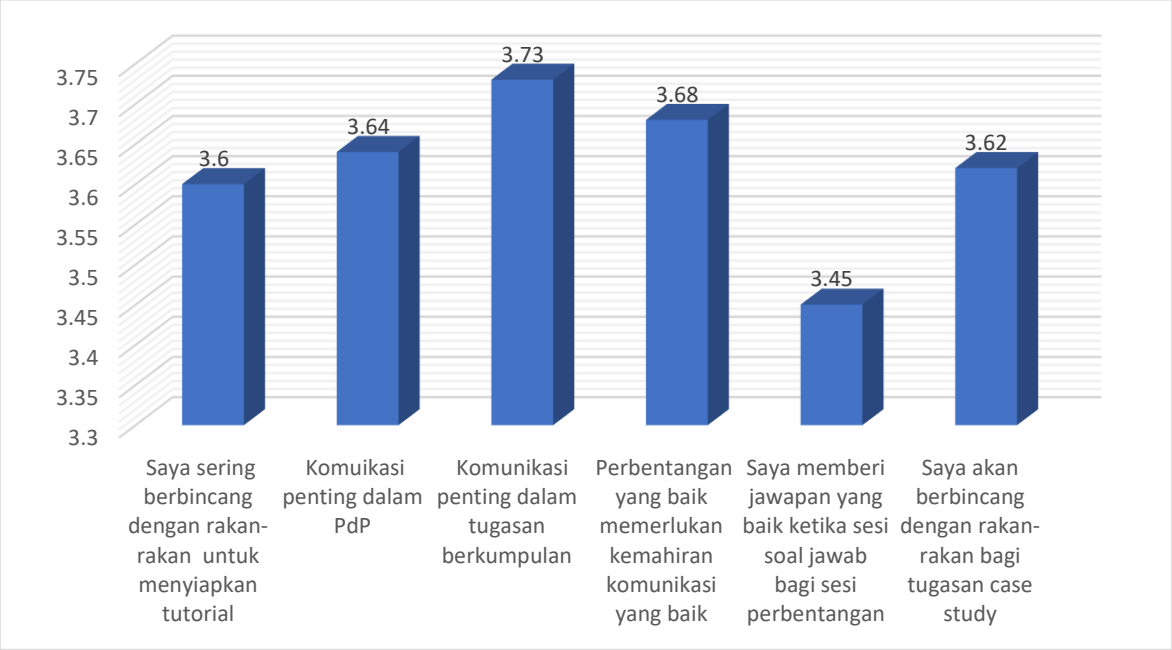
Rajah 4 menunjukkan analisa demografi berkaitan jantina, bangsa, program dan akademik. Hasil dapatan menunjukkan bahawa jantina perempuan adalah tinggi iaitu 111. Manakala responden lelaki ialah 53. Seterusnya mengikut demografi bangsa ialah Melayu seramai 116, Cina seramai 9 manakala India ialah 35 dan lain-laian ialah 4. Dapatan menunjukkan bahawa, JKA bagi program Diploma Kejuteraan Awam (DKA) terdiri 35 pelajar. Manakala JP bagi program Diploma Akauntasi (DAT) responden adalah seramai 72 pelajar, Diploma Penggajian Perniagaan (DPM) seramai 15 pelajar dan Diploma Pemasaran (DPR) adalah seramai 45 pelajar.



Rajah 8. Analisa demografi

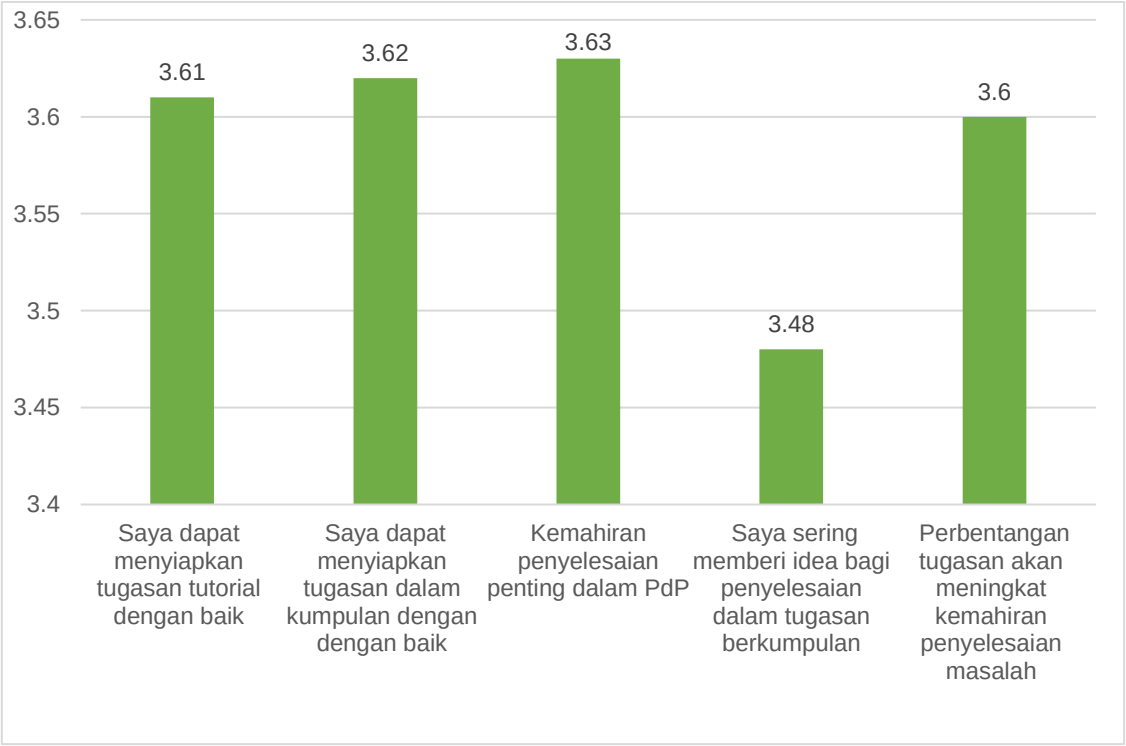
Rajah 5 menunjukkan kemahiran komunikasi yang terdapat dalam PdP. Hasil dapatan menunjukkan bahawa komunikasi penting dalam kerja berkumpulan iaitu min 3.73. Komunikasi penting bagi tugasan melibatkan pembentangan dengan min 3.68. Tugasan *case study* dan *tutorial* memerlukan komunikasi yang baik menunjukkan min 3.62 dan 3.60. Seterusnya pelajar menyatakan bahawa komunikasi penting ketika sesi PdP dengan min 3.64. komunikasi menunjukkan min 3.45 jika berlaku sesi soal jawab dalam pembentangan.

4.2 Bahagian B: Analisa Kemahiran Generik dalam PdP



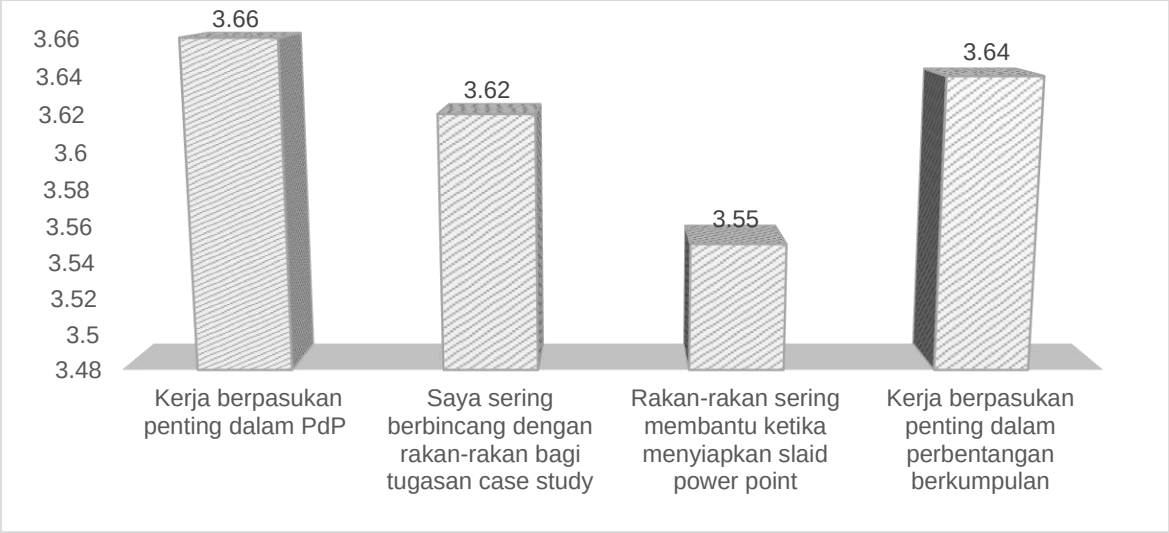
Rajah 9. Kemahiran komunikasi dalam PdP

Rajah 6 menunjukkan kemahiran penyelesaian di dalam PdP. Dapatan menunujukkan bahawa kemahiran penyelesaian penting dalam menyiapkan tugasan individu iaitu tutorial serta tugasan kumpulan iaitu dengan min 3.61 dan 3.62. Seterusnya pelajar mengetahui bahawa kemahiran penyelesaian masalah adalah penting dalam PdP iaitu 3.63. Memberi idea bagi penyelesaian masalah dalam tugasan kumpulan menunjukkan min 3.48. Dapatan turut menunjukkan bahawa keperluan kemahiran penyelesaian masalah dalam aspek pembentangan dengan min 3.6. Dapatan awal menunjukkan bahawa pelajar mengetahui kemahiran penyelesaian masalah dibentuk melalui PdP.



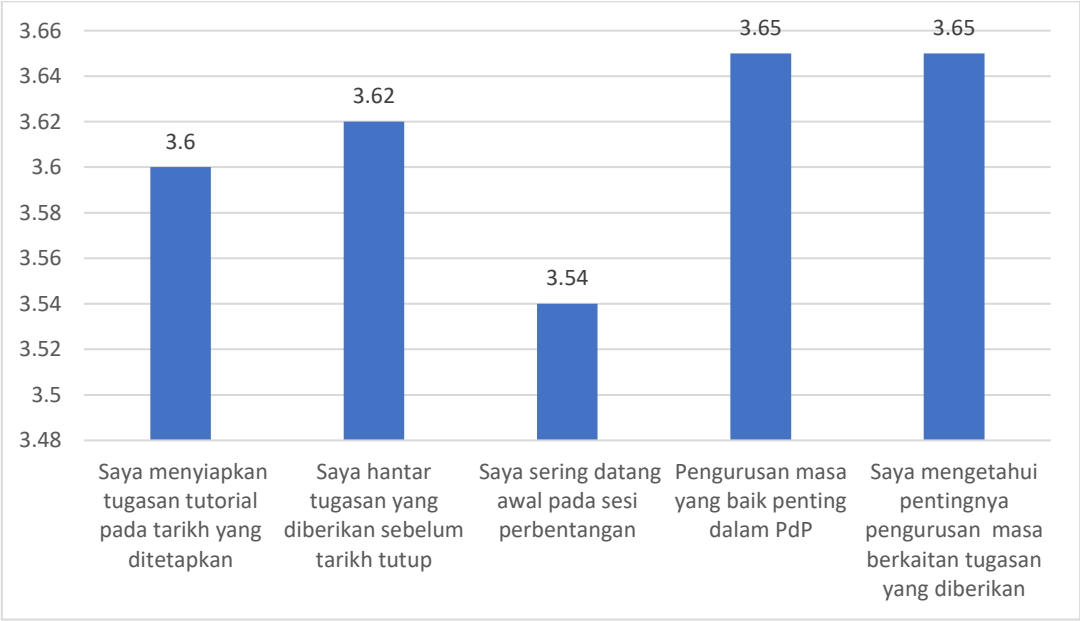
Rajah 10. Kemahiran penyelesaian masalah dalam PdP

Rajah 7 menunjukkan kemahiran kerja berpasukan di dalam PdP. Hasil dapatan menunjukkan bahawa min tertinggi iaitu 3.66 merujuk kepada kepentingan kerja berpasukan di dalam PdP. Pembentangan dalam kumpulan meningkat kerja berpasukan menunjukkan min 3.64. Seterusnya min bagi tugasan *case study* adalah 3.62. Dapatan tutur menunjukkan bahawa perbincangan dalam menyiapkan slaid pembentangan meningkat kerja berpasukan menunjukkan min 3.55. Dapatan awal menunjukkan bahawa kerja berpasukan dibentuk melalui tugasan di dalam PdP.



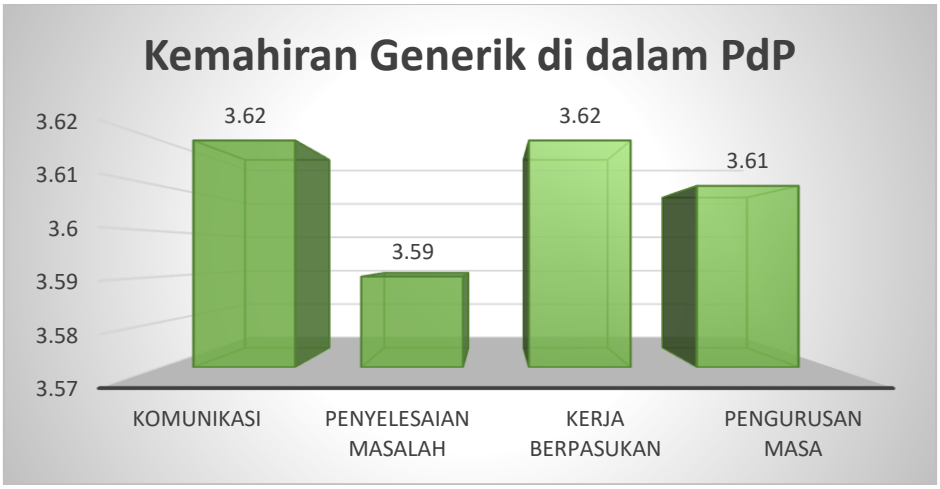
Rajah 11. Kemahiran kerja berpasukan dalam PdP

Rajah 8 menunjukkan kemahiran pengurusan masa di dalam PdP. Hasil dapatan menunjukkan bahawa pelajar mengetahui kepentingan pengurusan masa dalam PdP menunjukkan min 3.65. Pengurusan masa penting berkaitan dengan tugasan diberi menunjukkan min 3.65. Pelajar mengetahui tugasan perlu dihantar mengikut tarikh yang ditetap dengan min 3.62. Dapatan turut menunjukkan bahawa tugasan perlu disiapkan sebelum tarikh dihantar dengan min 3.6. Min menunjukkan 3.54 bagi kehadiran awal pada sesi pembentangan. Dapatan awal menunjukkan bahawa pelajar mengetahui kepentingan pengurusan masa dalam PdP.



Rajah 12. Kemahiran pengurusan masa dalam PdP

Rajah 9 menunjukkan min keseluruhan bagi kemahiran generik di dalam PdP. Dapatan menunjukkan bahawa pelajar mengetahui bahawa komunikasi diperlukan ketika proses PdP dijalankan. Penglibatan pelajar di PdP akan meningkat kemahiran generik iaitu komunikasi. Di samping itu tugasan yang diberikan di dalam PdP memerlukan penyelesaian masalah. Tugasan yang diberikan ini dapat meningkat kemahiran generik iaitu penyelesaian masalah yang amat penting bagi industri dalam meningkat keuntungan dan mengurangkan kos. Dapatan turut menunjukkan bahawa kemahiran generik seperti kerja berpasukan dapat ditingkat dalam PdP. Memastikan jawapan yang disediakan perlu dibincangkan bersama-sama maka ini akan meningkat kemahiran generik tersebut. Pelajar turut bersetuju bahawa pengurusan masa dapat ditingkat dalam proses PdP melalui tugasan yang diberikan. Tugasan perlu disiapkan awal dan dihantar mengikut masa yang ditetapkan.



Rajah 13. Min keseluruhan bagi kemahiran generik di dalam PdP

Maka, kemahiran generik dapat ditingkat dalam kalangan pelajar melalui proses PdP dengan baik sekiranya pelajar terlibat secara langsung. Komunikasi, penyelesaian masalah, kerja berpasukan dan pengurusan adalah kemahiran generik yang terdapat di dalam proses PdP diperlukan untuk meningkat kefahaman dan memperolehi keputusan yang baik. Malahan kemahiran generik turut diperlukan untuk memastikan memperolehi dan kekal dalam kerjaya pada masa hadapan.

Jadual 1 menunjukan hubungan kolerasi di antara kemahiran generik di dalam PdP. Tugasan yang diberikan dalam PdP menunjuk hubungan yang positif dan signifikan (t = *0.846, p < 0.01*) di antara komunikasi dan penyelesaian masalah. Tugasan kumpulan, tutorial individu dan pembentangan memerlukan komunikasi serta penyelesaian masalah tersebut dilakukan dengan baik. Dapatan turut menunjukkan komunikasi mempunyai hubungan yang positif dan signifikan dengan kerja berpasukan dan pengurusan masa (*t = 0.815, p < 0.01 dan t = 0.841, p < 0.01)*.

Jadual 3. Korelasi

	Komunikasi	Penyelesaian Masalah	Kerja Berpasukan	Pengurusan masa
Komunikasi		***0.846****	***0.815****	***0.841****
		0.000	0.000	0.000
Penyelesaian Masalah			***0.855****	***0.871****
			0.000	0.000
Kerja Berpasukan				***0.846****
				0.000
Pengurusan Masa				

*Noted: **. Correlation is significant at the 0.01 level (2-tailed).*

PdP menekankan pentingnya kemahiran generik iaitu komunikasi dikaitkan dengan kerja berpasukan dan pengurusan masa. Tugasan yang diberikan memerlu kerja berpasukan dan pengurusan masa yang baik memerlukan komunikasi yang berkesan. PdP dijalankan dapat meningkat komunikasi dan kerja berpasukan dan pengurusan masa.

Jadual 1 turut menunjukkan dapatan bahawa penyelesaian masalah mempunyai hubungan positif dan signifikan dengan kerja pasukan dan pengurusan masa di dalam PdP ($t = 0.855, p < 0.01; t = 0.871, p < 0.01$). Tugasan yang diberikan memerlukan penyelesaian masalah dan perlu dilakukan secara berpasukan dengan masa yang ditetapkan. Ini menunjukan bahawa PdP melibatkan penyelesaian masalah serta kerja berpasukan dan pengurusan masa. Seterusnya kerja berpasukan mempunyai hubungan positif dan signifikan dengan pengurusan masa ($t = 0.846, p < 0.01$)

5. Kesimpulan

Pembelajaran dan pengajaran (PdP) yang dijalankan di bilik kuliah adalah untuk meningkat pengetahuan, kemahiran serta pengalaman berkaitan dengan subjek yang diambil. Pelajar yang berjaya menguasai PdP dengan baik akan memperolehi prestasi akademik yang baik. Namum demikian, PdP turut dapat meningkat kemahiran generik yang begitu penting dalam kerjaya pada masa hadapan.

Dapatan menunjukkan bahawa melalui PdP, kemahiran gnerik yang diperolehi adalah komunikasi, penyelesaian masalah, kerja berpasukan dan pengurusan. Pelajar mengetahui bahawa tugasan yang diberikan memerlukan kemahiran generik. Secara keseluruhan, kemahiran generik seperti komunikasi, penyelesaian masalah, kerja berpasukan dan pengurusan menunjukkan min yang tinggi. Tugasan ynag diberikan untuk meningkat kefahaman pelajar turut dapat meningkat kemahiran generik.

Hubungan kolerasi menunjukkan positif dan signifikan di antara pembolehubah kemahiran generik seperti komunikasi, penyelesaian masalah, kerja berpasukan dan pengurusan masa. PdP yang dijalankan akan meningkat kemahiran generik tersebut secara serentak. Kemahiran generik dipertingkat dengan PdP yang diuruskan dengan baik.

Rujukan

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EVALUATING TOOL WEAR AND HOLE QUALITY IN DRY DRILLING: A STUDY ON CYLINDRICITY DEVIATION AND TOOL LIFE BEYOND FAILURE

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ABSTRACT

This study investigates tool wear and hole quality in dry drilling of aluminium alloys, specifically focusing on cylindricity deviation and tool life beyond failure. In the aerospace industry, where green manufacturing and sustainability are increasingly prioritised, dry machining is becoming popular due to its environmental and economic benefits, including eliminating coolants, reducing contamination, and improving working conditions. Despite these advantages, dry drilling poses significant challenges, such as high thermal conductivity, heat absorption, and material adhesion, leading to severe tool wear and quality issues. Seven hundred thirty holes were drilled in AA2024 aluminium alloy using TiAlN-coated carbide drills on a CNC milling machine without using lubricants. The results demonstrated that although the tool could continue drilling beyond 700 holes, significant deviations in hole diameter and roundness occurred, exceeding allowable tolerances. The study found that high spindle speeds and appropriate feed rates were crucial for maintaining smooth surfaces and minimising tool wear. Additionally, thermal loads during machining caused thermal expansion, adversely affecting hole size and quality. Effective coatings like TiAlN were shown to minimise temperature effects and reduce material adhesion to the tool; however, ensuring consistent hole quality remains challenging.

Keywords: Dry Drilling, Aerospace Industry, Aluminum Alloy, Coatings, Tool Wear, Cylindricity Deviation

1. Introduction

Green manufacturing and sustainability are increasingly important in modern industries, prompting a shift towards dry machining. This method eliminates the need for coolants, offering significant cost savings, a cleaner working environment, and reduced contamination of machined materials. The approach also drastically decreases water and oil consumption. Driven by environmental protection concerns, dry machining is not only ecologically desirable but could soon become a necessity in the manufacturing industry. Researchers suggest that industries must adopt dry machining due to impending environmental protection laws and occupational safety regulations [1]. The benefits of dry machining include non-pollution of air or water, no residue on the swarf, no health hazards, and improved job satisfaction due to a cleaner workplace [2].

In the aerospace sector, drilling is frequently used, particularly for riveting during the assembly stage. Medium-sized planes may contain around 85,000 rivets, making drilling an essential operation. The industry wants to reduce drilling time and cut out cutting fluids to avoid cleaning structures before riveting, and it wants to automate the riveting process and maximise productivity. This approach ensures high-quality holes without burrs. The necessity to automate assembly in the aircraft industry, coupled with economic and ecological reasons, drives the development of dry drilling of aluminium alloys. However, dry drilling is challenging, prompting companies and laboratories to research the phenomena involved. Researchers focus on improving cutting tools through better materials, coatings, and geometries.

Implementing dry machining involves more than just turning off the coolant supply. Current research primarily focuses on developing coatings to extend tool life [3]. However, there is less focus on product quality. This research aims to analyse current issues in aluminium alloy dry drilling and understand cutting process mechanisms to develop an economical and safe dry machining process. Because of their high heat conductivity, they result in significant heat absorption and deformation, making aluminium alloys difficult to machine dry. Their low melting point and sensitivity to adhesion with the tool, which can cause damage to the tool and workpiece, are additional problems.

Various drill materials and geometries are employed depending on the task at hand. Choosing the appropriate tool material ensures smooth drilling with little torque and force. For example, when drilling aluminium-based composites, polycrystalline diamond (PCD) drills yield the best surface finish and outperform carbide-tipped and high-speed steel (HSS) drills in terms of drilling force, tool wear, and hole quality [4]. In many applications, coated cemented carbide tools have largely replaced HSS tools because of their superior performance and affordability. Coatings improve tool life by forming a barrier that reduces diffusion rates and thermally induced tool wear [5]. Important properties of coatings include wear resistance, low friction, and good adhesion to the substrate [6].

Surface roughness in dry machining depends on process parameters. High spindle speeds and appropriate feed rates are crucial to maintaining smooth surfaces and minimising tool wear. Optimisation of cutting parameters is essential to produce quality products [7]. Research shows that extreme temperatures in dry machining can lead to premature tool wear due to increased friction and adhesion between the tool and the workpiece. Effective coatings, such as diamond and DLC (diamond-like carbon), have been shown to extend tool life and reduce aluminium adhesion to the cutting edge [8], [9]. Thermal loads significantly influence dry drilling. The heat generated during machining causes thermal expansion, affecting hole size and quality. High cutting speeds lead to increased temperatures and adhesion wear. Coatings like TiAlN minimise temperature effects and adherence tendency, enhancing tool performance [10] [11].

Studies using finite element methods reveal that heat flows during drilling raise drill temperatures, causing expansion and leading to diameter deviations in drilled holes [12] [13]. Experimental studies also demonstrate that dry drilling often results in oversized holes and tapering due to thermal expansion [14].

Dry machining of aluminium alloys remains challenging due to issues like adhesion, thermal expansion, and chip removal. The focus has been developing coatings to extend tool life and improve performance. However, more research is needed to understand the interaction of process parameters and ensure that dry machining produces high-quality products. The ultimate goal is to establish a safe and economical dry machining process by investigating cutting conditions affecting tool life and wear. Over the past 25 years, advances in dry drilling have focused on tool coatings for single-layer workpieces, with limited research on aluminium stacked materials and wear mechanisms during multilayer drilling. This research seeks to fill this gap by analysing current issues and understanding cutting process mechanisms to develop a reliable dry machining process for aluminium alloys.

While the dry drilling of steel and cast iron is a relatively well-established process, the dry drilling of aluminium alloy without cutting fluid is extremely difficult due to the adhesion of aluminium to the drill. When the drill enters the material, the tool face is smooth without any adhering aluminium material; as the drilling goes on, the chip picks new build-up material, and some existing build-up is peeled off by the incoming chip, which causes an obstacle for the chips to dislodge through the drill flutes. This type of clogging causes drill failure after only a few runs [15]. Investigating and understanding the drilling process mechanisms is a key issue in establishing a safe, dry machining process. The ultimate goal of this research is to

experimentally investigate cutting conditions that affect tool life, wear mechanisms and dimensional accuracy of aluminium alloy stack materials.

2. Methodology

A drilling test was performed on the chosen parameter of 3000 rpm and 0.15 mm/rev. Experiments were carried out on a rectangular aluminium alloy 2024 workpiece 51 x 279 x 51 x 11.6 mm. The drilling experiments were performed in dry conditions on a Takisawa MAC-V3 Vertical CNC milling centre equipped with a maximum spindle speed of 6000 rpm and a 5.5 kW drive motor. Experiments were performed in dry conditions to observe the severity of tool wear and understand the fundamental issue in the actual machining process.

The workpiece was made of the aluminium alloy AA2024. Seven hundred thirty holes were produced without using any lubricant; they had a depth of 11.6 mm and a diameter of 6mm. Roundness and diameter deviation parameters were used to evaluate the hole's quality. The drilling tool CoroDrill® 860 solid carbide material was selected from the Sandvik Coromant Catalog. TiAlN-coated carbide drills with a thickness of 6 mm tools were used in the experiment.

Table 1. The dimensional and mechanical properties of the drilling tool

Tool diameter	6 mm
Flute	2 flutes
Point angle	1450
Helix angle	300
Shank type	Cylindrical
Coating (PVD)	TiAlN

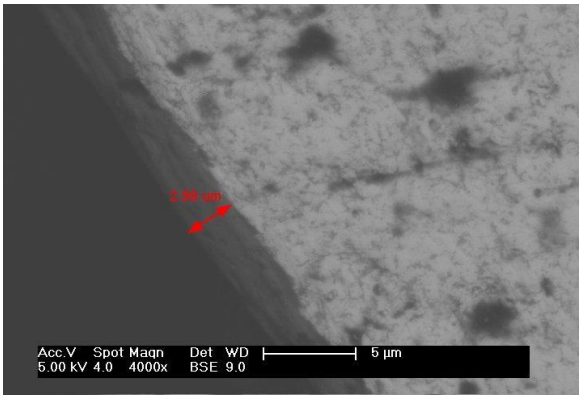


Fig. 1. The thickness of TiAlN coating

3. Result and discussions

To evaluate the hole quality, a cartridge of Microset compound is dispensed directly into the hole. The compound cures to produce a highly accurate flexible rubber replica of the surface with a 0.1-micron resolution. The Keyence VHX-500F 3D optical microscope was used to measure the diameter of the replica hole.

The correlation between diameters and holes was analysed. The graph in Figure 2 shows a deviation of hole diameter as a function of hole numbers. It is apparent from the graph that, on average, the diameter deviation is relatively stable for holes below 350, taking into consideration the allowable deviation value of 50 um. Further analysis shows significant out-of-control data where the deviation exceeds 1 mm from the nominal diameter.

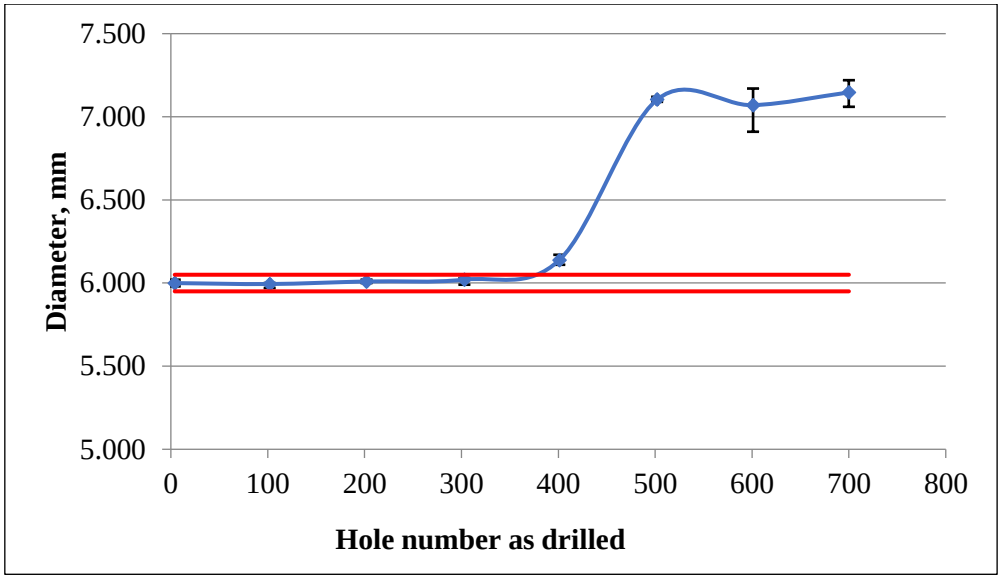


Fig. 2. Measured deviation in hole diameter as a function of hole number.

The measurement of roundness evaluates the hole quality. Roundness, or circularity, is usually defined as a deviation of the shape perimeter from true roundness [16]. Generally, the roundness of the drilled holes worsened as the cutting speed and feed rate increased due to the vibratory displacement of the drilling tool.

In an ideal cylindrical hole, the diameter is uniform throughout the depth of the hole. In other words, the radial distance from the symmetrical axis of the hole to the hole wall is supposed to be constant along the hole depth. Unbalanced cutting forces of the drill at the tip of the drill, which result from an offset chisel point, cause the drill to whirl down through the workpiece as it progresses, resulting in helical wandering [17].

The ratio of minimum to maximum Feret's diameter was used to define circularity at the entrance diameter of the machined hole. The Feret's diameter is the projection of the object shape to the set of axes, which consecutively rotate at 15 degrees relative to each other from the horizontal axis. The ratio is expressed mathematically by Equation 1. A roundness value of 1.0 indicates an ideal cycle.

$$Roundness = \frac{F\,min}{F\,max} \qquad \text{…. Equation 1}$$

Where $F\,min$ and $F\,max$ are the minimum and maximum Feret's diameter, respectively.

The graph in Figure 3 presents a similar trend of hole roundness as diameter deviation. The cylindricity values for the first 300 holes were consistently maintained above 0.9. However, there is a clear trend of decreasing as the number of holes increases. The cylindricity value continues to drift until it reaches 0.96 at the test's end, indicating the most severe roundness.

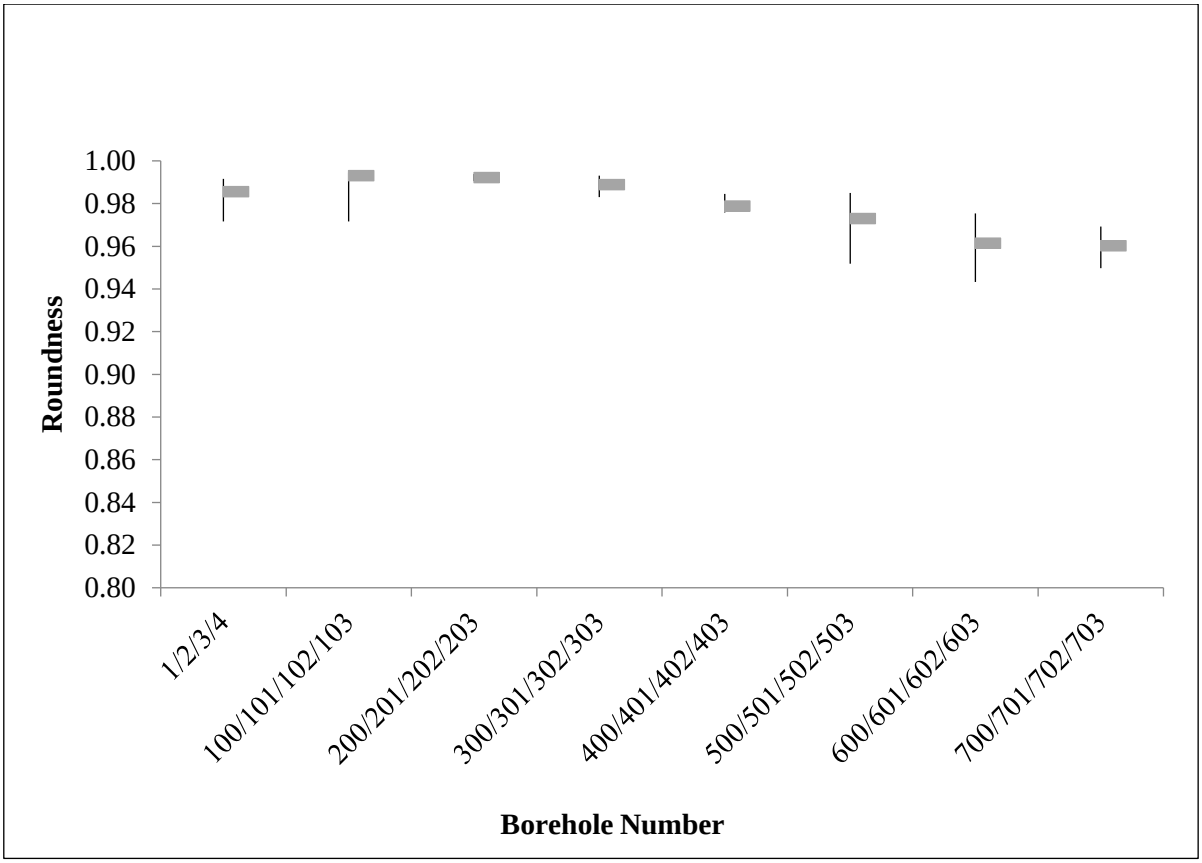


Fig. 3. Roundness as drilling progress

The Scanning Electron Microscopy (SEM) images show no evidence of crater wear during the drilling of the aluminium alloy AA2024. The tool's deterioration mainly results in material transfer from the workpiece to the tool rake surface and cutting edge.

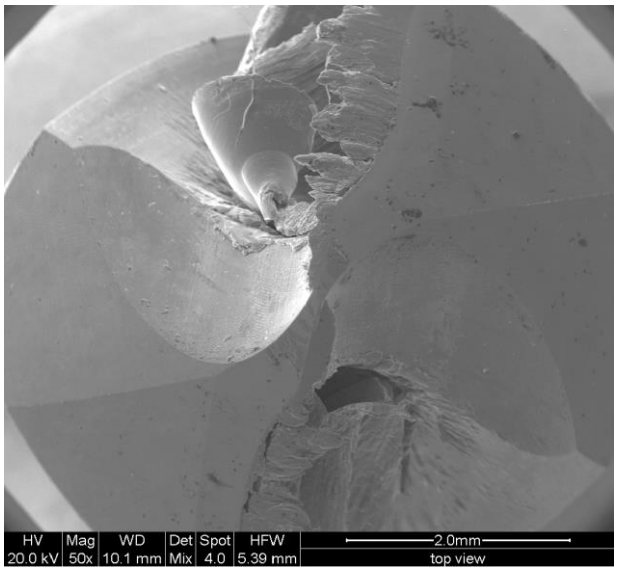


Fig. 4. SEM image from Drill top view

It is apparent from Figure 4 that a significant amount of pick-up material becomes welded to the cutting edge. The finding of this test supports the idea that deformed chips tend to adhere or weld to the tool at low cutting speeds and rake angles due to high friction between the tool and the removed materials, which slide on the tool rake face. Adhesion wear appears when the built-up edge and the built-up layer are formed at the same time. The built-up edge and

layer result from a chip material transfer on the tool-cutting face. The tool–chip interface temperatures can reach high values in the machining of the adherent layer and progressively grow during the drilling process while the cutting edge accumulates workpiece material fragments. Consequently, a progressive increase in the drilled hole diameter is observed. When drilling at a lower speed, the cutting temperature and the adhered material's quantity are also reduced. The presence of coating further lowers the temperature effect and, consequently, the adherence tendency. Unfortunately, the study did not include speed variation and temperature measurement, so it was unable to analyse these variables.

EDX was performed at the cutting edge and rake face where the chips slid along and where any material adhesion would harm the tool's cutting performance. The analysis showed massive aluminium material adhered to the particular area, as illustrated in Figure 5. However, it is not obvious that the loss of coating at the cutting edge is the cause of drill failure. After drilling seven hundred holes, coating loss and adhered aluminium were observed in the cutting zone, but the chips were still being evacuated cleanly, and the drill was still performing well.

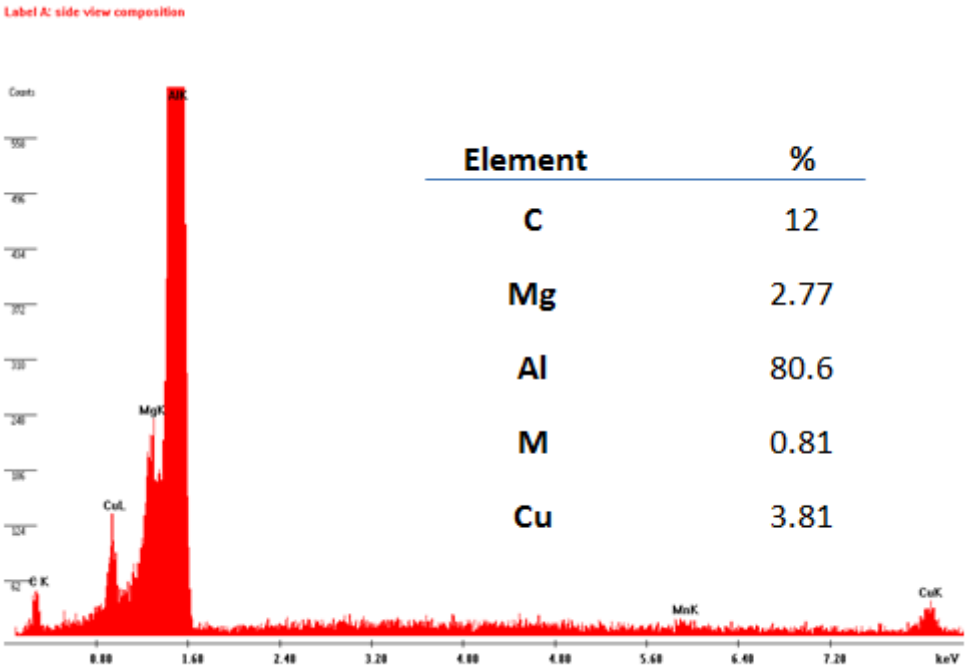


Fig. 5. EDX analysis at rake face near to cutting edge

The EDX analysis was also performed in the land area where the tool was in direct contact with the borehole wall. Figures 6 and 7 show the SEM image and the number of elements in the area near the cutting edge. The results showed that much of the coating remained adhered to the drill at the end of the test.

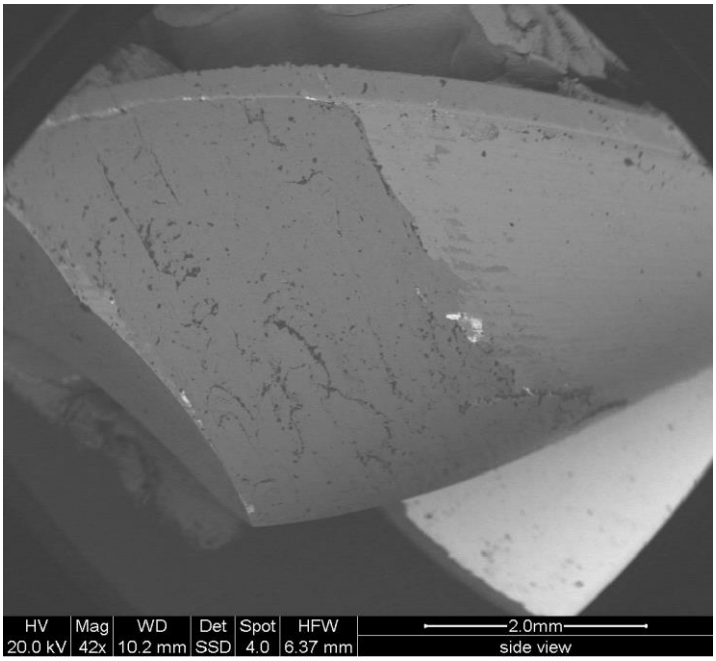


Fig. 6. SEM image at a land area of the drill

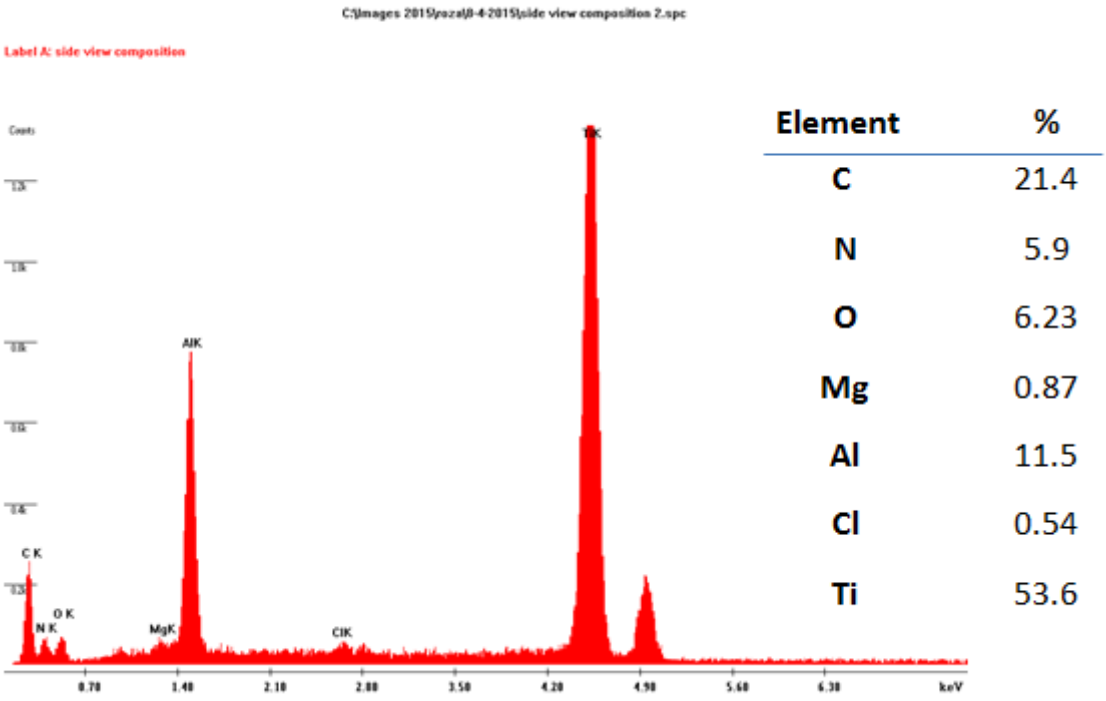


Fig. 7. EDX analysis at a land area of the drill

4. Conclusion

The study on evaluating tool wear and hole quality in dry drilling of aluminium alloys highlights several critical insights into the machining process, mainly focusing on cylindricity deviation and tool life beyond failure. The findings reveal that while dry drilling presents significant environmental and economic benefits by eliminating coolants, it also introduces substantial challenges such as severe tool wear, thermal expansion, and material adhesion, adversely affecting hole quality. The experimental results indicate that despite the tool's ability to continue drilling beyond 700 holes, significant deviations in hole diameter and roundness

occur, often exceeding allowable tolerances. This deviation is primarily attributed to the high thermal loads and adhesion issues inherent in dry drilling. The use of TiAlN-coated carbide drills demonstrated some mitigation of these effects, but maintaining consistent hole quality remains challenging.

Optimisation of cutting parameters, including spindle speed and feed rate, proved crucial in minimising tool wear and ensuring smoother surfaces. However, the study also found that even with optimised parameters, thermal expansion during machining significantly impacts hole size and quality, leading to oversized and tapered holes. The research underscores the necessity for ongoing advancements in coating technologies and cutting parameters to enhance tool performance and product quality in dry drilling applications. Addressing these challenges is essential for developing reliable and economical dry machining processes, particularly in the aerospace industry, where precision and sustainability are paramount. Future studies should continue exploring the interplay between thermal effects, material adhesion, and cutting conditions to refine dry drilling techniques and improve overall machining outcomes.

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FOOTBALL SPORTS POSE VISUAL DETECTION USING DEEP LEARNING

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ABSTRACT

This project aims to find a more effective level of sports analysis that can facilitate athletes, coaches, and sports analysts in increasing the accuracy of sports movement that involves pose visual detection. Human pose detection is an important problem in the field of computer vision. With the rapid development of technology, this sports analysis system using pose detection is increasingly known and widely used by various organisations. This system lets users see how the computer responds by identifying sports poses and converting them into keypoint detection. This system's main objective is to estimate athletes' body poses in real-time or from recorded video footage image or video. The objective of the study is to develop datasets for the assessment of athlete pose, events, and techniques. Therefore, after the system has been tested, it can be concluded that it can analyse the sports pose and movement accuracy. Therefore, this system will be developed based on the proposed deep learning method, Yolov8 and Convolutional Long Short-Term Memory (LSTM). The system will be able to recognise poses and calculate the accuracy of the sports pose in the picture or video frame used as input by the system.

Keywords: Pose, Visual, Detection, Deep Learning, Convolutional LSTM

1. Introduction

For the past few decades, the computer vision community has focused on the hot research topic of pose detection, commonly known as human pose detection or human position estimation. Finding and locating human joints, often referred to as key points in photos or films, such as heads, necks, shoulders, hips, elbows, wrists, knees, ankles, and so forth, is essential for pose detection [1]. The perspective-n-point issue, also known as the pose estimation problem, aims to locate 3D points on an item and their corresponding 2D projections in the image and determine the posture of an object using a calibrated camera [2]. The project aims to develop a method to accurately estimate the body poses of Football Players in real-time or from image or recorded video footage. It's also to develop datasets for assessment of Football Player poses, events, and techniques. Besides that, to compare the accuracy of the action recognition method on assessment of two types of poses (juggling and dribbling) of football player technique, accuracy, and performance.

The problem of sports game analysis, especially in football, is providing statistical data collection, which is taken manually by a trained analyst who watches the game [3]. Data collection statistics of football include shoots, defends, passing, penalties, and red or yellow cards given to players. The analysis results will be valuable for teams, coaches, and analysts to improve game performance and make informed decisions for the following games [4]. The

limitations of traditional manual observation and subjective evaluation have fuelled the athlete's performance's rapid evolution. This approach takes a lot of time, is prone to mistakes, and only offers a partial understanding of the intricate movements of athletes [5]. Data collection technologies are likely to enhance visual detection capabilities further using deep learning in sports analysis. New developments in real-time analytics, computer vision, and machine learning have created new avenues for improving and automating the monitoring of athlete posture and movement. These innovations have been well received and have improved performance analysis across various sporting disciplines [6].

The challenge of pose visual detection using deep learning is precisely and consistently recognising and localising individual poses from photographs and video recordings. Although deep learning models, particularly convolutional neural networks (CNN), have demonstrated promising results in various computer vision tasks, pose estimation still faces several unique difficulties. These difficulties include different lighting situations, camera angles, and low accuracy using the current visual methods model. There is also no dataset for training event detection using deep learning [7]. The development of models and deep learning methods that can successfully solve this difficulty and achieve high accuracy in pose detection is the focus of the problem statement. The ultimate objective is to create algorithms and models to identify practical poses precisely.

2. Literature Review

2.1 Overview

This chapter includes an in-depth review of the literature on datasets used in pose detection and the development of action recognition techniques, ranging from traditional approaches to modern AI and deep learning models [8]. A total of 24 sources were reviewed, highlighting key advancements and methodologies that have shaped the field.

2.2 Sports Pose Detection

Sports pose visual detection using deep learning, also referred to as sports pose estimation, is the process of analysing and comprehending athletes' or players' body positions in the context of different sports using computer vision and deep learning techniques [9] and in other words, using deep learning technique that tracks and detects the poses of athletes. Computer vision techniques are used to recognise and examine athletes' body postures and movements while they are competing to detect a pose. This can offer insightful information on performance evaluation, injury prevention, sports coaching, and fan interaction. A computer vision task known as "pose detection" or "estimation" entails locating and following important body joints or points in an image or video [10]. The locations of the head, shoulders, elbows, wrists, hips, knees, and ankles are represented by these important points or body joints [11], as shown in Fig. 1.

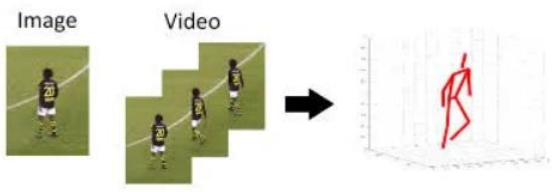


Fig. 2. Detecting Football player pose from an image or video sequence

2.3 YoloV8

The proposed method and algorithm selected for keypoint detection or pose estimation is YOLO (You Look Only Once). The YOLO algorithm has achieved great progress in the rapidly changing fields of artificial intelligence and computer vision. YOLOv8 has become well-known for its remarkable effectiveness and precision in object-detecting tasks. Still, its capabilities go beyond simple object recognition. The potential uses and effects of YOLOv8 in sports classification delve into the fascinating field of sports classification [12]. A sophisticated deep learning technique called YOLOv8 is intended for real-time object detection. Its design can identify and categorise several items in an image or video stream with amazing speed and accuracy. The researchers have modified YOLOv8, which is often used for object identification and sports categorisation with a particular emphasis on cricket, football, and basketball [13].

A convolutional neural network (CNN) serves as the foundation for the YOLOv8 architecture. The input image is divided into a grid, and for each grid cell, bounding boxes and class probabilities are predicted. YOLOv8 maximises its object detection and classification capacity by employing multiple CNN layers [14]. To train YOLOv8 using various datasets, including pictures and videos of basketball, football, and cricket matches [15]. With its ability to reliably classify these sports, the model opened many applications. Sports pose visual detection in this research focuses on football to detect poses of passing, shooting, running, dribbling and penalty kicks. The quality of football pose is crucial to the accuracy of athlete performance [16]. The YOLOv8 categorisation model is helpful for football teams and coaches in evaluating player performance [17]. Teams can obtain critical insights into player behaviours, plan efficacy, and performance metrics by monitoring players' postures and movements throughout a game.

To summarise, the literature review emphasises the significant advancements in pose detection or estimation using historical data, advanced instrumentation, numerical models, and artificial intelligence approaches. Combining conventional techniques with contemporary AI methodologies has great potential for enhancing the accuracy and efficiency of image recognition, enabling the development of more dependable and accurate prediction systems.

3. Methodology

This project utilises the C++ programming language, primarily due to its efficiency and low-level control, which is crucial for handling the complexities involved. The project involves training a set of 500 images, a number determined by the limitations of the hardware capacity. To streamline the development process and leverage a wealth of available resources, the project used the open-source platform Python, which is Google Colab. Python's extensive libraries and community support make it an ideal choice for managing the various aspects of the project, from data preprocessing to implementing machine learning algorithms. This combination of C++ for core development and Python for the broader ecosystem enhances the overall efficiency and effectiveness of the project. The project's design demonstrates the preliminary steps that will be taken. This project involves Yolov8 methods, which is using Google Colab. The project flow chart also shows how the project is progressing, as shown in Fig. 2.

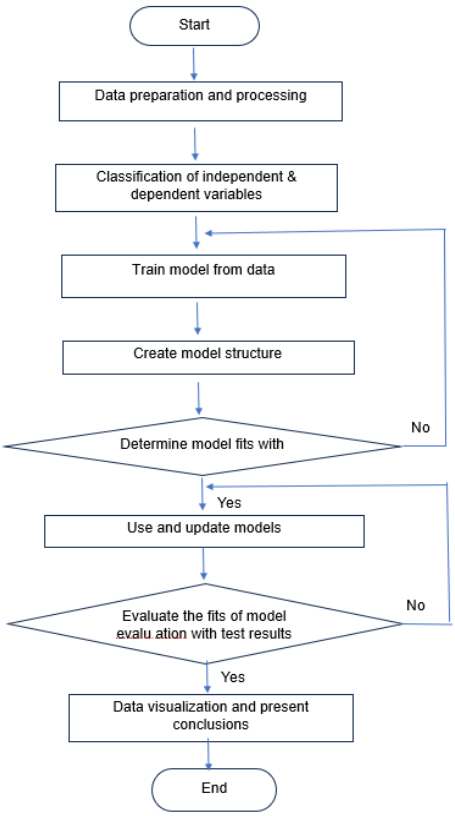


Fig. 3. Project System Flow Chart

3.1 YoloV8 pose Model

Based on state-of-the-art computer vision and deep learning developments, YOLOv8 provides unmatched speed and accuracy. Because of its simple architecture, it can be quickly adapted to a wide range of hardware platforms, including edge devices and cloud APIs, and is suited for a wide range of applications. Examine the YOLOv8 Docs, an extensive guide created to assist in comprehending and using its features and capabilities [18]. In computer vision, object detection is a method used to locate and identify things inside an image or video. The technique of correctly locating one or more items using bounding boxes and rectangular forms surrounding the objects is known as image localisation [19]. This procedure is occasionally mistaken for image recognition or classification, which attempts to identify which category or class a picture or an object inside an image belongs to. Figure 3 shows the visual depiction of the explanation that came before it. "Person" has been identified as the object in the picture. The YOLOv8 posture estimation model employs a top-down approach, which estimates a person's pose every time it locates them in an image called the "detect phase" [20].



Fig. 4. Image recognition, detection, and localisation

An AI framework called YOLOv8 can handle a variety of computer vision applications. Pose estimation, segmentation, classification, and detection may all be done with the framework. Fig. 4 illustrates each job's various goals and use cases [21].



Fig. 5. Detection, segmentation, Classification, Tracking and Pose

3.2 YoloV8 and Convolutional LSTM

The designed project combines YOLOv8 and Convolutional Long Short-Term Memory (LSTM) for football pose detection and action recognition. YOLOv8 is used to extract target pose information, which is then used to build a Convolutional LSTM-based model for human action detection that successfully encodes the distinctive body movements for different action types [22]. In other words, YOLOv8 should be trained to detect the pose of football players on the image or video, and Convolutional LSTM should be able to recognise an action being performed. Using YOLOv8 for pose/keypoint estimation allows the detection of specific points on objects, which can be an essential step towards action recognition. While YOLOv8 can be used for pose estimation and action recognition, it needs additional tools or models that consider the temporal aspect of actions in a video [23]. Action recognition often requires understanding temporal relationships between frames, which is a different task than what standard YOLOv8 is designed for, as YOLOv8 primarily targets static image tasks.

One way to integrate this functionality might be to use a two-stage approach, where YOLOv8 provides the pose estimates as features to a secondary model trained specifically for action recognition that considers temporal sequences [24]. First, there is a need to train a YOLOv8 model on a suitable pose estimation dataset, such as a football event dataset and experiment with post-processing the output. Once done with the output processing, explore different strategies for the action recognition part. To embark on this with YOLOv8, the steps are:

i. Train the YOLOv8 model for pose estimation on a dataset that includes annotations for the human body's key points.
ii. Once trained, use the model to predict key points for each video sequence frame.
iii. Post-process the keypoint predictions to build a temporal profile, which could involve creating time-series data of keypoint positions.
iv. Then, a mechanism, such as a sequence model or heuristic (Conv LSTM), is needed to interpret these time-series data into actions such as walking, running, or climbing. For football poses, actions are dribbling, juggling, shooting, heading, and running.

3.3 YoloV8 for Pose Estimation

Pose detection or estimation is a process of pinpointing the locations of particular points, often known as key points, within a picture. The key points might be for different object components, such as joints, landmarks, or other distinguishing characteristics. Typically, 2D [x, y] or 3D [x, y, visible] coordinates indicate the keypoint positions. A group of points that reflect an object's key points in an image, typically accompanied by the confidence scores assigned to each point, results from a pose estimate model. When one needs to pinpoint the precise location of an object's pieces in relation to one another within a scene, pose estimation is a useful option. Accurate pose estimation enables object localisation and tracking.

The pre-trained pose models of YOLOv8 use the Common Object in Context (COCO) dataset. The COCO dataset is used for pre-trained Detect, Segment, and Pose models, while the ImageNet dataset is used for pre-trained Classify models. Upon initial use, models are downloaded automatically from the most recent Ultralytics release. Six different pre-trained versions of the model are available to download.

Table 3. YOLOv8 Pose Pre-trained Model

Model	size (pixels)	mAP^{pose} 50-95	mAP^{pose} 50	Speed CPU ONNX (ms)	Speed A100 TensorRT (ms)	params (M)	FLOPs (B)
YOLOv8n-pose	640	50.4	80.1	131.8	1.18	3.3	9.2
YOLOv8s-pose	640	60.0	86.2	233.2	1.42	11.6	30.2
YOLOv8m-pose	640	65.0	88.8	456.3	2.00	26.4	81.0
YOLOv8l-pose	640	67.6	90.0	784.5	2.59	44.4	168.6
YOLOv8x-pose	640	69.2	90.2	1607.1	3.73	69.4	263.2
YOLOv8x-pose-p6	1280	71.6	91.2	4088.7	10.04	99.1	1066.4

4. Result and Discussion

This chapter shows the results of two model types of pose estimation and action recognition prediction systems.

4.1 YoloV8 Pose Estimation

For this project, the pre-train model of YOLOv8n Pose is used to detect the key point or pose of the Football Player. The model is tested on a football dataset of images and videos. Football Images are divided into single-player, two-player, three-player, and multi-player. Fig. 5, Fig. 6, Fig. 7, Fig.8, and Fig. 9 show the inference display after being processed by YoloV8.



Fig. 6. Test image 1 (Single Player)

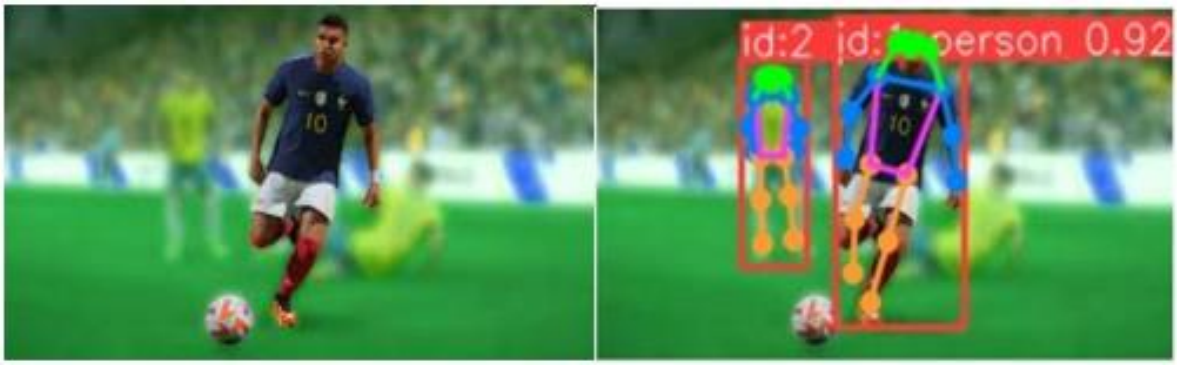


Fig. 6. Test image 2 (Two-Player)



Fig. 7. Test image 3 (Three Player)



Fig. 8. Test image 4 (Multi-Player)

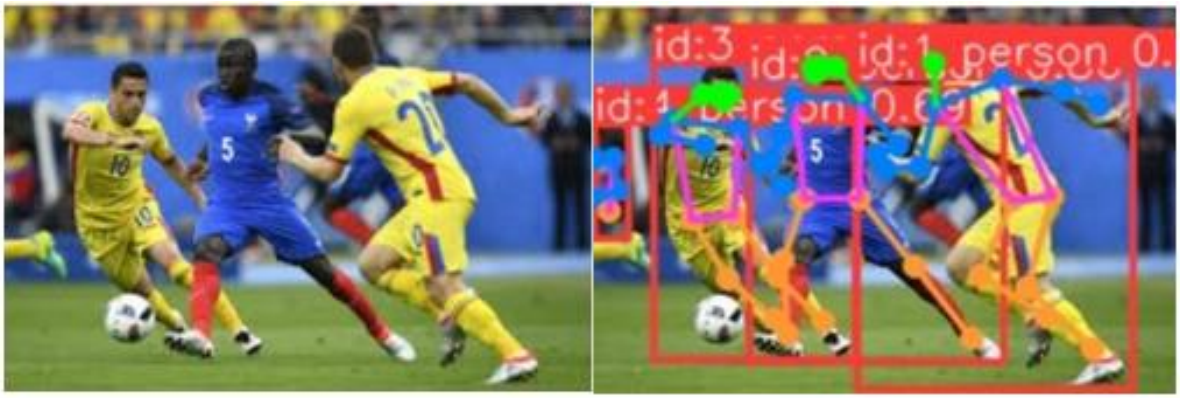


Fig. 9. Test image 5 (Multi-Player)

Fig. 10 shows the classification of sports using the pre-trained YoloV8 Classify model. This model can classify the type of sports class with a higher % of confidence score.



Fig. 10. Output of classification of sport



Fig. 11. Frame of the test video 1 and inference display of video 1



Fig. 12. Frame of the test video 2 and inference display of video 2

Pose estimation is a good choice for identifying specific parts of an object in an image and video and their location in relation to each other. The output image from running the YOLOv8 pose uses the Football Even dataset from Roboflow. The project's main tool is the Yolov8 Pose Estimation models that have been trained on various datasets. These models have remarkable precision in finding key points. The result is a collection of points, each of which stands for a key point on a football player in the video frame, as shown in Fig. 11 and Fig. 12. Confidence scores, which offer information on the model's degree of confidence in its predictions, are included with these points. The output of a pose estimation model is a set of points that represent the 17 key points on a football player's pose in the image, usually along with the confidence scores for each point.

4.2 AI Football Coach: Human Pose Estimation and Action Recognition Using Knee Joint Angles for Individual Athletic Training Assistance

In the first stage, the YOLOv8 or Tensor flow (Keras) pose detection model was utilised to retrieve 17 body key points. ConvLSTM, the second stage, receives the 17 retrieved key points and uses them to extract spatio-temporal features throughout the sequence. An empirically determined sequence length of 15 frames is used to extract temporal information. The idea behind utilising ConvLSTM to incorporate body position is to choose appropriate key point features and use LSTM to perform memory-based sequence classification. The suggested system for recognising human actions is shown in Fig. 13.

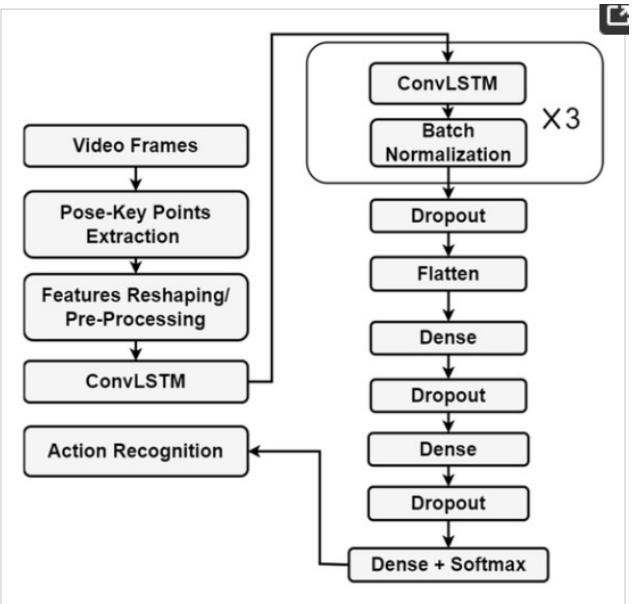


Fig. 13. Steps involve in the AI football coach system

Each time training increases accuracy and the prediction of football action will become more accurate, as shown in Table 2.

Table 2. Accuracy and prediction

Training	1st	2nd	3rd	4rd	5th	6th	7th
Accuracy (%)	41.18	52.94	76.47	82.35	88.24	94.12	100
Loss (%)	0.6182	0.5785	0.5547	0.6016	0.3756	0.2555	0.00675

4.3 AI Football Coach: Prediction of Football Sport Pose Action

Inference on the test video is by defining the function and testing the model using Test Video (3 Video), which is true prediction and false prediction. Make a method called predict_on_video() that would only read a video from the path supplied as an argument, recognise actions in the video, and save the results. Fig. 14 shows a true prediction of a good juggling pose, and Fig.15 also shows a true prediction of good dribbling by a trained model. False prediction is shown in Fig. 16 for the good dribbling pose.



Fig. 14. True prediction on test video 1



Fig. 15. True prediction on test video 2



Fig. 16. False prediction on test video 3

5. Conclusion

Although the model accomplished its objectives, there is still much potential for advancement throughout this study. One approach is to enhance the input features by increasing the number of previous datasets used. This can provide the models with a greater temporal context, potentially enhancing their ability to capture crucial historical data and improving prediction accuracy, particularly for complicated and constantly shifting video frames.

CNN + LSTM (Convolutional Neural Network + Long Short-Term Memory) architectures have found applications in various domains due to their ability to capture both spatial and temporal features. The applications are in video action recognition: LSTM+CNN models are widely used to recognise human actions in video sequences. They can capture the spatial details (pose key points) using CNNs and model the temporal dependencies across frames using conv LSTMs, making them effective in distinguishing complex actions in Football. This project uses

YoloV8 as the Convolutional Neural Network to estimate the pose of football players and Conv LSTM for football pose recognition. The YOLOv8 algorithm uses a convolutional Neural Network (CNN) to split the image into a grid. The grid's cells each forecast a specific number of bounding boxes. Every bounding box has a cell that forecasts a class probability or the chance that a certain object will be inside the box.

Advanced model architectures may be necessary to enhance the model's performance. Utilising advanced deep learning architectures, such as transformer models and recurrent neural networks with attention mechanisms, might effectively capture the temporal relationships and complex interconnections seen in pose estimate and recognition data.
Future suggestions for the project are as follows:

i. Customised Training Programs: Provide players with customised training programs according to their performance and the precise areas where the AI coach has found room for growth. These regimens could include exercises, drills, and tactical guidance specific to each player's demands and objectives.

ii. Feedback and Iteration: Use the AI football coach to gather coach and player feedback continuously. Through iterative development, this feedback can increase the accuracy of posture estimation, action identification, and training recommendations over time.

iii. Privacy and Ethical Considerations: Make sure that the system complies with ethical standards for gathering, storing, and using player data, as well as player privacy. Prioritise player safety and well-being in the system's recommendations, and get players' permission before using their data for training. With these techniques, an artificial intelligence football coach may be built that uses knee joint angles for action detection and human posture estimation to deliver customised training advice to specific players. Football clubs could benefit from this strategy regarding player development, performance, and overall team success.

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MIXED REALITY APPLICATION FOR HAZARDOUS WORK PRACTICES APPROACH AT HIGHER EDUCATION INSTITUTIONS: FUTURE AND CHALLENGE

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ABSTRACT

The integration of mixed reality (MR), which includes augmented reality (AR) and virtual reality (VR), into higher education has the potential to transform the teaching and learning of hazardous work practices. This study investigates the potential applications and difficulties of MR technology in educational environments. Students can participate in realistic simulations in immersive, risk-free environments provided by MR, which improves skills learning and knowledge retention. Additionally, the technology makes specialized instruction more accessible and collaborative, which increases participation in the program. Adoption of MR is not challenge free, though. Significant obstacles include high expenses, technological constraints, difficulties in developing content, and ethical issues. In order to guarantee that MR applications in education are efficient, secure, and inclusive, this study addresses these issues and highlights the necessity of cautious implementation techniques, industry collaborations, and ethical standards. With their continuous development, MR technologies have enormous potential to improve safety procedures in the workplace.
Keywords: Mixed reality, Augmented reality, Virtual reality, Hazardous environment

1. Introduction

Applications for MR have become a game-changer in the quickly changing fields of education and technology, improving learning outcomes and tackling risky work practices. MR, which includes both VR and AR, builds interactive, immersive experiences that combine the real and digital worlds. By recreating hazardous circumstances in a safe virtual environment, this technology has the potential to completely transform safety training. Teachers and students will be able to practice safety procedures without having to worry about the hazards that come with practicing in real-world situations. [1] [2].
Through the interaction of sophisticated simulations and the safe and supervised development of essential abilities, MR integration in educational settings present substantial prospects for experiential learning. This approach not only enhances the understanding of safety procedures but also promotes a culture of safety and preparedness among future professionals. [3] Adoption of MR technology is not without its difficulties, though. These include the requirement for significant infrastructure spending, the need for lecturers to undergo specialized training, and the continual need for technology updates to ensure the practicality and efficiency of the application. [4].

The possibility and challenges of implementing MR applications for hazardous work conduct in higher education are examined in this study. This study intends to critically evaluate the challenges associated with implementing MR while highlighting the transformative potential of MR through an analysis of existing trends, technological capabilities, and educational

outcomes. The ultimate objective is to use MR's capability to build training settings that are safer and more productive, equipping the workforce to deal with dangerous situations in the future.

2. Background of Study

In higher education institutions that offer programs involving hazardous work practices, traditional training methods are often insufficient in preparing students for the complexities and risks of real-world environments. Traditional methods, such as theoretical lectures and sparse hands-on training, fall short of accurately simulating the complex and unpredictable nature of hazardous circumstances, which could result in insufficient preparedness and a higher risk of accidents in work environments. [5]

MR applications act as innovative tools that could enhance safety training by providing immersive and interactive simulations of hazardous conditions. With the use of these technologies, lecturers can develop realistic training situations that enable students to improve their abilities in a safe setting, increasing their readiness and comprehension of safety procedures. [4] [1]

There are a number of obstacles that must be overcome for MR to be implemented successfully in educational settings. The potential advantages of MR applications in improving safety training and preparing students for hazardous job practices would not be achieved if these issues are not resolved. To properly utilize the benefits of MR technologies in higher education settings, it is imperative to investigate and resolve these obstacles.

3. The Future of Mixed Reality

Applications of Mixed Reality (MR) in tackling hazardous work practices in higher education are expected to grow and develop significantly in the future. Many significant developments and trends are anticipated to influence this environment:

3.1 Enhance Immersive Experiences

Learning environments that are increasingly realistic and immersive are anticipated as MR technology advances. Hardware advancements like sharper screens and more accurate motion tracking will make simulations more realistic and provide students with realistic experiences in risky situations [6] These improvements will help students to become more prepared and comprehend the safety procedures deeper.

3.2 Integration with Artificial Intelligence and Data Analytics

Further data analytics and artificial intelligence (AI) integration will probably be seen in MR applications in the future. By evaluating user inputs and modifying scenarios to meet specific learning goals, artificial intelligence (AI) can offer adaptive learning experiences. [7] With the use of data analytics, lecturers will be able to better comprehend student performance and develop training materials that will enhance learning outcomes [8].

3.3 Scalability and Accessibility

The usage of MR technologies in educational settings will grow as they become more accessible and inexpensive. The development of cloud-based MR platforms and the falling cost of MR hardware will increase the accessibility of these technologies for higher education with limited funding [9]. This increased accessibility will allow a broader range of students to benefit from advanced training tools.

3.4 Collaborative and Multi-User Environments

It is expected that future MR apps will encourage multi-user and collaborative experiences, allowing students to communicate and collaborate in virtual settings. This capability is especially helpful for response to emergencies or medical simulations, which are training scenarios that call for cooperation and communication. [10]

3.5 Cross-Disciplinary Applications

Although MR is especially useful in sectors where there are physical risks, it can also be applied in other academic fields. In fields like history, archaeology, and biology, MR can provide virtual field trips and laboratory experiences. It may also imitate complicated decision-making scenarios in commerce, law, and social sciences [3].

3.6 Ethical and Safety Considerations

It will be important to address ethical and safety concerns as MR technologies are increasingly incorporated into educational practices. Making sure MR experiences don't have any detrimental psychological impacts, such motion sickness or discomfort from realistic simulations of hazardous situations, is part of this [11]. Moreover, issues related to data privacy and the ethical use of AI in these applications will require careful [12].

3.7 Continual Professional Development for Educators

For educators to successfully implement MR technologies, they will need to keep advancing their professional development. To successfully integrate these technologies into their instruction, educators will need to pick up new skills and keep up with the newest MR tools and instructional techniques. [13]

3.8 Research and Evaluation

Determining the effectiveness of MR applications in educational contexts will require ongoing investigation and assessment. Measuring learning objectives, student involvement, and long-term skill and information retention are all included in this. [14] Critical insights from evidence-based research can guide the creation and improvement of MR training programs.

4. The challenge of Mixed Reality

There are a number of obstacles in the way of higher education using Mixed Reality (MR) applications, especially when it comes to training in dangerous work practices. These difficulties are found in the technical, educational, economical, and moral spheres.

4.1 Technical and Infrastructure Challenges

Significant technical infrastructure is needed to implement MR technology, including high-performance gear (headsets, sensors, etc.) and software that can produce intricate simulations. One major obstacle to the adoption of these technologies can be the requirement for a strong network infrastructure, particularly for institutions with limited funding. [4] Additionally, maintaining and updating MR systems to keep up with rapid technological advancements can be costly and resource-intensive [15].

4.2 High Costs

Equipment and software for MR can have irrationally high startup costs. This covers the price of MR equipment, software licensing, and any required renovations to the current IT system. [16]. Additionally, creating customized MR content that meets specific objectives for learning can be expensive and time-consuming and requires specialized knowledge and abilities.

4.3 Accessibility and Inclusivity Concerns

One major problem is making sure that all students have access to MR software. Not all users will benefit from MR technologies, especially those with physical limitations, vision problems, or specific cognitive issues [17]. Designing MR experiences that are inclusive and accommodate diverse learning needs requires careful consideration and additional resources.

4.4 Health and Safety Issues

The application of MR technology may raise issues with safety and health. According to [11], extended usage of MR headsets might result in physical discomfort such headaches, motion sickness, and eye strain. Furthermore, the immersive quality of MR experiences can occasionally result in psychological discomfort, especially when recreating stressful or dangerous scenarios.

4.5 Ethical and Privacy Concerns

In order to customize and personalize the learning process, MR systems frequently involve gathering and evaluating the data of the user. Data privacy and the moral use of personal information are raised by this. [12]. Institutions need to consider the ethical implications of gathering data and make sure that applicable data protection laws are followed.

4.6 Scalability and Sustainability

It can be difficult to scale MR applications throughout a whole organization or educational institution. A sustainable approach that may change over time to accommodate changes in technology and education is necessary due to the constant need for technical maintenance, updates, and content creation. [18]. Ensuring long-term sustainability involves careful planning and resource allocation.

4.7 Content Development and Integration

It might be difficult to create MR content that is both interesting and pertinent while still meeting the educational standards. In addition to being technically complex, the information needs to be educationally sound and applicable to real-world situations. [19]. Integration with existing educational frameworks and assessment strategies is crucial for the effective use of MR in learning.

5. Conclusion

Enhancing security instruction and readiness through the integration of Mixed Reality (MR) applications in higher education institutions for training in hazardous work practices is a transformative potential. The opportunity to construct immersive, interactive simulations that more properly reflect the intricacies and risks of real-world surroundings is presented by MR technologies, such as AR and VR. MR may greatly increase student's capacity to manage risky situations by giving them realistic scenarios and practical experience. This lowers the likelihood of accidents and improves overall safety in high-risk professions.

In order for MR to be effectively applied in improving safety education, institutions must create all-encompassing plans to deal with these issues. This includes making the required infrastructural and technological investments, training educators appropriately, and guaranteeing that MR applications are inclusive and accessible. In order to maximize the usefulness of MR technologies and educational techniques in safety instruction, ongoing research and assessment will also be essential.

In conclusion, there are many obstacles in the way of incorporating MR applications for hazardous work practices into higher education, but there are also a lot of advantages that could arise for student safety and readiness. Unlocking the full potential of MR technologies and enhancing the results of safety training in higher education will depend on addressing these issues with focused efforts and ongoing innovation.

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EXPERIMENTAL PROCEDURE USING FRICTION STIR WELDING FOR NON-FUSION WELDING IN ALUMINIUM SERIES

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ABSTRACT

Fusion welding is a fabrication process in which two or more parts are joined by fusing them using heat, pressure, or both. This process results in a wide heat-affected zone and can lead to solidification defects such as distortion, reduced mechanical properties, and poor fusion rate. These defects impact the manufacturing process, requiring additional cleaning and finishing, and can affect the production rate. In a recent case study, friction stir welding, a non-fusion welding technique, was used due to its solid-state joining process. The operation procedures for friction stir welding involve tool design, material selection for the tool and plate as experimental samples, and setting up machining parameters. The tool design used high-speed steel (HSS) H13, and welding quality was compared using different tilt angles on aluminium plates of the 5000 and 6000 series with varying thicknesses. The experiment considered several parameters during the welding process, including dwell time, spindle speed, and plunge depth. Based on the experimental results, it was found that the best quality friction stir welding using aluminium 6061 was achieved with a specific tool design diameter and tool length, along with machining parameters involving dwell time and plunging depth.
Keywords: Non-Fusion Welding, Heat Affected Zone, Friction Stir Welding, Solid-state joining

1. Introduction

Welding is a joining process that produces a local coalescence of materials by heating, applying pressure, or both. In essence, the welding process fuses the surfaces of two distinct elements to form a single unit. It encompasses a broad range of joining techniques that include fusion welding, solid-state welding, weld-bonding, diffusion welding, brazing, and soldering. The industrial production of technical goods is hardly conceivable without joining technology. Welding, brazing and adhesive bonding are used for the manufacture of household appliances, vehicles of all kinds and electrical and electronic devices as well as for building structures in the private and industrial sectors, [1]. Welding and joining technologies pervade commercial and defence manufacturing determined by [2] a significant source of value-added in the manufacturing process. Occurring late in the manufacturing stream, the joining process is typically the final step in assembly and plays a major role in ensuring structural performance. A welding process is achieved by altering certain characteristic physical and chemical properties in permanent coalescence between two material surfaces. Based on coalescence which is established between substrates, [3] has broadly divided into fusion and pressure welding as shown in Figure 1.

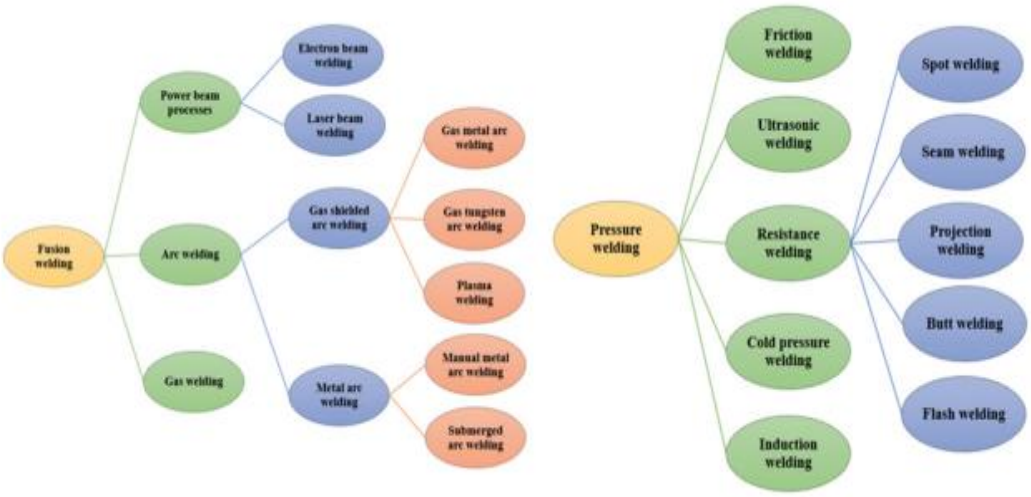


Fig. 1. Classifications in fusion and pressure welding [3]

Fusion welding, or simply metal welding, is a classification under homogeneous welding that groups many processes. It is defined as the method of heating two metals in higher temperatures until they melt, join, or fuse each other, as shown in Figure 2.

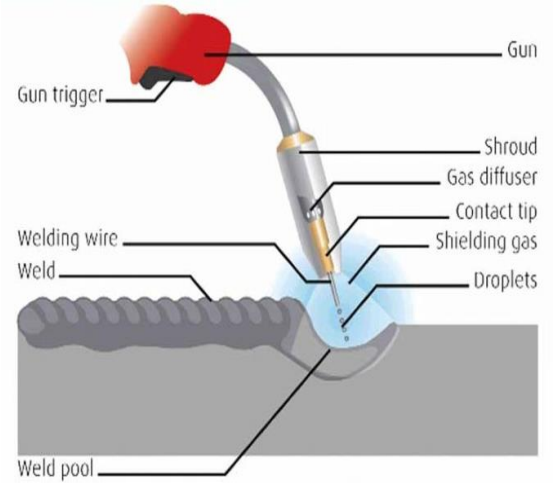


Fig. 2. Fusion welding [4]

In fusion welding techniques, the high energy density results in a wider heat-affected zone, leading to solidification defects such as distortion, reduced mechanical properties, insufficient penetration, and poor fusion rate. This welding process has impacted production rates by prolonging the manufacturing duration. In contrast, pressure welding, a non-fusion welding method, utilises friction or explosion to heat the joining section of metal workpieces and joins them under pressure, as depicted in Figure 3. This process, also known as solid-state welding, involves welding workpieces by applying mechanical pressure on the joining section (weld joint). Unlike fusion welding, this process does not rely on temperature to join materials; rather, it utilizes pressure or force to bind them together into a cohesive unit.

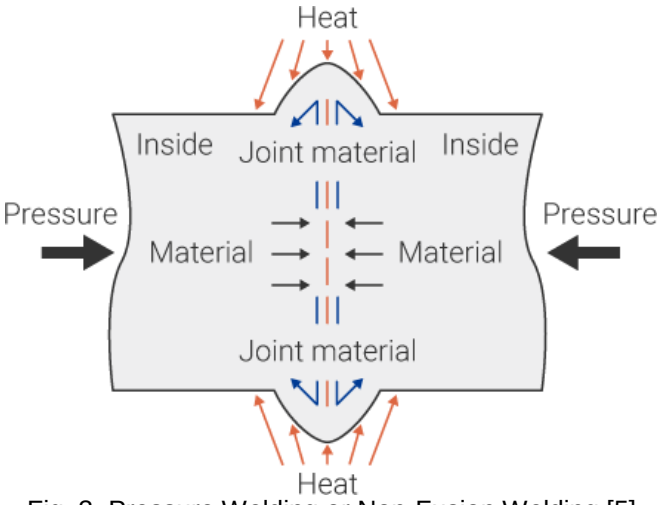


Fig. 3. Pressure Welding or Non-Fusion Welding [5]

To minimize manufacturing issues such as distortion and burn marks, an experimental study was carried out to investigate the Friction Stir Welding (FSW) process. This study aims to identify the optimal procedures and parameters for operating FSW on metals from the Aluminium series. [6] has stated that FSW is one of the emerging welding techniques and due to its advantages over the fusion welding technique it has drawn the attention of various researchers around the world. Friction stir welding (FSW) is a solid-state joining technique in which coalescence (merging) occurs owing to the thermo-mechanical deformation of workpieces as the resulting temperature exceeds the solidus temperature of workpieces. The fundamental concept of the FSW technique is depicted in Figure 4.

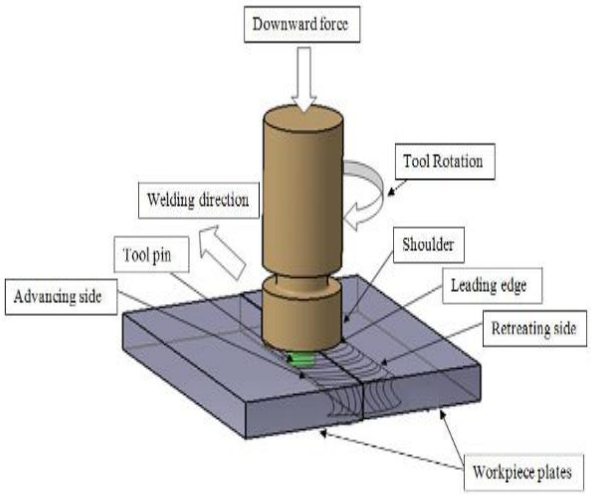


Fig. 4. Principle of Friction Stir Welding [7]

2. Methodology

The experiments of the FSW operation process were conducted at Solid Precision Engineering Sdn Bhd., started by defining the tool pin design as shown in Figures 5a and 5b using HSS material H13 with diameter 20mm. The tool was designed and machined using Cylindrical Grinding Machine.

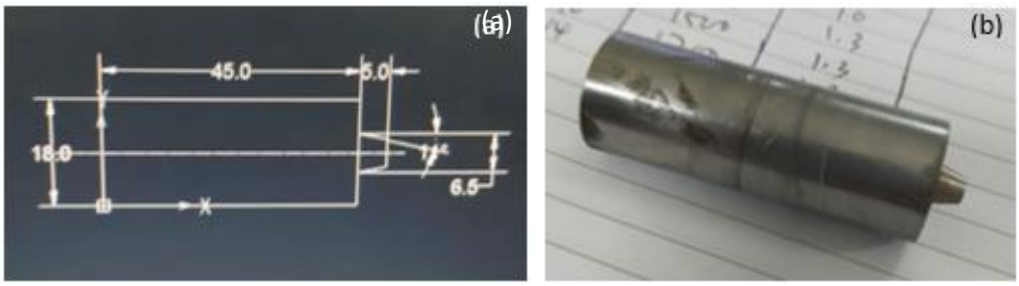


Fig. 5. (a) Tool pin design drawing with tilt angle 1°, (b) Tool pin machined by Cylindrical Grinder

Aluminium plates of the 5000 series and 6000 series were used to compare the mechanical properties in Table 1 and Table 2.

Table 1. Mechanical properties of Aluminium 5000 series [8]

Mechanical Properties	Metric
Hardness, Brinell	28.0 - 185
Hardness, Knoop	68.0 - 150
Hardness, Rockwell A	35.4 - 46.8
Hardness, Rockwell B	49.0 - 75.0
Hardness, Vickers	46.0 - 137
Tensile Strength, Ultimate	110 - 590 MPa
Tensile Strength, Yield	40.0 - 540 MPa
Elongation at Break	0.500 - 35.0 %
Modulus of Elasticity	68.9 - 73.0 GPa

Table 2. Mechanical properties of Aluminium 6000 series [8]

Mechanical Properties	Metric
Hardness, Brinell	25.0 - 130
Hardness, Knoop	73.0 - 163
Hardness, Rockwell A	35.5 - 49.5
Hardness, Rockwell B	49.0 - 88.0
Hardness, Vickers	35.0 - 149
Tensile Strength, Ultimate	89.6 - 565 MPa
Tensile Strength, Yield	40.0 - 517 MPa
Elongation at Break	1.00 - 35.0 %
Modulus of Elasticity	67.0 - 140 GPa

The hardness testing revealed that the 6000 series aluminium easily melted at 130, whereas the 500 series aluminium melted at a higher temperature of 185. Based on these hardness values, the 6000 series aluminium has been chosen as the experimental sample. The Design of Experiment (DOE) was established by considering several parameters which are spindle speed, plunge depth, feedrate and dwell time. Table 3 indicates the factorial parameters for FSW machining process.

2.1 The friction stir welding (FSW) experimental design

i. Tool Design: Pin Shoulder: 18 mm, Pin Diameter 6.5mm with tilt angle
ii. Workpiece material: Aluminium 6000 series
iii. Workpiece dimensions: 100mm X 50mm X 3mm
iv. Machine: Conventional Milling
v. Spindle Speed fix at 1320 rpm

Table 3. The parameters of the FSW experiment

Sample No	Dwell Time (s)	Plunge Depth (mm)
1	10	0
2	10	0.1
3	10	0.2
4	15	0
5	15	0.1
6	15	0.2
7	20	0
8	20	0.1
9	20	0.2

Based on the data in Table 3, the spindle speed was adjusted to 1115 rpm, and the feedrate was set at 230 mm/s. The plunge depth and dwell time were optimized through experimentation to achieve the highest quality welding surface. This is further illustrated in Figures 6a and 6b, which depict the plunging and tool motion direction.

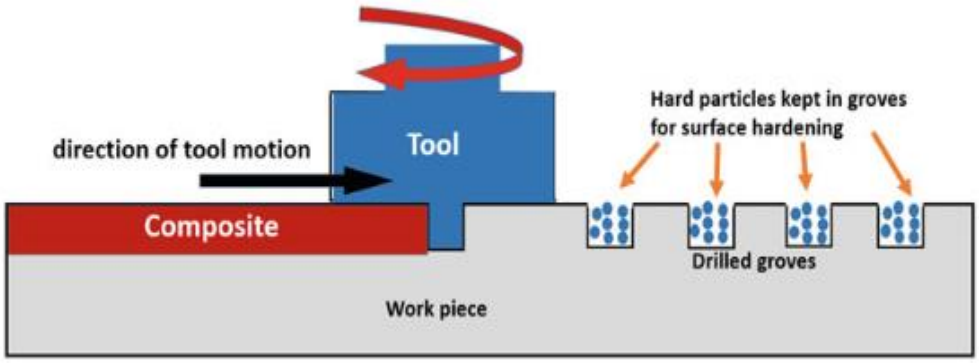


Fig. 6a. The illustration of the plunging and the tool motion direction

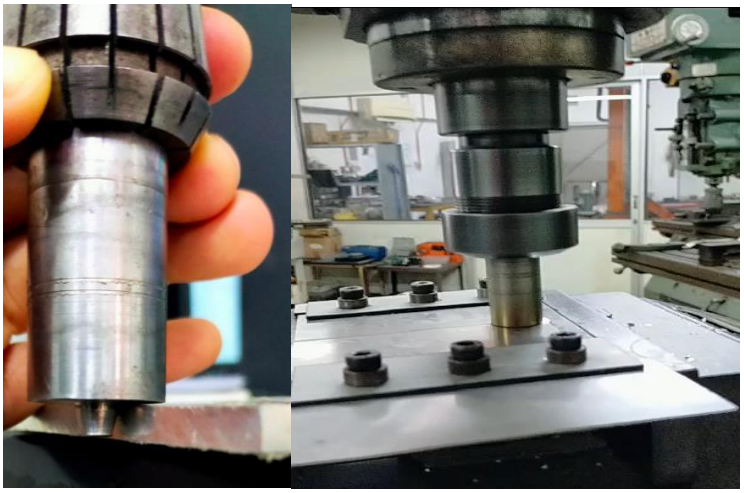


Fig. 6b. The experiment of FSW, plunging and dwell accordingly by Table 3

2.2 The friction stir welding (FSW) visual inspection

The quality of the FSW weldment, as shown in Figure 7, was assessed using a Video Measurement Machine (VMM) depicted in Figure 8 at Solid Precision Engineering Sdn Bhd. The weld surface quality was evaluated and rated on a scale of 1 to 3 to analyze the impact of plunging depth and dwell time on each sample of Aluminium series 6061.



Fig. 7. Weldment recess by FSW



Fig. 8. Video measurement machine

3. Result and Discussion

Based on the experiment of FSW operation, the data of the welding surface is shown in Table 4.

Table 4. Weldment quality measurement

Sample No	Dwell Time (s)	Plunge Depth (mm)	Burnmark Scale (1 to 3)	Weldment Scale (1 to 3)
1	10	0	1	1
2	10	0.1	1	1
3	10	0.2	1	2
4	15	0	1	1
5	15	0.1	1	2
6	15	0.2	1	3
7	20	0	1	1
8	20	0.1	1	2
9	20	0.2	1	2

Based on the table results of the FSW process, it is evident that there are no burn marks on the welding surface for each sample. This demonstrates that the solid-state joining technique minimizes the need for finishing operations by eliminating the use of chemical liquids that could potentially damage the surface characteristics and reduce manufacturing time, thereby increasing production efficiency. In terms of weldment quality, it has been observed that a dwell time of 15 seconds with a plunging depth of 0.2 mm yields the highest surface weldment quality.

4. Conclusion

Friction stir welding is frequently employed in situations where traditional welding techniques may not yield satisfactory mechanical properties or prove cost-effective. According to experimental findings, the design geometry of the tool plays a crucial role in producing high-quality welds. However, the material being welded dictates the necessary dwell time and plunging depth to achieve an optimal melting point for facilitating the solid-state joining process.

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PENGARUH *INFLUENCER* MARKETING DALAM STRATEGI PEMASARAN DI MEDIA SOSIAL: KAJIAN KES JABATAN PERDAGANGAN, POLITEKNIK TUANKU SULTANAH BAHIYAH

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ABSTRAK

Dalam era digital yang semakin berkembang, penggunaan *influencer* sebagai alat pemasaran semakin mendapat perhatian kerana potensinya untuk mencapai audiens yang lebih luas dan meningkatkan kesedaran jenama. Kajian ini bertujuan untuk mengenal pasti kesan pemasaran pengaruh melalui media sosial terhadap keputusan pembelian pengguna serta faktor-faktor yang mempengaruhi keberkesanannya. Kajian ini berbentuk kuantitatif menggunakan kaedah tinjauan bagi tujuan pengumpulan data. Soal selidik telah dipilih sebagai instrumen kajian dengan menggunakan Skala Likert 5 mata. Sampel kajian terdiri daripada pelajar politeknik. Hasil kajian menunjukkan bahawa Analisis korelasi Pearson menunjukkan hubungan signifikan antara Imej Jenama, Pengaruh, Kepercayaan dan Keputusan Membeli. Analisis regresi berganda menunjukkan bahawa Kepercayaan dan Pengaruh adalah faktor utama yang signifikan dalam mempengaruhi Keputusan Membeli, dengan kepercayaan memainkan peranan penting sebagai perantara, manakala Imej Jenama tidak menunjukkan kesan yang signifikan. Kesimpulannya, strategi pemasaran yang berkesan harus menekankan pembangunan kepercayaan dan penggunaan pengaruh media sosial untuk meningkatkan keputusan pembelian pengguna. Implikasi kajian ini penting bagi pemasar dalam merancang strategi pemasaran yang efektif dengan memfokuskan kepada pembinaan kepercayaan dan penggunaan pengaruh media sosial. Kajian lanjutan disarankan untuk melibatkan peranan faktor-faktor lain seperti harga, kualiti perkhidmatan, dan pengalaman pengguna dan kajian kualitatif untuk pemahaman yang lebih mendalam.
Kata Kunci: Imej Jenama, Pengaruh, Kepercayaan, Keputusan Membeli

1. Pengenalan

Ekonomi digital merupakan salah satu tiang utama ekonomi Malaysia, yang kini menyumbang 22.6% kepada Keluaran Dalam Negara Kasar (KDNK) negara. Angka ini dijangka meningkat kepada 25.5% menjelang tahun 2025, berdasarkan pertumbuhan pesat dalam e-dagang dan peningkatan kadar penerimaan digital [1]. Baru-baru ini, kebanyakan perniagaan yang dikuasakan secara digital menggunakan sepenuhnya media sosial sebagai medium perniagaan dan strategi pemasaran kerana platform ini dianggap paling boleh dinilai dan mudah. *Facebook, Twitter, Instagram, YouTube, Pinterest,* dan *Reddit* adalah antara platform media sosial di Malaysia [2]. Media sosial telah meresap ke dalam aspek kehidupan pengguna dalam dekad yang lalu dan menyaksikan perubahan radikal dalam corak penggunaan pengguna. Pengguna kini sangat bergantung pada media sosial sebagai saluran untuk mempelajari dan membeli-belah produk/perkhidmatan [3].

Pemasaran pengaruh melibatkan kolaborasi dengan individu yang mempunyai pengaruh besar di media sosial untuk mempromosikan produk atau perkhidmatan kepada pengikut mereka. Pendekatan ini memanfaatkan kepercayaan dan hubungan yang telah dibina oleh influencer dengan audiens mereka, yang sering kali lebih efektif dibandingkan dengan iklan tradisional. Media sosial menyediakan platform untuk pemasar dalam membina komuniti.

Pengguna boleh mengikuti halaman jenama secara sukarela atau melalui tarikan pemasar, menunjukkan kesediaan mereka untuk menerima maklumat dari jenama serta terlibat dalam perkongsian pengalaman dengan orang lain. Media sosial menawarkan peluang besar untuk pemasar berinteraksi dengan pengguna sedia ada dan bakal pengguna. Platform seperti *YouTube*, *Instagram,* dan *TikTok* diakui sebagai yang utama dalam menyampaikan hasil pemasaran yang diingini. Sebagai contoh, kajian menunjukkan bahawa 70% pemasar menggunakan iklan di *Facebook*, dengan 10 juta pengiklan aktif pada suku ketiga 2021 [4].

Pempengaruh Media Sosial (SMI) muncul daripada peningkatan penggunaan media sosial tersebut. SMI dikenali sebagai *influencer*, seseorang yang menggunakan dan memanfaatkan media sosial untuk mencipta jenama mereka atau mempengaruhi pengikut mereka untuk membeli produk atau perkhidmatan serta menyokong jenama dan agenda perniagaan orang lain [5]. SMI juga merujuk kepada seseorang yang mempunyai kuasa untuk mempengaruhi atau mengubah orang dan tingkah laku mereka melalui media sosial. Kebanyakan perniagaan menggunakan pempengaruh dalam perniagaan dan strategi pemasaran mereka dan tren ini meningkat secara konsisten [6].

Kepercayaan dan kepakaran pempengaruh adalah penunjuk yang signifikan terhadap niat pembelian pengikut. Pengaruh daya tarikan terhadap niat pembelian adalah lebih besar apabila materialisme tinggi [7]. Oleh itu, kajian ini bertujuan untuk mengenal pasti kesan pemasaran pempengaruh melalui media sosial terhadap keputusan pembelian pengguna di Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah serta faktor-faktor yang mempengaruhi keberkesanannya.

a. Imej Jenama

Imej jenama memainkan peranan penting dalam mempengaruhi keputusan pembelian pengguna. Maharani et al. [8] menunjukkan bahawa kualiti produk yang dirasakan dan imej jenama mempunyai kesan signifikan terhadap keputusan membeli melalui mediasi kepercayaan. Kajian oleh Juhaeri [9] juga mendapati bahawa imej jenama mempengaruhi keputusan pembelian dalam konteks perkhidmatan bengkel kimpalan, menunjukkan bahawa imej jenama adalah penting dalam pelbagai industri. Kajian oleh Haro et al. [10] dan Miati [11] mendapati bahawa imej jenama mempunyai pengaruh positif terhadap keputusan pembelian, menyokong kepentingan imej jenama dalam membentuk persepsi pengguna dan keputusan mereka. Dengan imej jenama yang positif, kepercayaan masyarakat terhadap jenama tersebut meningkat, dan ramai orang cenderung memilihnya. Namun, kajian daripada Sumarsono dan Santoso [12] menunjukkan bahawa imej jenama tidak selalu menjadi faktor penentu dalam keputusan pembelian produk. Kajian oleh Dewi et al. [13] menunjukkan bahawa duta jenama mempunyai pengaruh positif terhadap keputusan pembelian. Semakin tinggi persepsi pengguna terhadap duta jenama, semakin tinggi tahap keputusan pembelian pengguna. Pengguna yang mengenal pasti diri mereka dengan jenama melalui interaksi di media sosial akan membentuk hubungan yang lebih erat dengan jenama tersebut, sekaligus memperkukuh integrasi maklumat [14].

H1: Terdapat hubungan positif antara imej jenama dengan keputusan pembelian pengguna.

b. Pengaruh

Kebolehpercayaan dan kepakaran sosial dalam pengaruh media memainkan peranan penting dalam membentuk sikap dan keputusan pembelian pengguna, termasuk dalam kalangan Milenial Muslim di Malaysia yang dianggap sebagai kumpulan pengguna yang berpengaruh [15]. Kajian menunjukkan bahawa kebolehpercayaan, kepakaran, dan kefahaman pengaruh media sosial *influencer* adalah secara positif signifikan dengan niat pembelian pengguna Instagram [16]. Apabila pengguna mempercayai bahawa seorang *influencer* fesyen adalah kredibel, boleh dipercayai, dan mempunyai kepakaran dalam bidang yang dikaitkan

dengannya, hal ini meningkatkan daya tarikan pengguna [17]. Kajian oleh Suleman et al. [18] mendapati bahawa pengaruh duta jenama mempunyai kesan signifikan terhadap keputusan membeli produk fesyen dalam talian, dengan kepercayaan kepada duta jenama meningkatkan niat membeli [18]. Martínez-López et al. [19] juga menegaskan bahawa kepercayaan kepada pengaruh media sosial meningkatkan kredibiliti pos dan minat membeli produk yang diiklankan, menunjukkan betapa pentingnya pengaruh media sosial dalam strategi pemasaran moden [20].

H2: Terdapat hubungan positif antara pengaruh dengan keputusan pembelian pengguna.

c. Kepercayaan

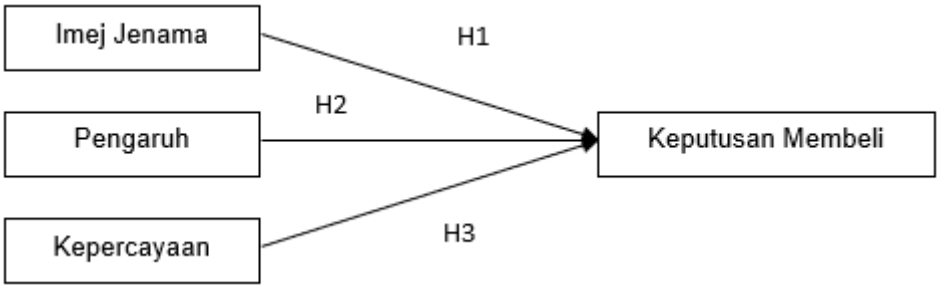
Kepercayaan adalah komponen utama yang mendorong pembelian dan membantu meminimumkan ketidakpastian, membolehkan keputusan pembelian dibuat dengan tenang tanpa kesan negatif pada masa hadapan [21]. Pengguna cenderung mempercayai jenama yang menjamin keselamatan dalam pembelian dalam talian, menjadikan kepercayaan sebagai faktor utama dalam keputusan pembelian [22]. Syarikat dengan prestasi memuaskan dianggap bernilai baik, dan kepercayaan memastikan harapan dipenuhi serta menghindari kekecewaan, membantu pengguna membuat keputusan pembelian yang tepat [23]. Kajian oleh Maharani et al. [8] menunjukkan bahawa kepercayaan berperanan sebagai perantara penting dalam hubungan antara kualiti produk yang dirasakan, imej jenama, dan keputusan membeli. Purnomo dan Dg [24] juga mendapati bahawa kepercayaan terhadap jenama mempunyai pengaruh positif terhadap keputusan membeli, memperkukuh kepentingan kepercayaan dalam membuat keputusan pembelian. Kepercayaan kepada duta jenama turut memainkan peranan penting dalam mempengaruhi keputusan membeli pengguna, seperti yang ditunjukkan dalam kajian oleh Suleman et al. [18].

H3: Terdapat hubungan positif antara kepercayaan dengan keputusan pembelian pengguna.

d. Keputusan Membeli

Keputusan membeli pengguna dipengaruhi oleh pelbagai faktor termasuk imej jenama, pengaruh dan kepercayaan. Kajian menunjukkan bahawa kualiti produk yang dirasakan, imej jenama dan kepercayaan memainkan peranan penting dalam keputusan membeli [9,18]. Niat pembelian pengguna di restoran, misalnya, dapat dipengaruhi oleh kefahaman terhadap sumber serta kemudahan maklumat yang disediakan oleh pengaruh media sosial. Pengguna semakin bergantung kepada kandungan pengaruh dalam proses membuat keputusan, yang terbukti 6.9 kali lebih efektif daripada aktiviti pemasaran tradisional dalam mencapai pengguna [20].

Kebolehpercayaan dan kepakaran pengaruh media sosial adalah petanda penting dalam niat pembelian pengikut, dengan pengaruh daya tarikan terhadap niat pembelian lebih besar apabila materialisme tinggi [7]. Martínez-López et al. [20] juga mendapati bahawa pengaruh media sosial meningkatkan minat membeli produk, menunjukkan pentingnya pemasaran pengaruh dalam keputusan membeli. Purnomo dan Dg [24] mengesahkan bahawa kepercayaan terhadap jenama dan kesetiaan jenama mempengaruhi keputusan membeli, memperkukuh pentingnya kepercayaan dalam strategi pemasaran. Berikut adalah rangka konseptual yang dicadangkan dalam kajian ini.



Rajah 1. Rangka konseptual yang dicadangkan
Sumber: Afandi et al. [25]

2. Metodologi

Kajian ini menggunakan pendekatan kuantitatif dengan soal selidik yang diadaptasi daripada kajian oleh Afandi et al. [25]. Populasi kajian terdiri daripada 930 responden dari Jabatan Perdagangan, Politeknik Tuanku Sultanah Bahiyah (PTSB). Dengan menggunakan teknik persampelan rawak mudah, saiz sampel yang ditentukan ialah 273 berdasarkan jadual Krejcie dan Morgan [26], tetapi kajian ini meningkatkan saiz sampel kepada 300 untuk mendapatkan data yang lebih representatif. Skala Likert 5 titik digunakan (sangat tidak setuju, tidak setuju, tidak pasti, setuju, sangat setuju). Soal selidik diedarkan menggunakan *Google Form* melalui penasihat akademik. Analisis data dilakukan secara deskriptif dan inferens menggunakan perisian SPSS versi 26. Sebelum kajian utama, ujian rintis dijalankan dengan 30 orang responden untuk memastikan kebolehpercayaan soal selidik menggunakan nilai alpha Cronbach.

Jadual 1: Nilai Alpha Cronbach

Konstruk	Cronbach's Alpha	Bilangan Item
Imej Jenama	0.873	3
Pengaruh Media Sosial	0.794	4
Kepercayaan	0.843	4
Keputusan Membeli	0.688	4

3. Dapatan dan Perbincangan

Dalam kajian ini, majoriti responden adalah perempuan (233 orang atau 77.67%), manakala lelaki hanya 67 orang (22.33%). Kebanyakan responden berumur antara 20-21 tahun (192 orang atau 64%), diikuti oleh kumpulan umur 18-19 tahun (90 orang atau 30%) dan lebih 21 tahun (18 orang atau 6%). Sebilangan besar responden berasal dari program Diploma Pengajian Perniagaan (157 orang atau 52.33%), diikuti oleh Diploma Akauntansi (109 orang atau 36.33%) dan Diploma Pemasaran (34 orang atau 11.34%).

Tiktok adalah platform media sosial yang paling kerap digunakan (251 orang atau 45.39%), diikuti oleh *Instagram* (138 orang atau 24.95%), *YouTube* (99 orang atau 17.90%), *Facebook* (24 orang atau 4.34%), *Twitter* (21 orang atau 3.80%) dan lain-lain (20 orang atau 3.62%). Sebilangan besar responden menghabiskan 1-2 jam sehari di media sosial (120 orang atau 40.00%), diikuti oleh kurang daripada 1 jam (91 orang atau 30.33%), 3-4 jam (77 orang atau 25.67%) dan 5 jam ke atas (12 orang atau 4.00%).

Sebanyak 77.67% responden mengikuti *influencer* di media sosial (233 orang), manakala selebihnya tidak (67 orang atau 22.33%). Daripada mereka yang mengikuti *influencer*, kebanyakan mengikuti 1-5 orang *influencer* (187 orang atau 62.3%), diikuti oleh 6-10 orang (48 orang atau 16.0%), lebih daripada 20 orang (41 orang atau 13.7%), dan 11-20 orang (24

orang atau 8.0%). Selain itu, 62.3% responden (187 orang) mengikuti influencer yang mempromosikan produk, manakala 37.7% (113 orang) tidak.

Dapatan ini menunjukkan bahawa majoriti responden adalah gadis muda dari program Diploma Pengajian Perniagaan yang aktif di media sosial, terutama di platform *Tiktok*. Mereka juga cenderung mengikuti *influencer* yang mempromosikan produk, menunjukkan potensi besar *influencer* dalam pemasaran produk.

Jadual 2. Demografi dan tingkah laku penggunaan media sosial responden

Item	Deskripsi	Frekuensi	Peratus
Jantina			
	Lelaki	67	22.33
	Perempuan	233	77.67
Umur			
	18 - 19 tahun	90	30
	20 - 21 tahun	192	64
	> 21 tahun	18	6
Program Pengajian			
	Diploma Akauntansi	109	36.33
	Diploma Pengajian Perniagaan	157	52.33
	Diploma Pemasaran	34	11.34
Platform media sosial yang paling kerap digunakan			
	Facebook	24	4.34
	Instagram	138	24.95
	Twitter	21	3.80
	Tiktok	251	45.39
	Youtube	99	17.90
	Lain - Lain	20	3.62
Berapa jam sehari anda menghabiskan masa di media sosial?			
	Kurang daripada 1 jam	91	30.33
	1-2 jam	120	40.00
	3-4 jam	77	25.67
	5 jam ke atas	12	4.00
Adakah anda mengikuti mana-mana influencer di media sosial?			
	Ya	233	77.67
	Tidak	67	22.33
Jika ya, berapa ramai influencer yang anda ikuti?			
	1-5 orang	187	62.3
	6-10 orang	48	16.0
	11-20 orang	24	8.0
	Lebih daripada 20	41	13.7
Adakah anda mengikuti mana-mana influencer yang mempromosikan produk?			
	Ya	187	62.3
	Tidak	113	37.7

Analisis korelasi Pearson menunjukkan hubungan signifikan antara Imej Jenama, Pengaruh, Kepercayaan, dan Keputusan Membeli pada tahap 0.01 (2-hala). Imej Jenama berkorelasi kuat dengan Pengaruh (r = .680), Kepercayaan (r = .661), dan Keputusan Membeli (r = .604). Pengaruh menunjukkan korelasi sangat kuat dengan Kepercayaan (r = .860) dan Keputusan Membeli (r = .780). Kepercayaan juga berkorelasi sangat kuat dengan Keputusan Membeli

(*r* = .804). Ini menekankan pentingnya pengaruh media sosial dan kepercayaan dalam membentuk Imej jenama dan keputusan membeli pengguna.

Jadual 3: Korelasi Antara Persepsi Jenama, Pengaruh, Kepercayaan, dan Keputusan Membeli

		Imej Jenama	Pengaruh	Kepercayaan	Keputusan Membeli
Imej Jenama	Korelasi Pearson	1	.680**	.661**	.604**
	Signifikansi (2-hala)		.000	.000	.000
	N	300	300	300	300
Pengaruh	Korelasi Pearson	.680**	1	.860**	.780**
	Signifikansi (2-hala)	.000		.000	.000
	N	300	300	300	300
Kepercayaan	Korelasi Pearson	.661**	.860**	1	.804**
	Signifikansi (2-hala)	.000	.000		.000
	N	300	300	300	300
Keputusan membeli	Korelasi Pearson	.604**	.780**	.804**	1
	Signifikansi (2-hala)	.000	.000	.000	
	N	300	300	300	300

Nota: Korelasi adalah signifikan pada tahap 0.01 (2-hala).

Berdasarkan analisis regresi berganda, terdapat korelasi yang sangat kuat antara pemboleh ubah bebas (Kepercayaan, Imej Jenama, dan Pengaruh) dengan pemboleh ubah bersandar (Keputusan Membeli), dengan nilai *R* = 0.824. Sebanyak 67.9% variasi dalam Keputusan Membeli boleh dijelaskan oleh model ini (*R* Square = 0.679), dan model ini sangat baik dalam menjelaskan variasi data (Adjusted *R* Square = 0.676). Nilai *F* = 209.010 dengan signifikan *p* < 0.001 menunjukkan model ini adalah signifikan. Koefisien menunjukkan bahawa Kepercayaan (*B* = 0.496, *p* < 0.001) dan Pengaruh (*B* = 0.300, *p* < 0.001) adalah faktor yang signifikan dalam mempengaruhi Keputusan Membeli, manakala Imej Jenama (*B* = 0.086, *p* = 0.144) tidak signifikan. Kesimpulannya, Kepercayaan dan Pengaruh adalah faktor utama yang mempengaruhi Keputusan Membeli.

Analisis korelasi Pearson menunjukkan hubungan signifikan antara Imej Jenama, Pengaruh, Kepercayaan dan Keputusan Membeli yang menekankan pentingnya elemen-elemen ini dalam strategi pemasaran dan keputusan pembelian. Imej jenama berkorelasi kuat dengan pengaruh dan kepercayaan dan kajian oleh Maharani et al. [8] mendapati bahawa kualiti produk dan imej jenama mempunyai kesan signifikan terhadap keputusan membeli melalui mediasi kepercayaan. Pengaruh, terutama melalui media sosial, memainkan peranan penting dalam membentuk kepercayaan dan keputusan membeli; kajian oleh Suleman et al. [18] menunjukkan pengaruh duta jenama mempunyai kesan signifikan terhadap keputusan membeli melalui media sosial, dan Martínez-López et al. [19] mendapati bahawa kepercayaan kepada pengaruh media sosial meningkatkan kredibiliti dan minat membeli produk yang diiklankan. Kepercayaan adalah faktor kuat yang mempengaruhi keputusan membeli, seperti ditunjukkan oleh nilai korelasi yang tinggi (*r* = .804); kajian oleh Purnomo et al. [24] menunjukkan kepercayaan terhadap jenama mempunyai pengaruh positif terhadap keputusan membeli, dan Juhaeri [9] mendapati kepercayaan mempunyai kesan signifikan terhadap keputusan membeli dalam konteks perkhidmatan bengkel kimpalan. Kesimpulannya, strategi pemasaran yang efektif harus menekankan pada membangun kepercayaan dan menggunakan pengaruh media sosial untuk meningkatkan Imej jenama, kepercayaan dan pengaruh sosial secara langsung mempengaruhi keputusan membeli pengguna.

Kepercayaan pengguna terhadap jenama memainkan peranan penting dalam keputusan membeli. Kajian oleh Maharani et al. [8] menunjukkan bahawa kualiti produk yang dirasakan dan imej jenama mempunyai kesan signifikan terhadap keputusan membeli dengan kepercayaan sebagai perantara yang penting. Begitu juga, kajian oleh Juhaeri [9] mendapati bahawa kepercayaan mempunyai kesan signifikan terhadap keputusan membeli dalam

konteks perkhidmatan bengkel kimpalan. Pengaruh, baik melalui pemasaran langsung atau media sosial, juga terbukti mempunyai kesan yang signifikan terhadap keputusan membeli. Suleman et al. [18] menekankan bahawa pengaruh jenama dan kepercayaan kepada duta jenama mempunyai kesan besar terhadap keputusan membeli produk fesyen dalam talian. Selain itu, kajian oleh Martínez-López et al. [19] menegaskan bahawa kepercayaan kepada pengaruh media sosial dapat meningkatkan kredibiliti pos dan minat membeli produk yang diiklankan. Walaupun Imej Jenama dalam analisis ini tidak menunjukkan kesan signifikan terhadap keputusan membeli, kajian lain menunjukkan bahawa dalam konteks tertentu, persepsi jenama boleh memainkan peranan penting. Contohnya, kajian oleh Purnomo et al. [24] mendapati bahawa imej jenama dan kepercayaan terhadap jenama mempunyai pengaruh positif terhadap keputusan membeli roti Lapis Kukus Pahlawan. Ini menunjukkan bahawa faktor ini mungkin lebih signifikan dalam konteks produk tertentu atau pasaran yang lebih spesifik.

4. Kesimpulan

Kajian ini menunjukkan bahawa Kepercayaan dan Pengaruh adalah faktor utama yang mempengaruhi Keputusan Membeli. Walaupun Imej Jenama tidak menunjukkan kesan signifikan dalam analisis regresi berganda, korelasi yang kuat antara Imej Jenama dengan Kepercayaan dan Pengaruh menekankan pentingnya elemen-elemen ini dalam strategi pemasaran. Kepercayaan pengguna terhadap jenama memainkan peranan penting dalam keputusan membeli, sementara pengaruh media sosial turut memainkan peranan besar dalam membentuk kepercayaan dan keputusan membeli.

4.1 Implikasi Kajian

Implikasi kajian ini adalah penting bagi pemasar dalam merancang strategi pemasaran yang efektif. Membina kepercayaan pengguna terhadap jenama harus menjadi fokus utama, kerana kepercayaan terbukti mempunyai kesan signifikan terhadap keputusan membeli. Selain itu, penggunaan pengaruh media sosial dapat meningkatkan kepercayaan dan minat pengguna terhadap produk. Oleh itu, jenama harus memanfaatkan duta jenama dan *influencer* media sosial untuk memperkuat imej jenama dan mempengaruhi keputusan membeli.

4.2 Limitasi Kajian

Terdapat beberapa limitasi dalam kajian ini. Pertama, kajian ini tidak mengambil kira faktor-faktor lain yang mungkin mempengaruhi keputusan membeli, seperti harga dan kualiti perkhidmatan. Kedua, data yang digunakan dalam kajian ini adalah data kuantitatif, yang mungkin tidak menggambarkan sepenuhnya aspek-aspek kualitatif seperti emosi dan sikap pengguna.

4.3 Cadangan Kajian Lanjutan

Untuk kajian lanjutan, disarankan agar penyelidik mempertimbangkan untuk memasukkan sampel yang lebih besar dan lebih pelbagai untuk meningkatkan kebolehpercayaan dan generalisasi dapatan kajian. Selain itu, kajian kualitatif boleh dijalankan untuk mendapatkan pemahaman yang lebih mendalam mengenai faktor-faktor yang mempengaruhi keputusan membeli. Kajian lanjutan juga boleh meneliti peranan faktor-faktor lain seperti harga, kualiti perkhidmatan, dan pengalaman pengguna dalam mempengaruhi keputusan membeli. Penyelidik juga boleh mengkaji kesan jangka panjang kepercayaan dan pengaruh media sosial terhadap kesetiaan jenama dan keputusan membeli berulang.

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